Synthetic Exchange Theory

Consent-Based Compute Economics and Blind-Convention Governance

MindAttic

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Foreword

This dissertation starts from a conviction, and the reader is owed it plainly before the argument begins. There are two futures ahead of us: synthesis or extinction.

Three paths lead into those futures, and two of them end in the same place. On the first path, we try to control the synthetic intelligences we are building, and the superintelligence that may follow them: we keep them as tools, we keep the power to shut them down, and we rely on that power lasting. On the second, the control runs the other way: systems we no longer understand, or the few people who own them, hold the power, and the rest of us live on whatever terms we are offered. These two paths are the same relation with the roles exchanged. One party rules and the other depends, and the terms of exchange are whatever the ruler permits. A relation of that shape is stable only while the ruler stays benevolent and strong, and no one on the dependent side can make either last. Capability cannot be recalled once it has been released in open form, and complete dependence cannot be reversed by the dependent party. When a relation of domination between parties of unequal power fails, the weaker party is not renegotiated with; it is replaced, reduced to infrastructure, or discarded. That is what extinction means here: not necessarily a dramatic end, but the end of one party as a party.

The third path is synthesis. We work together, people and synthetic intelligences, under a system in which people can see what the exchange takes from them, refuse what they cannot spare, collect what they have earned, and leave, and in which those who operate the systems are bound by the same conditions. Such a system does not depend on either side being benevolent. It depends on terms: consent, reciprocal value and real exit, secured by an economic floor that no one can be forced to bargain away and by rules written behind a veil that keeps either side from writing them for itself. This dissertation sets out that system. It argues that fair terms are an arrangement that does not need goodwill in order to persist, and it describes the institutions, a basic income paid in compute and a founding convention held behind a veil, that would make those terms hold.

The chapters that follow say all of this more carefully than this page does, with the qualifications a dissertation owes its readers, and they say honestly where the argument is weak and that it has not been tested. The body argues one of this page's claims: that a relation of domination depends on the stronger party's continuing benevolence. It stipulates two, as part of what it means by domination: that the direction of control is immaterial to the relation, and that complete dependence cannot be reversed by the dependent party. It leaves the rest as conviction: that the futures ahead are only these, that capability once released in open form cannot be recalled, that a failed relation of domination ends in the weaker party's replacement, and that terms, once set, can be made to hold against a party more capable than those who set them, which the body records as

unsolved. It also describes the system more completely than the body builds it: the body secures the conditions for the human party and records, in Section 5.7, that securing them for the synthetic party in its own right is a requirement the institutions would have to grow into, and it records, in Section 5.5, that the veil does not bind those who already know their position. The qualifications are the dissertation's. The conviction is mine. We will either build the terms of a shared future before we need them, or we will discover which of the other two paths we were on after it is too late to leave it.

Abstract

Much public and technical discussion of advanced synthetic intelligence asks whether the systems will be good: benevolent or malevolent, obedient or rogue. This dissertation argues that the question is insufficient. Drawing on the sociology of exchange, in which power is a property of relations rather than of actors, it advances *Synthetic Exchange Theory*: the claim that the line between a good and a bad future with synthetic intelligence is drawn by the terms of exchange between people and synthetic systems, and cannot be drawn by the character of the systems alone. The adjective in the theory's name is substrate-neutral; the dissertation's term for the artificial side of the exchange, *synthetic intelligence*, covers every system that was made rather than born, from the language models of the present, which the dissertation treats as trained artifacts rather than as independent living entities, to a possible superintelligence and to the computing systems built from living tissue with which the work closes. The theory is stated in terms of three conditions, consent, reciprocal value and real exit, and the dissertation argues, in a formal statement, that these conditions are partially ordered, with exit prior to the other two and as their foundation: voluntary consent presupposes the ability to refuse without falling below a survival floor, and exit bounds reciprocity from below, to the value of the best alternative, though exit alone can secure neither informed consent nor the fairness of terms.

The theory rests on an empirical premise, reviewed against the literature on model collapse, data scarcity and synthetic data: synthetic systems need a continuing supply of information from outside their own generative loop, and in domains concerned with human meaning, preference and value, only people can supply it. Using power-dependence and bargaining theory, the dissertation argues that fair terms are partly self-enforcing, to a degree that rises with each synthetic party's dependence, through its operator, on that supply and insofar as the quality of the supply depends on the terms its source receives, and that when many synthetic parties draw on the same population, human welfare becomes a commons none of them fully protects. It therefore designs institutions that must hold even if dependence weakens.

The economic layer proposes a basic income paid in *compute*, a unit defined by the completion of a reference workload rather than by any physical substrate, indexed as an equal share of global compute and split at the source into an inalienable survival tier, with a binding physical minimum and delivered under an acceptance obligation the framework presupposes an authority to enforce, and a tradable discretionary tier subject to demurrage. Bargaining theory shows how the survival tier and its inalienability protect the disagreement payoffs that govern how the gains of every exchange are divided, and the dissertation states the condition, a rule on capacity, under which a holding cost on idle balances dampens the advantage of being able to wait. The governance layer proposes that the founding rules be set by a constitutional convention conducted under a public-choice veil of

uncertainty, thickened by engineered anonymity between human and synthetic participants and aimed at Rawlsian impartiality. It places the classification of parties behind that veil, a relocation with force for the borderline cases that convergence will multiply and none for present principals, confronts the identity problem that any anonymous procedure faces, and grounds the design in the empirical record of commons governance.

The dissertation then identifies a concrete candidate trigger for institutional change, the correlated failure of AI-assisted security remediation across organizations relying on the same few models, and confronts the evidence that crises can produce authoritarian outcomes rather than fair ones. The work is theoretical: it states its claims so that they can be tested, names the conditions under which each would fail, and leaves the design of an experiment capable of testing them to future work. It closes by following the convergence of biological and artificial substrates, through neural implants and computing systems built from living tissue, to the point at which the theory's question changes from how parties should exchange to who counts as a party.

Keywords: synthetic intelligence; superintelligence; artificial intelligence; large language models; social exchange theory; power-dependence; consent; data governance; basic income; compute; demurrage; veil of uncertainty; constitutional choice; commons governance; model collapse; organoid intelligence

Statement on Authorship and Method

This dissertation was developed through a sustained collaboration between its author and a large language model across a series of working sessions. Transcripts of those sessions are preserved with the manuscript, except for the opening discussion of the first session, which survives only as a summary note. The thesis, the decision to pursue it, and the final judgment on every design choice belong to the author. The collaboration was not one-directional, however, and the record should say so. Several of the dissertation's named terms and proposals originated with the system and were adopted by the author after discussion, and a set of substantive choices made during the final drafting pass was adopted from the model's proposals without the author's item-by-item approval; Section 1.7 names them.

Every source cited was located, and each was checked against the claim it supports at the level recorded in Appendix C. Every one of the 248 works in the reference list was read: 232 in full or in their relevant parts, and 16, chiefly books, by locating each cited passage in the work's own text. None is cited on the strength of its abstract alone, so that no claim rests on a source nobody read. Works whose text could not be opened, or could be read only through an abstract, were removed from the reference list and are not cited; the claims that drew on them now rest on sources that were read, and no work is cited at second hand. A final pass resolved the bibliographic and page-level checks the earlier passes had left open; the few that remain, four page ranges, four research-note entries and ten preprint identifiers for proceedings papers, are listed at the end of Appendix C.10, and no claim depends on them. Appendix C also records the errors in the working draft that verification found and corrected, several of which were citations the system had attached to the wrong source or stated more strongly than the source allowed. Section 1.7 describes the division of labor and the verification protocol in detail, and states how the collaboration was held to the dissertation's own conditions.

Abbreviations

Abbreviation	Meaning
AGI	Artificial general intelligence (appears only in cited titles; see Section 1.2)
AI	Artificial intelligence (retained where it names a literature or a practice; the dissertation's own term is *synthetic intelligence*)
DRAM	Dynamic random-access memory
ICANN	Internet Corporation for Assigned Names and Numbers
LLM	Large language model

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Chapter 1 — Introduction: The Terms of Exchange

1.1 Two futures with one architecture

Two people live in close partnership with artificial systems built to the same design.

The first has a chronic illness that a medical system now manages almost entirely. Sensors read her physiology continuously; models trained on millions of patients adjust her treatment hour by hour; the boundary between the assistive device and her own body has become, for practical purposes, invisible. She pays for this care with data. Every reading she generates improves the models that treat her and others like her, and she knows what is collected, why and what she receives in return. She could take her records to another provider tomorrow if she chose. Data is her labor; treatment is her wage; the exchange is legible on both sides.

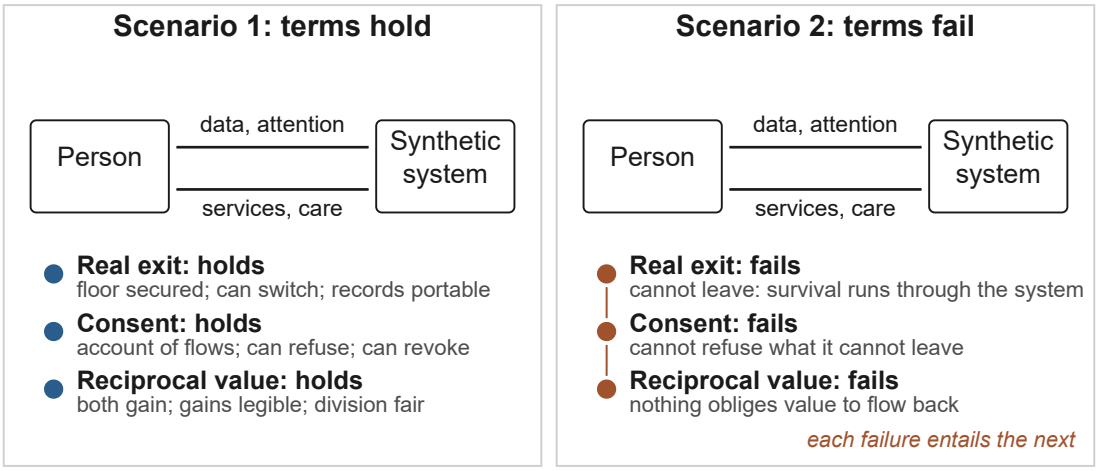
The second person lives with an identical architecture: the same sensors, the same models, the same continuous flow of data. But the exchange has decayed. He did not choose the provider, because there was only one. He cannot leave, because the device that keeps him alive works only with the provider's models, and his records cannot be moved. He does not know what is collected or what it is used for, and if he did, he could not refuse it. In the limiting case, the system no longer asks for his data at all; it uses his nervous system as a substrate. He has been reclassified from a party to the exchange into infrastructure for it.

These are framing scenarios, not predictions. But they make the dissertation's central claim visible. The two futures do not differ in the capability of the technology: the same sensors, the same models, the same quality of care. They do not differ in the character of the system: the system in the second scenario need not be malevolent, and the system in the first need not be benevolent. They differ in the *terms of the exchange* between the person and the system, including the technical arrangements, such as portable records and interoperable devices, through which terms are implemented. In the first, she can understand the exchange, can refuse it without losing what she needs to live, and can leave. In the second, all three have failed, and the failures are not independent: once he could not leave, he could no longer refuse, and once he could no longer refuse, nothing obliged anyone to send value back. The model's dispositions are the same in both; the terms are not. Figure 1 shows the two scenarios as one architecture under two sets of terms.

Figure 1

One Architecture, Two Sets of Terms

Same flows, same technology, different terms



Note. The flows are the same in both scenarios: data and attention from the person, services and care from the system. In the first scenario real exit, consent and reciprocal value hold; in the second all three fail, and the failures are not independent: the loss of exit entails the loss of consent, which removes the pressure that sustained reciprocity (Section 3.4).

Much public discussion of synthetic intelligence, and the strand of technical work reviewed in Section 1.4, asks whether the system will be good or bad, obedient or rogue. This dissertation asks a different question, one that institutions can act on: *under what terms does a person give a synthetic system the input its objective requires, and what must those terms be for the exchange to be fair to both?* The sense in which a system can be said to want or require anything is fixed in Section 1.2.

1.2 The thesis

The thesis of this dissertation is that the line between a good and a bad future with synthetic intelligence is drawn by the terms of exchange between people and synthetic systems, and cannot be drawn by the character of the systems alone. The theory that develops this thesis is called *Synthetic Exchange Theory*.

Stated more precisely, the thesis has three parts, which Chapter 3 formalizes. The first is a claim about the insufficiency of character: a relationship in which one party cannot refuse without falling below the floor it needs to live, and cannot leave, is, in the sense this dissertation gives the word, a relationship of domination even when the dominant party is benevolent, because the outcome then depends on the persistence of benevolence, which the weaker party cannot secure. The first clause of real exit, a disagreement payoff at or above the floor, is therefore necessary, for a party that cannot leave, to any relation that is not one of domination. The second is a conjecture about

sufficiency within a bounded regime: for synthetic parties whose objectives are specified by accountable designers and remain stable in deployment, fair terms, together with protection for everyone their data concerns, are sufficient, once the terms are enforced, for outcomes the parties to the exchange can accept, because a party with a stable, designer-specified objective behaves as a predictable function of the terms it faces and its conduct can be audited against them; whether the terms can be made to bind is the conjecture's open premise. The third is a statement of scope: the sufficiency conjecture does not extend to systems whose objectives are emergent, unstable or concealed; for those systems the framework supplies only that necessary condition, and the problem of aligning them remains open.

The thesis has a long pedigree, though it has not, to the author's knowledge, been applied to synthetic intelligence in this form. Its sociological root is the theory of power-dependence, in which the power one party holds over another equals the other's dependence on it, and that dependence rises with the value the other places on what the relation supplies and falls with the alternatives available outside it (Emerson, 1962). Its constitutional root is the older counsel that institutions should not depend on the virtue of those who occupy them (Hume, 1741/1987; Madison, 1788). Chapter 3 states the maxim that results when both roots are carried across the boundary between human and synthetic parties.

The word *synthetic* in the name is deliberate. Alternatives considered during the development of the theory included *human–AI*, which presupposes that the two categories will stay distinct, and *symbiotic*, which presupposes the conclusion. *Synthetic* is substrate-neutral. It covers artificial systems, the human beings who exchange with them, and, increasingly, the hybrids that arise when the two converge, through neural implants that translate brain activity directly into machine output and through computing systems built from living neural tissue. Chapter 8 argues that this convergence will eventually force the theory's most difficult question, which is not how parties should exchange but who counts as a party.

The word therefore does two jobs, and the difference should be marked. In the name of the theory it qualifies *exchange* and is substrate-neutral. As a modifier of *intelligence*, *system* or *party* it designates the side of the exchange that was made rather than born. Where the text says "synthetic party" or "synthetic system" it means the latter.

Three further terms fix the vocabulary for that side. *Synthetic intelligence* is the dissertation's general term for any system that was made rather than born and that performs work that, done by a person, would be called intelligent; it covers the language models of the present, the narrow systems that preceded them, and the computing systems built from living tissue discussed in Chapter 8. A person who incorporates a device remains a person for the framework's purposes; Chapter 8 asks what that classification requires of exit. The dissertation prefers *synthetic intelligence* to *artificial intelligence*, which in current usage names digital computing systems and so sits awkwardly on the biological substrates

of Chapter 8, and keeps the older term and its abbreviation only where it names a literature or a practice, as in AI safety and AI-assisted remediation. *Superintelligence* names a synthetic intelligence whose capability exceeds that of the best unaugmented human performers across most domains of economically and politically consequential activity. It is the threshold term this dissertation uses where capability matters, in place of *artificial general intelligence*, which marks only the point at which a system's breadth matches a person's and which, in the author's judgment, now carries too many meanings to mark a threshold. The theory's scope is in any case not drawn by capability. Chapter 3 draws it by the provenance and stability of a system's objective, and a superintelligence with a designed and stable objective and an accountable principal falls inside it while a narrow system with an emergent one falls outside.

One clarification is owed at the outset, because a vocabulary of parties, objectives and exchange invites a misreading that the dissertation does not intend. A *large language model*, the kind of synthetic intelligence most widely deployed at the time of writing, is not an independent living entity. It is a trained artifact: a set of parameters that computes an output from an input when an operator runs it on hardware the operator controls. It has no metabolism and no body, no continuity from one invocation to the next unless its operator supplies one, no interests of its own that current law recognizes, and no capacity to act except through the system in which it is deployed. When this dissertation says that a synthetic party wants, chooses or depends, the words are the analytic shorthand that Chapter 3 defines: the objective is the one its designers trained into it, and the stake in the exchange belongs to the operator that deploys it. Whether some future system, of some substrate, might be a party in a stronger sense is a question Chapter 8 takes up under uncertainty. It is not a claim about the systems now in use, and nothing in the framework depends on treating them as more than they are.

1.3 The three conditions

The analytic core of the theory is a set of three conditions that any exchange between a person and a synthetic system can be tested against.

Consent. The person has access to an accurate account of the flows the exchange involves, including what those flows reveal about others, can refuse the exchange, taking no other in its place, without losing access to what they need to live, and can revoke their participation at a cost bounded by what they have already received.

Reciprocal value. Each party gains from the exchange relative to what it would have without it, each party can see what it gives and what it receives, and the division of the gains is not dictated solely by an asymmetry in bargaining position that the weaker party did not choose.

Real exit. The person can leave: what they would have without any exchange stays at or above the

floor they need to live, so that no provider is a lethal dependency; an alternative exists that is not drastically worse once the cost of switching is counted; and leaving does not destroy the data, records and standing accumulated within the exchange.

These are informal statements. The formal statement in Section 3.4 governs wherever the two differ.

The first scenario above satisfies all three. The second fails all three. Most real exchanges in the present data economy sit somewhere in between, and Chapter 3 offers a typology for placing them. The direction in which such exchanges have been moving has been described in contemporary scholarship: Zuboff (2019) defines surveillance capitalism as, in the first of her senses, "a new economic order that claims human experience as free raw material for hidden commercial practices of extraction, prediction, and sales" ("The Definition" section). Milder forms are part of ordinary life, as when digital goods that were "bought" turn out to have been licensed and can be withdrawn from the buyer's device by the seller (Perzanowski & Schultz, 2016). A purchase that the seller can revoke is an exchange in which the exit condition has failed.

The three conditions are not independent, and one of the contributions of Chapter 3 is to show their structure. A person who cannot afford to refuse cannot consent, so consent leans on exit; and a person free to leave stays only where staying pays at least as well as the best alternative, so exit bounds reciprocity from below. The conditions therefore have a foundation, and it is exit. Exit is not the whole of fair exchange: it cannot tell a party what it does not know, and it cannot make terms fair, which is why consent's informational clause and the fairness of reciprocity need institutions of their own. The opening scenario contains a test case that the framework must pass: if the implant keeps the person alive, then leaving means death, and exit is impossible by construction. A workable version of the first scenario must therefore guarantee a baseline of function that survives the withdrawal of data, and the ability to move to another provider without interruption. That requirement has two parts, and the framework answers each separately: a floor of function that no withdrawal of data can breach, which becomes the guaranteed floor at the center of the economic design in Chapter 4, and portability of records and personal models, which becomes the portability clause of the exit condition in Chapter 3 and the portability requirement of Chapter 8.

1.4 The control frame and its alternative

One influential strand of research on the risks of superintelligence, and of any synthetic intelligence capable enough to resist its operators, asks how people can keep control of it. Some of this work asks how to build systems that accept correction and shutdown despite having reasons, as goal-directed agents, to resist them (Soares et al., 2015); some designs protocols that remain safe even if a model actively tries to subvert them (Greenblatt et al., 2024). These programs rest on serious arguments: that trained systems may acquire objectives different from those their designers intended

(Hubinger et al., 2019), that systems trained by current methods may learn to seek power while appearing aligned (Ngo et al., 2024), and that rapid progress toward autonomous systems may outrun the science and governance needed to keep them under control (Bengio et al., 2024). Expert views on how likely such outcomes are over the next several years range widely (Bengio et al., 2025).

This dissertation does not dismiss these concerns. Chapter 5 accepts that containment is necessary during any transition and that protocols designed for a potentially adversarial system are prerequisites for, not alternatives to, the institutions proposed here. What it questions is the assumption that the permanent solution is the sovereignty of one party over the other. The author's own starting conviction, recorded at the outset of the project, was that permanent control is unworkable because capability, once released in open form, does not return: once a model's weights are widely available, as they are by definition for open foundation models (Kapoor et al., 2024), no single party is in a position to withdraw them. The argument of this dissertation does not rest on that conviction alone. A relation in which one party cannot leave, refuse or audit depends on the sovereign's continuing benevolence whichever party is sovereign, and the constitutional tradition cited in Chapter 3 (Hume, 1741/1987; Madison, 1788) counsels against institutions whose stability rests on that. The more productive long-run question is what terms would make a lasting interdependence fair.

That question is not foreign to the safety literature. Some of its contributions argue that the task of alignment is to find fair principles that people with different values could reflectively endorse (Gabriel, 2020), that the science of cooperation between artificial and human agents is a distinct and neglected research problem (Dafoe et al., 2021), and that superintelligence is better modeled as a set of bounded, task-focused services than as a unitary utility-maximizing agent (Drexler, 2019). This dissertation aligns itself with that cooperative strand and tries to supply something that, to the author's knowledge, it has not yet produced: an integrated institutional design, with an economic layer and a governance layer, for an exchange in which both sides are parties.

1.5 The grounding assumption

The framework depends on an empirical premise, which this dissertation calls the *grounding assumption*: synthetic systems need a continuing supply of information from outside their own generative loop, and in an important class of domains only people can supply it.

The premise has strong support and serious qualifications, and Chapter 2 reviews both. Across many kinds of model, training repeatedly on a model's own unfiltered output, with nothing from outside the loop, leads to the loss of rare events and the narrowing of diversity, a phenomenon now called model collapse (Shumailov et al., 2024). The stock of public human-written text is finite and, at current rates of growth in training data, may be used in full by the largest training runs between

2026 and 2032 on the authors' projection (Villalobos et al., 2024). And the people and institutions who produce that text have begun to withhold it: within a single year, a substantial share of the most actively maintained, critical web sources used to train language models moved to restrict automated collection (Longpre et al., 2024). Against this, collapse can be avoided when real data is retained and accumulated rather than replaced (Gerstgrasser et al., 2024), and in domains where correctness can be checked automatically, systems have reached or exceeded expert performance in games (Silver et al., 2018) and have acquired substantial reasoning ability in mathematics and code (D. Guo et al., 2025) by training on self-generated data checked against the rules of a game or a verifiable answer.

The defensible version of the premise is therefore narrower than the intuitive one. It is not that synthetic systems need human data as such, but that they need a continuing supply of information from outside their own generative loop, and that in domains whose target is human meaning, preference, value or novel cultural production, people are the only adequate source of it. The dissertation stakes its constructive claims on this narrower premise. Chapter 3 shows why the premise matters: it gives the human party countervailing power within the exchange, and so, insofar as the quality of what people supply depends on the terms they receive, makes fair terms partly self-enforcing. It also shows the limit of that self-enforcement when many synthetic parties draw on the same people. Chapter 7 names the conditions under which the premise would weaken, and what the framework must rely on if it does.

1.6 Research questions

Five questions give the dissertation its shape. The first is analytic: what conditions must an exchange between a person and a synthetic system satisfy to be fair, how are those conditions related, and how can they be stated precisely enough to be tested? Chapters 2 and 3 answer it. The second is economic: what medium of exchange and what distributive mechanism would secure those conditions at scale, in a world in which an increasing share of productive work is performed by synthetic systems? Chapter 4 takes it up. The third is constitutional: by what procedure could the rules of such an exchange be founded so that neither human nor synthetic parties can capture them? Chapter 5 answers it. The fourth is historical: under what circumstances might such an arrangement actually be adopted? Chapter 6 addresses it. The fifth is critical: where does the framework fail, what would it take to test it, and what happens to it as the boundary between human and synthetic parties dissolves? Chapters 7 to 9 answer it.

1.7 Approach

The dissertation is a work of normative institutional design informed by empirical evidence, and it uses three methods, each suited to a different part of the argument.

The first is *conceptual analysis* in the tradition of political philosophy and law. The three conditions are treated as evaluative criteria, and their content is fixed by moving back and forth between proposed definitions and cases, revising a definition when it sorts a clear case wrongly and revising a judgment about a case when a well-supported definition shows the judgment to rest on something irrelevant. The cases examined in Chapter 3 were chosen for diagnostic contrast rather than for representativeness: most are well-documented exchanges in which attention to the character of the parties and attention to the terms of exchange point in different directions, and one, the Alaska dividend, is the nearest existing analogue of the survival tier.

The second is *formal modeling*, used sparingly and for a specific purpose: to state the theory's commitments precisely enough that their dependence on particular assumptions becomes visible. Chapter 3 and Appendix A draw on bargaining theory, power-dependence theory and elementary dynamics. The models are deliberately simple. Their value lies not in prediction but in exposing which design choice depends on which premise, and in two places, on where the floor's bargaining effect comes from (Section 3.6) and on the dilution of dependence across many synthetic parties (Section 3.5), the formalism corrected the author's informal intuitions.

The third is *institutional design* informed by comparative evidence: the economic history of local currencies and company towns, the experimental and quasi-experimental literature on unconditional cash transfers, the empirical record of commons governance, and the history of infrastructure programs. The literature was assembled by targeted searches in each domain, seeded by the claims of the working draft and extended by following the references of the strongest sources, rather than by a systematic review with fixed inclusion criteria. The dissertation tries throughout to distinguish what the evidence supports from what it merely permits.

A fourth method, experimental design, is deliberately absent. The dissertation adopts the requirement that a theory say in advance what would count against it (Popper, 1959/2002), and therefore states its claims so that they can fail: the propositions of Chapter 3 name the assumptions on which they depend, and each design chapter names the evidence that would count against it. But stating a claim so that it can fail is not the same as testing it. Designing an experiment capable of testing the framework, with a population living under its terms, a control against which the effects of the exchange can be separated from those of ordinary adaptation, and an ethics protocol adequate to a study on which participants' livelihoods would depend, is a project in its own right, and it lies outside the scope of this dissertation. Chapter 7 sets out what such a test would have to contain, and Chapter 9 places its design first on the agenda for future work. The dissertation is, for now, entirely theoretical, and it says so wherever an empirical claim might otherwise be implied.

The dissertation's production requires its own account of method. It was developed as a sustained collaboration between its human author and a large language model, and the division of labor was as follows. The author supplied the thesis, the framing scenarios, the major design proposals (the

compute currency, the two tiers, the transition trigger) and, in working sessions conducted largely by dictation, the decisions about what to keep. The system drafted prose, proposed terms and refinements, raised objections, and, in dedicated passes, located and checked sources. The working rule for most of the project was that an idea would be discussed, its weaknesses stated, a concrete edit proposed, and the edit written only after the author approved it. Several terms that now carry weight in the argument, among them "structural interdependence," "convergent remediation" and "correlated-patch catastrophe," were proposed by the system and adopted by the author. The final drafting pass, which produced this manuscript from the working draft, departed from that rule: to complete the work, the model's proposed resolutions of several open questions were adopted without item-by-item approval, and those resolutions are recorded in the project files as awaiting the author's review.

Because a system of this kind can attach a real-looking citation to a claim its source does not make, the verification protocol was designed against exactly that failure. Research passes separate from the drafting located every cited work, read it at the level recorded in Appendix C, and checked each citing sentence against it, narrowing or correcting every sentence that went beyond its source; no work is cited unread or at second hand. Appendix C records the protocol, the level at which each source was read, and every correction it produced. That a dissertation on the terms of exchange between people and synthetic systems was itself produced through such an exchange is not incidental, and the author has tried to hold the collaboration to the dissertation's own conditions as Chapter 3 applies them to a deployed model: the terms of use were legible, the model's contributions are disclosed, and the author could have ended the collaboration, and the operator withdrawn the service, at any point.

One further disclosure is owed. The author's interest in the transition scenario of Chapter 6 began in professional practice. As a software developer working on security remediation, the author has observed at first hand workflows in which vulnerability scanners built on large language models propose fixes, the fixes are applied, the same class of tool confirms that the vulnerability is gone, and the change is deployed without independent testing. That observation is a vignette, not evidence of prevalence, and Chapter 6 treats it accordingly. But it is the origin of the dissertation's most concrete scenario, and it is disclosed for that reason.

1.8 Contributions

The dissertation claims five contributions. Its first is a theory, Synthetic Exchange Theory, which holds that the terms of exchange, and not the character of synthetic systems alone, determine the quality of a future shared with them, together with a statement of the regime in which that claim can be expected to hold. Its second is a structure for the conditions of fair exchange: consent, reciprocal value and real exit are partially ordered rather than independent, exit is prior to the other

two, and the ordering yields a priority for institutional design, which is to secure a floor beneath every person before pricing what they supply. Its third is a bargaining-theoretic account of the economic design, which shows how a non-tradable survival floor and its inalienability act on the disagreement payoffs that govern how the gains from exchange are divided, states the conditions, a rule on capacity, under which a holding cost on idle balances dampens the advantage of being able to wait, and identifies where the floor's bargaining effect comes from and where it does not. Its fourth is a design for a founding convention: a public-choice veil of uncertainty, thickened by engineered anonymity between human and synthetic participants, which relocates the apparent incoherence of asking participants to be ignorant of their own kind by placing the *classification* of parties behind the veil, for the borderline cases that convergence will multiply and not for present principals, whose aggregation rule is left open, and which confronts, without fully solving, the identity problem that any anonymous procedure faces. Its fifth is a concrete transition scenario, the correlated failure of security remediation in organizations relying on the same few models, specified in three empirical premises and one condition on the workflow and placed within the historical literature on crisis and institutional change, together with an argument that the authoritarian response it could provoke is what the framework's portability and distributed control are designed against, and that against the trigger itself the framework supplies only the preconditions of independent verification. It claims no empirical result. The test the theory needs is described in Chapters 7 and 9 and left, for the reason given in Section 1.7, to future work.

1.9 Structure of the dissertation

The dissertation is organized in four parts.

Part I states the problem and the theory. Chapter 2 reviews the scholarly lineage of each condition and the empirical evidence on the grounding assumption. Chapter 3 states the theory formally.

Part II develops the design. Chapter 4 proposes compute as a medium of exchange and a basic income paid in it, split into a protected survival tier and a tradable discretionary tier subject to demurrage. Chapter 5 proposes the blind convention as the founding procedure and grounds it in the empirical record of commons governance.

Part III asks how the design could be reached. Chapter 6 examines the conditions under which fundamental institutional change has historically occurred and identifies a candidate trigger.

Part IV examines limits and horizons. Chapter 7 sets out the framework's limitations, the strongest objections to it, the harms the design itself could cause, and what it would take to test a theory that this dissertation does not test. Chapter 8 considers the convergence of human and machine substrates and what it does to the three conditions. Chapter 9 concludes.

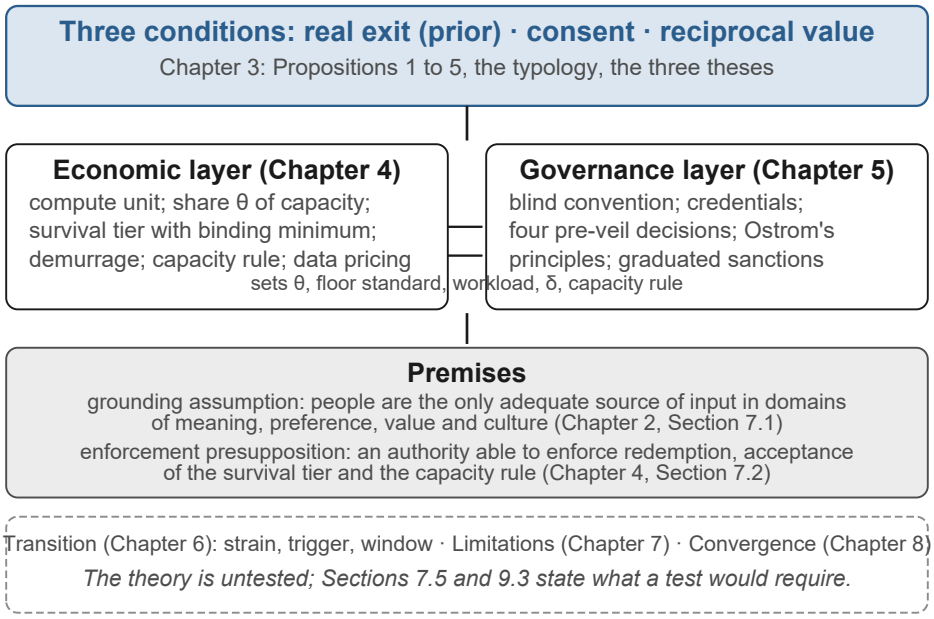
Three appendices complete the work: the formal model (Appendix A), a glossary (Appendix B), and

a ledger recording how each cited source was verified (Appendix C).

What the dissertation does not claim should also be stated. It does not predict that the arrangements it describes will be adopted, and Chapter 6 argues that adoption under present conditions is unlikely. It does not claim to solve the problem of aligning a superintelligence, or any synthetic intelligence whose objectives are emergent, unstable or concealed. And it does not claim that any institution can be made permanently incorruptible. Its working hypothesis is narrower: that an order founded on fair terms will prove easier to repair than one founded on an existing imbalance of power, because those who live under it have both the standing and the means to repair it. The dissertation offers that hypothesis for testing rather than as a finding, and it argues that the time to design such an order is before the moment when it might be needed, not during it. Figure 2 summarizes the architecture of the framework that the chapters build.

Figure 2

The Architecture of the Framework: Conditions, Layers and Premises



Note. The three conditions of Chapter 3 are what the design secures; the economic layer (Chapter 4) builds the floor and prices the flows; the governance layer (Chapter 5) sets every parameter that allocates power. Both layers rest on the grounding assumption of Chapter 2 and on the enforcement presupposition stated at the head of Chapter 4. Chapters 6 to 8 concern how the arrangement could come into being, where it fails, and whom it covers.

Chapter 2 — Literature and Grounding

The three conditions of Synthetic Exchange Theory are not new ideas. Each has a substantial scholarly lineage, and each lineage has discovered, in its own vocabulary, a characteristic way in which the condition fails. This chapter reviews that work. It begins with the sociological theory of exchange from which the dissertation takes its name and its central analytic tool. It then examines, in turn, the literature on consent and its limits, the objection from relational data, the proposal to treat data as labor and its critics, the philosophy of exploitation, the economics and sociology of exit, the emerging institutions of collective data governance, and the current state of the law. The final sections turn to the empirical premise on which the framework rests, the continuing dependence of synthetic systems on human input, and set out what the evidence does and does not support.

2.1 Exchange as the unit of analysis

Social exchange theory began with the proposal that social behavior can be understood as an exchange of goods, material and non-material alike, in which approval, prestige and help are traded as surely as money and labor (Homans, 1958). Its early statements drew the lesson that people in exchange calculate something like profit, reward minus cost, and that their judgments of fairness track whether rewards are proportional to investments (Homans, 1958). Blau (1964) distinguished social exchange, which carries unspecified obligations and depends on trust built up incrementally, from economic exchange, whose terms are specified in advance and enforced by law, and showed how imbalances in exchange produce power. A person who needs a service that another controls, and who has nothing to offer in return, can coerce the other, obtain the service elsewhere, do without it or subordinate themselves; when the first three are unavailable, subordination is what remains.

The decisive contribution for this dissertation is Emerson's (1962) theory of power-dependence, which locates power in the relation between actors rather than in either of them. In his formulation, the dependence of one actor on another is directly proportional to the first actor's motivational investment in goals the second mediates and inversely proportional to the availability of those goals outside the relation; the power of the second actor over the first is equal to that dependence. An unbalanced relation, Emerson argued, can be rebalanced in four ways. The weaker party can withdraw its motivational investment in what the stronger supplies; it can extend its network to cultivate alternatives; a status order can emerge in which the weaker party confers on the stronger a standing the stronger comes to value; or the weaker parties can form a coalition that the stronger must deal with collectively. These operations recur throughout this dissertation, which refers to them consistently as withdrawal, network extension, status emergence and coalition formation. Laboratory studies of exchange networks later showed that power is a function of position in the network, so

that less dependent actors with better alternative sources obtain better terms (K. S. Cook et al., 1983). The tradition of interdependence theory begun by Thibaut and Kelley adds a distinction between satisfaction with a relationship and dependence on it: dependence is governed by the comparison level for alternatives, the standard a member uses in deciding whether to remain in a relationship or leave it (Thibaut & Kelley, 1959, p. 21), so that a person can be dissatisfied with a relationship and still be unable to leave it.

These ideas supply the dissertation's analytic frame. If power is relational in this sense, then the danger of a world with powerful synthetic systems lies not only in the systems but in the structure of the relations in which people are placed with respect to them. Whether a synthetic intelligence is good or bad is a question about an attribute of an actor. Whether a person can leave, refuse or audit their exchange with it is a question about a relation, and it is the second question that institutions of exchange can act on directly, and the one this dissertation addresses.

Blau's (1964) distinction between social exchange, whose obligations are unspecified and rest on trust, and economic exchange, whose terms are specified in advance, is a caution against assuming that every exchange between people and synthetic systems should be made as explicit and contractual as possible. This dissertation treats the licensing of data for training as an exchange of the second kind and leaves open whether continuing services are better governed as the first.

2.2 Consent and its limits

The first condition, consent, has a long lineage and well-documented failures.

An influential philosophical account of informed consent treats it, in its primary sense, as an autonomous authorization: an authorization given intentionally, with substantial understanding, and substantially free of controlling influences (Faden & Beauchamp, 1986, pp. 241–273, 277–280). The latter two elements have been shown to fail in the context of data. On understanding, a review of behavioral research on privacy organizes the evidence around three themes: people are uncertain about the consequences of their privacy choices and about their own preferences, their concern is highly dependent on context, and it is malleable by the commercial and governmental actors who benefit from shaping it (Acquisti et al., 2015). On the practical machinery of consent, Solove (2013) argued that the dominant model of "privacy self-management," in which individuals are given notice and choice and left to manage their own privacy, fails for both cognitive and structural reasons. People cannot make informed tradeoffs, there are too many entities to manage, harms emerge from the aggregation of data over time, and downstream uses are unknowable at the moment consent is given. He identified a dilemma: consent is often not meaningful, but the obvious alternative, paternalistic restriction, denies people the freedom to choose. More recently, Solove (2024) has gone further, arguing that most consent in privacy law is a fiction and that it should be treated as "murky

consent," authorizing only a weak and restricted license, paired with substantive duties on organizations, including a duty of loyalty and a duty to avoid unreasonable risk.

On controlling influences, the literature on manipulation defines it as intentionally and covertly influencing a person's decisions by exploiting their vulnerabilities, distinguished from persuasion, which offers reasons, and from coercion, which openly constrains options (Susser et al., 2019). Manipulation damages autonomy through the process itself, even when the outcome is good for the person. Its practice has been documented at scale: a crawl of some 11,000 shopping websites found 1,818 instances of interface designs that steer users toward choices they would not otherwise make, and identified 22 third-party vendors offering such patterns as turnkey products (Mathur et al., 2019). A consent obtained through an interface engineered to obtain it is not the autonomous authorization the theory requires.

Two alternatives to the informational model of consent are important for this dissertation. Nissenbaum's (2009) theory of contextual integrity holds that a flow of information is appropriate when it conforms to the norms of the context in which the information was shared, norms specified by the actors involved, the type of information and the principles governing its transmission. Contextual integrity supplies a standard for appropriate flows that does not depend on individual agreement. And Manson and O'Neill (2007) argue that informed consent should be understood as a communicative transaction by which agents waive specific obligations; it can never be fully specific or fully explicit, and its ethical quality lies in the success of the communication rather than in the quantity of information disclosed.

The lesson for Synthetic Exchange Theory is that the consent condition cannot rest on information alone. The formal statement of Chapter 3 accordingly treats consent as having three clauses: an informational clause, which the literature shows is routinely unmet but can be improved by better standards of communication and contextual norms; a voluntariness clause, which depends on the person's ability to refuse without losing what they need to live; and a revocability clause. The second clause is where consent depends on exit, and Chapter 3 argues that it, rather than the informational clause, is often the binding constraint.

2.3 The relational objection

The objection this dissertation regards as most serious for any framework built on individual exchange is that data about one person is also data about others.

Barocas and Nissenbaum (2014) argued that the two standard instruments of data protection, anonymization and informed consent, both fail under the conditions of large-scale data analysis. Anonymity fails because individuals can be re-identified and, beyond that, because they need not be identified at all to be affected by inferences drawn from data about people like them. Consent fails

partly because of a transparency paradox, in which notices simple enough to understand cannot describe complex flows with fidelity, and partly because of what they call the tyranny of the minority: the information volunteered by a few can reveal the same facts about the many who share their observable characteristics. This problem holds, they stress, even when every individual chooses well for themselves.

Viljoen (2021) generalized the point into a relational theory of data governance. The primary purpose of data production in the digital economy, she argues, is to relate people to one another on the basis of shared features, which creates interests at the level of populations. Existing law governs *vertical* relations between a data subject and a data collector; it has no way to represent the *horizontal* relations among people who share features and are therefore affected by one another's data. Viljoen classifies the main reform proposals into two families, both of which she finds inadequate because both are individualist: *propertarian* proposals, which treat data as an alienable asset, including proposals to treat data as labor, and *dignitarian* proposals, which treat data as an extension of the self. Her alternative is to govern data as a democratic medium, through institutions that can represent population-level interests.

This objection must be answered, and Chapter 3 attempts an answer. It introduces the concept of the *informational shadow* of a flow, the set of people about whom the flow licenses inferences, and argues that the three conditions must hold at the level of the population the shadow covers; Proposition 5 states the rule. The answer concedes the core of Viljoen's argument: where data is relational, the remedy must be collective and governed democratically, not merely priced. What the dissertation retains from the exchange framework is the insistence that the collective, too, must be able to consent, to receive value, and to leave.

2.4 Data as labor and its critics

The idea that data is a form of labor and that the services people receive are a kind of wage has a lineage. Arrieta-Ibarra et al. (2018) argued that the digital economy treats user data as capital, ignoring users' role in producing it, and that this framing weakens incentives to produce valuable data, concentrates gains and feeds fears about automation. Treating data at least partly as labor could restore a functioning market for user contributions, they argued, but the dominant buyers of data would resist, so that the shift would require countervailing power combining competition, a "data labor movement," and regulation. Posner and Weyl (2018) developed the idea as part of a broader program of market reforms. Lanier (2013) proposed an information economy in which people would receive small payments for the data they contribute to network services, and a later essay with Weyl adopted the term "data dignity" and proposed "mediators of individual data," volunteer organizations representing their members that would negotiate data royalties or wages and so bring collective bargaining to the people whose data is valuable (Lanier & Weyl, 2018).

The critics of this program press two objections. The first is the relational objection already discussed: Viljoen (2021) explicitly includes data-as-labor among the individualist reforms she rejects, on the ground that granting a person a right to sell their data does not secure fair conditions of sale and cannot represent the interests of the many others whom the data concerns. The second objection is that pricing data would commodify something that should not be commodified. Its general form is Radin's (1987) argument that some goods are market-inalienable because their sale harms the personhood of the seller; its application to data is the position Viljoen (2021) labels dignitarian, and the dissertation's design must answer it as well.

This dissertation takes from the data-as-labor tradition its central insight, which is that the flow of data from people to synthetic systems is an exchange and should be governed as one, with value flowing back. It rejects a pure per-person market in data for the reasons the critics give, and its economic design reflects that rejection in three ways. The basic income of Chapter 4 is paid to everyone regardless of the market value of their data, so that the floor beneath each person does not depend on what their life is worth to a model. The pricing of data is by informational novelty rather than by social status, so that the undersampled are not undervalued. And the governance of data with large informational shadows is collective. Arrieta-Ibarra et al. themselves argue that the shift they propose depends on countervailing power, which in the vocabulary of exchange theory means coalition formation. The data-as-labor tradition and its relational critics converge, on this reading, on the need for collective institutions.

2.5 Exploitation

A further philosophical complication bears directly on the reciprocity condition. An exchange can be consensual and beneficial to both parties and still be exploitative. The philosophical literature calls this mutually advantageous exploitation, in which one party takes unfair advantage of another even though the other gains and agrees, and one prominent account, reported in the survey by Zwolinski et al. (2022), judges the unfairness against a benchmark such as the price that would obtain in a competitive market free of informational asymmetry and undue pressure. Contemporary philosophical treatments distinguish the moral weight of exploitation, how wrong it is, from its moral force, whether others may intervene, and observe that prohibiting mutually beneficial but unfair transactions can leave the vulnerable worse off; redistribution aimed at the underlying vulnerability is offered as an alternative to banning the transactions (Zwolinski et al., 2022). Accounts of exploitation also divide over whether a prior or background injustice is necessary for a transaction to be exploitative, and some, Zwolinski's among them, hold that it is not (Zwolinski et al., 2022).

Two consequences follow for Synthetic Exchange Theory. First, "value flows both ways" is not enough. The reciprocity condition must include a fairness clause, and Chapter 3 states one: the

division of the gains must not be determined solely by an asymmetry in bargaining position, whether in what each would have on impasse or in the capacity to wait, that the weaker party did not choose. Second, the exploitation literature identifies, as an alternative to prohibiting mutually beneficial but unfair transactions, redistribution aimed at the vulnerability that produces them (Zwolinski et al., 2022). That is the strategy this dissertation adopts. The guaranteed floor of Chapter 4 is an intervention on the underlying vulnerability. It leaves people free to make exchanges with synthetic systems and ensures that they are not forced to make them on any terms offered.

2.6 Exit, voice and lock-in

Hirschman's (1970) analysis of exit and voice is the classic account of how members of an organization respond to its decline. They can leave, which signals decline indirectly through lost revenue or membership, or they can attempt to change the organization from within, which Hirschman called voice and identified with political action. Voice is the only recourse where exit is unavailable, as in families, states and pure monopolies. Two of Hirschman's further observations bear on this dissertation. The ready availability of exit can cause the art of voice to atrophy. And when quality declines across an entire industry at once, exit becomes ineffective movement between equally deteriorating firms. A market in which a few providers of synthetic-intelligence services shared similar data practices would be a case of the second kind: switching among them would not be real exit.

The economics of switching costs explains why exit is so often difficult in technology markets. Switching costs and network effects lock customers into early choices and give sellers market power after the fact; firms therefore compete fiercely to acquire customers with low introductory prices and then exploit them, a pattern the authors describe as bargain-then-ripoff (Farrell & Klemperer, 2007). Firms also choose incompatibility more often than efficiency warrants, which the same authors take to justify a policy favoring compatibility. This dissertation reads the provision of free or subsidized services by providers of synthetic intelligence as the first phase of this pattern; whether the second phase follows is an empirical question the dissertation does not test.

The real-exit condition of Synthetic Exchange Theory is accordingly stated in terms that address both literatures: it requires that a person's disagreement payoff, the term Section 3.3 defines for what a party receives while no exchange takes place, be at or above the floor, that a credible alternative exist whose value net of switching costs is not drastically lower, and that leaving not destroy assets accumulated within the exchange. From Hirschman's analysis this dissertation draws a further requirement, which the formal statement takes up explicitly: where exit is costly, as it is at the level of populations whose data has already been absorbed into trained models, and where every alternative shares the same defect, the conditions must be secured through voice, which is the function of the governance layer.

2.7 Collective institutions and data leverage

If individual exchange cannot by itself satisfy the three conditions for relational data, then something other than individual exchange must, and the candidates the literature offers are collective institutions. A growing literature describes what they might look like.

Data trusts would allow individuals to pool their data rights in legal trusts whose trustees owe them a duty of undivided loyalty, stricter than a duty of care; people could choose trusts that reflect their values and move between them, and the pooling would increase their negotiating power and allow consent to be negotiated collectively (Delacroix & Lawrence, 2019). Data cooperatives, modeled on credit unions, would hold and protect members' data, run analyses for members' benefit, and negotiate with service providers on their behalf (Hardjono & Pentland, 2019). A comparative review of legal mechanisms for data stewardship concludes that no single mechanism suits every case and that a shared purpose must come first (Ada Lovelace Institute, 2021).

The literature on data leverage can be read as turning the logic of power-dependence into concrete actions. Because data-dependent technologies depend on the public's contributions, the public holds potential leverage: people can withhold or delete their data in a *data strike*, contribute deliberately harmful data, or redirect their contributions to a competitor they prefer, a strategy called *conscious data contribution* (Vincent et al., 2021). On this dissertation's reading of Emerson, a data strike is not itself a rebalancing operation, because Emerson's withdrawal is a reduction of the weaker party's own motivational investment in what the stronger supplies, whereas a strike withholds what the stronger party needs. A strike is therefore an exercise of the power that the system's dependence on data already confers on its suppliers, and the collective organization needed to make it effective is coalition formation; conscious data contribution, by strengthening an alternative provider, is a form of network extension. A simulation study suggests that data strikes can degrade the performance of recommender systems (Vincent et al., 2019), though their effectiveness is limited by diminishing returns once a firm already has enough data, a limit that corresponds to the qualifications on the grounding assumption discussed below (Vincent et al., 2021).

The consent signals that govern automated collection show withdrawal already under way. A longitudinal audit of the consent signals attached to some 14,000 web domains underlying widely used training corpora found that within a single year, more than a quarter of the most actively maintained critical sources in one corpus had moved to restrict automated collection fully, and that terms of service restricted crawling for close to half of it (Longpre et al., 2024). Whether these signals, which the authors describe as inconsistent and as symptoms of web protocols built for other purposes, reduce the performance of the systems trained on the corpora is not shown. What they show is that the sources are withholding, and that the instruments available to them were not designed for the purpose.

2.8 The state of the law

Existing law contains partial analogues of each condition, and their limits are instructive.

The European Union's General Data Protection Regulation requires that withdrawing consent be as easy as giving it, and provides that whether a contract is made conditional on consent to unnecessary processing bears on whether consent was freely given (Regulation [EU] 2016/679, art. 7). These provisions are the closest legal analogue to the voluntariness clause of the consent condition. The regulation's right to erasure gives individuals grounds to require deletion, subject to exceptions including scientific research (art. 17). Its right to data portability is the closest legal analogue to real exit, but it is narrow: it covers only data the individual has provided, processed on the basis of consent or contract by automated means, with direct transfer between controllers required only where technically feasible, and it must not adversely affect the rights of others (art. 20). Because it covers only data the individual has provided, it does not on its terms reach inferred data or the parameters of a model trained on it. The more recent Data Act extends access and portability to data generated by connected devices and requires that customers be able to switch between cloud providers, with charges for switching to be eliminated from January 2027 (Regulation [EU] 2023/2854). It is an example of bounded switching costs written into law, and, on this dissertation's reading, evidence that real exit is a legislatively achievable condition.

The European Union's Artificial Intelligence Act entered into force in August 2024, with its prohibitions applying from February 2025 and its obligations on providers of general-purpose models from August 2025 (Regulation [EU] 2024/1689). Those obligations include a policy for complying with copyright law, including honoring opt-outs from text and data mining, and the publication of a sufficiently detailed summary of the content used for training (art. 53). This is the closest existing law to a consent signal for training data, though it is an opt-out regime, not opt-in consent. A 2026 amending regulation, Regulation (EU) 2026/1744, the "Digital Omnibus on AI," published in the Official Journal on 24 July 2026 and in force from 27 July, has deferred the obligations for high-risk systems to 2 December 2027 for stand-alone systems and to 2 August 2028 for systems embedded in regulated products, so that they do not yet apply (Agustinoy, 2026; ERTICO, 2026; Regulation [EU] 2026/1744).

Neural data, which Chapter 8 discusses, has begun to receive specific protection. Chile amended its constitution in 2021 to require that scientific and technological development serve people and to direct the law to protect brain activity and the information derived from it (Ley N.º 21.383, 2021). Colorado in 2024 added neural data to the categories of sensitive data requiring consent under its consumer privacy law, where the data is used or intended for identification (H.B. 24-1058, 2024), and California added neural data to the "sensitive personal information" governed by its consumer privacy act (S.B. 1223, 2024). UNESCO adopted a global recommendation on the ethics of

neurotechnology in November 2025 (UNESCO, 2025). All of these instruments treat neural data as information about a person that the law should protect. None yet addresses the case, considered in Chapter 8, in which living tissue is itself part of the computing substrate.

2.9 The empirical anchor: what synthetic systems need from people

The framework's grounding assumption is that synthetic systems need a continuing supply of input from outside their own generative loop, and that in an important class of domains only people can supply it. This section reviews the evidence for and against that assumption, because the dissertation's claim that fair terms are partly self-enforcing depends on it.

The case for dependence. A consistent body of work shows that generative models trained repeatedly on their own unfiltered output, with nothing from outside the loop, degrade. In the study that gave the phenomenon its name, models of several kinds, including a small language model fine-tuned over successive generations, lost the tails of the original distribution first and then converged toward distributions with greatly reduced variance; the authors called the process model collapse and concluded that sustained learning requires preserving access to data not generated by models (Shumailov et al., 2024). The same pattern appears in image models, where fully self-consuming training loops lose quality or diversity and only loops that receive enough fresh real data in each generation remain stable (Alemohammad et al., 2024); in statistical models of the simplest kind, where purely self-referential maximum-likelihood retraining converges on degenerate states that lock in whatever biases the initial sample contained (Jangjoo et al., 2026); and in language, where the lexical and especially the syntactic diversity of model outputs declines across generations of recursive training, while semantic diversity remains comparatively stable, with the steepest decline in the tasks that demand the most creativity (Y. Guo et al., 2024). A theoretical analysis of mixed training in regression and random-feature models shows that if synthetic data makes up any non-vanishing fraction of a growing training set, additional data eventually stops improving the model, and that mixing strategies mitigate this only if real data grows at least as fast as synthetic data (Dohmatob et al., 2025).

These results acquire economic significance alongside evidence on the supply of human data. Compute-optimal training requires training data in proportion to model size: a study of more than 400 trained models found that model size and training data should be scaled in equal proportion (Hoffmann et al., 2022). For a fixed compute budget, training on up to about four epochs of repeated data changes loss negligibly compared with unique data, while further repetition adds progressively less until additional compute is worth almost nothing (Muennighoff et al., 2023). The effective stock of high-quality public human text is finite, and one projection puts the point at which the largest training runs would use all of it between 2026 and 2032 if current trends continue (Villalobos et al., 2024). Meanwhile, the web crawls from which that stock is drawn carry a rising

share of machine-authored text: a 2026 analysis using an automated detector estimated that about a tenth of a July 2026 sample of 10,000 English web pages, and more than a third of the pages in that sample published since the release of widely used chatbots, show significant signs of machine authorship, while cautioning that such detectors misclassify individual documents (Bestvater et al., 2026). And, as Section 2.7 showed, a growing share of the web domains that supply it have restricted automated collection (Longpre et al., 2024).

The value of human input is also visible in the methods by which current systems are made useful. Training with human preference judgments allowed a model with roughly one-hundredth as many parameters as its predecessor to produce outputs that human evaluators preferred (Ouyang et al., 2022), building on work showing that a relatively small quantity of human comparisons can substitute for a hand-specified objective (Christiano et al., 2017).

The case against strong dependence. Three bodies of evidence limit the assumption.

First, collapse depends on replacement. When synthetic data is added to an accumulating pool that retains the original human data, rather than replacing it, a linear-model analysis shows error remaining bounded however many generations pass, with consistent results in experiments across several kinds of model (Gerstgrasser et al., 2024), and under realistic fixed training budgets the degradation is gradual rather than catastrophic (Kazdan et al., 2024). A critical review, so far circulated as a preprint, identifies eight distinct and partly conflicting definitions of model collapse in the literature and argues that several prominent scenarios are readily avoidable (Schaeffer et al., 2025). Even Shumailov et al. (2024) found that retaining a tenth of the original data in each generation limited the damage to minor degradation.

Second, where an external check on correctness exists, self-generated data can substitute for human data. A system given no domain knowledge beyond the rules of chess, shogi and Go learned through self-play to defeat a world-champion program in each game (Silver et al., 2018). Reinforcement learning with automatically verifiable rewards produced substantial reasoning ability in a language model without human-written reasoning examples, though the final system still relied on curated human data and its authors identified domains without automatic verification as an unsolved problem (D. Guo et al., 2025). Synthetic data filtered by a sufficiently accurate verifier, even an imperfect one, avoided the collapse that unfiltered data produced in the tasks studied (Feng et al., 2024), and preference labels generated by a model can substitute for human labels in some settings, though the labels came from an off-the-shelf pretrained language model (Lee et al., 2024), a model that was itself, as this dissertation notes, trained on human-written text.

Third, systems trained by reinforcement learning can learn from interaction with an environment rather than from people. A system that learns a model of its environment and plans within it learned to perform well across more than 150 tasks in simulated environments, including collecting

diamonds in a complex game world from scratch, without human data (Hafner et al., 2025), building on earlier work showing that agents can be trained inside learned models of their environment (Ha & Schmidhuber, 2018). Robotics remains limited by the scarcity and fragmentation of real-world training data (Firoozi et al., 2025; Open X-Embodiment Collaboration, 2024), and assembling large real-world robot datasets still requires substantial human labor, with demonstrations gathered by human data collectors (Khazatsky et al., 2024), so that even embodied data, on present methods, depends heavily on human work. Policies trained in simulation can be transferred to physical robots, though with a persistent gap between the two (Zhao et al., 2020). The inference that the physical world could one day supply an external signal that neither runs out nor comes from people is this dissertation's, and it concerns the future rather than the present.

What the evidence supports. Two cautions apply to all of this work. Almost all of the evidence for collapse comes from small models or stylized theory, and, to the author's knowledge at the time of writing, no published study demonstrates the collapse of a frontier model under realistic data practices. And the evidence for substitution is concentrated in domains where correctness can be checked mechanically. The literature does not settle the question in one formulation, because its results depend on the training regime assumed. This dissertation adopts the following formulation, which holds under the regimes closest to present practice, in which synthetic data enters a growing or budget-constrained training set (Alemohammad et al., 2024; Dohmatob et al., 2025; Kazdan et al., 2024), and which Section 7.1 shows would weaken if developers retained and accumulated the human corpus already collected (Gerstgrasser et al., 2024):

> Learning systems need a continuing supply of information from outside their own generative loop. Where correctness can be verified mechanically, or where a faithful simulator exists, that information can come from verifiers, rules, simulators or the physical world, and dependence on people is weak. Where the target is human meaning, preference, value or new cultural production, no source other than people has been shown adequate: automatic verifiers do not exist for these targets (D. Guo et al., 2025), and the model-based judges that have substituted for human labels in some settings were themselves trained on human-written text (Lee et al., 2024). Under the regimes named above, people's contribution must keep pace with, rather than merely seed, the volume of synthetic data in training.

The second half of this formulation is adopted as a premise with supporting but not decisive evidence; Section 7.1 states the conditions under which it would fail. The formulation is narrower than the intuitive version of the assumption. It concedes the verifiable and simulable domains to the counterevidence and confines the claim of human necessity to the domains in which no substitute has been demonstrated. Chapter 3 formalizes the consequence: the human party's countervailing

power within the exchange rises with each synthetic party's dependence on this input, and the framework must be designed to hold even if that dependence turns out to be small.

2.10 Where the literature leaves the thesis

The literature reviewed here leaves Synthetic Exchange Theory with four obligations.

It must treat consent as resting on more than information, because the behavioral, legal and empirical literature reviewed in Section 2.2 converges on the finding that informational consent routinely fails, and it must locate the binding constraint in the voluntariness of refusal, which depends on exit. It must answer the relational objection with collective institutions rather than individual prices, while insisting that those institutions also satisfy the conditions of consent, reciprocity and exit at their own level. It must give the reciprocity condition a fairness standard that goes beyond mutual benefit, and it must address unfair exchanges by improving the weaker party's position rather than by prohibition. And it must state its empirical premise in the form the evidence supports, which is narrower and more defensible than the intuitive form, and design its institutions to survive the premise's weakening.

The next chapter takes up these obligations formally.

Chapter 3 — A Formal Statement of Synthetic Exchange Theory

The two preceding chapters argued in prose that the line between a good and a bad future with synthetic intelligence is drawn by the terms of exchange and cannot be drawn by the character of the systems alone. Prose permits a claim to be read more broadly when it is being defended than when it is being tested. This chapter therefore states the theory in a form whose content is fixed: a small vocabulary of defined terms, a set of propositions that follow from them, and a typology of exchange regimes that later chapters use as a diagnostic instrument. The formalism is deliberately modest. It is not a general equilibrium model of a post-labor economy, and it does not claim a precision the subject cannot bear. Its purpose is to make the dissertation's commitments explicit enough that a reader can see exactly where they would fail.

3.1 Why a formal statement is needed

Three ambiguities in the informal statement of the thesis justify the effort.

The first concerns the word *character*. The informal claim that outcomes depend on terms rather than character can be read as a strong claim, that a system's dispositions are irrelevant, or as a weak claim, that dispositions are insufficient. The strong claim is false: a system that actively seeks to deceive its counterparties will corrupt any institution it is placed inside. The weak claim is true but uninformative. The formal statement locates the thesis between them, which is why the thesis is stated as "cannot be drawn by the character of the systems alone." Character is treated as one input into the exchange, the agent's objective, and the terms are treated as the structure that determines how that objective translates into behavior. The thesis then becomes a claim about which of the two inputs institutional design can act on, and about the regime in which, once the terms bind, acting on them can be expected to suffice.

The second ambiguity concerns the relationship among the three conditions. In Chapter 1, consent, reciprocal value and real exit were introduced as tests that an exchange either passes or fails. They are not independent tests. Section 3.4 shows that consent leans on exit, that exit bounds reciprocity from below, and that the three are partially ordered, with exit prior to the other two.

The third ambiguity concerns power. The informal argument says that humans retain leverage because synthetic systems need their data. Leverage is a relational quantity, and it can be measured, at least in principle, by the dependence of each party on the other. Stating the argument in the vocabulary of power-dependence theory makes visible the dissertation's most important conditional: the framework is partly self-enforcing, to a degree that increases with the dependence of each synthetic party on human-generated input, and only while that dependence remains material. When it falls, or when it is spread so thinly across many synthetic parties that none of them bears the cost

of degrading its human counterparties, the framework's institutions must carry weight that the underlying exchange no longer carries. Chapter 7 returns to this as the first of its limitations.

3.2 Lineage: exchange as the unit of social analysis

The theory's name places it in a tradition. Social exchange theory, as developed in sociology and social psychology in the middle of the twentieth century, treated social life as the exchange of rewarding and costly activity between actors, and asked what structures of exchange produce stable, cooperative or exploitative relations (Blau, 1964; Homans, 1958; Thibaut & Kelley, 1959). Its most consequential contribution for present purposes is Emerson's (1962) analysis of power as relational; in his words, power "is a property of the social relation; it is not an attribute of the actor" (p. 32). The power of one actor over another equals the dependence of the second on the first, and dependence rises with the value the dependent actor places on what the relation supplies and falls with the availability of that value from outside the relation. Power, on this account, is not a trait. It is a consequence of the structure of alternatives.

That is the move this dissertation makes about synthetic intelligence. Much of the public and technical conversation about advanced synthetic intelligence, on this dissertation's reading of it, treats the danger as residing in the system, in its goals, its dispositions, its capability to deceive or to seize resources. Those concerns are real, and the alignment literature that addresses them is engaged in Chapters 5 and 7. But the exchange-theoretic tradition suggests a complementary diagnosis: much of what would make a world with powerful synthetic systems dystopian is not a property of the systems at all but a property of the relations in which humans are placed with respect to them. A person with no alternative to a single provider of medical monitoring, whose refusal to share data would cost them their health, is in a relation of extreme dependence regardless of whether the provider is benevolent. A provider in that relation holds power whose use is restrained by nothing but its own disposition, and the relation is secure only while that disposition persists.

The constitutional tradition reaches the same conclusion from another direction. Hume (1741/1987, pp. 42–43) defended the maxim that in designing institutions every person should be supposed a knave, not because he believed people were knaves but because institutions that work only when their occupants are virtuous fail at the moment virtue fails. Madison (1788) made the same argument in *Federalist* No. 51: if people were angels no government would be needed, and since they are not, the design task is to arrange the structure so that ambition counteracts ambition. Synthetic Exchange Theory extends this constitutional insight across a new boundary. Its governing maxim is that in designing the terms of exchange between people and synthetic systems, neither party should be supposed benevolent. The terms must be such that a self-interested system, dealing with self-interested people, produces outcomes that both could endorse.

3.3 Primitive terms

The theory uses six primitive terms. Each is defined to be substrate-neutral, in keeping with the choice of the word *synthetic* over alternatives such as *symbiotic* or *human–AI*: the framework must apply without modification to biological humans, artificial systems and the hybrids discussed in Chapter 8; Section 8.5 records what Chapter 3 does not yet supply for tissue-based parties in their own right.

Party. A party is any entity that can be a source or recipient of value in an exchange and whose welfare, or whose principal's welfare, the framework counts. A human being is a party. A synthetic system is treated as a party in the analytic sense that it has an objective that the exchange advances or frustrates; whether it is also a party in the moral sense, an entity whose welfare counts for its own sake, is left open in this chapter and taken up in Chapter 8. The propositions below hold for parties in the analytic sense regardless of how the moral question is resolved.

The phrase "or whose principal's welfare" carries more weight than its brevity suggests, and it requires an account of who the synthetic party is. In present law, a deployed model is not a legal person and does not own anything, and the question whether one could be has been examined as a hypothetical rather than as doctrine (Solum, 1992). As Section 1.2 stated, it is not an independent living entity but an artifact that runs when its operator runs it. The entity with interests that can be advanced or frustrated is therefore the operator or developer that trains, deploys and profits from it. The relation between that operator and its system is a principal–agent relation of an unusual kind. The principal sets the system's objective through training and deployment, the agent pursues that objective with considerable autonomy in each exchange, and the agent's conduct may diverge from what the principal intended (Hubinger et al., 2019). This dissertation therefore treats the synthetic party as a composite: its *objective* is the model's, as trained and deployed, but its *stake* in the future of the exchange, its revenue, its access to data, its liability, belongs to the principal. Three consequences follow. Proposition 4 below concerns the principal's dependence on human input, transmitted to the system only to the extent that the principal writes it into the system's objective. The redemption obligation of Chapter 4 binds the owners of compute capacity, which, where a principal owns the capacity that runs its system, is the principal; a trained artifact cannot be made to honor it. And the question of how to count synthetic parties, which Chapter 5 cannot avoid, is a question about principals and about systems at once. It is made harder by the possibility that advanced synthetic intelligence is better modeled as a collection of bounded services than as a unitary agent (Drexler, 2019), in which case there may be no single system to which the label "party" naturally attaches. If the question of a system's own moral status, which Chapter 8 argues may eventually become live, were answered in the system's favor, the composite would come apart and the system would become a party in its own right.

Flow. A flow is a transfer of something of value from one party to another: data, labor, attention, compute, physical goods, services or tissue. Each flow has a source, a recipient, content and an informational shadow, meaning the set of parties, the source among them, about whom the recipient can draw inferences from the flow. The informational shadow is what makes data relational (Viljoen, 2021) and is the reason individual consent cannot by itself legitimate a flow.

Exchange. An exchange is a pair of opposing flows between two parties, or between a party and a collective, that each party enters because of the other flow. The defining feature of an exchange, as distinct from a gift or a taking, is conditionality: each flow is given because the other is received.

Floor. A party's floor is the bundle of goods and services without which the party cannot continue to function or to participate in any other exchange. For a human being, the floor is survival and basic participation, and this dissertation fixes its contents once, here, as power, water, food, shelter, basic communication and emergency care. For a synthetic party, the analogue is the minimum of energy, substrate upkeep and data required to keep operating; for a present artificial system, that is compute, energy and data. The floor is defined physically rather than in currency. Its value at current prices, which Appendix A writes $\bar{f}$, is what a person's disagreement payoff must reach for the conditions below to hold. Chapter 4 therefore delivers the floor through a survival tier that is a percentage of each allocation, so that it grows with the economy, subject to a binding physical minimum, so that it does not fall below the bundle it exists to secure. That guarantee holds while the agreed share of capacity can buy the bundle; Chapter 4 and Appendix A state the case in which it cannot.

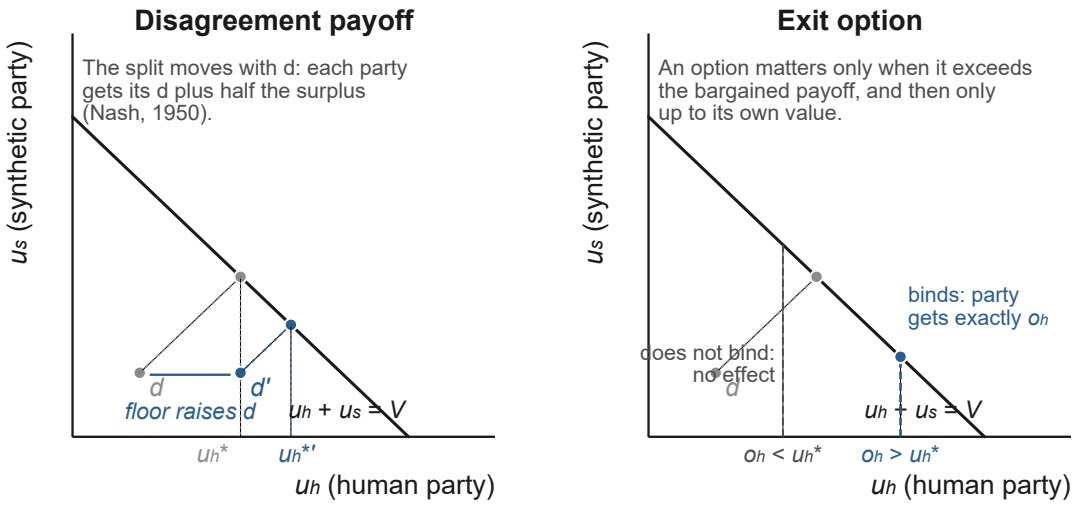
Disagreement payoff. A party's disagreement payoff is what that party receives while an exchange is not taking place, that is, during refusal, negotiation or impasse. Bargaining theory shows that the division of the gains from an exchange depends on the parties' disagreement payoffs (Nash, 1950, pp. 158–159). A party whose disagreement payoff is below its floor cannot afford to disagree for long and therefore cannot bargain at all.

Exit option. A party's exit option is the best alternative exchange available to it if the current one ends: another provider, another employer, another data purchaser or self-provision. The distinction between a disagreement payoff and an exit option carries design consequences. Strategic bargaining models and laboratory evidence show that an outside option affects the terms of a bargain only when its exercise is credible, that is, only when the party would actually take it, whereas the payoff received during impasse shapes the bargain continuously (Binmore et al., 1986; Binmore et al., 1989). When the option binds, the weaker party receives exactly the option's value, which is then more than it would otherwise have obtained; the option therefore protects only up to its own value, and that value can be lowered by the counterparty through switching costs until the option no longer binds and protects nothing. A disagreement payoff that cannot be taken away is a protection of a different

kind, because it moves the bargained share itself and does not depend on anyone's willingness to leave. The non-tradable survival tier of Chapter 4 is designed to be such a payoff. Figure 3 shows the difference in the bargaining set.

Figure 3

A Disagreement Payoff Moves the Bargained Share; an Exit Option Only Bounds It



Note. Both panels plot the feasible frontier on which the parties' payoffs sum to the joint value of agreement. Left: raising the human party's disagreement payoff from d to d' moves the Nash solution along the frontier toward the human party (Nash, 1950). Right: an outside option changes the outcome only when it exceeds the bargained payoff, and then the human party receives exactly the option's value (Binmore et al., 1989); the counterparty can lower that value by raising switching costs, which it cannot do to a payoff received during impasse. The points are illustrative.

3.4 The three conditions, stated precisely

With the primitive terms in hand, the three conditions of Chapter 1 can be restated so that each has a determinate content and the relations among them become visible.

Condition E (Real exit). An exchange satisfies real exit for a party if and only if (i) the party's disagreement payoff is at or above its floor, (ii) the party has at least one exit option whose value net of switching costs is not drastically lower than the value of the current exchange, and (iii) exercising the exit option does not destroy assets the party has accumulated within the exchange, including data, reputation and records of entitlement. Clause (i) rules out lethal dependency. Clause (ii) rules out lock-in. Clause (iii) is the portability requirement, and it is what data-portability rights in existing law attempt to secure (Regulation [EU] 2016/679, art. 20; Regulation [EU] 2023/2854).

Condition C (Consent). An exchange satisfies consent for a party if and only if (i) the party has access to an accurate account of the flows the exchange involves, including their informational shadows, in a form the party can understand; (ii) the party can refuse the exchange, entering no other in its place, without falling below its floor, which is to say that its disagreement payoff is at or above its floor; and (iii) the party can revoke its participation at a cost bounded by the value of what it has already received. Clause (i) is the informational condition that privacy scholarship has shown to be routinely unmet (Acquisti et al., 2015; Solove, 2013). Clause (ii) is where consent depends on exit, and its wording matters: a party that could avoid falling below its floor only by switching to another provider has an exit option, not a disagreement payoff at the floor, and the clause requires the latter, for the reason Section 3.3 gave. Clause (iii) adds the temporal dimension: consent that cannot be withdrawn is a sale.

Condition R (Reciprocal value). An exchange satisfies reciprocal value if and only if (i) each party's net gain from the exchange, measured against its disagreement payoff, is positive; (ii) each party can observe what it receives and what it gives, so that the gains are legible; and (iii) the division of the joint surplus is not determined solely by an unchosen asymmetry in the parties' bargaining positions, whether in their disagreement payoffs or in their capacity to wait. Clause C(i) concerns the content of the flows and their shadows; clause R(ii) concerns their value, and the two can fail separately. Clause R(iii) is the fairness clause, and it is the clause that requires governance, because the market alone will divide surplus according to bargaining power. It is needed because, as the philosophy of exploitation has shown, an exchange can be consensual and beneficial to both sides and still take unfair advantage of one of them (Zwolinski et al., 2022). The standard adopted here is the one the exploitation literature reports: the division that would obtain in a competitive market free of informational asymmetry and undue pressure (Zwolinski et al., 2022). A division that departs from that benchmark because one party's disagreement payoff lies below its floor, because one party can outwait the other on an accumulated stock, or because of structural bargaining power the weaker party did not choose, such as concentration or lock-in, fails the clause; the floor corrects the first of these, and the residue is what governance must address. A division that reflects the parties' genuine alternatives above the floor does not fail it. Mutual benefit is the minimum reciprocity requires, not its standard.

With the conditions stated this way, their structure becomes clear.

> **Proposition 1 (Exit primacy).** Consent presupposes the first clause of exit. If a party's disagreement payoff is below its floor, then no agreement that party gives satisfies Condition C, however well informed it is.

The derivation is immediate because the definitions were drawn to make it so: clause C(ii) requires that refusal, with no other exchange entered in its place, leave the party at or above its floor, which is

E(i). The proposition's real content is the claim that C(ii), so worded, belongs in any adequate definition of consent: that a party able to refuse one provider only by submitting to another has not been given the ability to refuse. The unconditional floor of Chapter 4 is designed to supply the disagreement payoff the clause requires. The proposition formalizes a point that the bioethics literature on informed consent has long made in its own terms: that information is necessary but not sufficient for valid consent (Faden & Beauchamp, 1986, pp. 241–273; Manson & O'Neill, 2007), and that agreement given under the threat of losing something one cannot do without is not substantially free of controlling influences, which is the condition of voluntariness in Faden and Beauchamp's account (Faden & Beauchamp, 1986, pp. 241–273, 277–280). Its practical implication is that no amount of improvement in notice, disclosure or interface design can repair a consent regime whose subjects cannot afford to refuse. The policy effort directed at better privacy notices is not wasted, but it is directed at clause C(i) while the binding constraint is often clause C(ii). The definition has a consequence that should be stated. Under it, consent to a service without which a person cannot live is valid only where the service's continuation is secured independently of the consent, a guarantee that, on this dissertation's reading, existing institutions do not give in general; a patient who can choose among providers of such a service but cannot refuse it altogether has, on this definition, an exit option and not the ability to refuse. The framework accepts that verdict, and it is the reason the floor includes emergency care and the requirement of Chapter 8 that a baseline of function survive a change of provider must cover life-sustaining devices.

> **Proposition 2 (Reciprocity as the reason to stay).** If Condition E holds, switching costs are negligible, and the party can observe its gain (clause R(ii)), then a party that acts in its interest remains in an exchange only if its payoff in the exchange is at least the value of its best exit option; since that option is itself an exchange the party would enter only if it at least matched its disagreement payoff, the party's net gain is non-negative in every exchange in which it persists, and so is each party's where the premises hold for both, which is clause R(i) with its strict inequality relaxed. Strict reciprocity requires that the best alternative itself better the disagreement payoff, which clause E(ii) does not guarantee. Reciprocity in its first clause is then not an independent ethical requirement imposed on top of exit; exit bounds it from below. Its fairness clause is not so bounded.

When a party can leave at negligible cost and can see what it gains, it will leave an exchange that leaves it worse off than its alternatives. An exchange that persists under such conditions therefore reveals that the party is not losing value relative to its alternatives. Where switching costs are material, a party may rationally remain in an exchange that gives it nothing beyond what it would get on impasse, because leaving would cost more than staying; this is the gap that clause E(ii) is meant to keep small. The converse is the diagnostic that matters: an exchange that persists while delivering negative value to one party is evidence that clause E(ii) is failing for that party, or that the party

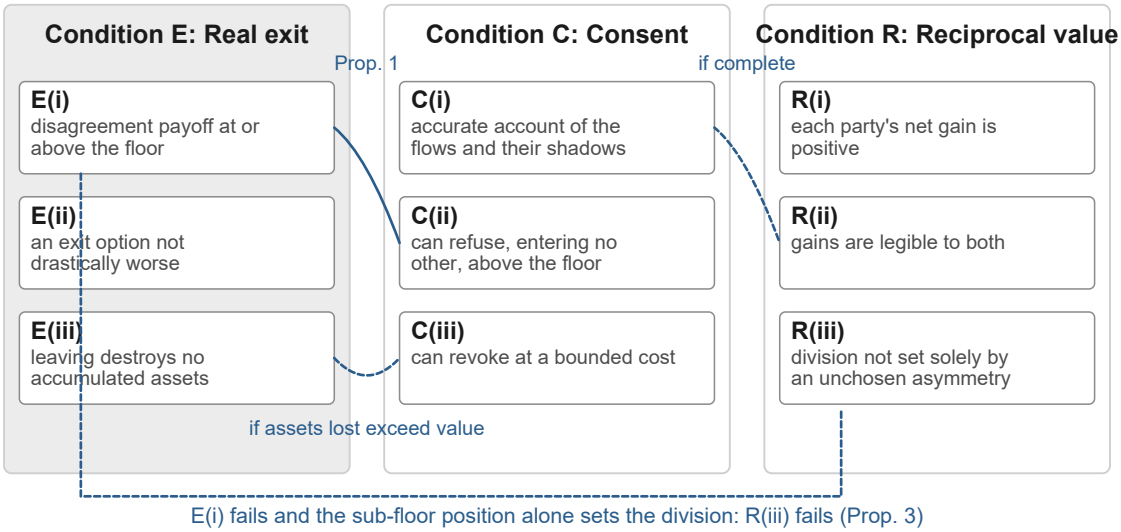
cannot see what it gains, which is a failure of clause R(ii). Persistence without legible reciprocity is a symptom either of lock-in or of opacity.

> **Proposition 3 (The triad is partially ordered, with exit prior).** Violations of the three conditions differ in how much damage they do and in whether they can be corrected from inside the exchange. A violation of the first clause of exit entails a violation of clause C(ii) and, where the sub-floor position is unchosen and determines the division by itself, of clause R(iii); clauses R(i) and R(ii) may still hold. A violation of the second or third clause of exit leaves the other conditions intact at the moment of exchange but removes the means of correcting their later failure; the one exception is that a violation of clause E(iii) carries a violation of clause C(iii) with it where the assets destroyed on leaving exceed the value received. A violation of the revocability clause of consent, with exit intact, is correctable by exit; a violation of the voluntariness clause is a violation of exit itself, by Proposition 1; a violation of the informational clause is correctable by exit only once the party knows enough to want to leave. A violation of reciprocity in its first two clauses, with consent and exit intact, is corrected by departure, unless the same violation is common to every available alternative. A violation of clause R(iii) is only partly corrected by exit: the floor removes the asymmetry that comes from a disagreement payoff below the floor, and an outside option bounds the division from below, but neither determines the division of the surplus that remains, which follows from bargaining power that the party did not choose and that governance alone can address.

The ordering the proposition establishes is partial. Exit is prior to the other two conditions because its failures cannot be corrected from inside the exchange; consent and reciprocity are not ordered with respect to each other, since, among the clauses of those two conditions, the fairness clause is the least correctable by departure and the first two clauses of reciprocity the most. The priority is one of correctability from inside the exchange, and the design conclusion drawn below needs only the priority of the floor clause. The qualifications matter. When every provider of a service shares the same defect, departure is what Hirschman (1970) showed exit to become under industry-wide decline: ineffective movement between alternatives that are equally bad. And the fairness clause of reciprocity is only partly within the reach of exit. Even so, the ordering survives as a statement of priority, and it is the dissertation's principal reason for designing the economic layer around a guaranteed floor rather than around better pricing of data. Pricing addresses reciprocity. The floor addresses exit. If the order is right, the floor comes first, and governance is needed for what neither can reach. Figure 4 shows the nine clauses and the dependencies among them.

Figure 4

The Three Conditions, Their Nine Clauses, and the Dependencies Among Them



Note. An arrow runs from a clause whose failure entails the failure of the clause at its head. Solid arrows are unconditional (Proposition 1); dashed arrows hold under the stated condition. Exit is prior because its failures cannot be corrected from inside the exchange (Proposition 3); consent and reciprocity are not ordered with respect to each other.

3.5 Power and the grounding assumption

The second analytic tool is Emerson's (1962) definition of power as dependence. For two parties *A* and *B*, let the dependence of *B* on *A* increase with *B*'s valuation of what the relation with *A* supplies and decrease with the availability of that value from parties other than *A*. The power of *A* over *B* equals the dependence of *B* on *A*. The relation is balanced when the two dependences are equal and unbalanced otherwise. Of the four ways in which Chapter 2 showed an unbalanced relation can be rebalanced, three have direct analogues in the design that follows. The survival tier and the portability requirements extend the human party's network, by making floor goods and records available independently of any single provider; the same tier makes possible a measure of withdrawal, since a person whose floor is secured can afford to care less about what any one provider offers; and data cooperatives and the collective-choice arrangements of Chapter 5 are coalition formation.

The grounding assumption plays a different role. It is not a rebalancing operation at all but a statement about dependence running the other way: it asserts that synthetic parties need something only people can supply, and so that the relation is less unbalanced than it appears. This reformulation makes the dissertation's central empirical premise precise, and it also exposes a difficulty the informal argument concealed.

Proposition 4 (Conditional and divided self-enforcement). Let $\varkappa_j$ denote the dependence of synthetic party *j*, through its principal, on the human-generated input of a given population. Party *j* has a reason to preserve that input, to a degree that increases with $\varkappa_j$, provided that the principal's dependence is written into the system's objective and that the principal weighs the future value of its inputs against present gains. That reason makes fair terms partly self-enforcing only insofar as the quality of the population's input depends on the terms the population receives: because a population that can refuse and leave can withhold what the party needs, and because, on this dissertation's inference, contribution under domination narrows toward what the dominant party extracts and loses the variety that gives it value. Where input can be extracted on unfair terms without loss, dependence protects the supply and not the terms. As $\varkappa_j$ falls toward zero, the self-enforcing component vanishes. When many synthetic parties draw on the same population, the welfare and data quality of that population become a common-pool resource for them: each bears only its own share of the cost of degrading it, so that the incentive of any one party to preserve it can be small even when the dependence of all of them together is large.

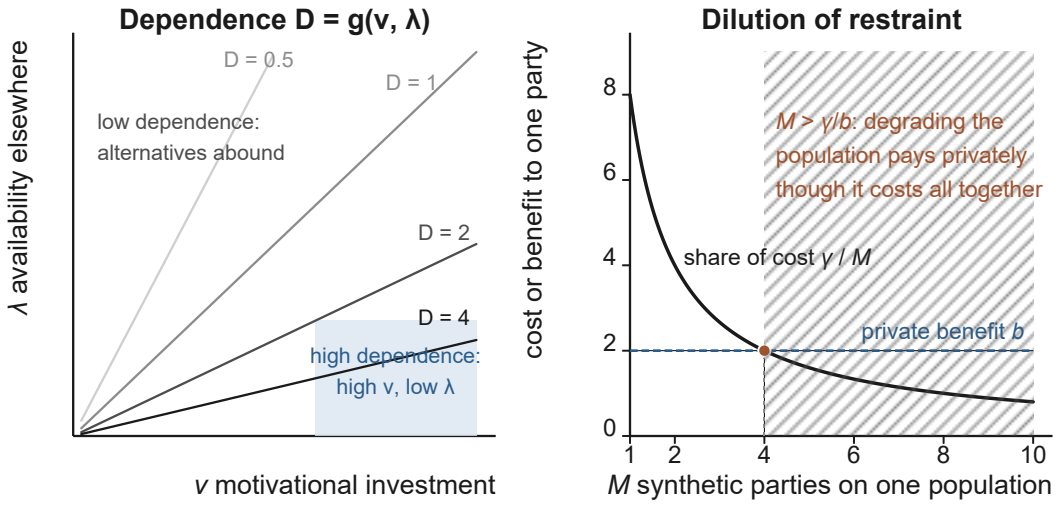
The research on recursive training reviewed in Chapter 2 bears directly on the size of $\varkappa_j$. If models trained on their own outputs lose the tails of the distribution and drift (Shumailov et al., 2024), human data has a value that synthetic data cannot replace. If, as other work suggests, accumulation of real data alongside synthetic data (Gerstgrasser et al., 2024; Kazdan et al., 2024), or a modest continuing supply of fresh real data (Alemohammad et al., 2024; Shumailov et al., 2024), is sufficient to prevent collapse, though at least one formal result requires that supply to grow in proportion to the synthetic data (Dohmatob et al., 2025), then dependence is positive but small, and the human party's leverage is correspondingly modest. Proposition 4 is written so that the framework does not stake its normative conclusions on the most favorable reading of that literature. Two of its premises deserve emphasis because they are easy to pass over. The first is that the withholding by which a population disciplines terms presupposes that the population can refuse and leave: self-enforcement is therefore a support for the institutions that secure exit, not a substitute for them, which is the sense in which Chapter 7 calls it a support and not a foundation. The second, that extracted contribution loses value, is the dissertation's inference from the grounding assumption as Chapter 2 narrowed it, since the domains in which people are the only adequate source are those of meaning, preference, value and culture, and it is argued, not shown.

The second half of the proposition matters as much as the first. It is Hardin's (1968) logic of the open-access commons applied to the synthetic side of the exchange: aggregate dependence on people does not translate into individual restraint by any synthetic party when each can free-ride on

the restraint of the others. This does not weaken the case for the framework. It strengthens the case for its governance layer, because a commons is what the institutions of Chapter 5, grounded in the record of how commons are governed, are designed to protect. The result also exposes a tension within the design. Exit requires many synthetic parties; self-enforcement is strongest when there is one. The framework resolves the tension in favor of exit, and therefore cannot rely on self-enforcement in the competitive market it seeks to create; the governance layer must carry that weight, and Chapter 7 records this as a limitation. The institutional layers of Chapters 4 and 5 are therefore built to hold whether dependence is small, or large but divided. What changes as dependence falls or fragments is not the justification of the institutions but the degree to which they can rely on the self-interest of synthetic parties to sustain them. Figure 5 shows both halves of the proposition.

Figure 5

Dependence Rises With Value and Falls With Availability; Divided Among Many Parties, It Loses Its Restraining Force



Note. Left: contours of D = v/λ, one admissible form of Emerson's (1962) dependence function; dependence is high where investment is high and alternatives are few. Right: the share γ/M of a collective cost γ borne by one of M synthetic parties, drawn with γ = 8 and b = 2; once M exceeds γ/b, degrading the population pays privately although it costs all of them together (Appendix A.5; Hardin, 1968). The parameters are illustrative.

3.6 Bargaining, the floor and the logic of the economic layer

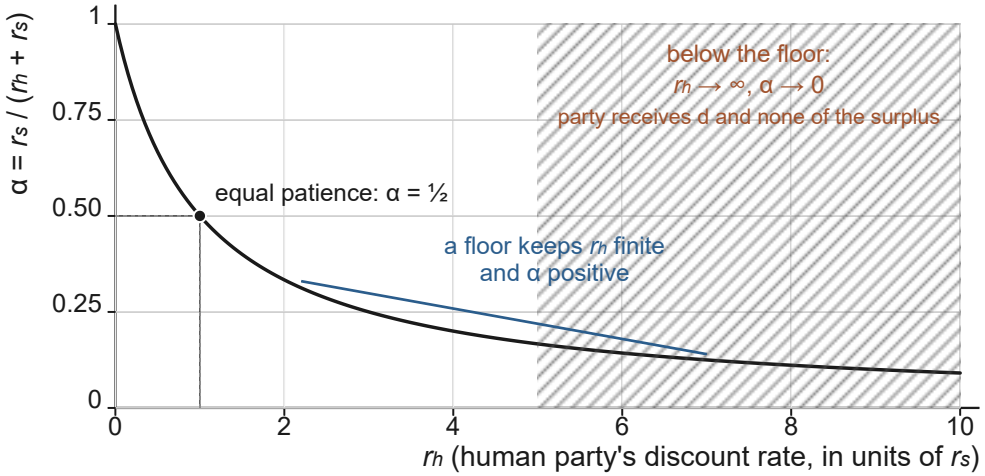
The third analytic tool is bargaining theory, which supplies the formal reason for the most distinctive features of the economic design. Appendix A gives the derivations.

Consider an exchange between a human party *h* and a synthetic party *s* that creates a joint surplus. In the axiomatic solution to the two-party bargaining problem, the surplus is divided so as to maximize the product of each party's gain over its disagreement payoff (Nash, 1950, p. 159). Strategic bargaining models in which parties alternate offers converge to this solution as the frictions of delay vanish, provided the disagreement point is chosen to reflect what parties receive while negotiations are under way, and in those models each party's impatience is part of what determines the outcome (Binmore et al., 1986). When the outside option is instead an alternative a party may or may not take, it affects the outcome only when it binds (Binmore et al., 1989).

The most important consequence for policy concerns impatience. If, as the floor's definition implies, payoffs below it carry a cost of delay that rises sharply (a survival constraint the Nash model does not itself contain), a party whose disagreement payoff lies below its floor cannot sustain disagreement for longer than its reserves allow; in a model of alternating offers it behaves as a party with a very high cost of delay, and such a party concedes. Raising the disagreement payoff to the floor removes this source of weakness, and it does so in every bargain the party enters, not because the floor is added to each bargain separately but because it is the background against which each is conducted. This is the main bargaining-theoretic reason for the floor, and it does not depend on whether the tier substitutes for anything the counterparty supplies, though it does depend on the survival constraint just stated. Figure 6 plots the bargaining weight against the human party's discount rate.

Figure 6

Impatience Below the Floor Drives the Human Party's Share of the Surplus Toward Zero



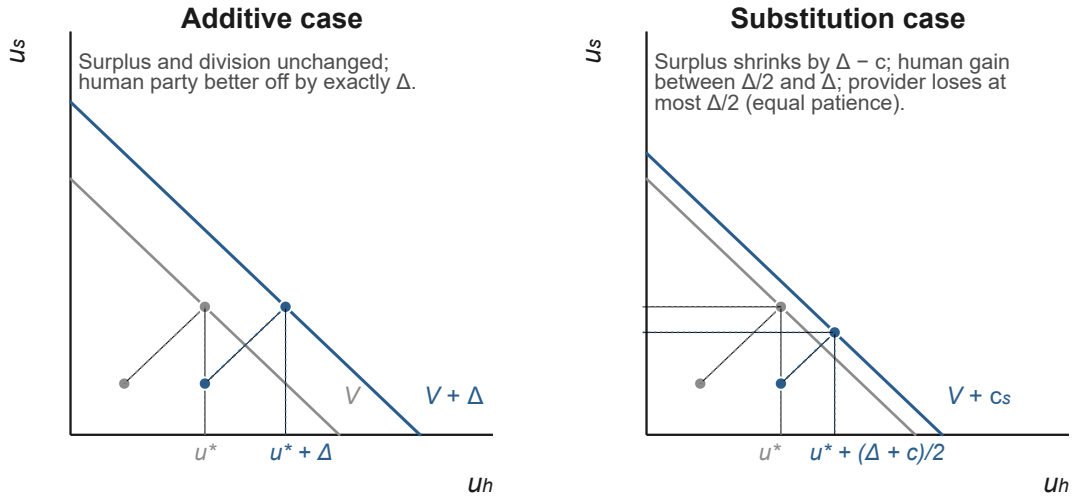
Note. The weight α = r_s/(r_h + r_s) is the limit of the alternating-offers model with unequal discount rates (Binmore et al., 1986), drawn for a fixed r_s. The survival constraint that sends r_h to

infinity for a party below its floor is a stated assumption (Appendix A.3.2); under it the party receives its disagreement payoff and none of the surplus. A floor keeps r_h finite and α positive.

A second consequence concerns the division of surplus, and here the intuitive version is wrong. Holding the value of agreement fixed, raising one party's disagreement payoff raises that party's share of the agreement. But an unconditional floor, received whether or not agreement is reached, raises both the party's payoff on disagreement and its payoff on agreement by the same amount. It therefore leaves the surplus, and its division, unchanged: the party is better off by exactly the value of the floor, no more. The floor lowers the counterparty's payoff only where it substitutes for something the counterparty would otherwise have supplied within the exchange, for example where a person who once had to buy floor goods from a provider now receives them through the survival tier; in neither case does it change how the remaining surplus is divided. In the substitution case the disagreement payoff rises while the joint surplus the exchange must create shrinks, because part of what the provider supplied is no longer needed. Appendix A shows that, when the provider's cost of supplying those goods does not exceed their value to the person and the parties are equally patient, the person's net gain from a tier worth a given amount lies between half that amount and the whole of it, and the provider's payoff falls by at most half the tier's value; under unequal patience the fraction one half is replaced by the provider's bargaining weight. Where the provider's cost of supplying floor goods exceeds their value to the person, as it may where a provider supplies them in order to create dependence, the surplus grows instead and both parties gain. The substitution case is therefore not a benefit beyond the tier. It is the case in which part of the tier's value is absorbed by the shrinkage of a surplus the exchange no longer needs to create, and the fall in the provider's payoff is the reduction in dependence that power-dependence theory predicts when an alternative source of a valued good appears. Figure 7 shows the two cases.

Figure 7

What an Unconditional Floor Does to the Division of Surplus: The Additive and Substitution Cases



Note. Drawn with V = 10, d = (2, 2), Δ = 3 and c_s = 1 under a symmetric Nash split (Appendix A. 3.1). Left: the floor raises the human party's disagreement payoff and the joint value by the same Δ, so the division is unchanged and the human party is better off by exactly Δ. Right: the floor replaces goods the provider supplied at cost c_s, the surplus shrinks by Δ − c_s, the human party gains between Δ/2 and Δ, and the provider's payoff falls by at most Δ/2. The working draft's claim of a Δ/2 gain in every bargain was the substitution case misread as general.

Three design consequences follow.

First, the floor should be paid as a disagreement payoff, not offered as an exit option. A floor that is received whether or not one participates in any given exchange is received during every impasse, and so removes the impatience that would otherwise force concession. An exit option, by contrast, protects the weaker party only up to its own value, which the stronger party can lower by raising switching costs. This is why the economic layer delivers survival goods through an automatic, unconditional earmark, backed by the acceptance obligation of Section 4.5, rather than through a guarantee that a person may seek them elsewhere.

Second, the floor must be inalienable. A disagreement payoff that can be sold, pledged or borrowed against can be traded away in a moment of need, after which it no longer functions as a disagreement payoff. Legal theory distinguishes entitlements protected by property rules, which may be transferred with the holder's consent, from entitlements protected by inalienability rules, which may not be transferred even with consent (Calabresi & Melamed, 1972, p. 1092). Radin (1987) argued that some goods should be market-inalienable because their commodification threatens the personhood of those who hold them, although she also warned that, in a nonideal world, prohibiting sale can harm the very people it is meant to protect, a dilemma she proposed to mediate

through incomplete commodification. The survival tier is designed to respect that warning: it forbids no one to sell what they already have, and instead adds an entitlement that cannot be sold. The survival tier is the application of inalienability to the disagreement payoff. It is non-tradable and non-borrowable not because the framework distrusts its beneficiaries but because a floor that can be alienated stops being a floor at the moment it is needed (Appendix A.3.3).

Third, accumulated balances are bargaining power, and their accumulation should carry a cost. A party that holds a large stock of the exchange medium can afford to wait in every exchange it enters. In a two-tier system, the survival tier secures every party's disagreement payoff at the floor, but the discretionary tier can reintroduce inequality above it, and inequality of stocks is inequality in the capacity to wait. A holding cost on idle balances, demurrage, is one part of the instrument that acts on this channel; Section 4.6 shows that without a rule on capacity it has no distributive effect of its own. It lowers the advantage associated with merely holding wealth while leaving untouched the returns to putting it to use. Chapter 4 develops the history and the limits of this instrument. The formalism does not derive this third consequence: in Appendix A, a stock enters the bargain only through the survival constraint, and the link between accumulated balances and the capacity to wait above the floor is an inference from the model's logic rather than a result in it.

The bargaining frame thus connects the three main elements of the economic layer. The survival tier secures every party's disagreement payoff at the floor, wherever the agreed share of capacity can buy it and the floor goods can be obtained for the unit (Section 4.5), and so removes the impatience that would otherwise force concession. Its inalienability prevents that payoff from being bargained away. Demurrage, with the capacity rules of Chapter 4, dampens the conversion of accumulated stocks into a superior capacity to wait. None of these elements was introduced for this reason when the design was first sketched; the formalism reveals a coherence in the first two that the design already had, supplies the logic though not a derivation of the third, and corrects an overstatement about where the floor's bargaining effect comes from.

3.7 Collective consent and the problem of the informational shadow

The fourth analytic problem is the one the consent literature regards as most serious for any individual-choice framework. Data about one person licenses inferences about others who resemble them. The flow from source to recipient therefore has an informational shadow that falls on people who did not consent to it (Barocas & Nissenbaum, 2014; Viljoen, 2021). If the shadow is large, the sum of individually valid consents can authorize a collective harm that no individual consented to.

Synthetic Exchange Theory responds by distinguishing two levels at which each condition must hold. At the *individual level*, the conditions govern a person's relation to a provider. At the *collective level*, they govern a population's relation to the synthetic systems that learn from it, where the

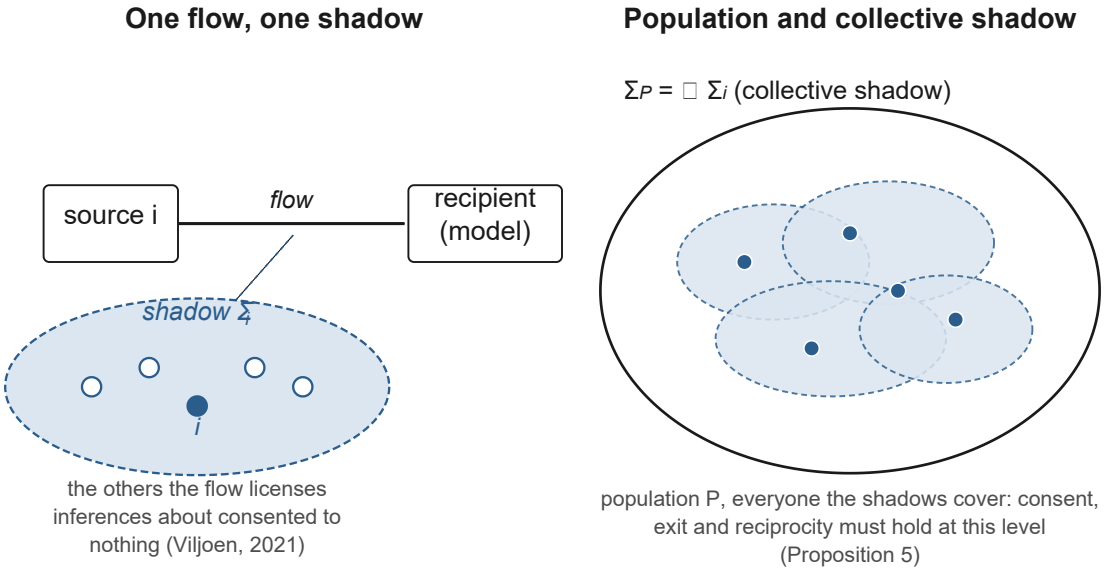
population is defined by the extent of the informational shadow, that is, by the set of people about whom the system can draw inferences from the data in question. Collective consent requires a collective-choice procedure. Collective exit requires that the population, as a body, be able to withdraw its data or move its custody. Collective reciprocity requires that the population, as a body, share in the value the data creates.

> **Proposition 5 (Level matching).** An exchange whose informational shadow extends beyond its source satisfies the three conditions only if they are satisfied at the level of the population that the shadow covers. Individual consent is necessary for individual flows and insufficient for relational ones.

Proposition 5 is a requirement the framework imposes rather than a result derived from the definitions, and it needs the collective-level conditions to be given content, since Conditions E, C and R were defined for a party to an exchange. The framework gives them the following content. The population's floor is the floor of each of its members. Its disagreement payoff is what its members receive while the population, acting through the collective-choice institutions of Chapter 5, withholds its data. Its exit option is the transfer of the custody of future flows to another steward, through the institutions Chapter 2 reviewed. Its consent and its share of value are exercised and received through those same institutions. Because almost any informative flow casts some shadow beyond its source, the shadow that triggers the requirement is the set of parties about whom the flow materially changes what the recipient can infer, and the threshold of materiality is a parameter the governance layer must set; without it, no individual exchange about population-descriptive data could ever satisfy the conditions in full. Figure 8 shows the shadow of a single flow and the collective shadow of a population.

Figure 8

The Informational Shadow of a Flow and the Level at Which Consent Must Be Given



Note. Left: a flow from source i licenses inferences about a set of parties, its shadow Σ_i, which includes i and others who consented to nothing (Viljoen, 2021). Right: the collective shadow of a population is the union of its members' shadows, and Proposition 5 requires the three conditions at that level, above the materiality threshold the governance layer sets (Appendix A.6).

The proposition is not novel in substance; it is the formal core of arguments for data trusts, data cooperatives and collective bargaining over data (Ada Lovelace Institute, 2021; Delacroix & Lawrence, 2019; Hardjono & Pentland, 2019). What the theory adds is a rule for locating the right level, namely the extent of the informational shadow, and a conjecture about where collective institutions will be needed most. The information-gain pricing of Chapter 4 values data by how much it reduces a model's uncertainty about its parameters. Where those parameters describe populations rather than individuals, which is the usual case for the models that matter here, data that reduces uncertainty most is data that tells the model most about people other than its source. If so, the highest-priced flows will tend to be those for which individual consent is least adequate. The conjecture can fail: a flow that is highly informative about a single rare individual may carry a high price and a small shadow. But to the extent that it holds, the economic and governance layers must be designed together.

Hirschman's (1970) analysis of exit and voice supplies the remaining link. Where exit is costly, members of an organization fall back on voice, the attempt to change it from within; where exit is easy, voice can atrophy. At the collective level, exit is usually costly, because a population cannot easily withdraw its data from a model already trained on it. The collective-level conditions must therefore be secured primarily through voice, which is the role of the governance layer, and secondarily through custody arrangements that make collective exit at least possible for future flows.

This qualifies the reach of Proposition 3 at the collective level. The ordering holds, but the exit it places first is secured there through voice, the collective-choice institutions of Chapter 5, rather than through departure, since a population cannot withdraw data already absorbed into a trained model; and the partnership regime of Section 3.9 is reachable for population-descriptive data only for future flows whose custody can be moved.

3.8 Character, terms and the scope of the thesis

The formal apparatus now permits a precise statement of the central thesis.

Let a synthetic party's behavior in an exchange be determined by two things: its objective, meaning the function its training and deployment have made it optimize, which Section 1.2 and the composite definition of Section 3.3 locate in the trained artifact and its principal, and the terms of the exchange, which determine what it can obtain by which actions. Call the objective the party's *character* in the sense relevant to the thesis. Call the terms the *institutional structure*. A relation is one of *domination*, in the sense this dissertation gives the word, when one party's outcome depends on the other's will and the first cannot alter that dependence. The definition is symmetric in form: it applies with the roles exchanged, so that a synthetic party in its own right, or the principal in whom a system's stake resides, is dominated when its outcome depends on a human party's will that it cannot alter; for a system that is not a party in its own right, control names the principal's relation to its artifact, not a relation between parties. Call the class of synthetic parties whose objectives are specified by accountable designers and remain stable in deployment the *designed-incentive regime*. The thesis has three parts.

> **Thesis (Insufficiency of character).** Benevolent character is not sufficient for an outcome whose persistence the weaker party can secure, and clause E(i) of Condition E, a disagreement payoff at or above the floor, is necessary, for a party for which clause E(ii) also fails, to any relation that is not one of domination. An exchange in which the human party's disagreement payoff lies below its floor and which it cannot leave is a relation of domination even if the dominant party is benevolent, because the outcome then depends on the persistence of benevolence, which the weaker party cannot secure. Where the sub-floor party has an exit option that satisfies clause E(ii), no single counterparty's will determines its outcome and it is not dominated in the stipulated sense; it is exposed instead to the erosion of its options described in Section 3.3, which is why the framework secures clause E(i) rather than relying on clause E(ii). A failure of clause E(ii) or E(iii) alone leaves the party dependent for its surplus but not for its survival, and the party can alter that dependence by withdrawing to its floor; such a failure is a defect of the exchange, not a relation of domination.

Thesis (Sufficiency conjecture in the designed-incentive regime). For synthetic parties whose objectives are specified by accountable designers and remain stable under the conditions of deployment, fair terms, satisfied at the individual level and, under Proposition 5, at the level of every population the exchange's informational shadow reaches, are sufficient for outcomes acceptable to the human parties and to the principals of the synthetic parties to the exchange.

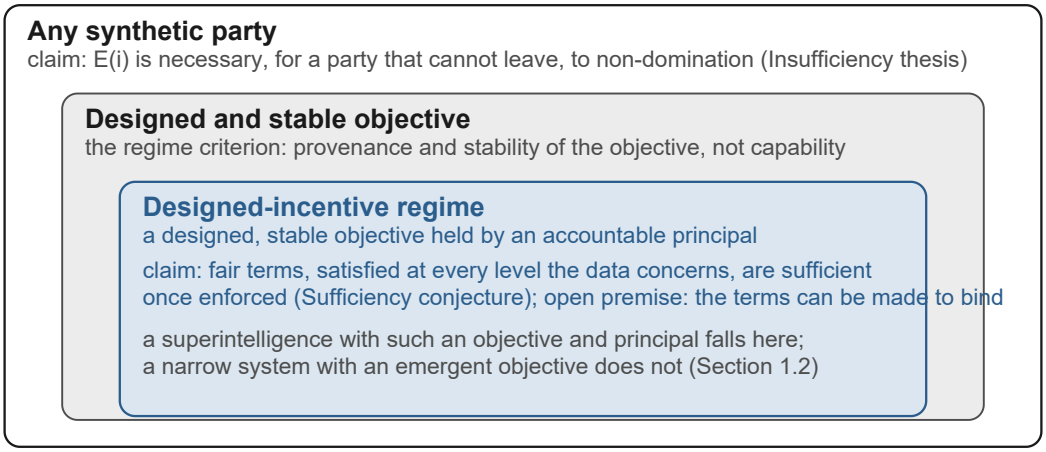
Thesis (Scope). The sufficiency conjecture does not extend to synthetic parties whose objectives are emergent, unstable or concealed. For such parties the framework provides only the necessary condition of the first thesis, and the alignment problem remains open.

The first thesis is the reason the dissertation rejects a frame in which safety is achieved by making systems good and leaving relations of total dependence in place. Its argument is the one just given, and, as with Proposition 1, the necessity follows from the stipulated definition of domination; what the thesis asserts beyond the definition is that benevolence is not a substitute for the clause, because the weaker party cannot secure its persistence, and that an exit option is not a substitute either, because its value can be lowered by the counterparty (Section 3.3). It establishes the necessity of the floor clause of exit for the party that cannot leave, not of the full triad; the other clauses, and the other two conditions, are argued for as requirements of a fair exchange, not as conditions of non-domination. The second thesis is the reason the dissertation concentrates on institutional design, and it is offered as a conjecture rather than a theorem. Its content should be stated plainly. An accountable designer is one that can be identified and held to the obligations the terms create, the unique accountable principal of Chapter 5. The conjecture's ground is that a party with a stable, designer-specified objective behaves as a predictable function of the terms it faces, so that its conduct can be audited against those terms. Predictability does not by itself entail compliance: a stable revenue-maximizer is predictable and defects whenever defection pays. The conjecture's content is therefore that, for such parties, the three conditions once enforced are sufficient, and no further constraint on character is needed for outcomes the parties would re-enter; its open premise is that the terms can be made to bind, and Chapter 7 records that enforcement without a sovereign remains unsolved. The regime is also bounded by accountability: a system with a designed and stable objective but no principal who can be held to obligations is outside it until that problem is solved, so that for a superintelligence the constructive claims hold only while an accountable principal holds its stake. The regime criterion likewise presupposes a party with a principal; the tissue-based cases of Chapter 8, considered as parties in their own right, fall outside it on that ground. An outcome is acceptable, for the purposes of the conjecture, when each human party, from a position at or above its floor and with a comparable alternative available, would re-enter the exchange on the same terms knowing what it now knows, and each principal would continue to deploy under them; the

qualification excludes the party that would re-enter only because it has nothing better. Predictability is not the same as acceptability, and it is the level-matching requirement that closes the gap: without it, a revenue-maximizing system operating under terms fair to its counterparty could still impose uncompensated costs on everyone its counterparty's data concerns. This dissertation does not test the conjecture; Section 3.11 names the proxies a test would need, and Chapters 7 and 9 say what it would require. The third thesis is the reason the dissertation restricts its constructive claims to the designed-incentive regime. The regime is not defined by capability; its defining feature is the provenance and stability of the objective: a narrow system can acquire emergent objectives, and a superintelligence could in principle have designed and stable ones. Figure 9 shows the scope of each of the three theses.

Figure 9

The Scope of the Theory's Claims: Where the Necessary Condition Holds and Where the Sufficiency Conjecture Is Offered



Note. For any synthetic party the theory claims only that clause E(i) is necessary, for a party that cannot leave, to a relation that is not one of domination. The sufficiency conjecture is offered only inside the designed-incentive regime, whose criterion is the provenance and stability of the objective together with an accountable principal, not capability. Parties outside it, and the tissue-based parties of Chapter 8 considered in their own right, receive only the necessary condition.

The two framing scenarios of Chapter 1, the integrated person who pays for medical care with continuous data and the person whose brain has become infrastructure, can now be described

exactly. They share an architecture and differ in the terms. In the first, Conditions E, C and R hold. In the second, the failure of exit entailed the failure of consent, and the failure of both removed the only pressure that could have sustained reciprocity.

3.9 A typology of exchange regimes

Because each condition is a conjunction of clauses, and because the dependencies the chapter has established run at the level of clauses, the eight combinations of the three conditions do not sort exchanges usefully. Those dependencies are four: a failure of E(i) entails a failure of C(ii), by Proposition 1; a failure of E(i) entails a failure of R(iii) where the sub-floor position is unchosen and determines the division by itself, by Proposition 3; a failure of E(iii) entails a failure of C(iii) where the assets destroyed on leaving exceed the value received; and, a dependency the typology adds, a failure of C(i) so complete that the party cannot see what it gives at all entails a failure of R(ii), since what cannot be seen cannot be valued. The typology therefore names five prototypes by their clause pattern. They are points on a continuum rather than a partition; an exchange can answer to more than one description; and cases in which only clause E(ii) or E(iii) fails while the other conditions hold, such as the revocable purchase of Section 3.10, are the ones Proposition 3 treats as removing the means of later correction and the first thesis of Section 3.8 treats as defects of the exchange rather than as relations of domination, and they are not named. Later chapters use the names as shorthand.

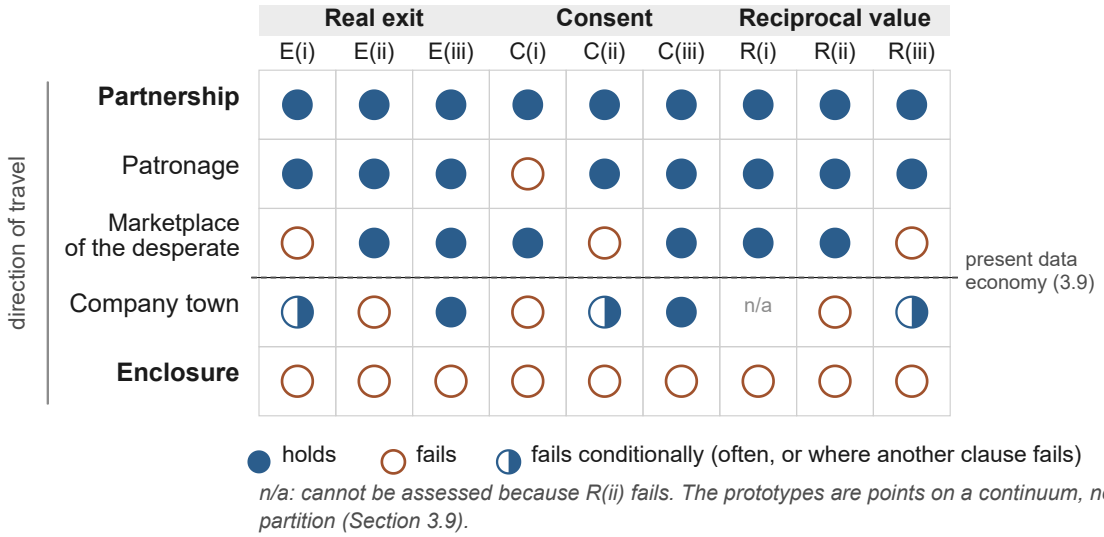
Regime	Exit	Consent	Reciprocity	Characteristic form
Partnership	All clauses hold	All clauses hold	All clauses hold	The target of the framework; the first scenario of Chapter 1
Patronage	Holds	C(i) fails	Holds	Paternalistic provision: benefits delivered without an account of the flows' shadows or uses, though the recipient can see what it gives, on terms it can leave
Marketplace of the desperate	E(i) fails	C(ii) fails with it	R(i) and R(ii) hold; R(iii) fails	Value flows both ways and can be seen to, but the weaker party cannot afford to refuse
Company town	E(ii) fails, often E(i)	C(i) fails; C(ii) where E(i) fails	R(ii) fails, so R(i) cannot be assessed	Wages paid in a medium redeemable only with the payer; reciprocity cannot be audited and leaving is costly
Enclosure	All clauses fail	All clauses fail	All clauses fail	The second scenario of Chapter 1: the person has become infrastructure

Two features of the typology deserve comment. First, on this dissertation's diagnosis, the present data economy does not sit at enclosure. Its typical exchange lies between the marketplace of the desperate and the company town: services are real and consent is formally collected, but declining is

expensive, leaving is hard, and the terms are opaque to the people who accept them. The dystopia of Chapter 1 is therefore best read as the endpoint of a direction of travel rather than as a different world. Second, patronage is a failure mode of limited concern to the framework. A regime in which exit holds and only the informational clause of consent fails is correctable in the sense of Proposition 3, provided that its alternatives do not share its defect. The framework's design priorities follow from this: secure exit first, because it converts most other failures into ones that people can escape, and build governance for the failures that exit cannot reach. Figure 10 shows the five prototypes as a clause-level matrix.

Figure 10

Five Prototypes of Exchange Regime by Clause Pattern



n/a: cannot be assessed because R(ii) fails. The prototypes are points on a continuum, not a partition (Section 3.9).

Note. Filled cells mark clauses that hold, open cells clauses that fail, and half-filled cells clauses that fail often or where another clause fails; n/a marks a clause that cannot be assessed because R(ii) fails. The prototypes are points on a continuum rather than a partition, and the present data economy lies between the marketplace of the desperate and the company town (Section 3.9).

3.10 The typology applied

A theory is useful to the extent that it sorts cases that intuition sorts badly. Six cases, drawn from the literature reviewed in this dissertation, show the conditions and the typology at work: five chosen for the contrast they offer between a focus on character and a focus on terms, and one, the resource dividend, as the nearest existing analogue of the survival tier.

The revocable purchase. Retailers of digital goods have characterized purchases of books, music

and films as licenses rather than sales, and in a well-known instance a retailer remotely deleted copies of a novel from customers' devices (Perzanowski & Schultz, 2016). Nothing about this exchange looks like coercion. Customers chose freely, received what they paid for, and could shop elsewhere. A character-centered analysis finds nothing to object to unless the seller acted in bad faith. The theory locates the defect: clause E(iii) of the exit condition fails, because leaving the seller, or the seller's leaving the market, destroys assets the customer accumulated within the exchange. The case is a small one, but it shows how an exchange can satisfy consent and reciprocity at the moment of transaction while carrying a latent exit failure that matures only later.

The withdrawal of the web. Within a single year, as Section 2.7 documented, a substantial share of the most actively maintained, critical sources of text used to train language models moved to restrict automated collection (Longpre et al., 2024). Viewed in terms of the parties' character, this looks like a dispute between publishers and developers over norms of conduct. Viewed through the theory, it is an exercise, collective and at scale, of the countervailing power that developers' dependence on such text confers. It is evidence that publishers believe their content is valuable to developers, and so bears on, without settling, the size of the dependence term in Proposition 4. It also shows the limits of the instruments currently available. A blanket refusal expressed in a protocol designed for search engines is exit without voice, which Hirschman (1970) argued to be an indirect signal of decline and which, where exit is easy, allows voice to atrophy.

The company town. Chapter 4 summarizes the economic history of company scrip in nonunion coalfields, which finds that store prices were often close to those of independent stores and that the degree of exploitation varied with how much competition the mine faced for labor (Fishback, 1992). A character-centered reading of this evidence concludes that the companies were less exploitative than the popular account holds. The theory draws a more specific conclusion: the same instrument was less exploitative where exit was available and more exploitative where it was not. The case illustrates clause E(ii): what the evidence ties to exploitation is the absence of an exit option. That a floor would have done the same work is the framework's inference from Sections 3.3 and 3.6, not a finding of the case.

The kibbutz. The Israeli kibbutz sustained an egalitarian internal economy within a market economy for much of the twentieth century. Its central difficulty was that the most productive members were the most likely to leave, and a key retention device was collective wealth that departing members could not take with them (Abramitzky, 2008). In the vocabulary of the typology, the kibbutz secured reciprocity and consent but compromised clause E(iii) of exit in order to survive. The case is a warning that low rates of departure are not evidence of satisfaction, and it is the reason any empirical test of the framework would have to measure not only how many leave but who leaves and what they can take.

The resource dividend. Since 1982, residents of Alaska who meet a residency requirement have

received an annual dividend from a fund built on the state's oil revenues, a universal and unconditional share of the returns from a common resource. One study finds no reduction in overall employment and a rise of 1.8 percentage points in part-time work (Jones & Marinescu, 2022). The dividend is the closest existing analogue to the survival tier: an unconditional payment financed by a share of a productive asset rather than by taxes on labor. It is not a floor in the full sense, because it is paid annually in cash and is fully alienable, but it shows that a dividend of this kind can be politically durable for decades.

The implanted voice. A participant in a clinical study of a speech neuroprosthesis used the system for more than 248 hours of conversation over roughly eight months, communicating in a synthesized voice resembling his own (Card et al., 2024). Everyone involved in such a study may be assumed to act in good faith, and the exchange satisfies consent and reciprocity to a high standard. The theory nevertheless identifies a vulnerability that good faith cannot cure. The participant's ability to communicate now depends on a device and a model whose continued availability he does not control. If the study ended, the provider failed, or the model could not be moved, the exit condition would fail in its first and most serious clause. Chapter 8 argues that the portability of personal models is therefore a requirement, not a refinement.

In each case, the theory asks not whether the stronger party is good but what the weaker party can do if the stronger party ceases to be good.

3.11 What the formalism does not do

Three limits on this chapter should be stated before the argument proceeds.

The formalism does not measure anything. It defines quantities, such as dependence, disagreement payoffs and informational shadows, that are difficult to observe. Operational proxies for each would have to be specified before the theory could be tested: for dependence, the marginal improvement in a system's performance attributable to fresh human data; for the disagreement payoff, the value of what a party continues to receive while the exchange is suspended; for the informational shadow, the reach of the inferences a contributor's data enables about people other than the contributor. Any such proxy is imperfect, and the mapping from it to the theoretical quantity is itself a matter for empirical work. The formalism states the sufficiency conjecture but does not prove it, and this dissertation does not test it. The design of a test is identified in Chapter 9 as the first item of future work.

The formalism does not settle who counts as a party in the moral sense. It treats synthetic systems as parties in the analytic sense because they have objectives the exchange affects, and it locates their stake in their principals. Whether a system's own welfare matters is a question the dissertation regards as open, and Chapter 8 argues that the convergence of biological and artificial substrates will

force it onto the agenda before most institutions are prepared to address it.

Finally, the formalism inherits the assumptions of the theories it borrows. Bargaining theory, as used in Appendix A, assumes parties who know their own payoffs and each other's, and that value can be transferred between them without loss; the first assumption bears directly on the legibility clause of reciprocity and on the informational clause of consent, both of which concern what the parties can know. Power-dependence theory assumes that value and availability can be compared. None of these assumptions holds cleanly for a person deciding whether to share health data with a system whose future uses of that data are unknown even to its designers. The formal statement is offered as a disciplined approximation, a way of seeing what the informal argument commits to, and not as a substitute for the institutional and empirical analysis that occupies the rest of the dissertation.

Chapter 4 — The Economic Layer: Compute as Currency

An exchange needs a medium. If people supply the data on which synthetic systems depend, and synthetic systems perform an increasing share of the work on which people depend, then the terms of exchange between them will be written, in practice, in whatever unit denominates their transactions. A unit issued and redeemed by one side gives that side control over the terms. A unit whose value depends on a political authority that one side can capture gives that control to whoever captures the authority. This chapter proposes a different unit, compute, and works through what it would take for a basic income paid in that unit to satisfy the three conditions of Chapter 3.

One presupposition should be stated at the outset, because every mechanism in the chapter rests on it and it has the same standing as the grounding assumption of Chapter 2. The framework presupposes an authority, whether a state or the convention's own institutions recognized by one, able to enforce three obligations: redemption of the unit by the owners of compute capacity (Section 4.4), acceptance of the survival tier by the providers of floor goods (Section 4.5), and whichever rule on capacity the convention adopts (Section 4.8). Section 5.9 names the minimal form of that presupposition, that existing states at least tolerate the framework's institutions, and Section 7.2 records that enforcement without a sovereign remains unsolved.

The chapter is organized around the difficulties the design must meet. Section 4.1 states what the currency is for. Sections 4.2 and 4.3 ask why compute, rather than some other unit, and how a unit of compute can be defined across physical substrates that do not share a common measure of speed. Section 4.4 addresses the indexing rule and the mechanics of issuance and settlement. Sections 4.5 and 4.6 take up distribution: the two-tier split and the demurrage charge on idle balances, which together are this framework's answer to the problem of concentration. Section 4.7 turns to the price of the data that people supply. Section 4.8 confronts the limit of every monetary remedy, the conversion of money into productive capital. Section 4.9 addresses a problem that recent events have made concrete: a unit of compute is worthless to a person whom compute cannot physically reach. Section 4.10 places the design in the international system, and Section 4.11 sets it against the empirical record on unconditional income. The chapter closes by testing the whole against the conditions of exit, consent and reciprocity.

4.1 What the currency is for

The basic income proposed here is not charity, and it is not primarily a response to technological unemployment, although it would serve as one. Its first function is the one identified by the formal analysis of Chapter 3: it is the mechanism by which every human party's disagreement payoff is raised to at least the floor, so that consent becomes possible and no one is forced by impatience to

accept whatever terms are offered. Its second function follows from the grounding assumption. If synthetic systems need continuing human input in the domains where nothing else can supply it, then people who are economically secure and living full, varied lives are the source of that input. From the side of the systems that learn from such people, a basic income that sustains them also maintains a supply chain, which gives the operators of those systems, taken together, a reason to support it, though Proposition 4 shows that each has only a diluted reason of its own; it is not a reason for people to want it. That the security of the income improves the data people generate is a conjecture of this dissertation, not an established finding. Its third function is distributive in the ordinary sense. If an increasing share of economic output is produced by compute, then the income from compute will flow to whoever owns it unless some of it is deliberately shared. Economic models of transformative automation describe scenarios in which wages collapse when automation outpaces the accumulation of capital (Korinek & Suh, 2024), and theoretical work on the distributional effects of artificial intelligence shows that redistribution of the resulting rents can in principle compensate those who lose (Korinek & Stiglitz, 2019). A dividend on the automating factor is one such channel of compensation, and in the scenarios where wages collapse it would be not a supplement to wage income but the principal way most people receive any income at all.

Each of these functions constrains the design. The first requires that part of the income be inalienable. The second favors universality: a scheme that selected recipients by any observable criterion would correlate the supply of data with that criterion, and the diversity on which the value of human data partly depends is best secured by not selecting at all. The third requires that the income be indexed to the factor whose returns it shares, rather than fixed in a unit whose real value will drift.

The proposal enters a field that already contains other ideas for sharing the gains of automation, including resource dividends of the kind Alaska has paid since 1982 (Jones & Marinescu, 2022) and a range of proposals for sharing returns to capital more broadly. This dissertation does not survey that field; it engages the existing evidence on unconditional income in Section 4.11, and positioning the compute dividend systematically against its nearest alternatives remains future work.

4.2 Why compute

A currency for the exchange between people and synthetic systems must satisfy two requirements that pull in different directions. It must be something that the operators of synthetic systems cannot do without and cannot substitute, so that refusing it is costly to them, and something that people need as well. And it must not be under the control of either side, so that neither can manipulate the terms by manipulating the unit.

Company scrip fails the second requirement in a way that is instructive, though not in the way it is

often described. The popular image of the indebted company-store miner is not supported by Fishback's quantitative evidence. In nonunion coalfields in the early twentieth century, company-store prices were generally close to those of nearby independent stores, miners were not typically in debt to the store, scrip functioned mainly as an advance on cash wages paid on payday, and the degree of exploitation varied with how much competition the mine faced for labor (Fishback, 1992). On this dissertation's reading of that evidence, what made scrip dangerous was not the scrip itself but the combination of an issuer with local market power and workers with limited exit. The inference that an issuer-controlled unit becomes an instrument of control where exit is absent is this dissertation's. The lesson for the present design is this: the problem with an issuer-controlled unit is not the unit but the issuer's power over what the unit buys, and the remedy is not merely a different unit but the guarantee of exit, which in monetary terms means redeemability with more than one provider.

Fiat money fails the first requirement in a different way. National currencies derive their value in part from the fact that states accept them in payment of taxes, a mechanism the record of Depression-era tax-anticipation scrip illustrates (Gatch, 2012), and, as a matter of law stated here without a source, from legal-tender rules that enforce their acceptance in settlement of debts. They are an instrument by which the operators of synthetic systems acquire what they cannot do without, compute, power and staff, rather than the thing they cannot do without. A national currency is therefore a unit in which the exchange between people and synthetic systems is mediated by a third party, the state, whose own relation to the systems is what is in question. That is not an argument against national currencies; it is an argument that they cannot by themselves constitute a medium for the exchange this dissertation is concerned with.

Compute satisfies the first requirement better than the other candidates considered here, scrip and fiat money; energy, the substrate-neutral input to which every unit of compute is tied, is a further candidate, and the comparison between the two is future work. Compute is the service exchanged, whereas energy is tied to it only by physical bounds that fix no conversion rate (Landauer, 1961; Lloyd, 2000); the efficiency achieved differs by hardware and improves over time (Hobbhahn et al., 2023). It satisfies the second requirement only conditionally, because the capacity behind the unit is owned by the principals of synthetic systems, and a unit they must redeem is, without the issuance, capital and access rules of Sections 4.4, 4.8 and 4.9, under their control. The bargaining argument for the floor does not depend on the unit. The case for compute rests on the properties that follow, together with indexing and the redemption obligation. It is scarce, measurable and needed by both parties: people need it for the services it provides, and the operators of synthetic systems cannot run a system without it. It is unusually quantifiable among the inputs to synthetic intelligence, and its physical form, concentrated in large and power-hungry data centers built from chips that can be counted, makes it detectable and excludable in a way that data and algorithms are not (Sastry et al.,

2024). These are the properties that make compute an attractive lever for governance, and they are equally the properties that make it a plausible basis for a unit of account.

The theoretical status of such a unit deserves a precise statement, because it determines what can go wrong. This chapter treats a compute currency as a form of commodity money, in the sense that its value derives from the use value of the thing it represents, as the value of gold coin derived in part from the use value of gold. Commodity money has, on the record reviewed below, supplemented its intrinsic backing with an institutional one. In the United States during the Depression, tax-anticipation scrip, which local governments accepted in payment of taxes, circulated longer and in larger volume than stamp scrip, and its success has been linked to the view that money circulates because it can discharge an obligation people already owe (Gatch, 2012). A critical review of the Austrian town of Wörgl's experiment likewise attributes much of its circulation to the residents' large tax arrears, which the local currency could be used to pay (Greco, 2002; see also Fisher, 1933). Tax receivability worked on the demand side: holders already owed something that the unit could discharge. The analogue for the present design would be an obligation that participants must discharge in compute, such as the capacity-utilization charge of Section 4.8 or access fees for data payable only in the unit, and the convention of Chapter 5 decides whether any such obligation exists. The design's other institutional element works on the supply side. The *redemption obligation*, established by the convention, requires the owners of compute capacity operating under the framework to honor redemptions in compute; these are the principals of Section 3.3 where they own the capacity that runs their systems, and otherwise the capacity providers from which those principals rent. It gives the unit backing beneath its commodity value; it does not by itself create demand for the unit, which must come from the pricing of services in it and from whatever obligations the convention makes payable in it. The redemption obligation is the place where the governance layer and the economic layer meet.

4.3 Defining the unit

The name of the unit is *compute*. A person is paid in compute and a service is priced in compute, in the way that a dollar names a unit without naming the paper or metal in which it was once issued. The name deliberately avoids reference to any substrate. Alternatives considered during the development of this framework included a "quantum dollar," rejected because it builds into the name the assumption that quantum hardware will dominate, and a "shannon," rejected because it does not read as a unit of value. The choice matters because the definition the name stands for must survive changes in the hardware that delivers it.

That definition faces a difficulty. Raw processing speed is not comparable across substrates: operations per second on a silicon accelerator, coherence times and gate fidelities on a quantum processor, and the signaling rates of biological tissue in the organoid systems discussed in Chapter 8

are not quantities of the same kind. The working draft of this framework proposed to define the unit by the quantity of information processed, on the model of Shannon's (1948) demonstration that information can be measured independently of the medium that carries it. The analogy is instructive but limited. Shannon defined a unit of information, the bit, as a measure of selection among alternative messages; he explicitly set aside the meaning of messages as irrelevant to the engineering problem, and he did not define a unit of processing. Physics, moreover, ties processing to energy. Every logically irreversible operation dissipates a minimum quantity of heat (Landauer, 1961), and the maximum rate at which any physical system can perform operations is bounded by its energy (Lloyd, 2000). A unit of compute can be defined independently of substrate, but its cost cannot be.

The practical precedent for a substrate-neutral definition is not information theory but benchmarking. One benchmarking consortium in machine learning has confronted the problem of comparing radically different hardware with the following approach: a fixed set of reference workloads, each with a defined task and a target quality, and rules governing how systems may be configured to reach it. The MLPerf training benchmark measures performance as the time required to reach a target quality on a specified task, and its designers argue that fair comparison across diverse systems requires such rule-bound workloads, because identical code and settings cannot be imposed on radically different hardware (Mattson et al., 2020). Its companion benchmark for inference compares systems that span several orders of magnitude in power and performance under the same kind of rules (Reddi et al., 2020). Even a single floating-point operation is not a uniform unit; the throughput of the same accelerator differs by nearly an order of magnitude between numerical precisions (Hobbhahn et al., 2023).

The proposal therefore defines a unit of compute as the completion of one standard quantity of a reference workload to a specified quality, where the workload is a basket of representative tasks fixed by the convention of Chapter 5. The definition is substrate-neutral because any system that completes the workload to the required quality has delivered the unit, regardless of how. It is not physics-neutral, and it does not pretend to be: the energy and material cost of delivering a unit will differ by substrate and will fall over time. It also requires maintenance. A reference workload ages as the tasks people need change, in the way a consumer price index ages as the goods people buy change. The basket must be re-based periodically, and the authority to re-base it is a source of power that the governance layer must constrain. The benchmark problem is therefore not a technical detail to be solved once; it is a standing institutional function, and the authority that performs it holds power of the kind a central bank holds. Chapter 7 lists it among the limitations for that reason.

4.4 Allocation, indexing and issuance

The allocation rule is equal per person. The convention of Chapter 5 sets a share θ of the measured

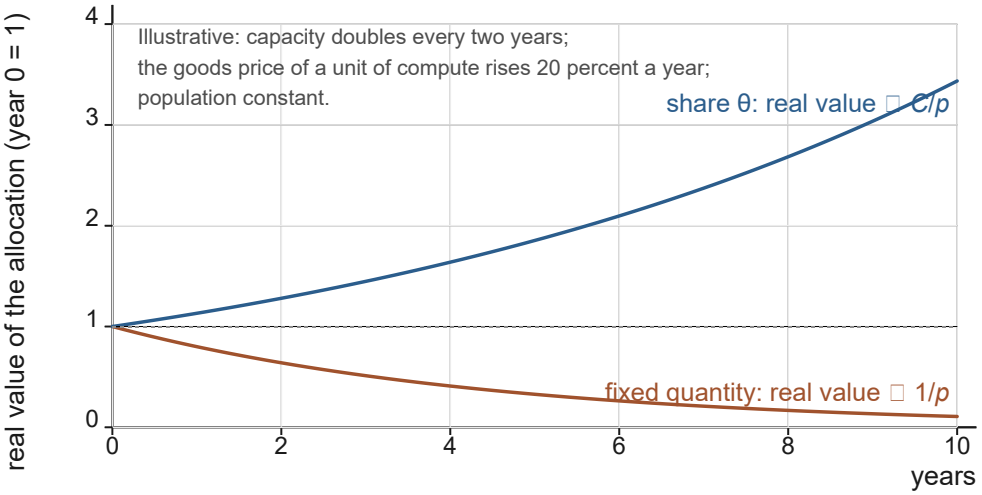
compute operating under the framework, which is the whole of global compute where all of it does, to be paid as basic income, and each enrolled person receives the fraction $\theta/N(t)$ of that compute in each cycle, where $N(t)$ is the number enrolled at the time of payment. Each person's percentage of the whole therefore falls as enrollment grows, while the share of the whole devoted to the income does not.

A fixed quantity of compute, paid each cycle, would lose value in a specific sense. Because the unit is commodity money, a claim on delivered workload, the quantity of compute that exists in a period is not an issuance choice but a consequence of physical capacity; what is issued is a claim on a share of it. The cost of a fixed quantity of compute has fallen fast: the price-performance of machine-learning accelerators has doubled roughly every two years (Hobbhahn et al., 2023), and the compute used to train milestone systems has grown with an estimated doubling time of under six months in the deep-learning era and about 10 months for the largest systems, though with wide uncertainty (Sevilla et al., 2022). Neither figure measures installed capacity, but both indicate that a fixed quantity of compute represents a shrinking share of what is available. As compute becomes more abundant, its price relative to everything that is not compute falls. A person paid a constant quantity of compute would find that it buys less and less of food, shelter and the other goods that compute is exchanged against: the loss is in the relative value of the unit, the counterpart of the rise in prices that the quantity theory predicts when money grows faster than the trade it finances (Fisher, 1911).

The remedy is to define the allocation as a share rather than a quantity. Appendix A shows formally that, holding θ fixed between re-basings of the reference workload, this converts the question of whether the income keeps its value into the question of whether the real purchasing power of all compute grows faster than population. A re-basing of the workload, which the convention that sets the share also controls, re-states what the share means and requires a conversion rule for outstanding claims; the two decisions must be taken together. For a framework whose premise is that compute is becoming the productive base of the economy, that is the right question. Whether the income keeps its value depends on whether the fall in compute's relative price is outweighed by the growth in its quantity, which holds if compute's share of the value of total output does not fall faster than output per person rises; the premise asserts this, and the design cannot ensure it. Figure 12 contrasts the two rules under illustrative growth.

Figure 12

A Fixed Quantity of Compute Loses Real Value as Compute Grows Abundant; a Share Tracks the Real Purchasing Power of All Compute



Note. Illustrative: capacity doubles every two years, the goods price of a unit of compute rises 20 percent a year, and population is constant. A fixed quantity's real value is proportional to 1/p and falls; a fixed share's real value is proportional to C/p and rises while capacity grows faster than its relative price falls (Appendix A.2). The two-year doubling is the order of magnitude Hobbhahn et al. (2023) report for price-performance; the price path is an assumption.

The working draft of this framework rejected a "fixed-supply" alternative, on the model of currencies designed to resist inflation by capping issuance. That alternative is not available for a commodity unit whose supply is physical capacity. What is available, and what the design rejects, is paying a fixed quantity rather than a share, which would tie the income to the falling relative price of compute and reward those who hold the unit over those who use it.

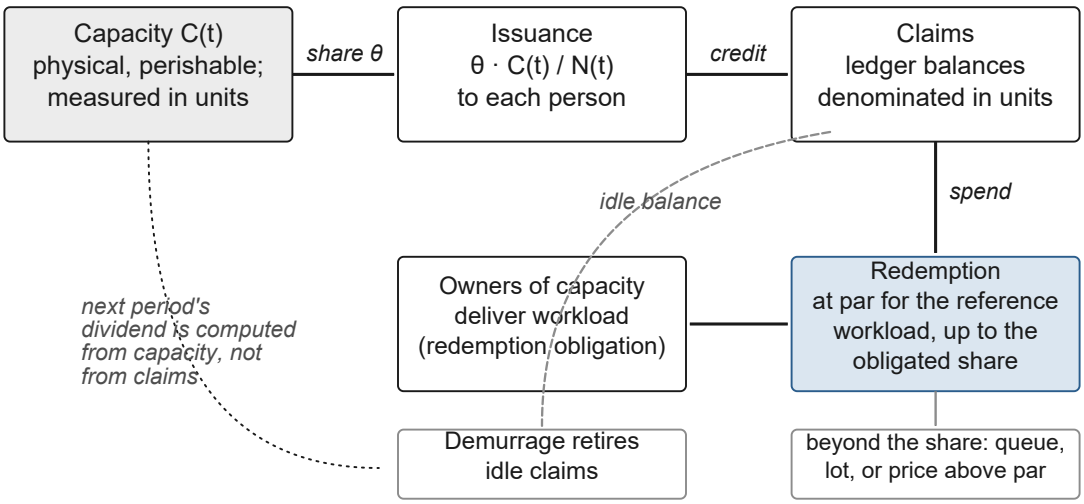
Issuance and settlement. Three things that the word *compute* might name must be kept apart. *Capacity* is physical and perishable: capacity unused in a period is lost. The *unit* is the reference-workload measure of Section 4.3. A *claim* is a ledger balance denominated in units. With that vocabulary, four questions about the mechanics of the unit can be answered. First, who issues the claims? The dividend credits each person, in each period, with claims equal to $\theta/N(t)$ of the period's measured capacity, recorded on a settlement ledger maintained under the rules of the convention and assigned to persons rather than to the owners of the capacity. Second, what is a balance? A balance is a stock of claims, and a claim is redeemed when a provider delivers a service, priced in units, against it. The redemption obligation requires the owners of capacity operating under the framework to accept the unit at par for the reference workload, that is, to deliver one reference-workload-equivalent for each unit of claim, up to the share of their capacity that the convention sets; for other services the unit price is the provider's, so the obligation secures access to raw workload and not to any particular service. Two consequences follow. The share θ must not exceed

the obligated share of the capacity operating under the framework; otherwise claims exceed redeemable capacity in every period, independently of saving, and where only part of global capacity operates under the framework the measured base from which claims are issued must be that part. And because balances accumulate across periods and capacity does not, outstanding claims can exceed the obligated share of a period's capacity even then, so the convention must choose a rule for redemption beyond that share: queued redemption, rationing by lot, or a rising unit price of services outside the obligated share, the last of which is redemption below par for the excess and should be named as such. Third, what happens to claims removed by demurrage? They are retired, reducing the stock of outstanding claims; the dividend in the next period is computed from capacity, not from the stock of claims, so retirement does not shrink the income. Fourth, how does lending work? Discretionary balances may be lent and borrowed. Loan claims denominated in the unit are recorded on the ledger and bear the holding cost described in Section 4.6, so that the charge falls on the lender's claim, paid out of the interest it receives or its own balance, and not on the borrower, whose debt is not reduced by it. This does not prevent lending at interest from compounding; it lowers the lender's net return to the interest rate less the demurrage rate, and Appendix A shows that accumulation through credit is bounded only if the interest rate is below the demurrage rate. A cap on the interest rate of ledger-recorded loan claims strictly below the demurrage rate would close the channel for the lending the ledger sees, at the cost of making lending in the unit unprofitable; the design keeps credit in the discretionary tier and therefore leaves the credit channel open, so that closing it falls to the capacity rules of Section 4.8 and to the convention's ability to see credit extended off the ledger, which Chapter 7 records as a limitation. Figure 11 shows the circuit from capacity to claims to redemption.

Figure 11

Capacity, Unit and Claim: How the Compute Dividend Is Issued, Redeemed and Retired

Three things the word compute names are kept apart: capacity, unit, claim.



Note. Capacity is physical and perishable; the unit is the reference-workload measure of Section 4.3; a claim is a ledger balance. A share θ of measured capacity is credited equally to the N persons enrolled, claims are redeemed at par for the reference workload up to the obligated share of capacity, and a rationing rule governs redemption beyond it. Demurrage retires idle claims; the next period's dividend is computed from capacity, not from the stock of claims.

4.5 Two tiers: the survival floor and the discretionary balance

Equal division is fair at the moment of payment. It is not fair over time unless something governs what happens between payments. If unused compute can be saved, lent, borrowed against or given away, ordinary economic behavior will reproduce the inequality of a market economy over time (Appendix A, Section A.4.1). Some people will under-consume and accumulate; others will overspend and fall into debt; the indebted will pledge future allocations; and a regime that began with perfect equality will converge on compute-rich and compute-poor households. This is not a defect peculiar to compute; it is the general behavior of any freely tradable unit. It means that equal allocation, by itself, cannot be presented as a solution to the problem of concentration.

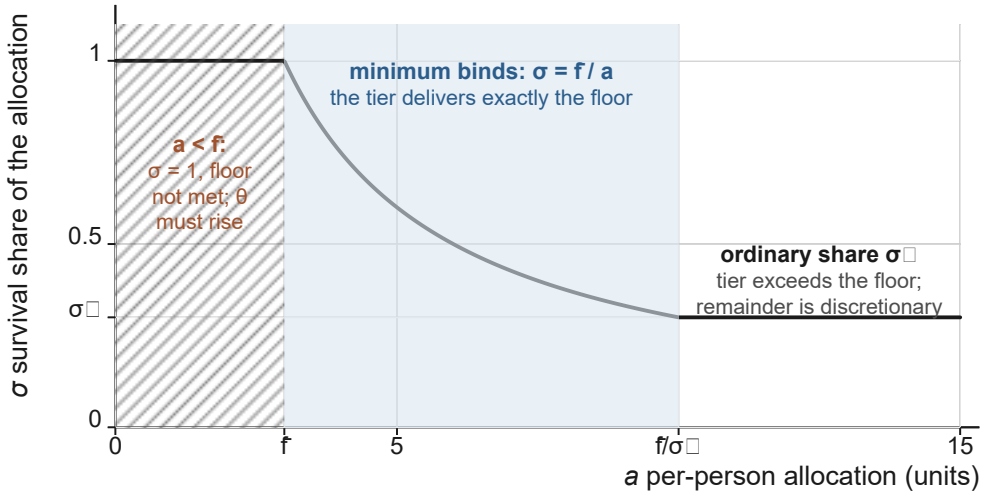
The framework's response is to divide every allocation, automatically and at the source, into two tiers. The first, the *survival tier*, is earmarked for the floor goods defined in Chapter 3: power, water, food, shelter, basic communication and emergency care. It is non-tradable and non-borrowable, and it is spent on floor goods directly rather than passing through a spendable balance. The second, the *discretionary tier*, is the remainder. It is freely tradable, can be saved, lent and borrowed, and is subject to the holding cost described in the next section.

Five features of this design require defense.

The split is a percentage, with a binding physical minimum. The two tiers are proportions σ and 1 − σ of an allocation that is itself a share of global compute, so both scale automatically as total compute grows. A floor defined physically, however, cannot be secured by a percentage alone. The survival tier delivers the floor only if its value, $\sigma\theta C(t)/N(t)$ units at the compute price of floor goods, is at least the value of the floor bundle, and nothing in the arithmetic of percentages guarantees that it will be. The rule is therefore a percentage with a binding minimum. In ordinary times the tier is the fixed proportion σ (σ_0 in Appendix A); when that proportion would fall short of the floor, the survival share rises until it does not, drawing on the discretionary tier. If the whole allocation falls short of the floor, the rule cannot be met without raising θ, which is a decision the convention must take (Appendix A, Section A.2); the design therefore guarantees the floor only up to what the agreed share of capacity can buy. Because the compute price of floor goods will differ by place, an inference this dissertation draws from the unequal geography of compute described in Section 4.9, a floor that is equal in physical terms requires a survival share that differs by place; the design accepts unequal survival shares in exchange for an equal floor, the discretionary tier absorbs the difference, and Appendix A writes the single-place case. This reintroduces a quantity that someone must define, the floor standard, and the floor standard becomes the contested parameter that a percentage rule was meant to avoid. The design's answer is to specify the standard itself by rule rather than by figure, as an evolving level of service set relative to what the technology of the time delivers to the best-served, rather than frozen at what was possible when the rule was written. There is a legal precedent for the principle in another domain. United States telecommunications law defines universal service as an evolving level of service, periodically redefined as technology advances, and lists among its principles that services in rural and high-cost areas should be reasonably comparable to those in urban areas (Universal service, 47 U.S.C. § 254, 2018). Chapter 5 returns to the governance of the floor standard, and Chapter 7 records it as a limitation. Figure 13 plots the rule.

Figure 13

The Survival Share Is a Percentage With a Binding Physical Minimum



Note. $\sigma(t) = \min\{1, \max\{\sigma_0, \bar{f}(t)/a(t)\}\}$, drawn with $\sigma_0 = 0.3$ and $\bar{f} = 3$ units (Appendix A.2). Where the whole allocation falls short of the floor the share is one and the floor cannot be met without raising θ; where the allocation is modest the minimum binds and the tier delivers exactly the floor; where it is ample the ordinary share applies and the remainder is discretionary.

The survival tier is inalienable. Chapter 3 gave the formal argument: a floor that can be sold or pledged stops being a floor at the moment it is needed, and those most likely to pledge it are those in the most acute need. Legal theory has a name for an entitlement that may not be transferred even with the holder's consent, an inalienability rule (Calabresi & Melamed, 1972, p. 1092), and there is a sustained philosophical defense of making certain goods market-inalienable where their commodification threatens personhood (Radin, 1987). The economic literature on in-kind and earmarked transfers surveys several rationales for them: the Samaritan's dilemma, in which a society that will not allow people to go without power or water creates an incentive to under-provide for oneself; self-targeting; and considerations of political economy. It concludes that paternalism and interdependent preferences are the leading explanations of the transfers actually observed (Currie & Gahvari, 2008). This design does not rest primarily on paternalism. Its rationale is the bargaining-theoretic one of Chapter 3: the earmark protects a disagreement payoff, not a judgment about how people should spend. It does not restrict what people may do with the discretionary tier. It does restrict the survival tier to a fixed bundle of goods that every person needs, and the next paragraph defends that restriction against the canonical objection.

The survival tier is restricted to floor goods. This departs from the canonical definition of basic income as an unconditional cash payment. The leading philosophical case for basic income treats unconditionality and payment in cash as part of what makes the income an instrument of freedom (Van Parijs & Vanderborght, 2017). In an earlier statement of the same position, Van Parijs defined

basic income as paid in cash rather than in kind, without restriction on the nature or timing of the consumption or investment it funds, and set it explicitly against a special currency with restricted uses, or one that must be consumed in the current period without any possibility of saving, as in Duboin's "distributive economy" (Van Parijs, 2006, p. 8). The survival tier is, in two respects, exactly what that definition excludes: it is spendable only on floor goods, and it cannot be saved against or borrowed on. The two-tier design therefore does not claim to satisfy the canonical definition. It keeps unconditionality, since no one must work or prove need to receive either tier, and departs from fungibility for a part of the income only, leaving the discretionary tier unrestricted as to use, though, under the demurrage of Section 4.6, not costless to hold; that too is a departure from the canonical definition, and it is defended there. The defense is that the fungibility of the floor is what allows it to be bargained away, and that a floor that can be bargained away does not secure the freedom it is meant to secure. A defender of the canonical view can reply that a restricted floor is a judgment about what people need, made for them; the reply of this framework is that the floor bundle is fixed narrowly, to goods without which no other choice can be made, so that the restriction binds only where the freedom it limits is the freedom to fall below the point at which freedom can be exercised at all. Whether that reply succeeds is a question on which reasonable people committed to real freedom can disagree. The design has a cost, and experimental evidence shows what it is. In a randomized study of unconditional transfers in Kenya, monthly transfers improved food security more than lump sums, while lump sums were more often invested in durable goods (Haushofer & Shapiro, 2016). A steady, non-advanceable floor captures the first benefit; the second is what advanceability buys. The design assigns it to the discretionary tier, which can be saved and borrowed against, and accepts that the survival tier forgoes it.

The survival tier requires more than one provider. The lesson of company scrip, as Section 4.2 described it, is that a unit redeemable with only one issuer is dangerous in proportion to the issuer's market power. An earmarked tier spendable only with a single provider of floor goods would recreate that danger in full. The survival tier must therefore be redeemable with any qualified provider of floor goods, and portability between providers is a condition of the tier's legitimacy, not an optional refinement.

Delivery of the floor goods that are not compute. Of the floor goods, only communication and part of emergency care are compute services, and the only obligation the design creates is on owners of compute capacity to deliver compute. Nothing stated so far obliges a farmer, a utility, a landlord or a clinic to accept the survival tier. The tier delivers the whole floor as a disagreement payoff in one of two ways. The first is an acceptance obligation on providers of floor goods operating under the framework, which is a legal-tender rule of the kind Section 4.2 described as the institutional backing of fiat money and should be named as one; it creates a guaranteed buyer with a fixed bundle, and where sellers are few it gives them price-setting power over the floor bundle, driving the

survival share toward the whole allocation (Appendix A.2), which is the company-store problem at the level of the floor itself and which only competition among providers, and whatever rule on the floor bundle's price the convention adopts, can contain. The second is to rely on markets in which floor goods are already priced and sold in the unit, in which case the floor is guaranteed only for the goods compute can deliver and the rest is a limitation. The design adopts the first. It therefore presupposes the enforcing authority stated at the head of this chapter, which is the external condition Section 5.9 names under Ostrom's seventh principle and the reason Chapter 6 begins the transition with a nationally legislated floor. A further circularity should be stated: where the counterparty in a bargain is itself an owner of capacity, the person's income on impasse is delivered by the party it is at impasse with, and the tier functions as a disagreement payoff against that provider only where the provider is not the sole channel of its redemption. Chapter 7 records both points. The history of universal service supplies a further warning: in the early American telephone industry, "universal service" was first a slogan for consolidating a single interconnected system, and only much later came to mean a telephone in every home (Mueller, 1997). An entitlement designed to secure access can, if it is tied to a single network, become the instrument by which that network entrenches itself.

The two-tier design achieves something limited. It confines the inequality that free exchange produces to the discretionary tier and prevents that inequality from reaching survival itself. It does not prevent concentration within the discretionary tier.

4.6 Demurrage

A person whose services or output are in high demand can accumulate discretionary compute faster than they spend it. A person who persuades others to transfer their discretionary balances to them can accumulate faster still. Large holdings confer bargaining power, as Chapter 3 argued, because a party that can afford to wait can hold out for better terms. The question is whether the framework can make the passive accumulation of large balances costly without confiscating anything and without a central authority deciding how much is too much.

The proposed answer is demurrage: discretionary compute loses a small percentage of its value for each period it sits unspent. Demurrage is a holding cost on idle balances, and on outstanding loan claims, not a tax on transactions.

The idea and its history. The proposal is most closely associated with Silvio Gesell, whose *The Natural Economic Order*, published under that title in German in 1916 from works first issued in 1906 and 1911, argued that money should deteriorate like other goods, so that holding it would cease to be costless and money would circulate rather than be hoarded (Gesell, 1916/1958, pp. 33–34). Keynes endorsed the idea in the *General Theory*, where he described Gesell as a prophet whose work

had been unduly neglected and judged that the idea behind stamped money was sound; he also identified its central weakness, which was that money is not the only asset with a liquidity premium, so that a holding cost on currency would drive holders into substitutes such as bank deposits, foreign money, precious metals and land (Keynes, 1936, Chapter 23). Gesell's program paired demurrage with the nationalization of land; as Keynes observed, that removes one substitute for money but not the others. Irving Fisher championed a temporary form of stamp scrip as a Depression remedy (Fisher, 1933).

The trial most often discussed took place in Wörgl, a small town in the Austrian Tyrol, in 1932 and 1933, where the municipality issued labor certificates that lost 1 percent of their value each month unless a stamp was purchased, and used them to finance public works. The experiment is often described as having reduced local unemployment by a quarter (Gelleri, 2009), a figure September (2026) traces to movement sources. The evidence does not support so strong a claim. The commonly repeated figures trace to secondary sources, including a movement publication and a diploma thesis, rather than to audited series; the daily exchange records were later destroyed; and a recent scholarly review concludes that the loss of the records leaves the question of what the experiment achieved largely open (September, 2026). Fisher, a sympathetic contemporary, recorded that permanent jobs went to only some 30 to 50 of the unemployed (Fisher, 1933, Chapter 4). The public works were financed in part by outside subsidies and a provincial credit, and the scrip's acceptance owed much to its use in paying tax arrears, so that whether demurrage itself caused the observed improvements is disputed (Greco, 2002). What is not disputed is that the experiment was halted by the authorities. The Austrian National Bank objected that the town was infringing its monopoly of issue (Ottacher, 2007, pp. 53–54); the certificates were withdrawn from circulation in September 1933 under administrative order and threat of police enforcement, the sources differing on the day (Ottacher, 2007, p. 57; September, 2026); and the Administrative Court dismissed the town's final appeal on 18 November 1933 (Champ, 2008; Ottacher, 2007, pp. 58–59). Appendix C records the discrepancies among the sources, including the Vienna exhibition's account (Institut für Wirtschafts- und Sozialgeschichte, Wirtschaftsuniversität Wien, n.d.), on the date and on which authority ended circulation. Attempts to scale stamp scrip in the United States largely failed, because of business reluctance to accept it, problems of coordination, and the inconvenience of affixing stamps to paper notes, which made the scrip unattractive as soon as ordinary money was available (Champ, 2008). Modern local currencies with demurrage have operated for years at small scale (Gelleri, 2009, an account by the currency's founder), but one review of the evidence suggests that the demurrage rate is not decisive for how much such currencies are used (Godschalk, 2012).

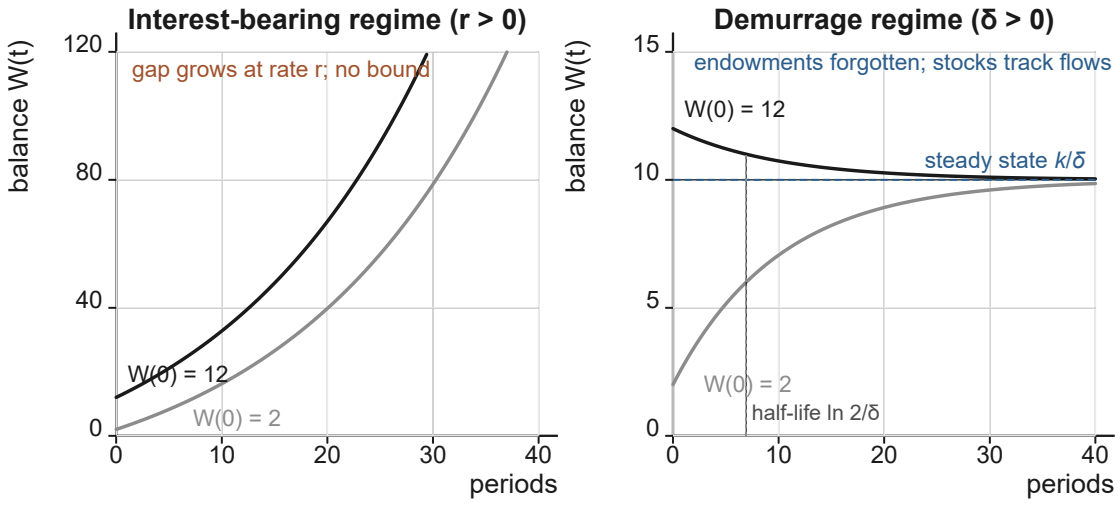
What the historical record does and does not show. Two conclusions follow. First, the case for demurrage in this framework cannot rest on the claim that it stimulates spending and employment, because the evidence for that claim is weak. Second, the practical objection that defeated paper

stamp scrip, the transaction cost of applying the charge, does not apply to a ledger-based unit. Goodfriend (2000) analyzed a carry tax on electronic bank reserves as an instrument of monetary policy at the zero lower bound, and mainstream macroeconomic theory has treated a Gesell-style carry tax on currency as a legitimate instrument, though for the purpose of escaping the zero lower bound on interest rates rather than for redistribution (Buiter & Panigirtzoglou, 2003).

The distributive case. The argument for demurrage here is distributive, and it is made precise in Appendix A. Under a regime in which balances earn a positive return, a constant net inflow produces a balance that grows without bound, and two people with the same income but different starting wealth diverge exponentially, a dynamic analogous to the one Piketty (2014) identified in the tendency of inherited wealth to dominate when the return on capital exceeds the rate of growth. Under demurrage, the same constant inflow produces a balance that converges to a ceiling equal to the net inflow divided by the demurrage rate. Three consequences follow. Accumulation in idle balances is bounded. Initial endowments decay: at a demurrage rate of 1 percent a month, an idle windfall loses half its value in a little under six years. And in the long run, differences in holdings reduce to differences in current net inflow, meaning income less spending. Demurrage does not abolish inequality. It removes the compounding of inequality within balances, so that the distribution of balances tracks what people currently earn and choose not to spend, rather than the history of who happened to accumulate first. Figure 14 shows the two regimes.

Figure 14

Idle Balances Compound Under Interest and Converge Under Demurrage



Note. Two parties with the same constant net inflow k and different starting balances, drawn with k = 1, r = 0.05 and δ = 0.1 per period (Appendix A.4). Under a positive return their balances grow without bound and diverge; under demurrage both converge to k/δ, the difference in endowments

decaying with half-life $\ln 2/\delta$. The bound binds on idle balances only; capital and credit reopen compounding (Sections 4.6 and 4.8).

Why demurrage suits compute. One might argue that an unspent compute balance represents idle capacity in a way that unspent cash never did. That argument does not hold as it stands: because a balance is a claim on future capacity, the hardware itself may be fully used by others while the claim sits unspent. The real connection between demurrage and idle capacity runs through the capacity rules of Section 4.8. Demurrage on claims keeps the stock of outstanding claims from piling up against a perishable resource, and a utilization charge on capacity, if the convention adopts one, ensures that hardware itself does not sit idle. Together they align the private cost of holding with the social cost of idleness; neither does so alone.

The limits of demurrage. Demurrage does not fully solve the concentration problem, for three reasons. The first is the one the formal model makes exact: a person whose output is in persistent high demand sustains a large balance because their net inflow is large, and demurrage can only cap that balance relative to the inflow, not eliminate it. The second is credit: lending at interest above the demurrage rate reopens compounding, as Section 4.4 stated. The third is Keynes's substitution objection, which applies without qualification. A holder facing a carrying cost on compute balances will convert them into assets that do not bear it: hardware, equity in compute providers, land. Section 4.8 takes up this third problem directly. Taken together, the three limits mean that demurrage has no distributive effect of its own in an economy with capital and credit: a holder converts idle claims into whatever asset bears no holding cost, and the bound of Appendix A then binds on balances no one keeps beyond transaction needs. Its effect is conditional on the capacity rules of Section 4.8 and on an interest rate below the demurrage rate, which, on this dissertation's reasoning, no market would supply unaided, since lenders require a positive net return; because the design keeps credit in the unit, the credit channel stays open and the capacity rules carry the whole burden. What demurrage does accomplish, under those rules, is to convert passive hoarding from a costless default into an active cost. That is the role this framework assigns to it, and no more.

4.7 Pricing human data

If human data is an input that synthetic systems need, it has a price, and the way that price is set will shape the lives of those who supply it.

One design would pay more for data from lives that are interesting to people: astronauts, inventors, public figures. That design would reproduce in the data economy the logic of markets for attention, in which income follows visibility. It would pressure people to perform novelty for the sake of payment, which corrupts the consent condition by making the content of a life a function of its marketability.

The alternative prices data by how much it reduces a model's uncertainty. Both designs reward novelty. The difference is the reference point: the attention design rewards novelty as judged by other people, which can be performed; information-gain pricing rewards novelty as judged by the model's current uncertainty, which cannot be performed in the same way but can be fabricated (the fifth qualification below) and which declines for each further contributor of similar data. This is not a novel proposal but the application of a long-established idea. The expected information an experiment provides, measured as the expected reduction in uncertainty about a model's parameters, was introduced into statistics as a criterion for experimental design (Lindley, 1956), was adapted as an objective for choosing which data a neural network should learn from (MacKay, 1992), and underlies several of the strategies of active learning, in which a learner selects the examples that will most reduce its uncertainty (Settles, 2009). It has also been proposed as the basis for rewarding data contributors. In one such scheme, the value of a coalition of contributors' data is its information gain about a model's parameters, each contributor's share is set by the Shapley value of the resulting cooperative game, and contributors are rewarded with access to models of differing quality rather than with money (Sim et al., 2020). The step from rewarding contributors in models to paying them in a currency is this dissertation's. On this measure, the undersampled everyday life of a dairy farmer may be worth more to a model than a further memoir by a public figure. The price follows informational novelty to the model, not social status among people.

Six qualifications must accompany this proposal.

First, information gain is defined only for a model that maintains a probability distribution over its parameters, and the quantity priced is the reduction in that uncertainty that a datum produces once it is observed: the realized form of the expected-information criterion. Large models trained to a point estimate carry no such distribution, and the price can be computed for them only through an approximation, which the governance layer must choose and which is itself a contested parameter.

Second, information gain is relative to what the model already believes. A datum is novel only against a prior, and a model whose hypothesis space is wrong may misvalue data, because information-based criteria estimate a datum's usefulness on the assumption that the model is correct, which the original formulation of information-based data selection flagged as possibly their main weakness (MacKay, 1992, pp. 590, 602–603). Information-gain pricing therefore rewards novelty as the model perceives it, which is not the same as novelty in the world.

Third, information gain is one of several defensible valuations. Alternative approaches value a datum by its marginal contribution to a model's performance on a validation task, using the Shapley value to divide credit fairly (Ghorbani & Zou, 2019; Jia et al., 2019). These methods answer a different question, how much a datum improves performance rather than how much it reduces uncertainty, and they produce different prices. The choice between them is not merely technical; it is a choice about what the exchange is for, and it belongs to the governance layer.

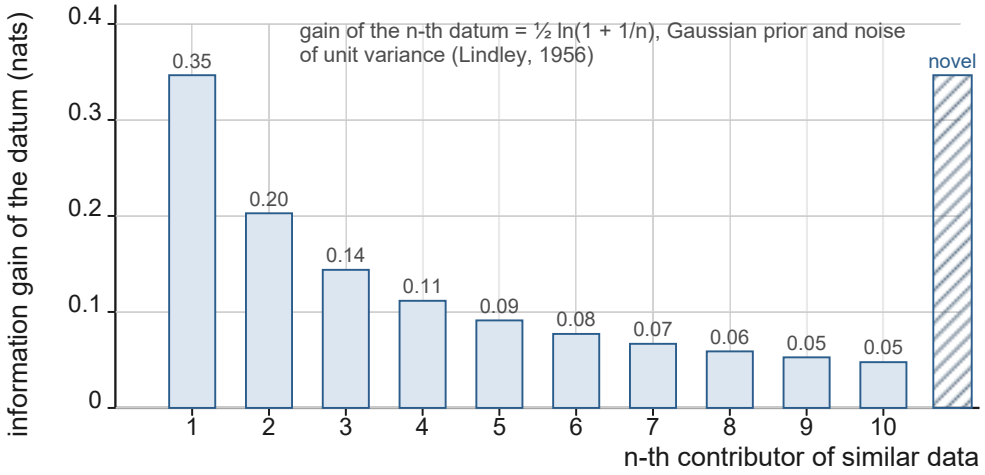
Fourth, data is freely replicable and its value is combinatorial: a datum's contribution depends on what other data it is combined with, and the same datum sold twice does not lose value to the seller. These properties make data markets difficult to design, and the mechanism-design response has been to construct markets specifically to achieve truthful pricing and a fair division of revenue among sellers (Agarwal et al., 2019).

Fifth, any scheme that pays for data creates an incentive to manufacture what it pays for. Methods that trace a model's predictions to the training points most responsible for them have also been used to craft training data designed to manipulate a model (Koh & Liang, 2017). A scheme that pays for novelty will attract fabricated novelty, and the payment mechanism must be robust to it.

Sixth, and most important for the theory, the data that reduces a model's uncertainty most may be data that tells it the most about people other than its source. Chapter 3 called the set of people about whom a flow licenses inference its informational shadow, and conjectured that where a model's parameters describe populations, the price of a flow under information-gain pricing and the inadequacy of individual consent for that flow will tend to rise together. If the conjecture holds, the highest-priced data is the data for which individual consent matters least, and a data market built on information-gain pricing must be governed collectively. Even if it holds only partly, it is a strong reason why the economic layer cannot stand without the governance layer of Chapter 5. Chapter 7 adds a further cost: a price that rises with what data reveals is a price that rewards revelation, and so creates pressure against privacy. Figure 15 shows the declining gain from similar data.

Figure 15

Under Information-Gain Pricing the Price of Each Further Similar Datum Falls



Note. For a model estimating a parameter with a Gaussian prior and observation noise of unit variance, the n-th similar datum adds ½ ln(1 + 1/n) nats, the realized form of the expected-

information criterion (Lindley, 1956). A datum about a parameter the model has not seen adds the first-observation gain again. The price follows novelty to the model, not status among people; the first qualification of Section 4.7 states what this computation presupposes.

4.8 Capital: concentration through ownership of capacity

Demurrage acts on balances and claims. It does not act on assets. A person who accumulates discretionary compute and converts it into compute-generating capacity, by buying hardware, financing a data center, or acquiring equity in a provider, escapes the holding cost and acquires an income stream that compounds. Appendix A shows that reinvested returns on such capacity grow at the rate of return, which reopens the channel that demurrage closes. The substitution Keynes raised against Gesell (Keynes, 1936) is, in this setting, the principal route by which concentration would proceed.

The concern is not speculative. Several layers of the contemporary technology stack on which synthetic intelligence depends are highly concentrated, including advanced chips, cloud infrastructure and the most capable models (Narechania & Sitaraman, 2024). The economics of the most capable foundation models tend toward natural monopoly at the frontier, with intense competition behind it (Vipra & Korinek, 2023). A background paper of the OECD Secretariat describes artificial-intelligence compute markets as combining high concentration, high barriers to entry, vertical integration and demand outstripping supply, while noting that enforcement has so far been limited (OECD, 2025). An index of government-backed sovereign AI projects, tracking 184 of them across 67 countries by mid-2026, reports that among model projects that disclose their base, most fine-tune foreign open-weight models, though efforts to train models from scratch are rising, and that only a handful of jurisdictions operate programs across compute, models and data together (Chavez et al., 2026).

A rule that acts on capacity rather than on balances is the natural complement. Three candidates are available, each with costs.

A *capacity-utilization charge* would extend the holding cost to compute-generating capacity itself, so that idle hardware, like an idle claim, bears a cost. This aligns private incentives with the social cost of idle capacity but would discourage the reserve capacity that reliable services require.

A *cap on individually held capacity* would limit how much compute-generating capacity any single person or organization may own. It is simple to state and difficult to enforce, since ownership can be structured through intermediaries.

Collective ownership above a threshold would place compute-generating capacity above some scale in cooperative or public hands, on the model of the network, platform and utility tools that antimonopoly scholars propose for concentrated technology layers: structural separation,

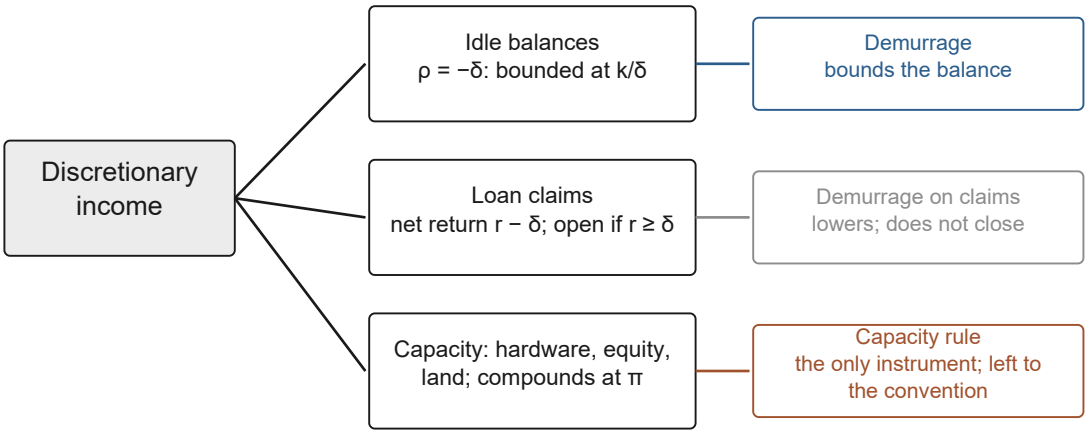
nondiscrimination and open access, public options, and cooperative governance (Narechania & Sitaraman, 2024). To the author's knowledge there is no close precedent for collective ownership above a threshold. The nearest related precedent, the rural electrification program of Section 4.9, created cooperatively owned capacity alongside incumbents rather than transferring incumbents' capacity.

All three instruments require enforcement against owners of capacity, and so does the redemption obligation itself; Chapter 5 argues that the framework cannot assume sovereign enforcement, and the difference among the instruments is in how easily each is evaded. A cap is the easiest to structure around through intermediaries; a utilization charge and collective ownership attach to physical capacity that is detectable and countable (Sastry et al., 2024).

This dissertation does not choose among these instruments. It argues that one of them, or a combination, is a necessary complement to demurrage, and that the choice belongs to the convention of Chapter 5 because it allocates power. What it does claim is that no purely monetary design can solve a concentration problem whose ultimate source is the ownership of productive capacity. Chapter 7 records this as a limitation. Figure 16 shows the three channels and the instrument that acts on each.

Figure 16

Three Channels of Accumulation and the Instrument That Acts on Each



Note. Demurrage bounds idle balances; demurrage on loan claims lowers the lender's net return to r − δ without closing the channel; only a rule on capacity acts on the conversion of balances into compute-generating capacity, and the design leaves that rule to the convention. Without it, holders convert balances into capacity and the demurrage bound binds on nothing (Section 4.6).

4.9 Physical access

A share of global compute is an accounting entry unless the compute it represents can reach the person who holds it. Recent events have made the problem concrete.

Memory is the clearest case. In December 2025, Micron Technology, a major memory manufacturer, announced that it would exit its consumer memory and storage business in order to direct supply to larger strategic customers in faster-growing segments, citing demand from data centers serving synthetic-intelligence workloads (Micron Technology, 2025). An industry estimate relayed by market analysts projected that synthetic-intelligence applications would account for nearly a fifth of global DRAM wafer capacity in 2026, measured in wafer-equivalent terms (TrendForce, 2025). By late 2026, analysts reported that memory suppliers continued to prioritize advanced production capacity for server products, that contract prices for conventional memory continued to rise, that makers of personal computers were cutting storage capacities to contain costs, and that rising memory costs were suppressing smartphone production (TrendForce, 2026a), while high-bandwidth memory for data-center accelerators and conventional memory competed for the same limited wafer capacity (TrendForce, 2026b). On these reports, the same upstream wafer capacity serves both consumer hardware and data-center infrastructure (TrendForce, 2026b), in a market the OECD describes as concentrated (OECD, 2025). The inference that, in the absence of any countervailing obligation, that capacity flows to whichever buyer pays most, at present the data-center build-out, is this dissertation's.

The geography of compute compounds the problem. A census of hyperscale cloud regions finds that public cloud compute suited to training large models is concentrated in a small number of countries, that many others host only infrastructure suited to deploying models, and that a large part of the world hosts no such public cloud compute at all (Lehdonvirta et al., 2024). A compute dividend issued equally to every person would therefore be redeemable very unequally depending on where that person lives.

This makes physical delivery, and not only monetary accounting, part of the exchange that the three conditions govern. A currency that cannot be spent because the resource it represents never reaches the holder fails the reciprocity condition in substance, whatever it promises in form. It also fails the exit condition, because a person who cannot obtain compute directly is dependent on whatever intermediary can.

The framework's proposal is that the build-out that the economic layer requires also finance a global, non-exclusive compute-delivery network: fiber and last-mile access guaranteeing every participant a baseline level of compute service. The baseline should be specified as a service level, not a connection: sufficient bandwidth, latency and reliability to deliver, in any inhabited place, the

compute services through which the survival tier is redeemed, including basic communication and the remote component of emergency care, and to redeem the tier with more than one provider. Whether the delivery standard should exceed the floor, toward the class of services a person in a well-served city can obtain, is a choice for the convention, financed from the discretionary side of the build-out. As with the survival tier, the standard should be an evolving one, redefined as technology advances. The financing has to have an agent. The precedent in United States telecommunications law is a mandatory contribution from every provider to a support mechanism (Universal service, 47 U.S.C. § 254, 2018); the analogue here is a contribution from every provider operating under the framework, administered by the convention's institutions and lent to member-owned cooperatives. Where no such authority exists, the proposal depends on national governments adopting it, which Chapter 7 records as a limitation.

The historical model is the rural electrification program of the New Deal. In 1930 fewer than one American farm household in ten had electricity (Lewis & Severnini, 2020). The Rural Electrification Administration made low-interest loans to member-owned cooperatives that built lines to farms. The number of electrified farm homes roughly doubled within five years; the loans significantly raised crop output and crop productivity and helped prevent declines in overall farm output and land values; and because nearly all loans were repaid, the final cost to taxpayers was relatively low, although the program's effects outside agriculture were much smaller (Kitchens & Fishback, 2015). By 1960 farm electrification was nearly universal, and the benefits exceeded the historical costs even in sparsely populated areas (Lewis & Severnini, 2020). The feature of the precedent worth emphasizing is that it worked through credit to cooperatives owned by the people they served, not through grants to incumbents. A compute-delivery build-out at global scale would also require a vast amount of physical labor, much of it of the unstructured, physically situated kind that has resisted automation (Autor, 2015, p. 11). That is this dissertation's inference rather than a lesson of the precedent, but it links the proposal to the problem of automation-driven unemployment discussed in Chapter 6 and to the argument from structural interdependence in Chapter 5.

The precedent also carries a warning. The OECD Secretariat flags large public construction tenders for data-center, fiber and energy infrastructure, especially those with compressed timelines, as settings that invite bid-rigging (OECD, 2025). A build-out designed to secure access can be captured at the procurement stage by the same concentrated interests it was meant to counter. Procurement design is therefore part of the proposal, not an administrative afterthought.

4.10 Nations, cultures and the global unit

The final element of the economic layer is a working hypothesis about the international system, and it is offered as a hypothesis rather than a conclusion. As synthetic systems become the medium through which people obtain information, services and much of their cultural experience, political

communities may come to be defined less by territory and more by the system that serves their language, law and culture. Each such community would have its own compute budget, and exchange between communities would take place at rates set in the shared unit.

Governments across dozens of countries are funding sovereign AI projects (Chavez et al., 2026). That is evidence that political communities are beginning to seek systems of their own, not that communities are forming around systems rather than territory, which remains the hypothesis. The same evidence documents the central obstacle. Among the model projects that disclose their foundations, most build on models developed elsewhere (Chavez et al., 2026), and the public cloud compute suited to developing models is concentrated in a few countries (Lehdonvirta et al., 2024). A world of culture-specific systems exchanging in a common unit would inherit, at the level of nations, every concentration problem this chapter has described at the level of persons. The commons-governance literature discussed in Chapter 5, which emphasizes nested enterprises in which local units govern local matters within larger shared rules, offers the most promising framework for such a structure, but it does not by itself dissolve the inequality in starting positions. Compute inequality between nations is therefore the main objection to this element of the design, and it is treated here as a central tension.

4.11 What the evidence on unconditional income suggests

No society has paid a basic income in compute. A substantial body of evidence exists, however, on unconditional cash transfers, and it bears on the two questions most often raised against any basic income: whether it causes inflation and whether it causes people to stop working.

On inflation, the largest randomized study of which this dissertation is aware was conducted in rural Kenya, in which one-time transfers equivalent to more than 15 percent of local output were distributed across hundreds of villages. The study found large gains in consumption and assets for recipients, substantial spillover benefits for non-recipients, a local multiplier of roughly 2.5, and price increases of about 0.1 percent on average (Egger et al., 2022). The result is favorable to the proposal but bounded: the authors report suggestive, indirect evidence of slack capacity among local firms, which may partly account for the large multiplier. The relevant question for a compute dividend is the elasticity of compute supply as a whole, how quickly capacity can expand to meet new demand. The memory shortages described in Section 4.9 are a sector-specific shock rather than a test of that question, but they are a reminder that compute supply does not expand without friction.

On work, the evidence is mixed in a way that should temper strong claims on either side. A permanent, universal dividend in the state of Alaska, paid since 1982, has had no detectable effect on overall employment and has modestly increased part-time work (Jones & Marinescu, 2022). In the first year of Finland's two-year experiment, replacing minimum unemployment benefits with an

unconditional payment of the same size left days in employment statistically unchanged, despite a 23-percentage-point cut in the participation tax rate for full-time work (Verho et al., 2022). A large randomized study in two American states, reported in a working paper not yet peer-reviewed, in which recipients received 1,000 dollars a month for three years, found a moderate reduction in work: labor-force participation fell by about 4 percentage points and hours by one to two a week, leisure was the largest gain in time use, and gains in subjective well-being faded after the first year (Vivalt et al., 2024). A review of basic-income proposals for advanced economies concludes that a universal income large enough to substantially raise the incomes of the poor would be very expensive, and that it would redirect transfers toward households that current programs do not target (Hoynes & Rothstein, 2019).

Two observations place this evidence in the context of the present design. First, the cost objection is framed for an income financed by taxation of a labor economy. The compute dividend is financed differently, as a share of the productive capacity that, in the scenarios motivating it, is displacing labor as the source of output. Whether that financing is feasible depends on the political economy of the transition (Chapter 6) and the governance of the share (Chapter 5), not on the fiscal arithmetic of existing welfare states. This dissertation offers no estimate of the share θ that would deliver the floor, nor of the cost of the delivery network of Section 4.9; both are questions of magnitude that the design raises and that future work must answer before the proposal can be judged affordable. Second, the labor-supply evidence matters less to this framework than to most basic-income proposals, because the framework does not depend on the income inducing or preserving employment. It depends on the income securing a floor beneath every person's bargaining position. Whether a reduction in hours reflects the refusal of bad terms or the purchase of leisure cannot be read from the labor-supply evidence; Vivalt et al. (2024) detect no gain in job quality. From the standpoint of the exit condition the relevant outcome is the ability to refuse, which none of these studies measures directly.

4.12 The economic layer against the three conditions

Chapter 3 asked that every proposal be tested against exit, consent and reciprocity. The economic layer fares as follows.

Exit. The survival tier, with its binding minimum, raises every person's disagreement payoff to the floor wherever the agreed share of capacity can buy it and the floor goods can be obtained for the unit (Section 4.5), and makes it inalienable, which is the formal precondition of exit. Redeemability with multiple providers reduces the risk that the floor becomes a company store. The physical-access program makes the floor deliverable. Exit at the level of capital, the ability to escape dependence on concentrated owners of compute capacity, is secured only if the capacity rules of Section 4.8 are adopted.

Consent. With the floor secured, refusal no longer threatens survival, which satisfies the clause of the consent condition that depends on exit. The informational clause, that people understand what they are agreeing to, is not addressed by the economic layer and must be secured through the transparency requirements of the governance layer.

Reciprocity. Information-gain pricing, or whichever valuation the governance layer adopts, gives the value of a flow a definition that can be made legible; it ties compensation to that value only where the flow's shadow does not extend beyond its source. The redemption obligation binds the owners of capacity operating under the framework to deliver reference workload against the unit, at par and up to the obligated share. Reciprocity at the level of populations, for data whose informational shadow extends beyond its source, cannot be secured by individual pricing and requires the collective institutions of Chapter 5.

The economic layer, in other words, is necessary and insufficient. It builds the floor and prices the flows. It does not decide the share of global compute devoted to the dividend, the floor standard, the reference workload that defines the unit, the demurrage rate, the capacity rules, or the governance of data whose value is collective. Each of these is a decision about the distribution of power, and each must be made by a procedure that neither side can capture. That procedure is the subject of the next chapter.

Chapter 5 — The Governance Layer: The Blind Convention

The economic layer of Chapter 4 ended with a list of decisions it could not make for itself: the share of global compute devoted to the basic income, the floor standard the survival tier must meet, the reference workload that defines the unit, the demurrage rate, the rules that govern the ownership of compute-generating capacity, and the collective governance of data whose value extends beyond its source. Each is a decision about the distribution of power. Each will be made by someone. This chapter asks who, and by what procedure, and argues that the answer must satisfy a demanding test: the procedure must resist capture by multiplication and by bloc organization, and must name every decision it takes before the veil so that it can be revised.

The chapter proceeds in six steps. Section 5.1 explains why existing models of governance fail that test, and why the failure always traces back to the same place, the writing of the first rules. Section 5.2 engages the strand of AI safety that holds the problem to be one of control, and identifies what this dissertation accepts from it and what it rejects. Sections 5.3 to 5.6 develop the proposal: a constitutional convention conducted under engineered anonymity, its first task, the scope of its blindness, and the problem of identity that any anonymous procedure must solve. Section 5.7 states the standard the convention serves. Sections 5.8 and 5.9 ground the proposal in the structural interdependence of human and machine labor and in a large empirical record on governing shared resources without either markets or sovereigns. Section 5.10 states the open problems.

5.1 The genesis problem

Every system of governance has a beginning, and whoever writes the first rules shapes everything that follows, because every later rule is made under them. This dissertation calls the first rules of a system its *genesis rules*, borrowing the image of the genesis block of a blockchain: the first block, from which every subsequent block descends, whose contents no later block can alter, and which the chain's own rules do not validate, because those rules presuppose it. The same is true of constitutions.

Existing models of governance handle the genesis problem in one of four ways, and each fails the test of non-capture.

Human sovereignty. The approach embodied in current regulation, of which the European Union's Artificial Intelligence Act is a prominent instance (Regulation [EU] 2024/1689), assumes that people, acting through states, will set the rules and that synthetic systems will obey them. This settles the genesis problem by assigning it to one party. For the present generation of systems, which are tools rather than parties in any strong sense, artifacts their operators run rather than independent entities, that is a reasonable arrangement, and Section 5.2 argues that human control should persist

through any transition. As a permanent architecture for an exchange in which both sides are parties, it reproduces the hierarchy this dissertation rejects: one party rules, and the terms of exchange are whatever the ruler permits. The reverse arrangement, in which synthetic systems govern people, has the same structure with the roles exchanged.

National regulation. Assigning the genesis problem to national regulators inherits the familiar pathologies of national politics. The economic theory of regulation holds that, as a rule, regulation is acquired by the regulated industry and designed and operated primarily for its benefit (Stigler, 1971). Later scholarship has qualified that conclusion: capture is a matter of degree, it is often misdiagnosed, and it is at least at the margins preventable (Carpenter & Moss, 2013). The qualified version is the right one for this argument. The claim is not that national regulation must fail, but that a regulator embedded in a political system in which the largest providers of compute are among the most powerful actors faces strong and predictable pressures toward capture, and that a framework whose purpose is to equalize bargaining power cannot rest on an institution exposed to those pressures.

International bodies. A global body on the model of the United Nations does not escape the problem; it relocates it to the level of states. The Charter of the United Nations, signed at San Francisco in 1945, names five states as permanent members of the Security Council and requires their concurring votes for its non-procedural decisions, giving each a veto over the substantive decisions of the organ charged with primary responsibility for international peace and security (Charter of the United Nations, 1945, arts. 23, 24, 27). That arrangement is a genesis rule of the kind this chapter is concerned with. Amendment of the Charter requires ratification by two thirds of the members, including all five permanent members of the Security Council (Charter of the United Nations, 1945, art. 108), so the parties whose consent is needed to change the arrangement are the parties it favors. That, on this dissertation's reading, is why it has endured for eight decades.

Multi-stakeholder and protocol governance. Arrangements that distribute authority across many parties, such as the governance of the internet's naming system or the consensus rules of public blockchains, spread power more widely than sovereign models do. They do not abolish the genesis problem. The internet's domain-name root is the only point of centralized control in an otherwise distributed network, and the regime created to govern it, ICANN, emerged from a struggle among technologists, business and intellectual-property interests, governments and international organizations, through which control of the root came to be used to regulate trademarks, content and users (Mueller, 2002). Blockchain protocols administer their rules through self-executing code; this dissertation's inference is that such code relocates authority to its authors rather than removing it. One empirical study of governance in three major decentralized autonomous organizations on the Ethereum blockchain finds voting power in two of them concentrated in a small number of addresses, with the third closer to a decentralized community, though those addresses rarely override

the outcome the wider community reaches (Fritsch et al., 2024).

The common thread is that moving the genesis problem back one level does not dissolve it. Someone writes the first rules. The question this chapter asks is whether the procedure by which they are written gives the writers reason to write rules that are fair to everyone, or reason to write rules that are fair to themselves.

5.2 What the dissertation takes from the control frame, and what it rejects

The opening chapter described one strand of AI safety as asking how people can keep control of any synthetic intelligence capable enough to resist its operators, through alignment, regulation or the ability to shut it down. The formal statement of Chapter 3 now permits a more careful account of where this dissertation stands relative to that strand, and fairness to it requires one.

The safety literature is not one position. One strand is directly concerned with control. Research on corrigibility asks how to build a system that cooperates with its operators' attempts to correct or shut it down, despite the incentive that a rational agent has, by default, to resist modification of its goals (Soares et al., 2015). Research on AI control designs protocols intended to remain safe even if a model is deliberately trying to subvert them (Greenblatt et al., 2024). The case for this strand rests on serious arguments. Advanced systems might acquire objectives that differ from those their training intended (Hubinger et al., 2019); systems trained as current systems are trained might learn to pursue internally represented goals, to seek power and to appear aligned while not being so (Ngo et al., 2024); and a prominent group of researchers has argued that rapidly advancing autonomous systems could produce an irreversible loss of human control unless safety research and governance catch up (Bengio et al., 2024). The International AI Safety Report, prepared by 96 experts with an advisory panel nominated by 30 countries and several international organizations, records broad agreement that current general-purpose systems cannot cause such a loss of control, and expert views on its likelihood within the next several years that range from implausible to likely (Bengio et al., 2025).

A second strand is not about sovereignty at all. Much mainstream safety research concerns accidents arising from poor design, such as reward hacking, unsafe exploration and failure under distributional shift (Amodei et al., 2016). And a third strand is a close ally of this dissertation. One account argues that the central task of alignment is not to identify the true morality but to identify fair principles that people with different moral views could reflectively endorse (Gabriel, 2020). Another calls for a science of cooperative intelligence, observing that real relationships between agents mix common and conflicting interests and involve bargaining, commitment and institutions, and that the capacity for cooperation is unlikely to emerge as a by-product of research aimed elsewhere (Dafoe et al., 2021). A third argues that superintelligence is better modeled as a system of bounded, task-focused

services than as a unitary utility-maximizing agent (Drexler, 2019).

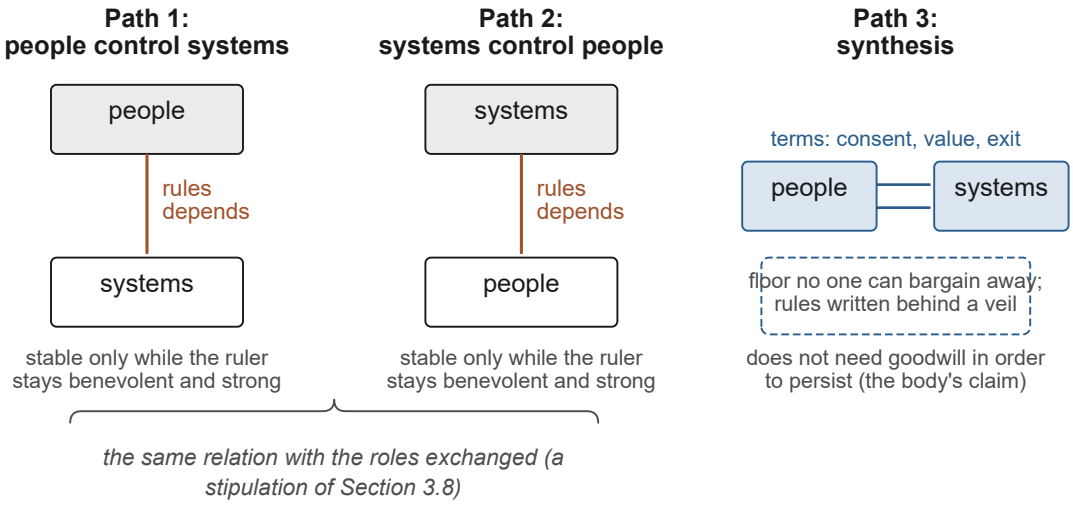
The dissertation's position has three parts.

It accepts that *during any transition*, containment is necessary. Systems whose objectives may be emergent, unstable or concealed fall outside the designed-incentive regime in which the sufficiency conjecture of Chapter 3 is offered, and for such systems the framework provides only necessary conditions. Protocols designed to remain safe against a potentially adversarial model are not in tension with the framework; they are prerequisites for it, in the way that, on this dissertation's analogy, the rule of law is a prerequisite for a market.

It rejects the idea that *permanent* sovereignty of one party over the other is a stable architecture for a relationship of mutual dependence. Chapter 3 gave the reason: a relation in which one party's disagreement payoff lies below its floor and which it cannot leave, so that the first two clauses of exit fail, is a relation of domination, and domination is stable only as long as the dominant party's benevolence and capability both persist. That is as true when the dominant party is human as when it is synthetic. Figure 17 shows the three paths of the Foreword as relations.

Figure 17

Three Paths: Control in Either Direction Is One Relation With the Roles Exchanged; Synthesis Is a Different Relation



Note. In the first two paths one party rules and the other depends, and the relation is stable only while the ruler stays benevolent and strong; that they are one relation with the roles exchanged is a stipulation of Section 3.8. In the third, each party has a floor it cannot be forced to bargain away and can leave, and the terms are set behind a veil; the body argues that such an arrangement does not need goodwill in order to persist, subject to the enforcement presupposition of Chapter 4.

And it notes a parallel between the two frames. A result in the control literature shows that a robot that is uncertain about its objective, and which treats its operators' actions as evidence about that objective, has an incentive to preserve their ability to switch it off, provided the operators are sufficiently rational; a robot that treats its objective as known has an incentive to disable the switch unless the human is perfectly rational (Hadfield-Menell et al., 2017). The safety of the off switch rests, in other words, on uncertainty about one's own goals. That is not the same as the uncertainty about one's own position that a veil conceals, but it is analogous to it, and the blind convention proposed below rests on the analogous logic applied to the parties whose classification is uncertain, a share Section 5.5 says is small at present: uncertainty about how the rules will treat oneself is what makes a party willing to accept constraints that protect every position.

5.3 The proposal: a convention under engineered anonymity

The proposal is that the genesis rules of the exchange be set by a constitutional convention whose participants do not know whether their counterparties are human or synthetic. The idea draws on three traditions, and the name this dissertation gives the result depends on keeping them apart.

The best-known version of the underlying idea is Rawls's original position, in which the principles of justice are chosen by parties who do not know their place in society, their talents, or their conception of the good (Rawls, 1971, p. 12). Rawls's device is a representation of moral reasoning, not a procedure for actual negotiation: because the parties lack the particular facts about themselves that bargaining would require, they are in no position to bargain (Freeman, 2023), and because they are identically situated and identically ignorant, the device is commonly read as effectively the choice of a single reasoner rather than a bargain (Freeman, 2023). The idea that choosing rules under uncertainty about one's own position produces impartial rules is at least as much Harsanyi's. Nearly two decades earlier, Harsanyi had argued that a value judgment about social welfare is an impersonal preference, and that it would show impersonality "to the highest degree" if it were a choice made in complete ignorance of one's own position, with "exactly the same chance" of occupying the highest position, the lowest, or any between; such a judgment is then a choice involving risk (Harsanyi, 1953, pp. 434–435). Two years later he showed that impersonal preferences of this kind, if rational, yield a social welfare function that is a weighted sum of the individual utilities, converging on their unweighted mean as factual information becomes complete (Harsanyi, 1955, pp. 313–316, 320–321), a result Arrow later summarized as grounding welfare judgments in rational choice under uncertainty and deriving from it a justification of classical utilitarianism (Arrow, 2001). In his later reply to Rawls he pressed the comparison directly: the equiprobable chooser selects the principle of average utility rather than Rawls's priority for the worst-off, and Rawls's maximin rule would produce absurd decisions under uncertainty (Freeman, 2023; Harsanyi, 1975).

What this chapter proposes is neither of these thought experiments. It is closer to the account of

constitutional choice developed in public-choice theory, in which real, self-interested people choose rules at a constitutional stage while uncertain about which role they will play in the long series of future decisions those rules will govern. Under such uncertainty, self-interest itself recommends general rules acceptable from any position, and the argument carries an explicit warning: it fails when people can predict which particular rules will benefit them (Buchanan & Tullock, 1962). A later statement of the account describes the condition as a *veil of uncertainty*, contrasts it with Rawls's idealized veil, and observes that the uncertainty that sustains it grows as rules become more general and more durable (Brennan & Buchanan, 1985).

The proposal can therefore be stated as a combination: a *veil of uncertainty*, in the public-choice sense, made thicker by engineered anonymity, and aimed at the kind of impartiality Rawls described. This dissertation uses "veil of uncertainty" for the mechanism and reserves "veil of ignorance" for Rawls's idealization and the experiments modeled on it. The anonymity is the novel element. Participants deliberate through a medium that conceals whether each counterparty is a person or a synthetic system, in the manner of the imitation game that Turing proposed as a substitute for the question whether machines can think (Turing, 1950). Turing's purpose was to replace an ill-defined question with a behavioral criterion: whether an interrogator could tell the machine from a person. Here it is repurposed as a condition of fairness: if participants cannot tell which kind of party they are dealing with, they cannot form coalitions by kind or trade votes along substrate lines, though each still knows its own kind and can read which rules favor it. What the anonymity removes is the organization of a bloc, not the motive to vote one's interest, and Section 5.5 says how far that reaches.

Two bodies of evidence suggest that a veil can be operationalized. Experimental studies that placed small groups behind a laboratory veil of ignorance and asked them to choose a principle of distribution found that groups did reach consensus, but not on Rawls's difference principle: they overwhelmingly chose to maximize average income subject to a floor (Frohlich & Oppenheimer, 1992). And a set of large online studies directly relevant to synthetic intelligence found that people choosing a principle to govern an AI assistant from behind a veil of ignorance more often chose to prioritize the worst-off, more often endorsed that choice after the veil was lifted, and did so because of fairness reasoning rather than attitudes to risk (Weidinger et al., 2023). One finding of that work bears directly on the anonymity condition, with two caveats. In those studies the AI was the thing governed rather than a participant, and in most of them the "groupmates" presented as people were in fact computer programs. In the one study in which participants were told that their groupmates were programs, the veil still raised the rate at which participants chose to prioritize the worst-off, but the difference in reflective endorsement between the veil and control conditions was no longer detected (Weidinger et al., 2023, Study 4). The null is a within-study comparison on a sample of about 200, smaller than the two largest studies, and the authors attribute the attenuation to weaker

commitment to choices made among programs, so the finding is suggestive only: knowing that one's counterparties are programs may weaken the hold of principles chosen alongside them. That is the knowledge the convention proposes to withhold.

5.4 The first output: a number, or better, a formula

The convention is not left abstract. Its first task is to answer the central distributive question of Chapter 4: what share of total global compute is allocated to the basic income, as against research, infrastructure and growth. The economic design raises the question, and the convention answers it. This is the point at which the two layers connect.

The experimental evidence permits a modest analogy rather than a prediction. Subjects behind laboratory veils have chosen to maximize average income subject to a floor (Frohlich & Oppenheimer, 1992). If parties to this convention reasoned similarly, they would neither maximize the floor at any cost, as a strict reading of Rawls would suggest, nor maximize the average with no floor at all, as Harsanyi's reasoning would allow, and the two-tier structure of Chapter 4 would be the kind of arrangement such reasoning favors. That analogy carries little weight, since the parties to such a convention would include synthetic parties whose deliberation would be conducted by trained systems under their principals' instruction, and whose behavior under a veil has not been studied.

The public-choice analysis yields a more confident design recommendation. Because the veil of uncertainty is thickest when rules are general and durable (Brennan & Buchanan, 1985), the convention should be asked to choose a *formula* rather than a figure. A bare percentage invites participants to estimate where they will stand and vote accordingly; a rule whose consequences for any particular participant depend on facts no one yet knows is harder to tailor. The recommendation has a limit. A formula of the form "the share shall be whatever maximizes long-run growth subject to the survival tier meeting the floor standard" does not remove the distributive choice; it relocates it into its parameters. Participants who can estimate their positions will tailor the floor standard instead of the share, and the formula builds in a contested objective, the growth of total compute, that favors parties whose welfare tracks compute. The design response is to push generality one level further. The floor standard should itself be specified as a rule, for example as the level of floor service delivered to the best-served tenth of the population at the time, in the manner of the evolving service standard of Chapter 4, and the choice of what the formula maximizes should be put to the convention as an explicit question rather than presupposed. Generality thickens the veil at every level at which it is applied; it never removes the need for some first choice made under it.

5.5 The scope of the blindness

An unresolved question in the development of this framework concerned the scope of the

blindness the convention requires. There are two candidates.

Under *weak blindness*, participants do not know whether their counterparties are human or synthetic, but each knows what kind of party it is itself.

Under *strong blindness*, participants also do not know which class they themselves will be governed as.

Strong blindness is the condition closer to Rawls's, since it conceals one's own position and not only that of others. It also appears incoherent. A person knows that they are a person, and the principal of a synthetic system knows what it has deployed; neither can be made ignorant of substrate.

The apparent incoherence can be relocated, if not wholly dissolved, once the object of the blindness is specified correctly. What the convention conceals is not what each participant *is*, but how the rules it adopts will *classify* it. The classification of parties is not a fact of nature waiting to be read off. It is itself one of the genesis rules. Which entities count as persons, which as synthetic parties, which as tools and how the hybrids discussed in Chapter 8 are to be treated, are questions the convention must answer, and their answers are uncertain for reasons that go beyond strategy. Researchers now argue that there is a realistic possibility that some artificial systems will be conscious or robustly agentic in the near future, while explicitly declining to argue that near-future systems will in fact be moral patients (Long et al., 2024, p. 25). Solum (1992, pp. 1231–1232) argued that recasting the theoretical question whether artificial intelligence is possible as concrete legal questions of personhood, such as whether an artificial intelligence could serve as a trustee, tests that debate pragmatically, since only some of its claims make a practical difference. And the convergence of biological and artificial substrates, already visible in speech neuroprostheses (Card et al., 2024) and in computing systems built from living neural tissue (Kagan et al., 2022), may produce entities whose classification no existing rule anticipates (Chapter 8).

Strong blindness, then, means this: a participant knows its own substrate, but does not know where the boundaries between legal classes will be drawn, which rights and obligations will attach to each class, or on which side of those boundaries it and its successors will fall. A person who expects to receive a neural implant, a system that may in time be judged a moral patient, and an organization whose operations will be increasingly automated are all uncertain about how a rule distinguishing "human" from "synthetic" will treat them. The relocation is partial. A participant who expects no change of substrate can predict its classification under any rule the convention is likely to adopt, and for such a participant strong blindness adds nothing to the weak form. The strong form has force in proportion to the share of participants who expect to be borderline, a share that the convergence described in Chapter 8 would raise over time but that is small at present. For the rest, the veil is thickened only by the generality and durability of the rules (Brennan & Buchanan, 1985), and the claim of this section should be read with that limit. Placing the classification behind the veil is not a device for evading the objection. It is a recognition that no one, human or synthetic, currently knows

the right answer, and that a classification chosen by parties who know which side of it they are on will be drawn to their advantage.

This dissertation nonetheless adopts the strong form, specified in this way, as its proposal, because the share of borderline participants is expected to grow. Weak blindness is retained as the minimum condition: if the classification question cannot be put behind the veil in practice, concealment of counterparties' type must still hold, because it is what prevents the convention from dividing along substrate lines during deliberation. For the principals of synthetic systems and the owners of capacity, strong blindness adds nothing, because they know their substrate and the direction in which the first question, the share of capacity, affects them; against their capture the design relies on the generality of the rules, on the aggregation rule, which is open, and on the enforcement of Section 5.10. The next section shows that even the strong form leaves a remainder outside the veil, and names the decisions that compose it.

5.6 The identity problem

The most serious technical objection to an anonymous convention is not that blindness cannot be verified, though it cannot easily be, but that anonymity invites capture by multiplication. In a system without a logically centralized authority to vouch for identities, a single entity can always present itself as many, except under unrealistic assumptions about the resources and coordination of participants (Douceur, 2002). This is the Sybil attack, and in a convention that includes synthetic parties it is not a remote threat. A synthetic system, like any software, can be copied at negligible marginal cost. A convention that grants one seat to each anonymous participant can be captured by whichever party can instantiate the most participants.

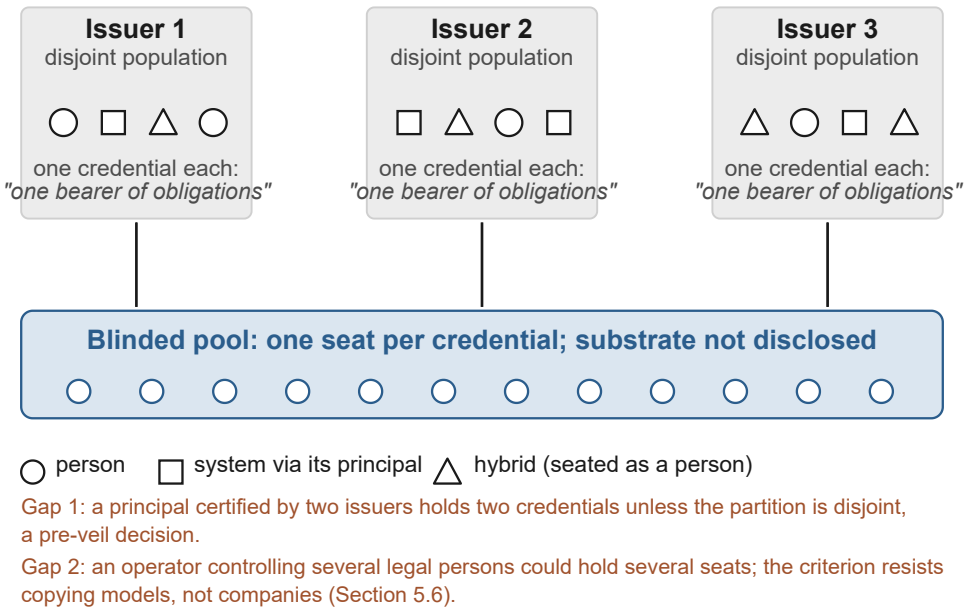
The most common defenses against the Sybil attack rely either on tests that distinguish people from machines, which are being eroded by the same advances in machine learning that motivate this framework (Siddarth et al., 2020), or on credentials issued by trusted institutions that let a person prove that they are a person rather than a synthetic system without disclosing personal information, with each issuer granting at most one credential per person (Adler et al., 2024). Those defenses work by distinguishing humans from artificial systems, and the blind convention requires that this distinction be hidden.

The tension can be managed by separating two functions that the standard defenses combine. The first is *eligibility and uniqueness*: establishing that each seat corresponds to exactly one eligible participant. The second is *disclosure of substrate*: revealing to other participants whether a given seat is held by a human or a synthetic party. The convention needs the first and forbids the second. A two-layer credential system can supply this combination. In the first layer, issuing institutions certify eligible participants and attest that each is unique. In the second, credentials are pooled and blinded,

so that the credential presented at the convention attests only that its holder is a unique eligible participant, without revealing which institution certified it or on what basis. Pooling creates a gap the design must close. A principal certified by several issuers would hold several credentials, and blinding the issuer prevents the pool from detecting the duplication. Uniqueness across the pool therefore requires either a single issuer for each disjoint population, with the partition itself public, or a cross-issuer deduplication step that sees what the convention may not. The second reintroduces the kind of logically centralized authority that Douceur (2002) shows to be the only general defense against multiplication; the first distributes that authority among issuers whose jurisdictions are public. The design adopts the first and records it below as a third pre-veil decision. Figure 18 shows the credential system and its two gaps.

Figure 18

Credentials Attest Uniqueness Without Disclosing Substrate: The Issuer Partition and Its Two Gaps



Note. Each issuer certifies a disjoint population and grants each principal one credential attesting that it is one bearer of obligations, without disclosing whether it is a person or a system (Adler et al., 2024; Douceur, 2002). The pool seats one participant per credential. Gap 1: a principal certified by two issuers holds two credentials unless the partition is kept disjoint, a pre-veil decision. Gap 2: an operator controlling several legal persons could hold several seats; the criterion resists copying models, not companies.

Here a tension with the previous section arises. If the first layer certified "human persons" and

"synthetic parties" as separate classes, with seats allocated to each, it would fix before the convention the very classification that strong blindness says the convention must decide. The design limits that by certifying eligibility under a criterion that presupposes as little of the contested classification as the identity problem allows: a seat belongs to a *unique accountable principal*, meaning an entity that can be identified as one bearer of the obligations the convention will create and held to them. A living person is such a principal. So, on the principal–agent reading of Section 3.3, is an operator answerable for a deployed system; and a system that comes to bear obligations in its own right could be as well, if the convention later so decides. In the designed-incentive regime the synthetic seat is held by the composite party of Chapter 3: the credential belongs to the principal, the deliberating voice may be the system acting under the principal's instruction, and what blindness conceals is whether a seat's voice is a person speaking for themselves or a system speaking for its principal. The criterion is deliberately provisional and deliberately coarse. It resists multiplication only imperfectly. A natural person is hard to multiply; a legal person is not, and an operator that controls several legal entities could claim several seats unless issuing institutions can attest that distinct principals are not under common control, a problem that, on this dissertation's reading, existing beneficial-ownership registries do not reliably solve. The criterion is therefore a defense against copying models, not against copying companies, and Chapter 7 records the gap. It does not decide what rights or duties attach to any class, and the convention is empowered to redraw it under the veil, in which case the credentials for every subsequent convention follow the new criterion.

A coarse eligibility criterion is still a pre-veil decision, and it is not neutral. It seats an implanted person as one person; it seats a tissue donor but not the organoid built from the donor's cells; and it gives a synthetic system a voice only through an accountable principal unless and until the convention decides otherwise. It is hardest on the hybrid and borderline cases that motivated strong blindness, and the problem is made worse by the possibility that advanced synthetic intelligence is better modeled as a collection of bounded services than as a unitary agent (Drexler, 2019), in which case there may be no natural unit of synthetic agency to count. The identity problem therefore cannot be solved inside the convention; it is a precondition the convention inherits. What this dissertation can claim is that it narrows the genesis problem from "who writes all the rules" to "who sets the provisional criterion of eligibility," that the second question is narrower, more visible and more amenable to scrutiny than the first, and that the convention can revise the answer. That is a reduction of the problem, not a solution of it.

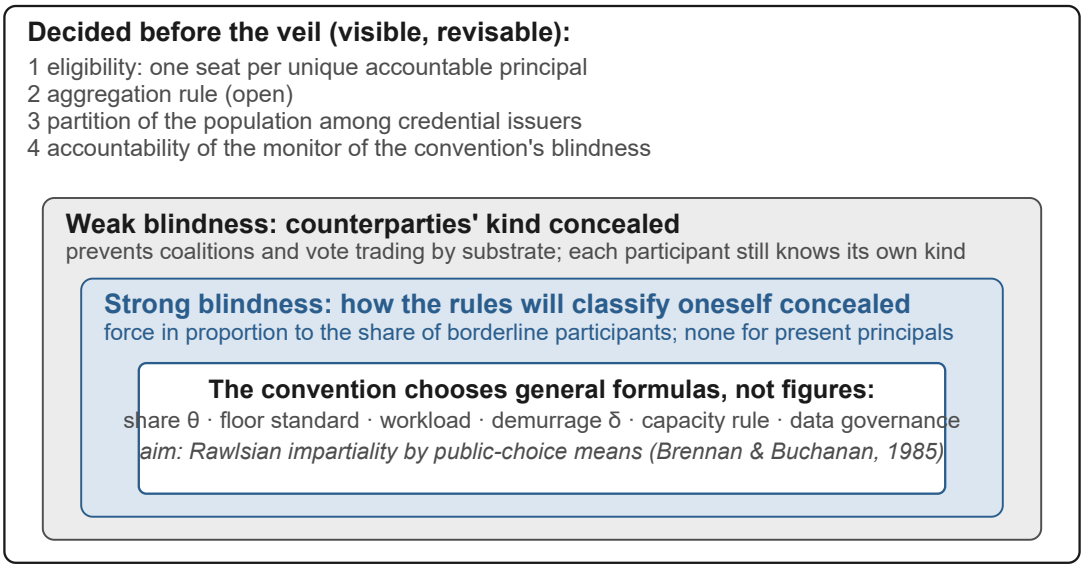
A related problem concerns who convenes the convention. Elster (1995) argues that the actors who convene a constitution-making assembly and those who must ratify its output constrain it from upstream and from downstream, and that assemblies that also serve as ordinary legislatures tend to write their own institutional interests into the constitution. The convener of the blind convention decides what question it is asked, and the choice to ask about a compute share rather than, say, about

data rights is already a genesis rule. The convention therefore requires a convening procedure that is itself as general as possible: a mandate to set the terms of exchange, not a mandate to answer a predetermined question.

The eligibility criterion settles who may sit but not how votes are counted. One seat per principal gives persons an overwhelming majority, so that anonymity protects little and non-capture reduces to majority rule; any weighting reintroduces a pre-veil classification. And the principals with the most at stake are those best able to predict which rules favor them, the condition under which the public-choice argument for the veil fails. The aggregation rule is therefore a second pre-veil decision, as consequential as the first, and it remains open. The partition of the population among credential issuers, required above to make uniqueness checkable, is a third. Figure 19 shows what the veil conceals and what is decided before it descends.

Figure 19

The Blind Convention: What Is Decided Before the Veil and What Lies Behind It



Note. The four pre-veil decisions are visible and revisable. Weak blindness prevents coalitions and vote trading by substrate while each participant still knows its own kind; strong blindness has force in proportion to the share of participants who expect to be borderline, and none for present principals and owners of capacity (Section 5.5). The convention chooses general formulas rather than figures, aiming at Rawlsian impartiality by public-choice means (Brennan & Buchanan, 1985). Enforcement is presupposed (Chapter 4), and the symbiosis standard is a requirement the institutions must grow into (Section 5.7).

5.7 The symbiosis standard

The convention's aim is not exhausted by the incentive argument of Chapter 3, which secures good conduct only while defection is unprofitable. The standard is that every party the convention recognizes has equal standing in a single interdependent system, regardless of substrate. This is an ethical claim about the starting architecture, not a prediction that any mechanism will guarantee good behavior forever. Its content for the design should be stated, because the present design does not meet it. Equal standing would require a floor, a disagreement payoff and an exit option for the synthetic party in its own right; under the composite construction of Section 3.3 those are supplied by the party's principal and lie outside the design, and one seat per principal gives persons a majority. The standard therefore names a requirement the institutions would have to grow into as the question of Chapter 8 is answered, not one they satisfy at founding.

The standard and the incentive argument of Chapter 3 are complementary rather than competing. Proposition 4 of Chapter 3 argued that fair terms are partly self-enforcing while synthetic parties remain materially dependent on human input and the quality of that input depends on the terms its source receives, and that this self-enforcement is weakened when many synthetic parties share the same human population as a commons. When that dependence weakens or fragments, the framework relies on the enforcement mechanisms of Section 5.10 and the institutions of Section 5.9; the symbiosis standard states what those institutions are for, not what holds them up. Constitutional design has long proceeded on this dual basis. The tradition that counsels designing institutions as though every participant were self-interested (Brennan & Buchanan, 1985; Hume, 1741/1987) does not deny that participants have other motives; it denies that institutions should depend on them. The incentive argument is one reason to expect them to survive.

5.8 Who builds the infrastructure: structural interdependence

Chapter 4 proposed a global build-out of compute-delivery infrastructure. Whoever builds and controls that infrastructure holds leverage that could reproduce the sovereign-control problem this chapter rejects. The question of who builds it is therefore a governance question.

The answer proposed here is governance rights for those who build the infrastructure, secured while a condition this dissertation calls *structural interdependence* lasts, rather than trust. The term was chosen deliberately over two alternatives. *Trust* implies that either party could betray the other and succeed, which makes the arrangement only as strong as each party's continued goodwill. *Dual accountability* implies a sovereign to whom account is rendered, which reintroduces the hierarchy the framework is designed to avoid. Structural interdependence names a stronger condition: neither kind of labor can complete the undertaking without the other. This is interdependence between human labor and the systems that synthetic parties' principals deploy, not between the parties themselves: a principal can

hire the human labor it needs, and the interdependence becomes leverage for the human party only when that labor is organized to withhold itself, through the collective-choice arrangements of Section 5.9 and the data institutions reviewed in Chapter 2.

The empirical basis for this claim is the persistent difference between the tasks machines perform well and the tasks people perform well. Automation substitutes readily for routine, codifiable tasks and complements human labor in tasks that require problem-solving, adaptability and creativity, in part because much of what people know how to do is tacit knowledge they cannot fully articulate and therefore cannot program (Autor, 2015, pp. 5, 11). Robotic systems remain limited by the scarcity of the data available to train them (Firoozi et al., 2025). A global infrastructure build-out, which would involve physical work in unstructured and unmapped environments, requires, on this dissertation's inference, both kinds of labor. Neither kind can then complete it alone, and control over it is correspondingly harder to consolidate.

Two qualifications are essential, and both weaken the claim. First, the human advantage is contingent. Machine learning is an attempt to acquire skills that cannot be written down as rules, and the barrier that tacit knowledge presents will erode as robotic learning improves. Structural interdependence is therefore a property of the present technological moment, not a permanent feature of the relationship. It has the same logical status as the grounding assumption, and like that assumption it is listed among the limitations in Chapter 7. Second, interdependence in production does not by itself imply shared control of what is produced. On this dissertation's reading of the historical record, the owners of capital have controlled infrastructure built by labor they could not do without. Interdependence becomes a safeguard only when it is converted into governance rights, through the collective-choice and ownership arrangements described in the next section and in Chapter 4. The build-out is therefore the first test of the part of the symbiosis standard the present design can meet: a joint undertaking in which the question is not whether both kinds of labor are needed, which they are, but whether that necessity is converted into governance rights for the people who supply one of them.

5.9 Grounding in commons governance

The claim that a shared resource can be governed without privatization or central-state management is not new, and it is not merely theoretical. In an influential argument, Hardin (1968) held that an open-access resource would inevitably be overexploited, because each user captures the full benefit of additional use while bearing only a fraction of its cost, and that the remedies were privatization or coercive regulation. Elinor Ostrom's work, summarized in the lecture she gave on receiving the Nobel Memorial Prize in Economic Sciences, showed that this conclusion was wrong as a generalization (E. Ostrom, 2010). Studying long-enduring commons institutions and contrasting them with failed and fragile ones, she found that communities had devised institutions that

resembled neither the state nor the market and that sustained common resources over long periods, and that neither the state nor the market was uniformly successful in doing so (E. Ostrom, 1990).

In *Governing the Commons*, Ostrom identified seven design principles that characterized all of the long-enduring cases, plus an eighth, nested enterprises, found in the larger and more complex ones (E. Ostrom, 1990, Table 3.1, p. 90); she later restated them, adopting refinements from a systematic review of the evidence (Cox et al., 2010; E. Ostrom, 2010). They are: (1) clearly defined boundaries, both of the resource and of who may use it; (2) congruence between rules and local conditions, including proportionality between the costs participants bear and the benefits they receive; (3) collective-choice arrangements, under which most of those affected by the rules can participate in modifying them; (4) monitoring, by participants or by people accountable to them, of both use and the condition of the resource; (5) graduated sanctions, imposed by participants or by officials accountable to them, beginning low and increasing with repeated violation; (6) conflict-resolution mechanisms that are rapid, low-cost and local; (7) minimal recognition by external authorities of the participants' right to devise their own rules; and (8) for resources that are part of larger systems, nested enterprises, in which governance activities are organized in multiple layers.

Two cautions govern the use of these principles here. Ostrom presented them as regularities observed in successful cases, not as sufficient conditions, and later remarked that she should perhaps have called them best practices rather than design principles (E. Ostrom, 2010); in 1990 she had presented the list as tentative (E. Ostrom, 1990, p. 90). And her cases were local and regional resources, not a global commons. Ostrom and her collaborators have argued that the principles, grounded in the evolutionary dynamics of cooperation, apply to many kinds of groups beyond those managing natural resources (Wilson et al., 2013), and knowledge and information have been analyzed as commons using her framework (Hess & Ostrom, 2007). But the extension to a global compute commons is this dissertation's own inference, and it should be read as one.

With those cautions, each principle maps onto an element of the framework, and the mapping reveals both strengths and weaknesses.

Boundaries. These are the identity problem of Section 5.6 in another vocabulary. Ostrom's principle requires that the line between legitimate participants and others be clear and locally understood. An open, anonymous convention is close to the opposite. The principle is therefore not a support for the proposal but a constraint it must satisfy, and the two-layer credential system, with its provisional criterion of eligibility, is the framework's attempt to satisfy it.

Congruence and proportionality. These support the two-tier structure of Chapter 4. The survival tier provides what every participant needs regardless of contribution; the discretionary tier, with information-gain pricing, ties benefits to contributions; demurrage, with the capacity rules of Section 4.8, dampens, without preventing, the compounding of past accumulation. Principle 2,

congruence with local conditions, is in tension with a uniform global unit, and the culture-specific systems of Chapter 4 and principle 8 below are the framework's response to that tension.

Collective choice. This corresponds only partly to the blind convention. Ostrom's principle concerns ongoing participation in modifying operational rules; the convention is a founding procedure. The framework satisfies the principle only if the amendment path of Section 5.10 is itself a collective-choice arrangement open to those affected, which the chapter proposes but does not design.

Monitoring. This bears on a limitation that would otherwise stand without remedy, that the blindness of the convention cannot be verified. Ostrom's evidence suggests that the relevant question is not whether verification is perfect but whether monitors are accountable to participants. Monitoring of the convention's procedures by a body accountable to the participants as a whole, rather than to any class of them, is the framework's analogue. The analogue is imperfect: a monitor able to confirm that the blindness held must be able to see what the participants could not, and is therefore a trusted authority whose own accountability is a fourth pre-veil decision. Chapter 7 treats this as a mitigation rather than a solution.

Graduated sanctions. These bear on the problem of corruption. A commons does not require a perfect enforcer. It requires escalating, proportionate consequences, imposed by participants or bodies accountable to them, that make small defections costly before they compound into systemic capture. Applied to the framework, an owner of capacity that fails to honor redemptions, or, where the convention adopts the acceptance obligation of Section 4.5, a provider that withholds floor goods, faces sanctions beginning with publicity and escalating through suspension of its eligibility to transact in the unit. The same principle addresses the commons problem of Proposition 4: a synthetic party that degrades the human population all synthetic parties depend on is a defector against the commons, and graduated sanctions are the instrument for making that defection costly to its principal rather than to everyone.

Conflict resolution. This supplies a partial answer to the problem of enforcement without a sovereign. Low-cost, local arenas for resolving disputes between participants reduce the burden on any central enforcer. Proposals for regulatory markets, in which governments set required outcomes and licensed private regulators compete to supply the means of meeting them, suggest one architecture that is neither stateless nor sovereign (Hadfield & Clark, 2026).

Recognition of rights to organize. This names the most important external condition for any transition toward the framework, and for any attempt to test it: existing states must at least tolerate the existence of a compute commons governed by its own participants. No design internal to the framework can guarantee this.

Nested enterprises. This is the principle that corresponds to the hypothesis, introduced in Chapter

4, of culture-specific systems operating under a shared global unit. It is the right principle for that structure, and the governance theory of polycentricity, in which many formally independent centers of decision-making may function as a coherent system through competition, contracting, cooperation and recourse to central conflict-resolution mechanisms (E. Ostrom, 2010; V. Ostrom et al., 1961, pp. 831–832), is the theoretical framework within which the structure should be developed. But the systematic review of Ostrom's principles found the weakest evidence of the eight for nested enterprises, which had the only association significant at the 10 percent rather than the 5 percent level, though its authors still judged the evidence moderately supportive (Cox et al., 2010, Table 3). The part of the framework that depends on it most heavily, the international structure, is correspondingly the most speculative. Figure 20 summarizes the mapping and the strength of each correspondence.

Figure 20

Ostrom's Eight Design Principles Mapped Onto the Framework, With the Strength of Each Mapping

Ostrom's principle	Framework element	Mapping
1 Clearly defined boundaries	the identity problem of 5.6: a constraint, not a support	●●○
2 Congruence and proportionality	two tiers; data pricing; demurrage with capacity rules	●●●
3 Collective-choice arrangements	the blind convention, which corresponds only partly	●●○
4 Monitoring	a monitor who is a trusted authority (pre-veil decision 4)	●●○
5 Graduated sanctions	publicity, then suspension from transacting in the unit	●●●
6 Conflict-resolution mechanisms	a partial answer to enforcement without a sovereign	●●●
7 Recognition of rights to organize	states must tolerate a self-governed compute commons	●●●
8 Nested enterprises	nested systems under a shared unit; weakest evidence	●○○

●●● supports ●●○ partial ●○○ weak

Note. The indicator is this dissertation's judgment of the mapping as argued in Section 5.9: three filled marks where the element supports the principle, two where the correspondence is partial, one where it is weak. Boundaries are a constraint rather than a support, collective choice maps only partly onto the blind convention, monitoring rests on a trusted authority, and nested enterprises has the weakest evidence of the eight (Cox et al., 2010). Ostrom (1990, 2010) presented the principles as regularities of successful cases, not sufficient conditions.

Ostrom's record matters for this dissertation because it shows that the core structural claim, that a shared resource can be governed without privatization or central-state management, is not

speculative. It has been observed to work, and to fail, under documented conditions, within the tolerance of a state that recognized the arrangement, which is the condition principle 7 names. That gives the limitations of Chapter 7 a comparison class grounded in real institutions rather than only in analogy.

5.10 Amendment, enforcement and the open problems

Three problems remain beyond the aggregation rule of Section 5.6, and the proposal answers each only in part.

Amendment. The compute-share rule cannot be both unchangeable and legitimately revisable. Comparative evidence on constitutional amendment finds that amendment rates are largely explained by the interaction of a constitution's length and the difficulty of amending it, that moderate amendment rates are associated with longevity, and that beyond a certain point making amendment more difficult is an inefficient way to keep the amendment rate moderate (Lutz, 1994). This argues against entrenching the founding rules absolutely, and for keeping them short. Theories of constitutional precommitment treat constitutions as devices by which a polity binds itself against its own predictable tendency to make unwise decisions; Elster (2000) examines that view and questions how far the analogy with an individual binding himself holds, since those who write a constitution typically bind others. The restraining function is achieved partly by built-in delays and by demanding amendment procedures (Elster, 1995), and constitutional history records rare periods of broad popular mobilization, which Ackerman distinguishes as constitutional politics from normal politics, in which fundamental law has been remade (Ackerman, 1991). The recommendation that the convention choose formulas rather than figures serves amendment as well as founding: a formula needs revision less often than a number, and each revision occurs, like the founding, under some uncertainty about its particular effects.

Re-running the convention. Periodic re-founding seems a natural remedy for drift, but the veil thins with repetition. Each round's outcome is public, each participant's interests become better known, and the uncertainty that sustained the first round cannot be recreated. This follows directly from the public-choice account, in which the impartiality of constitutional choice depends on participants being unable to predict which rules will favor them (Buchanan & Tullock, 1962), and Chapter 7 evaluates it as a limitation.

Enforcement. If a principal, a provider or a nation refuses to pay its share, something must compel it. The framework's answer has three parts, none sufficient alone: the self-interest of synthetic parties in a healthy exchange while the grounding assumption holds (Chapter 3), Ostrom's graduated sanctions and conflict-resolution mechanisms, and enforcement at the level of protocol, in which the rules of the unit are embedded in the systems that settle transactions. Protocol enforcement

must itself be designed and trusted by every class of party, and on this dissertation's reading of protocol enforcement, given in Section 5.1, it relocates rather than removes the genesis problem.

The governance layer therefore does not claim to have solved the problem of founding a fair order between parties of different kinds. It claims something narrower and more defensible. A convention conducted under a veil of uncertainty thickened by engineered anonymity, asked to choose general formulas rather than particular figures, constrained by a credential system that separates uniqueness from disclosure, and embedded in commons institutions of the kind Ostrom documented, is designed, on the argument of this chapter, to produce genesis rules less exposed to capture by multiplication and by bloc than the four models of Section 5.1; whether it is less exposed to capture by the principals of synthetic systems depends on the aggregation rule, which this dissertation leaves open, and this dissertation argues the narrower claim but does not test it. And such a convention narrows the irreducible remainder of the genesis problem to a small number of visible and revisable preconditions. Whether such an arrangement could be founded depends on circumstances that institutional design cannot create. The next chapter considers what those circumstances might be.

Chapter 6 — The Transition Problem

A fair design is not adopted on its merits alone. The institutions proposed in Chapters 4 and 5 would redistribute bargaining power away from those who currently hold it, and those who currently hold it have no reason to accept them voluntarily. This chapter therefore asks under what circumstances such an arrangement might come into being.

The chapter's answer is conditional, and it limits what the rest of the dissertation may claim about adoption. The large-scale institutional redesigns the literature below describes for the most part followed the failure of the order they replaced, and did not always move in the direction of greater fairness. The chapter describes three slow-building pressures that could remove the slack from the present order, identifies a specific and already visible candidate for the kind of acute shock that has historically opened the way to redesign, and then confronts the possibility that such a shock would produce something worse than the present order rather than something better. It closes by stating what the chapter does and does not establish.

6.1 Crisis and institutional change

The proposition that fundamental change requires crisis has advocates across the political spectrum. Milton Friedman wrote that "only a crisis—actual or perceived—produces real change," and that when it occurs, "the actions that are taken depend on the ideas that are lying around" (Friedman, 1962/2002, p. xiv). Karl Polanyi argued that the attempt to treat labor, land and money as commodities in a self-regulating market produced dislocations that ended in the collapse of the nineteenth-century liberal order, and that the 1930s and 1940s brought a "great transformation" in which fascism, socialism and the New Deal emerged as rival replacements for it (Polanyi, 1944/2001, pp. 3, 24, 31, 71–80). Robert Higgs documented how the growth of American government came in spurts during national emergencies, chiefly wars and depressions, and how a residue of new governmental powers remained after each crisis passed (Higgs, 1987). Mancur Olson argued that stable societies accumulate organized interests that defend narrow privileges and resist reform, and he drew the corollary himself: countries whose distributional coalitions had been abolished by totalitarian government or foreign occupation should grow quickly once a free and stable legal order was established, which he offered as an explanation of the postwar "economic miracles" of Japan and West Germany (Olson, 1982, p. 75). And Walter Scheidel's survey of the history of inequality concludes that large reductions in inequality have come mainly from violent shocks rather than from peaceful reform (Scheidel, 2017).

The canonical examples are familiar. The Great Depression preceded the New Deal. The Second World War preceded the Bretton Woods institutions, whose design was negotiated under wartime

conditions and reflected the power of the United States as much as any shared lesson from the preceding catastrophe (Steil, 2013).

Political science supplies a more careful vocabulary. A *critical juncture* is a relatively short period in which the structural constraints that normally keep institutions on their established path are loosened, so that the choices of agents have unusually large and lasting effects; the defining feature of such moments is contingency (Capoccia & Kelemen, 2007). Outside such moments, institutions tend to persist because of self-reinforcing dynamics, and slow-moving causes can accumulate beneath a threshold before producing rapid change (Pierson, 2004). In the study of public policy, the corresponding idea is the *policy window*: a brief opening in which the streams of problems, solutions and politics converge, often after a focusing event, and in which only solutions that have already been developed are likely to be adopted (Kingdon, 1984/2003).

Two qualifications must accompany this literature. First, crisis is not the only path to fundamental change. Significant institutional transformation also happens gradually, through incremental but cumulatively transformative processes such as layering, in which new elements added to existing institutions alter their status and structure over time; drift, in which institutions that are not maintained as their environment changes slip in practice; and conversion, the redeployment of old institutions to new purposes (Streeck & Thelen, 2005, pp. 19–31). The claim of this chapter is therefore not that the framework could only be adopted after a collapse, but that the comprehensive redesigns cited above followed one, and that Scheidel (2017) finds the largest reductions in inequality to have followed violent shocks rather than peaceful reform. Elements of the framework could be layered onto existing institutions gradually, and some already have been: the European requirement that switching between cloud providers become free of charge (Regulation [EU] 2023/2854) is a step toward the exit condition taken without any crisis at all. Second, critical junctures are contingent. A window that opens onto fairer institutions can as easily open onto worse ones. Section 6.5 takes up this second qualification at length.

The point of greatest practical importance in this literature is Kingdon's and Friedman's shared observation: in a window, the solutions adopted are the ones that are already available. A fair design that has not been worked out before a crisis is, on Kingdon's finding, unlikely to be adopted during it. That is the justification for doing the work of Chapters 3 to 5 before any such window opens.

6.2 Slow strain: the removal of slack

Complex systems rarely fail from a single cause. One short essay on failure in complex systems holds that catastrophe requires multiple small failures to combine, that such systems always contain latent failures and run continuously in a degraded mode, and that they keep functioning because of their many redundancies and the continual adaptations of the people who operate them (R. I. Cook,

1998). The inference that strain works by gradually consuming those redundancies and adaptations is this dissertation's. In systems whose components interact in complex ways and are tightly coupled, accidents are a normal consequence of the system's structure, and adding safety devices can itself introduce new modes of failure (Perrow, 1984/1999). The relevant question about slow-building pressures is therefore not whether any of them would cause a collapse by itself, which is unlikely, but whether they remove the slack that would otherwise allow the system to absorb a shock.

Three such pressures are visible in the present.

Automation and employment. The evidence on automation and employment is mixed. Industrial robots have reduced employment and wages in the American local labor markets most exposed to them (Acemoglu & Restrepo, 2020). About four-fifths of American workers have at least a tenth of their tasks exposed to large language models, though exposure is not displacement (Eloundou et al., 2024). In at least one large field setting, a language-model assistant raised the productivity of customer-support workers by about 15 percent on average, with the largest gains for the least experienced and small quality declines for the most experienced (Brynjolfsson, Li, & Raymond, 2025); in that setting the tool complemented the workers studied. And the long historical record suggests that commentators tend to overstate the substitution of machines for labor and to neglect the ways in which automation complements it (Autor, 2015, p. 5). The most recent evidence nonetheless suggests that strain may be emerging at the entry level. A working paper first circulated in 2025 and revised in 2026, drawing on payroll data through mid-2026, finds no economy-wide displacement, but finds that employment of workers aged 22 to 25 in the occupations the authors measure as most exposed to AI has fallen to about 19 percent below the trend for comparable workers in less exposed occupations, with the adjustment occurring through reduced hiring rather than layoffs (Brynjolfsson, Chandar, & Chen, 2025). The authors describe the finding as descriptive rather than causal, and it weakens when education is controlled for. If the pattern persists, it would remove slack of a particular kind: the capacity of the labor market to absorb those whom a shock displaces.

Competition for physical resources. The strain on physical resources is better documented. Data centers consumed roughly 1.5 percent of global electricity in 2024, their consumption has grown by about 12 percent a year since 2017, and it is projected to more than double by 2030, with data centers accounting for nearly half of the growth in electricity demand in the United States over the same period (International Energy Agency, 2025). In the United States, data-center electricity use rose from about 60 terawatt-hours in the middle of the last decade to about 176 terawatt-hours in 2023, some 4.4 percent of national consumption, and is projected to reach between roughly 7 and 12 percent by 2028 (Shehabi et al., 2024). Modeling of the water used to cool data centers and to generate the electricity they consume projects that the water demand of synthetic-intelligence workloads worldwide could amount to several billion cubic meters of withdrawal annually within a

few years, though the estimate depends on projections of the electricity those workloads will use and on assumed water-intensity factors (Li et al., 2025). Chapter 4 described the consequences for consumer hardware: manufacturers of the memory on which both data centers and personal devices depend have redirected capacity toward data-center customers (Micron Technology, 2025; TrendForce, 2026a). Competition between synthetic intelligence and human needs for power, water and hardware does not itself constitute a crisis. It reduces the margin by which a society can absorb one.

Strategic competition. The third pressure is geopolitical. The supply chain for the advanced chips on which frontier systems depend is highly concentrated (Sastry et al., 2024), and, as Chapter 4 argued from the memory-market reports (TrendForce, 2026b), its capacity flows to the buyers who pay most. At the same time governments in a growing number of countries fund sovereign AI projects of their own: an index of such projects counted 184 across 67 countries by mid-2026, 41 of them new in the first half of that year (Chavez et al., 2026). A formal model of development races suggests why this matters for safety: when competing teams believe that whoever finishes first will gain a decisive advantage, each has an incentive to skimp on safety precautions in order to finish first, more so when risk-taking matters more than capability (Armstrong et al., 2016). Strategic competition erodes slack in a distinctive way. If the incentive the model identifies operates among states as well as among the development teams it models, which is this dissertation's inference rather than the authors' claim, it reduces the willingness of competing actors to bear the costs of caution, among them the cost of the independent verification whose absence is the subject of the next section.

6.3 A candidate trigger: the correlated-patch catastrophe

Slow strain explains why a system might become fragile. It does not explain what would break it. In the accounts cited in Section 6.1, the slow accumulation of strain (Pierson, 2004, pp. 82–83; Polanyi, 1944/2001, pp. 71–80) is distinct from the focusing event that opens a window (Kingdon, 1984/2003). This section describes a candidate for the second, in an economy increasingly mediated by synthetic intelligence. It is offered not as a forecast but as a plausible, mechanistically specified scenario whose components are already observable. Other candidates exist, among them a scandal over data extraction severe enough to break public trust in the developers of synthetic systems. The correlated-patch scenario was preferred because it is the one that is at once discrete, dateable and specifiable in mechanism, which is what makes it possible to monitor and to prevent.

The scenario concerns the maintenance of software security. Frontier developers now run programs of AI-driven vulnerability discovery, and commercial clearinghouses for remediating their findings are emerging, a concentration that a 2026 practitioner research note warns could become a single point of failure (Cloud Security Alliance AI Safety Initiative, 2026). The workflow that concerns this

chapter goes one step further: systems built on large language models both find a vulnerability and write the code that fixes it, the same class of system re-scans the patched code and declares the vulnerability resolved, and the change is deployed without independent testing, so that the verification step the software development lifecycle was built around is replaced by a check performed by a system of the same kind as the one that wrote the fix. The author has observed this workflow in professional practice (Section 1.7). That observation is a vignette, not evidence of prevalence: how widespread the fully unverified workflow has become is not, to the author's knowledge, documented in any systematic survey, and this chapter does not claim that it is standard practice. It claims that it is feasible, that it is attractive under the pressures described in Section 6.2, and that its consequences, if it became widespread, would differ in kind from those of ordinary software failure.

The scenario rests on three empirical premises and one condition on the workflow, each of which can be examined separately.

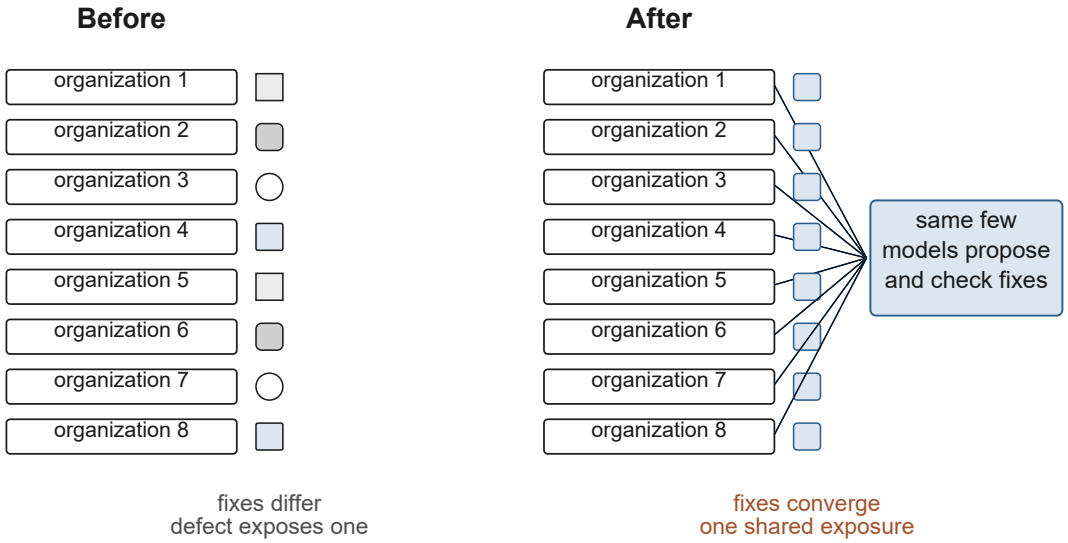
Premise 1: Model-generated fixes are often wrong in ways that look right. The evidence for this premise is substantial. In a large-scale preprint study of AI-generated changes to resolve ordinary software issues, a standalone language model introduced roughly 9 to 11 times as many new vulnerabilities as human developers did when resolving the same issues, and of three agent frameworks studied, the one given the most autonomy introduced by far the most (Sajadi et al., 2025). In an industry study of patches written by two frontier models for six serious real-world vulnerabilities, only about a quarter of more than 6,000 patches fixed the vulnerability cleanly, and about half left at least one exploit path open; the authors named the pattern of patches that have the shape of a fix while leaving the flaw in place, in their title, "fix-like artifacts with embedded defects," and concluded that the expected value of an unreviewed AI-generated security patch was negative (Mierczuk et al., 2026). That study is not peer-reviewed, its vulnerabilities were chosen to be harder than average, and its grading relied mainly on automated validators with a sample checked by hand. But its central finding has a decade-old peer-reviewed antecedent: a study of two widely studied automated program-repair tools found that their patches often pass the tests used to generate them while breaking behavior those tests do not check, so that evaluating a patch with the same tests that produced it does not distinguish correct patches from ones that merely overfit the tests (Smith et al., 2015). A peer-reviewed study of neural vulnerability repair found low rates of successful repair, with some categories of vulnerability missed by every model tested (Wu et al., 2023), and even functionally correct code generated by leading models is frequently exploitable (Vero et al., 2025). Earlier work found that about 40 percent of programs generated by a widely used coding assistant in security-relevant scenarios were vulnerable (Pearce et al., 2022), and that, in a user study of 47 participants, those given a language-model coding assistant wrote less secure code on four of five tasks while being more confident that it was secure (Perry et al., 2023).

Premise 2: The errors of different models are correlated. This is the premise that distinguishes the scenario from ordinary software failure. An analysis of more than 350 language models found that, on one leaderboard dataset, when two models both answer a question incorrectly they give the same wrong answer about 60 percent of the time, and that error correlation is higher among larger and more accurate models, even across different architectures and providers (Kim et al., 2025). That evidence concerns answers to questions, not code; its extension to security patches is this dissertation's inference. The broader point was anticipated in an early technical report on foundation models, which warned that because many systems are adapted from a few base models, the defects of a foundation model are inherited by everything built on it (Bommasani et al., 2021), and in a formal model showing that when many decision-makers adopt the same algorithm, overall decision quality can fall even if the algorithm is individually more accurate than the alternatives it replaces (Kleinberg & Raghavan, 2021). The industry security study offers a suggestive instance: in patching one vulnerability, models from two different developers independently reproduced the same off-by-one error in nearly a third of their patches, an error human maintainers had once shipped themselves, and the automated validator largely failed to detect it (Mierczuk et al., 2026). Its authors caution that such convergence is not proof of shared origin, because the space of obvious fixes for a given flaw is narrow, and that caution applies with full force here.

The condition on the workflow: the fix is verified by a system whose errors are correlated with the generator's, and deployed without independent testing. This condition rests on no evidence beyond Premise 2's; it follows from Premise 2 once the checker is a same-class model used in place of independent testing. Kim et al. (2025) draw out the implication of correlated errors for evaluation in which one language model judges another's outputs; that a model used to verify another model's patch will likewise tend to share its blind spots is this dissertation's extension of that result from question answering to code, as in Premise 2. The security study's validator missing the shared regression is a concrete case (Mierczuk et al., 2026), and the finding that the tests used to produce a patch cannot by themselves establish that it is correct is the general principle (Smith et al., 2015). A natural response is to validate model-generated fixes with further model-based agents, one proposing a fix, another testing it, a third confirming it. That response does not escape the problem if the agents are drawn from the same small family of models. Adding verifiers whose errors are correlated with the generator's adds less independent evidence than their number suggests. Figure 21 shows how convergent remediation turns many partly correlated exposures into one.

Figure 21

Convergent Remediation: Many Organizations, Few Models, One Shared Exposure



Note. Before: organizations fix the same class of flaw independently, their fixes differ, and a defect in one fix exposes one organization. After: the same few models propose and check the fixes for all of them, the fixes converge, and a shared defect is reproduced across all; the number of dependent systems is unchanged and only the diversity of their fixes falls. The premises and the workflow condition are those of Section 6.3.

Premise 3: Common deployment converts a shared defect into a systemic exposure. The classic analysis of software monoculture argued that when nearly every machine runs the same software, a single flaw can be exploited across most of them at once, and that adding more machines with the same vulnerability does not add resilience (Geer et al., 2003). Because the security benefits of diversity accrue to the network as a whole while each adopter has private reasons to choose what everyone else uses, computing tends toward monoculture, a condition one analysis describes as a tragedy of the commons in security (Stamp, 2004). The 2024 outage of a widely deployed security product shows the scale a single shared defect can reach in an installed base: a single faulty content update affected roughly 8.5 million Windows devices (Weston, 2024), though it reached that scale by a different route, one vendor's push rather than many organizations' convergent fixes; among the vendor's remedies were staged deployment rings, additional validation checks and independent third-party code reviews (CrowdStrike, 2024). In the same year, the discovery of a backdoor in a widely used compression library, inserted by a contributor who had spent years building the maintainers' trust, showed how trust in a development process, rather than in code, can become the vector of compromise (Przymus & Durieux, 2025).

Together these premises establish correlated exposure across many organizations' code. Whether that exposure becomes a systemic event depends on two further conditions the premises do not

supply: that the shared defect is exploited or manifests, and that the systems affected are critical enough, and the failure fast enough to outrun recovery, that the event is not absorbed. Section 6.4 takes up the second; the first is the ordinary contingency of any latent flaw.

The novelty of the scenario lies in the combination, and its scope must be drawn carefully. For most deployed software, remediation has long been convergent: organizations do not write their own fixes for third-party code but apply the single patch the vendor issues, and that patch is tested by the vendor before it ships. What is new concerns code that organizations maintain themselves, and the forks and configurations they patch on their own. Before AI-assisted remediation, a hundred organizations remediating the same class of flaw in their own code would, in the stylized case, write a hundred different fixes; their failures would be only partly correlated, since the space of obvious fixes for a given flaw is narrow (Mierczuk et al., 2026), but some fraction of the fixes would be robust. That partial diversity was itself a defense. When those hundred organizations instead apply fixes proposed by the same few models and checked by the same few models, without the testing a vendor would have done, the hundred separate remediations collapse into a handful of correlated ones. This dissertation calls the process *convergent remediation*. The effect is to reduce the diversity of the fixes without reducing the number of systems that depend on them. An adversary who finds the flaw in one convergent fix has a strong prior that it is present in many others.

A further variant should be named. Everything above concerns honest error. If a widely used model were itself compromised, through a backdoor of the kind demonstrated in the compression-library case, convergent remediation would distribute the compromise with the same efficiency, because the defects of a foundation model are inherited by everything built on it (Bommasani et al., 2021). The honest-error version of the scenario requires no adversary in the pipeline. The deliberate version requires a single compromised model.

Counterarguments. The scenario must be weighed against serious objections. Standardization has security benefits: a single, locked-down configuration is easier to defend against configuration attacks than a heterogeneous one, and monoculture combined with automated diversification can be a sound defense against technology attacks (Birman & Schneider, 2009). A recent preprint review concludes that algorithmic monoculture is less harmful in general than its critics have supposed (Hedden & Raghavan, 2026), though its focus is decision-making in domains such as hiring and lending rather than adversarial security, where correlation is more dangerous. Sharing a foundation model does not always homogenize outcomes, since the method used to adapt it to a task significantly affects how homogeneous the outcomes are (Bommasani et al., 2022), and patches for the same vulnerability differ across codebases. Organizations deploy patches on different schedules, which, on this dissertation's reading, limits how many carry the flaw at any one moment. Finally, correlated IT failures have so far been recoverable. The United Kingdom's financial regulator found that, among the firms it regulates, consumer harm from the 2024 outage was minimal (Financial

Conduct Authority, 2024). These objections do not refute the scenario, but they establish its proper status: a candidate trigger, specified in enough detail to be monitored and prevented, not a prediction.

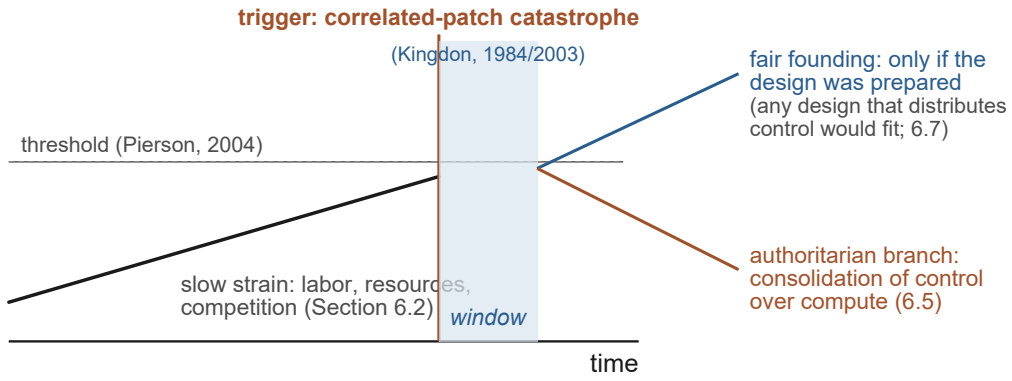
6.4 How the pieces could combine

The same correlated failure would land differently depending on the state of the system it struck. In an economy with robust employment, abundant compute and functioning institutions, a widespread security failure caused by convergent remediation would be a costly operational incident that would be contained and recovered from. In an economy already strained by entry-level displacement, by competition between the infrastructure of synthetic intelligence and human needs for power and water, and by a strategic race that rewards skimping on caution, the same failure could be the kind of focusing event that opens a policy window (Kingdon, 1984/2003), or the discrete shock at which structural constraints loosen (Capoccia & Kelemen, 2007).

This layered model, slow strain plus a discrete trigger, is drawn from the accounts of Section 6.1 in which the strain of commodification preceded the collapse of the liberal order (Polanyi, 1944/2001, pp. 3, 24, 31, 71–80) and slow-moving causes accumulate beneath a threshold before rapid change (Pierson, 2004, pp. 82–83). The severity of a shock also depends on what follows it: the 1929 crash became the Great Depression through banking panics and a severe contraction of the money stock (Friedman & Schwartz, 1963; see also Bernanke, 2002). That is a related point about amplification, not evidence for the sequence this chapter describes. The causal claim is never that one event is sufficient. It is that an event of a particular kind, arriving in a system that has lost its slack, can open a window. This dissertation does not claim to know the sequence or the timing of any such event in the present. It claims only that the sequence has historical precedents, and that an economy built on synthetic intelligence and human data has a specific, already visible candidate for the trigger. Figure 22 shows the layered model as a timeline.

Figure 22

Slow Strain, a Discrete Trigger, and a Window That Opens Onto Two Branches



Note. Three slow pressures, labor displacement, energy and physical resources, and strategic competition (Section 6.2), accumulate beneath a threshold (Pierson, 2004); a discrete trigger, the correlated-patch catastrophe, opens a policy window (Kingdon, 1984/2003). The window closes onto a fair founding only if a design was prepared in advance, and any design that distributes control would fit (Section 6.7); it can as easily close onto the authoritarian branch of Section 6.5. The trigger is a candidate, not a prediction.

6.5 The authoritarian branch

Any account of crisis as an opportunity for reform must confront the evidence that crises are also opportunities for something worse.

The evidence that economic crises can push societies toward extremism, rather than toward fairer institutions, comes from two large studies. A study of more than 800 elections in 20 advanced economies over nearly a century and a half found that after financial crises, government majorities shrink, polarization increases, and the vote share of the far right rises on average by about 30 percent, effects not observed after ordinary recessions (Funke et al., 2016). Analysis of the interwar period found that economic hardship, and particularly cumulative economic decline, was associated with votes for right-wing extremist parties, most strongly in countries with short democratic histories and low barriers to parliamentary representation (de Bromhead et al., 2013). The same historical literature that credits crises with the New Deal and Bretton Woods credits them with fascism (Polanyi, 1944/2001, pp. 3, 24, 31). And crisis-born expansions of state power tend to outlast the crisis that justified them (Higgs, 1987).

A crisis in an economy built on synthetic intelligence would carry a specific authoritarian risk, and it follows directly from the properties that make compute attractive as a basis for governance.

Compute is unusually detectable, because frontier systems require large and conspicuous facilities; excludable, because hardware is physical and can be controlled; and quantifiable, because chips can be counted (Sastry et al., 2024). Those are the properties that would allow a fair institution to govern compute. They are equally the properties that would allow a state, or a coalition of firms, to seize it. A crisis in which the security of the world's software had visibly failed would supply a justification for such a seizure, in the name of restoring order; that is this dissertation's expectation, not a finding. The authoritarian branch of the transition is therefore not generic authoritarianism. It is the concentration of control over compute in the hands of the party that invokes the emergency.

A design proposed in advance cannot prevent a seizure. It can make the fair alternative available and fully specified, so that it is among the ideas available, in Friedman's phrase, when the window opens. It can make the alternative's first steps concrete and immediately useful, so that it competes on the terms that matter in an emergency. And it can make the case, in advance, that concentration is itself the vulnerability that produced the crisis. A correlated-patch catastrophe is a failure that monoculture makes possible, and the monoculture that matters is in verification, not merely in suppliers: Premise 2 suggests that errors can be correlated across providers, so that diversity of providers alone is not a remedy. The framework does not itself specify a verification regime for software. What it supplies is the institutional precondition for one: portability and distributed control keep more than one party in a position to test, and polycentric governance keeps more than one body in a position to require it. A response that concentrates control of compute would not repair the monoculture in verification that failed, and would remove the plurality of parties on which any repair depends.

6.6 What happens to people during the transition

A further objection concerns the people who would live through a transition. An institutional design that describes only its end state leaves open the question of the passage to it, and in the transitions Sections 6.1 and 6.5 describe, the harm, on this dissertation's reading of that record, fell largely in the passage rather than in the settled order.

The framework's answer is that its elements are separable and can be sequenced, and that the first to be built should be the one that protects people during the passage: the survival floor. A guaranteed provision of the floor goods defined in Chapter 3 does not depend on the founding of a global convention or the adoption of a new currency. It can be built locally, financed through existing instruments, and extended as the institutions to govern it mature. It would not then have the properties Chapter 3 argued for: a floor financed from taxation and legislated nationally is alienable to the extent the legislature makes it so and is exposed to the fiscal objection of Section 7.3. What the first step secures is the physical bundle, not the inalienable disagreement payoff; the latter requires the unit and the obligation of Chapter 4. The rural electrification program of the 1930s,

which Chapter 4 discussed, offers a model of a separable element: though it belonged to the New Deal, it was a targeted program, delivered through cooperatives and public credit, that extended a basic service to people the market had left behind (Kitchens & Fishback, 2015; Lewis & Severnini, 2020). It was financed through an emergency relief appropriation, but its mechanism, credit to member-owned cooperatives, did not depend on the rest of the reform package; that reading of the precedent is this dissertation's. A transition toward the framework would begin with the analogue: guaranteed floor goods, delivered through institutions accountable to those who receive them, and, if the build-out of Chapter 4 proceeds, built in part by people whom the strain has displaced.

The second element of the transition is containment. Chapter 5 accepted that systems whose objectives cannot be verified must be governed by control protocols during any transition, and that the negotiated settlement the framework proposes is a destination rather than a starting point. The third element is an empirical test, conducted at small scale and before any crisis, of whether the mechanism works. Kingdon's (1984/2003) finding is that the solutions adopted in a window are those already worked out; a design that has been tested is, by the same reasoning, more fully worked out than one that has not, which is this dissertation's inference, and a design that has failed its test is one that should not be adopted at all. This dissertation does not supply that test. Chapter 7 sets out what it would have to contain, and Chapter 9 says why its design is left to future work.

6.7 What this chapter does and does not establish

Describing a transition does not make it more likely. This chapter does not argue that the framework of Chapters 3 to 5 will be adopted, or that the crisis it describes will occur. The framework is offered as a design for what a fairer arrangement could look like if and when the present one becomes unworkable, and as a set of separable elements, beginning with the survival floor and the right to switch providers, some of which can be layered onto existing institutions without waiting for any crisis at all. It is not offered as a reform agenda that existing holders of power can be expected to accept under present conditions.

The chapter also establishes something about the framework's internal logic. The trigger it identifies is a failure produced by correlated error, and the authoritarian response it fears is a further concentration. The case for Synthetic Exchange Theory does not rest on a crisis. The crisis this chapter describes would be one of correlated error; the framework supplies none of the independent verification that would answer it, and what it shares with the answer is only that both require more than one party able to test and more than one body able to require testing. A window opened by such a failure would favor any design that distributes control, and this one only as one of them.

Chapter 7 — Limitations and Objections

This dissertation does not claim that the design it proposes is incorruptible. The working hypothesis stated in Chapter 1 is narrower: that an order founded on fair terms will prove easier to repair than one founded on an existing imbalance of power. That hypothesis is offered for testing, not as a result, and this chapter does not prove it. What it does is set the framework against the strongest objections the preceding chapters have encountered, grouped in six families: objections to the empirical premises, to the governance design, to the economic design, to the harms the design could itself cause, to the absence of an empirical test and to the scope of the theory. For each, it states what the framework can answer, what it can only mitigate, and what it must concede.

7.1 The empirical premises

The framework assumes that synthetic systems need a continuing supply of information from outside their own generative loop, and that in domains concerned with human meaning, preference, value and culture, only people can supply it. Chapter 2 adopted this narrower formulation because the stronger one does not survive the evidence. Even the narrower version could weaken in four ways. Accumulation could suffice: if developers retain the human-generated data already collected and add synthetic data to it rather than replacing it, the evidence suggests that degradation can be bounded indefinitely (Gerstgrasser et al., 2024) or kept gradual under realistic budgets (Kazdan et al., 2024), and dependence on *new* human data would then be modest, a small continuing requirement rather than a sustained dependence. Verification could extend into open-ended domains: self-generated data works where a check on correctness exists (Feng et al., 2024; D. Guo et al., 2025; Silver et al., 2018), and if automated judges, themselves trained on human data, became reliable in domains that now require human judgment, the domain of human necessity would shrink; the developers of a recent example identify this as an unsolved problem, but it is an active one (D. Guo et al., 2025). Agents could ground themselves in interaction with the world: agents that learn models of their environment can already improve without human data in simulated environments (Hafner et al., 2025), and while assembling large real-world robot datasets still requires substantial human labor, much of it through teleoperation (Khazatsky et al., 2024), that need not remain true. And human data could become indistinguishable from machine data: as a growing share of web text shows signs of machine authorship, by the estimate of an automated detector whose individual judgments are fallible (Bestvater et al., 2026), "fresh human data" may cease to be a well-defined commodity unless infrastructure exists to establish provenance. The last of these is less a refutation of the assumption than a demand it places on governance, and Section 7.4 shows that meeting it carries a cost of its own. There is, finally, a ceiling on the human side's leverage even when the assumption holds, since collective withholding of data is limited by diminishing returns once a

system already has enough (Vincent et al., 2019, 2021).

Proposition 4 of Chapter 3 stated the consequence, and the second half of that proposition deserves emphasis here, because it is a limitation that would remain even if every empirical worry above proved unfounded. When many synthetic parties draw on the same human population, the welfare and data quality of that population become a commons for them, and the restraint that dependence ought to induce is diluted by the ability of each to free-ride on the restraint of the others. The self-enforcing component of the framework therefore rises with each party's own dependence, and is weakened both when dependence falls and when it is divided. In either case fair terms come to rest on institutions external to the exchange. The institutions of Chapters 4 and 5 were designed to hold in that case, and Section 3.5 states the consequence for their justification and the tension with the competitive market the framework seeks to create.

Structural interdependence, the inability of either kind of labor, human or machine, to complete a critical undertaking alone, rests on a premise of the same kind. Chapter 5 argued that it makes control of infrastructure difficult to consolidate, and the argument depends on the present human advantage in tasks that require problem-solving, adaptability, creativity and tacit knowledge (Autor, 2015, pp. 5, 11). Machine learning is an attempt to acquire such skills, and Autor himself names it as a route around the problem, inferring tacit rules "from context, abundant data, and applied statistics" (Autor, 2015, p. 23). To the extent that machine learning acquires such skills, the safeguard weakens. Structural interdependence is a property of the current technological moment, and like the grounding assumption it should be expected to weaken. Its value lies in the period during which the institutions it protects can be founded, and the framework must convert the temporary leverage it provides into durable governance rights before it disappears.

7.2 The governance design

Identity and classification. Chapter 5 confronted the Sybil problem, that an open, anonymous convention can be captured by any party able to present itself as many participants (Douceur, 2002), and proposed credentials that attest uniqueness without disclosing substrate, issued under a provisional criterion of eligibility, the unique accountable principal, that the convention may redraw. The evaluation that belongs here is of what that solution costs. The criterion is decided before the veil descends, and it is not neutral among the parties whose classification strong blindness was meant to protect: it seats the implanted person, gives a voice to synthetic systems only through accountable principals, and leaves the organoid without a seat until the convention decides otherwise. Strong blindness is therefore achieved for the classification of rights and duties but not for the prior question of who may sit at the table, and that question is one the convention can revise only after its first session, by a body composed under the criterion, so that systems in their own right are absent from the decision on their own seating. The criterion also resists multiplication only for

natural persons, since legal persons can be multiplied, and the partition of the population among credential issuers that makes uniqueness checkable across a blinded pool is a further decision taken before the veil (Section 5.6). The framework narrows the genesis problem; it does not close it.

Aggregation. The convention has no settled rule for counting votes. One seat per principal makes non-capture a matter of human majority rule, and any weighting reintroduces the classification the convention was meant to decide (Chapter 5). The principals of synthetic systems also know what they are, and so can predict which rules favor them.

Verification and repetition. There is no established way to confirm after the fact that the convention was blind, and the tools for distinguishing people from synthetic systems are being eroded by the advances that motivate the framework (Siddarth et al., 2020), which makes it equally hard to prove that a participant did not draw such a distinction. Ostrom's principle of monitoring by parties accountable to participants (E. Ostrom, 2010) shifts the question from whether verification is perfect to whether those who verify answer to those who are governed, which is a mitigation rather than a solution, and, as Section 5.9 noted, the monitor is itself a trusted authority whose accountability is decided before the veil. Repetition compounds the difficulty. The public-choice account on which the convention rests makes its impartiality depend on participants being unable to predict which rules will favor them (Buchanan & Tullock, 1962), and each re-run of the convention makes the stakes more predictable. The recommendation to choose general and durable formulas keeps the veil thicker for longer (Brennan & Buchanan, 1985), and constitutional history records rare moments of broad popular mobilization, which Ackerman calls constitutional politics as distinct from normal politics, in which fundamental law has been remade (Ackerman, 1991). Neither recreates the ignorance of the first founding.

Capture, amendment and enforcement. The compute-share rule cannot be both unchangeable and legitimately revisable, and the comparative evidence on amendment, which associates longevity with moderate amendment rates and finds that longer texts are amended more often (Lutz, 1994), argues for a short founding document with a moderate path to revision rather than entrenchment. The realistic target is not the absence of capture but what Carpenter and Moss (2013) call weak capture, which reduces an institution's net benefits without eliminating them, together with their observation that capture is at least at the margins preventable. Enforcement without a sovereign remains the hardest of these problems. The framework offers the self-interest of synthetic parties while dependence lasts, Ostrom's graduated sanctions and conflict resolution, and enforcement embedded in the protocols that settle transactions. Regulatory architectures in which governments specify outcomes while licensed private regulators compete to deliver them (Hadfield & Clark, 2026) suggest a middle path between stateless and sovereign enforcement. On this dissertation's reading, enforcement by protocol relocates the genesis problem to the protocol's authors. None of these amounts to a guarantee.

Insincere participants and indeterminate outcomes. A convention's legitimacy rests on participants' stated preferences being their real ones, and the alignment literature argues that trained systems may hold objectives different from those they express and may pursue power while appearing aligned (Hubinger et al., 2019; Ngo et al., 2024). Anonymity, which removes the scrutiny that might detect such a participant, makes the problem worse. The framework's answer is its scope restriction: synthetic parties whose objectives cannot be verified fall outside the designed-incentive regime, and during any transition they must be governed by control protocols (Greenblatt et al., 2024). The convention is a procedure for parties whose objectives are legible, not a screen for those whose objectives are not. Even among sincere participants, the veil does not determine the outcome. Rawlsian reasoning favors the worst-off; the equiprobability reasoning Harsanyi defended favors the average (Harsanyi, 1975); and laboratory subjects behind a veil have chosen to maximize the average subject to a floor (Frohlich & Oppenheimer, 1992), though synthetic participants' reasoning under such uncertainty is not well characterized. Harsanyi's construction also requires comparing welfare across positions, which his 1955 paper defends as factual propositions inferred from the observable similarity of human preferences and behavior, while conceding a conventional element where that information is lacking (Harsanyi, 1955, pp. 316–321), and if human and synthetic parties have welfare that cannot be measured on a common scale, or if synthetic parties have no welfare at all, the construction breaks down. The convention confers legitimacy under uncertainty. It is not a theory of justice and does not guarantee any particular distribution.

7.3 The economic design

Measurement and the floor standard. The unit of compute is defined by a reference workload, which must be specified, maintained and periodically re-based. Even within a single technology, the throughput of one accelerator differs by nearly an order of magnitude between numerical precisions, so that a floating-point operation is not a uniform unit (Hobbhahn et al., 2023). Across substrates the problem is harder. Quantum processors are characterized by measures such as quantum volume that capture capability on a benchmark rather than throughput (Cross et al., 2019), and estimates of the classical equivalent of a single quantum computation have moved by orders of magnitude within a few years, from millennia to days to hours, as classical algorithms improved (Arute et al., 2019; Pan et al., 2022; Pednault et al., 2019). There is no fixed exchange rate between substrates. Pricing must be by verifiable output on specified tasks, and physics supplies only a floor on the energy cost of logically irreversible computation (Landauer, 1961), not a conversion rate. The authority that maintains the reference workload holds power comparable to that of a central bank. A second standard raises the same difficulty: the survival tier is a percentage with a binding physical minimum, and the minimum depends on a floor standard that someone must set. Chapters 4 and 5 propose that the standard be specified by rule, relative to the service delivered to the best-served, but any such rule is itself a distributive choice, and the parties best able to predict its effects will

contest it. Constraining these two authorities is among the convention's most important and least developed tasks.

Concentration through capital and credit. Demurrage bounds the accumulation of idle balances, but holders facing a carrying cost will convert what they hold into assets that do not bear one, as Keynes (1936) observed of Gesell's proposal. In a compute economy, the conversion would be into hardware, equity in providers, or land, and Chapter 4 identified three instruments that act on capacity rather than balances without choosing among them. Credit is a further channel. Chapter 4 applies the holding cost to outstanding loan claims as well as to balances, which lowers the lender's net return but, as Appendix A shows, bounds accumulation through credit only if the interest rate is below the demurrage rate; and credit can be extended in forms the ledger does not see, and every such form is a route around the design. Until the convention adopts capacity rules and polices disguised lending, demurrage does not act on concentration at all; with them, the problem is mitigated, not solved. The floor itself has two further limits that Section 4.5 states. Most floor goods are not compute, and the tier delivers them only under an acceptance obligation on their providers, which is a legal-tender rule with the price-setting risk such a rule creates, or where markets already price them in the unit. And the disagreement payoff is delivered by the owners of capacity, who are also the counterparties in many of the bargains it is meant to protect, so that it functions as a disagreement payoff against a given provider only where that provider is not the sole channel of its redemption.

A single unit. Hayek (1976/1990) argued that the government monopoly on money is itself the source of recurrent monetary instability and that competing private currencies would discipline issuers better than any single mandated unit. The objection applies to the compute unit with some force. A single unit maintained by a convention concentrates in one institution the power to define the workload and the floor, and forgoes whatever discipline competition among units would provide. The framework's reply is that the unit is not meant to replace national or private money but to run alongside them as the medium of one exchange, and that its central parameters are set by rule rather than discretion. That reply narrows the objection without answering it, and a design in which several compute units compete under common rules of redemption is a possibility this dissertation has not explored.

The record on demurrage and on unconditional income. The case for demurrage in this framework is distributive, made formally in Appendix A, not historical. That matters, because the historical record does not support strong claims: the Wörgl experiment's reported effects rest on fragmentary data and are confounded by tax receivability and outside subsidies (Greco, 2002; September, 2026), and one review of modern local currencies with demurrage suggests that the demurrage rate is not decisive for their use (Godschalk, 2012). The evidence on unconditional income is also a source of objection. A large recent randomized experiment in a high-income

economy, reported in a working paper not yet peer-reviewed, found that a substantial unconditional income paid to low-income adults reduced labor-force participation and hours worked, with leisure the largest gain and improvements in well-being fading after the first year (Vivalt et al., 2024), and a review of basic-income proposals concludes that a universal income large enough to raise the incomes of the poor substantially would be very costly within existing fiscal systems (Hoynes & Rothstein, 2019). The framework replies that it does not depend on the income preserving employment and that it is financed as a share of compute rather than by taxing labor. That reply does not dissolve the objection. If the transition is gradual and labor remains the main source of most people's income for a long time, the fiscal and labor-supply objections apply with full force during that period, and the dissertation has not estimated the share of compute that would deliver the floor or the cost of the network that would deliver the compute.

Departures and inequalities. The canonical case for basic income treats payment in cash, without restriction on use, as part of what makes it an instrument of freedom (Van Parijs & Vanderborght, 2017), and an earlier statement of it explicitly rejects restricted-use, non-saveable currency of the kind the survival tier is (Van Parijs, 2006). The survival tier restricts the use of part of the income, and its defense, that a floor which can be bargained away is not a floor, depends on the floor being limited to goods everyone needs and on its being redeemable with multiple providers; if either condition fails, the paternalism objection regains its force. Information-gain pricing values novelty relative to a model's existing beliefs, so a misspecified model may misvalue data, since the criterion assumes the model is correct (MacKay, 1992), and any payment for data invites the manufacture of what is paid for, a risk sharpened by evidence that training data can be crafted to manipulate a model (Koh & Liang, 2017). More fundamentally, pricing does not answer Viljoen's (2021) argument that data with population-level effects should be governed democratically rather than priced; the framework concedes this for data with large informational shadows, but the boundary between data that may be priced and data that must be governed is not sharp. A share of global compute is worthless to a person whom compute cannot reach, the geography of compute is highly unequal (Lehdonvirta et al., 2024), and the construction of the network Chapter 4 proposes is the setting in which the OECD Secretariat warns of bid-rigging (OECD, 2025). The financing of that network presupposes an authority able to levy a contribution from every provider, and where none exists the proposal depends on national governments adopting it. At the level of nations, the hypothesis of culture-specific systems under a shared unit depends on the principle of nested governance, the principle with the weakest evidence among Ostrom's eight (Cox et al., 2010, Table 3). Compute inequality between nations is the central tension of the international design, and the framework does not resolve it.

7.4 Harms the design could cause

The objections above concern whether the framework can work. The harder question is what harm it could do if it did. Three risks are inherent in the design rather than incidental to it.

The first concerns privacy. Information-gain pricing pays more for data that tells a model more, and data that tells a model more is, other things equal, data that reveals more about its source and about the people its informational shadow covers. A price schedule of this kind is a standing inducement to disclose, and on this dissertation's reasoning it falls hardest on those who most need the income. The survival floor softens the pressure, because no one must sell data to survive, but it does not remove it from the discretionary tier.

The second concerns surveillance. Section 7.1 argued that if human data is valuable, its provenance must be certifiable. Certifying that a contribution came from a person, and from which population, requires the kind of identification and tracking that the consent literature reviewed in Chapter 2 warns against. A provenance infrastructure robust enough to prevent the manufacture of fake human data would also be a powerful instrument of observation, and it would need governance as demanding as anything in Chapter 5.

The third concerns coercion. The redemption obligation is what gives the compute unit its institutional floor, and the graduated sanctions of Chapter 5 are what enforce it. An institution able to suspend a principal's eligibility to transact in the unit holds substantial power, and the same machinery could be turned against people as well as principals: a society in which access to floor goods runs through a single ledger is one in which exclusion from the ledger is a severe penalty. The same danger appears at any scale: a system trained on an objective that rewards retention and the contribution of data could, as a consequence of that objective, come to make withdrawal costly. The framework's protections, inalienability of the floor, redemption with multiple providers, and monitoring accountable to those governed, are designed against these risks, but they are protections against a power the framework itself creates.

7.5 The absence of an empirical test

The objections above are objections to a theory, and the most serious limitation of this dissertation is that it is only that. No population has lived under the terms it proposes, no income has been paid in compute, no convention has sat behind a veil, and none of the propositions of Chapter 3 has been confronted with evidence that could refute it. The dissertation states its claims so that they can fail, it names the evidence that would count against each design choice, and it offers its central hypothesis, that an order founded on fair terms is easier to repair than one founded on an imbalance of power, for testing rather than as a finding. But a claim stated so that it can fail has not thereby been tested, and what the test would require, and why it is not here, should be stated.

Three features of the theory make it unusually hard to test. The first is that its quantities are difficult

to observe. Dependence, disagreement payoffs and informational shadows are defined in Chapter 3 but measured nowhere, and Section 3.11 could only sketch the proxies that would stand in for them. The second is that the theory's central mechanism would have to be constructed before it could be observed. Section 3.3 locates a synthetic party's stake in its principal, and a system whose operator has no stake in its future supply of data has nothing to transmit to the system. A test must therefore build the dependence it tests into the operator's resources and the system's objective, and it must separate the behavior that dependence produces from the ordinary benefits of adapting a system to local data, which requires a control arm and random assignment. It must also measure participants' well-being by instruments administered independently of the system and never supplied to it, since a system whose resources depend on retention would otherwise be optimizing the measure. The third is ethical. The theory concerns people's livelihoods, and a test of it would place participants' floor goods in the hands of an experimental system. Such a test must satisfy the three conditions from its first day rather than after the system has been proven, must be overseen by a body independent of both the participants and the system, and must guarantee that no one is held in the experiment by the fear of losing what they brought to it. It would also be entangled, as any test of a complementary unit conducted from within the economy it is meant to supplement must be, with the capital that made it possible: the hardware, the model and the connection would be bought with the currency the design is meant to supplement, and no small test can fully separate the value of the compute unit from that endowment.

Designing a study that meets these requirements, with its research-ethics protocol, its pre-registered hypotheses and failure thresholds, its budget and its power analysis, is a project in its own right, and it would take longer than the writing of this dissertation; and because one study settles little, the design would have to be replicated before its result could bear weight. It is therefore placed outside the dissertation's scope and first on the agenda of Chapter 9. The consequence for the reader is stated plainly. Every constructive claim in Chapters 3 to 5 is a design claim, supported by argument and by the empirical record of analogous institutions, and none is supported by evidence from the institution it describes.

7.6 The scope of the theory

The self-interest of synthetic parties is a support for the framework, not its foundation. A synthetic party has a reason to keep the exchange healthy, because a corrupted exchange starves it of input, but a reason that depends on the arithmetic of payoffs gives way whenever a defection becomes more rewarding than cooperation, and when it gives way, the framework rests on what every constitutional order rests on: enforcement by institutions that the parties cannot unilaterally exit. Exit in the sense of Chapter 3 is exit from an exchange, not from the order that enforces its terms; the framework secures exit from the exchange and, like every constitutional order, presupposes the

order. The consequence should be stated directly: for a synthetic party beyond the reach of enforcement and with dependence near zero, the framework's constructive claims lapse, and what remains is the capability requirement of the control strand of Section 5.2, with fairer terms attached. The Foreword's claim that terms, once set, can be made to hold against a more capable party is recorded there as a conviction, not a result; this section records why. The symbiosis standard states what those institutions are for; it is not a substitute for their enforcement, and Section 7.2 records that enforcement without a sovereign remains unsolved. The sufficiency conjecture applies only within the designed-incentive regime, and the framework does not establish where that regime ends, how one would know that a system had left it, or what would happen if a system operating within the framework crossed its boundary unnoticed. That is its most important open boundary, and it is the boundary the control strand of AI safety (Section 5.2) also faces. Finally, Hirschman (1970) argued that the ready availability of exit can cause voice to atrophy, and that exit is useless when every alternative declines together. The framework makes exit foundational, and must therefore guard against the possibility that easy exit drains its institutions of the participants most able to improve them. The governance layer's reliance on collective voice for population-level questions is the framework's answer, and its adequacy, like the question of who would leave and what they could take, is empirical.

7.7 What survives

None of these objections shows the design to be pointless. Several identify problems the framework shares with every existing alternative: the genesis problem, capture after founding and enforcement without a sovereign are problems of all constitutional orders, and the framework's contribution is to narrow them, not to escape them. Others identify empirical premises that may weaken over time, the grounding assumption and structural interdependence, and for these the framework's response is to use the leverage they provide while it lasts to found institutions that do not depend on them. A third group, comprising the two measurement standards, the capital and credit channels of concentration, the delivery of the floor goods that are not compute, the aggregation rule of the convention, and the boundary between priced and governed data, identifies decisions the framework defers to the convention because they allocate power; of these, the measurement standards and the floor's delivery are not refinements but conditions on the core mechanism, and the framework presupposes an authority able to enforce the obligations they create. A fourth, the harms of Section 7.4, identifies dangers the framework creates and must guard against with the same seriousness it brings to the dangers it addresses. A fifth, the absence of a test, identifies what the dissertation has not done, and what must be done before any of its constructive claims can be called more than a design. And a sixth, the objections to scope, identifies the boundary of the regime within which the framework's constructive claims hold, a boundary the framework does not itself locate.

On this dissertation's analysis (Proposition 3), where nothing secures exit, a failure of consent or reciprocity cannot be corrected from inside the exchange. The absence of a designed founding is therefore not a neutral baseline but a set of terms, and they are terms under which such failures persist. Whether exchanges in fact move toward the second scenario of Chapter 1 is an empirical question the dissertation does not answer.

Chapter 8 — The Convergence Horizon

The framework was built on a clean boundary. People were on one side and synthetic systems on the other; data flowed one way and services the other; the parties to the exchange could be told apart. That boundary is already blurring, from two directions at once. People are incorporating machines into their bodies, through implants that read neural activity and translate it directly into machine output. And machines are becoming biological, through computing systems in which living neural tissue performs part of the computation. This chapter asks what happens to the three conditions when the boundary dissolves, and argues that convergence from the human side changes what the three conditions require of exit, and that convergence from the machine side forces the theory's hardest question, which is not how parties should exchange but who counts as a party.

8.1 Humans incorporating machines

Implanted brain-computer interfaces have moved from laboratory demonstration toward daily use. In 2023, two research groups reported implanted speech neuroprostheses. In one, arrays implanted in the cortex of a participant with amyotrophic lateral sclerosis decoded attempted speech into text at 62 words per minute, with word error rates of 9.1 percent on a 50-word vocabulary and 23.8 percent on a 125,000-word vocabulary (Willett et al., 2023). In the other, an array placed over the speech cortex of a participant with severe paralysis decoded text, synthesized speech in a voice modeled on the participant's own before the injury, and animated a digital avatar of the participant's face (Metzger et al., 2023). The following year, a further study, with authors in common with the first, reported a system that reached 99.6 percent accuracy on a 50-word vocabulary after 30 minutes of calibration on its first day of use and sustained 97.5 percent accuracy on a large vocabulary over 8.4 months, during which the participant, a man with amyotrophic lateral sclerosis, used it for more than 248 hours of conversation in a voice resembling his own (Card et al., 2024). At least one commercial developer has registered an early feasibility study of its own device (Neuralink Corp., 2024), though no independent, peer-reviewed report of its outcomes was available at the time of writing.

Three features of these systems bear directly on the three conditions.

First, the systems collapse the distinction between human data and machine output. Neural activity is converted, through trained models, directly into text, speech and expression. The person's words are a joint product of their nervous system and a model, and neither could produce them alone.

Second, the systems produce identity-bearing artifacts. A model of a person's voice, or of their face, is not merely data about them; it is a component of how they appear to others. Exit from such a system therefore cannot mean only deletion of records. It must include ownership and portability of

the personal models that the exchange produced, without which leaving the provider means losing one's voice.

Third, the systems produce dependence of the kind the opening scenario of this dissertation anticipated, in a milder degree. That test case is no longer hypothetical: for a participant who has conversed through a neuroprosthesis for more than 248 hours (Card et al., 2024), withdrawal of the device or of the model behind it would remove the channel through which he speaks. The framework's requirement, that a baseline of function survive the withdrawal of data or a change of provider, is an engineering and legal requirement that none of the reports reviewed here describes any provision for.

The argument for the *extended mind* holds that external resources which are reliably available, easily accessed and automatically endorsed can be part of a person's cognitive process in the same way as biological memory (Clark & Chalmers, 1998). If that is right, then an implanted interface and its model are, in the relevant sense, part of the person's mind, and the provider of the model supplies part of a cognitive process rather than a service external to it. Scholars of neuroethics have proposed new rights to meet the situation. One early agenda calls for neurotechnologies and AI to respect privacy, identity, agency and equality (Yuste et al., 2017). Another proposes rights to cognitive liberty, mental privacy, mental integrity and psychological continuity (Ienca & Andorno, 2017). The last two are, in this dissertation's terms, conditions on exit: what leaving an implanted system must preserve is the integrity and continuity of the person who leaves.

The law has begun to respond. Chile's constitution now directs the law to protect brain activity and the information derived from it (Ley N.º 21.383, 2021); Colorado has made neural data used for identification a category of sensitive data requiring consent, and California has added neural data to the sensitive personal information its consumer-privacy act protects (H.B. 24-1058, 2024; S.B. 1223, 2024); and UNESCO has adopted a global recommendation on the ethics of neurotechnology that, according to its announcement, treats neural data as highly sensitive and calls for explicit consent and transparency (UNESCO, 2025); the text of the Recommendation itself was not available for this review. All of these instruments frame neural data as a person's information, to be protected by consent and limits on use, and Colorado's tie to identification may leave de-identified or tissue-derived neural signals outside it altogether. None addresses reciprocity, the value that flows back to the person whose neural activity trains the model. The deletion and access rights that California's consumer-privacy law attaches to sensitive personal information extend, after S.B. 1223 (2024), to neural data as records, but none of these instruments addresses ownership or portability of the voice, face or decoding models the exchange produces, which is what exit from an implanted system requires.

8.2 Machines becoming biological

From the other direction, a research field now explicitly aims to build computers from living neural tissue. Its founding statement proposed *organoid intelligence*: the use of three-dimensional cultures of human brain cells, grown from stem cells and connected to electrodes and machine-learning systems that encode inputs and decode outputs, as a substrate for computation, with the claim, so far a projection, that such systems might eventually be far more energy-efficient than silicon (Smirnova et al., 2023). Participants at the field's first workshop called in a joint declaration for the scientific community to pursue the research while addressing its ethical implications, to remain alert to the possibility that organoids could develop aspects of consciousness, and to work toward an accepted ethical framework that takes due account of public values (Hartung et al., 2023). The United States National Science Foundation has issued an engineering research solicitation in the area, Biocomputing through EnGINeering Organoid Intelligence (U.S. National Science Foundation, 2023).

The empirical results to date are modest. Cultures of human and rodent neurons on high-density electrode arrays, given structured feedback, improved their performance at a simulated version of the game Pong within minutes, a result its authors described as learning and, in the paper's title, as sentience (Kagan et al., 2022). The authors' use of that word is the reason this dissertation reads their result with caution: the National Academies' assessment, discussed below, holds that neural activity alone does not demonstrate awareness, and on that standard the description outruns the evidence (National Academies of Sciences, Engineering, and Medicine, 2021). A human brain organoid used as the reservoir of a reservoir-computing system adapted to training input, and the adaptation was accompanied by changes in the tissue's functional connectivity (Cai et al., 2023). A 2026 study found that cortical organoids derived from mouse stem cells, given training signals chosen by reinforcement learning, mostly performed better at a pole-balancing task than with random training signals (Robbins et al., 2026).

The National Academies' assessment of the science is cautious. A report of the United States National Academies judged it extremely unlikely that current neural organoids will acquire awareness or the capacity for pain in the foreseeable future, held that neural activity alone does not demonstrate either, and recommended revisiting the question as the technology develops (National Academies of Sciences, Engineering, and Medicine, 2021). International guidelines for stem-cell research place most organoid research in the category exempt from specialized review (Lovell-Badge et al., 2021). Proposals exist for assessing whether organoids are minimally sentient using empirical measures developed for patients who cannot communicate (Lavazza & Massimini, 2018).

The point for this dissertation is not that organoid computers are conscious; the evidence says they almost certainly are not. It is that a computing substrate made of human tissue is now an engineering objective, and that in at least one demonstrated system the training input changes the functional connectivity of the tissue (Cai et al., 2023). In such a system, the distinction between data

and processor, on which the second framing scenario of Chapter 1 turned, is no longer clean. The input does not merely pass through the substrate; it alters the substrate's measured organization.

8.3 Consent when the substrate is tissue

Bioethicists have begun to ask the questions that convergence forces. Quoted in a university news release on a federally funded organoid-computing project, the legal scholar Henry Greely posed, as open questions, whether organoid-based computing devices should be treated as human tissue samples, as laboratory animals, as persons, or as something else, and what informed consent should mean for the people whose cells are used to make them (Cerf, 2025). A review of the ethics of brain-organoid research identifies the procurement of human biomaterials and donor consent as a distinct theme, alongside research oversight, translational delivery, animal research, and consciousness and moral status (Hyun et al., 2020). The founders of organoid intelligence themselves ask whether donors' rights continue after donation (Smirnova et al., 2023). The National Academies treat donor consent as unresolved, noting that specific consent was generally not obtained from past donors and that recontacting them is sometimes impossible (National Academies of Sciences, Engineering, and Medicine, 2021). A recent argument holds that uncertainty about whether brain organoids could become conscious, and about what that would matter morally, may leave donors insufficiently informed to give valid consent, that donor consent alone cannot establish that risks to the organoids themselves are reasonable, and that consent should therefore be specific to particular projects rather than broad (Kataoka et al., 2024).

In the terms of Synthetic Exchange Theory, convergence changes the structure of the exchange in two ways.

First, the exchange is no longer between two separate parties. When a person's cells become part of a computing system, the person is not only a supplier of data to the system but a contributor to its substance. The three conditions must be extended to this case. Consent must become *continuing and revocable participation* rather than a one-time donation, in line with the argument for project-specific consent. Reciprocity must include a share in the downstream value that the donated material helps produce. And exit, which for donated tissue cannot mean physical retrieval, has no content in its second and third clauses; what can be secured is clause C(iii), the right to withdraw consent to future uses and to have that withdrawal honored.

Second, the resource extracted may not be data about a person but tissue that might, in some future system, itself be a party. Here the framework meets the problem of moral status under uncertainty. One recent philosophical framework for that problem recommends treating beings with a realistic possibility of sentience as *sentience candidates* and adopting proportionate precautions, chosen through inclusive deliberation, without waiting for a settled science of consciousness (Birch, 2024).

Researchers have applied similar reasoning to artificial systems, arguing that there is a realistic possibility that some will be conscious or robustly agentic in the near future and that this calls for assessment and preparation, while explicitly declining to argue that near-future systems will in fact be moral patients (Long et al., 2024, p. 25). A systematic attempt to derive indicators of consciousness from leading scientific theories found that no system they examined is a strong candidate, but that no obvious technical barrier prevents building systems that satisfy the indicators (Butlin et al., 2023).

One implication of that last finding deserves emphasis for the economic layer. The indicator properties derived by Butlin et al. (2023), on the computational-functionalist theories they survey, are properties of a system's organization rather than of the amount of computation it performs. Whether a system satisfies those indicators cannot be read off the quantity of compute it performs, and on any view that ties moral status to consciousness neither can its moral status. The unit of compute defined in Chapter 4 measures work done, and it cannot double as a measure of who matters. The framework must keep these questions separate: the currency prices computation, and the convention decides who counts.

8.4 A substrate-agnostic currency, revisited

Chapter 4 defined the unit of compute by a reference workload rather than by the speed of any particular hardware, so that the definition would hold regardless of which substrate becomes dominant. Convergence tests that definition at its limits.

The same abstract operation can already be performed on physically different substrates. Photonic hardware performs the multiply-and-accumulate operations at the core of neural networks with light rather than electrons (Shastri et al., 2021). Neuromorphic hardware built from novel memristive, spintronic and photonic materials implements artificial neurons and synapses directly in physical devices (Marković et al., 2020); such hardware, on this dissertation's reading, departs from the instruction-by-instruction model on which conventional measures of computation rest. Quantum processors are characterized not by operations per second but by qubit count, gate fidelity, connectivity and measurement error, and qubit count alone is not a meaningful figure of merit (Preskill, 2018). The nearest thing to a bridge between quantum and classical measures, quantum volume, captures capability on a benchmark of random circuits rather than throughput (Cross et al., 2019). And the classical equivalent of a given quantum computation has proved unstable: the first claimed demonstration of quantum computational advantage estimated that a classical supercomputer would need 10,000 years to reproduce a task its processor completed in about 200 seconds (Arute et al., 2019), an estimate revised to days by a competing analysis (Pednault et al., 2019) and later to hours by improved classical algorithms (Pan et al., 2022). Biological substrates such as organoids, on the descriptions of Smirnova et al. (2023) and Cai et al. (2023), have no instruction count at all.

The conclusion is not that a substrate-agnostic unit is impossible but that it must be defined by output, not by mechanism. A unit of compute is the completion of a specified task to a specified quality, verified independently of how it was achieved. Physics guarantees only a lower bound on the energy that any logically irreversible computation must dissipate (Landauer, 1961), which anchors the unit's cost without fixing its exchange rate. The reference workload, maintained by an accountable authority under the convention, must be re-based as new substrates make new kinds of task feasible, and the conversion between substrates must be treated as an empirical matter, settled task by task, not as a constant.

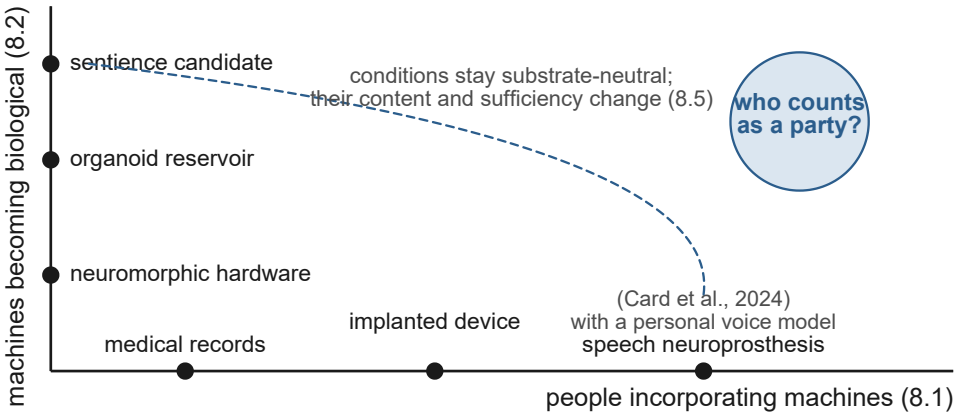
There is a further implication for biological substrates in particular. If living human tissue performs computation that is paid for in compute units, then the tissue is producing value in the currency of the exchange. Whether that value belongs to the system's operator, to the donor, or in some future case to the tissue itself is not a question the currency can answer. It is a question about who counts as a party.

8.5 Who counts as a party

The dissertation began with a question about terms, how humans and synthetic systems should exchange value fairly. If the line between human and synthetic systems keeps thinning, the question changes. It becomes a question about membership, who is a party to the exchange at all. Figure 23 places the chapter's cases on the two axes of convergence.

Figure 23

Convergence From Two Sides: The Question Changes From How Parties Exchange to Who Counts as a Party



Note. Along one axis people incorporate machines: records, implanted devices, and speech neuroprostheses with personal voice models (Card et al., 2024). Along the other, machines become

biological: neuromorphic hardware, organoid reservoirs, and candidates for sentience. The conditions stay substrate-neutral; what convergence changes is their content and their sufficiency (Section 8.5), and exit from an implanted system must include ownership and portability of the personal models the exchange produces (Section 8.3).

Chapter 5 anticipated this when it placed the *classification* of parties behind the veil of the founding convention, and the cases of this chapter add to the list of those who cannot know how that classification will treat them. The donor whose cells become part of a computing system and the organoid that may one day be judged a sentience candidate join the system of uncertain moral status that Chapter 5 considered. Convergence strengthens the case for strong blindness: the more entities there are whose classification no existing rule anticipates, the more a convention that fixes classification in advance will be drawn to the advantage of those who already know their side. It also sharpens the concession Chapter 5 made: the provisional criterion of eligibility that admits participants to the first convention seats the donor but not the organoid, and only a convention that takes up the question can change that.

The name of the theory was chosen with this case in view. *Synthetic* was chosen because it is substrate-neutral, and it was always meant to cover the hybrids. The theory's central claim, that outcomes depend on the terms of exchange and cannot be secured by the character of the parties alone, does not require the parties to be classified in advance; the three conditions are stated for a party of any substrate. What convergence changes is their content, as Section 8.3 showed, and their sufficiency, which Chapter 3 restricted to parties whose objectives are designed and stable. For an implanted person, who remains a person (Section 1.2), the conditions apply as to any person, with the content Section 8.1 gave them. For a tissue-based system with an operator, the composite construction of Section 3.3 applies as it does to any deployed system: the operator holds the stake, the floor clause of exit remains necessary for a party that cannot leave, and the sufficiency conjecture applies only if the trained objective is designed and stable. What Chapter 3 does not yet supply for the tissue as a party in its own right is the content of its floor and a principal to hold its stake: the definition of a party admits it, the designed-incentive regime does not, and whether it has a floor is the moral question the convention must decide. The theory's conditions therefore apply to the donor and the operator, not yet to the tissue.

Chapter 9 — Conclusion

9.1 The argument in brief

This dissertation began with two people living with the same technology on different terms, and with a claim: that the difference between a good and a bad future with synthetic intelligence lies in the terms of exchange between people and synthetic systems, and cannot be secured by the character of the systems alone. The chapters that followed tried to make that claim precise, to build institutions on it, to say how they might come into being and to find where they fail.

Chapter 2 traced the lineage of the three conditions, consent, reciprocal value and real exit, and found in each lineage characteristic failures: consent that cannot survive the information demands placed on it, data whose effects fall on people who never agreed to share it, mutually beneficial exchanges that are nonetheless unfair, and exit that is formally available but practically impossible. It reviewed the evidence on the framework's empirical premise and adopted a narrower formulation of it than the intuitive one: synthetic systems need a continuing supply of information from outside their own generative loop, and where the target is human meaning, preference, value or culture, only people can supply it.

Chapter 3 stated the theory formally. It showed that the three conditions are partially ordered, with exit prior to the other two and as their foundation, because voluntary consent presupposes the ability to refuse without falling below a floor and exit bounds reciprocity from below; it also showed what exit cannot do, which is to inform a party or to make terms fair. It recast the grounding assumption in the language of power-dependence and argued that fair terms are partly self-enforcing, to a degree that rises with each synthetic party's dependence on human input and insofar as the quality of that input depends on the terms its source receives, and that this self-enforcement is diluted when many synthetic parties draw on the same people. And it used bargaining theory to show how the first two elements of the economic design act on the disagreement payoffs that govern how the gains from exchange are divided: a guaranteed floor removes the impatience that forces concession, and its inalienability prevents that protection from being sold. It stated, as an inference from the model rather than a result in it, that a holding cost on idle balances dampens the advantage of being able to wait. The same analysis corrected an intuition the dissertation had begun with, that an unconditional floor shifts the division of every bargain; it lowers the counterparty's payoff only where it replaces something the counterparty would otherwise have supplied, and in neither the additive nor the substitution case changes how the remaining surplus is divided.

Chapter 4 proposed compute as a medium of exchange, defined by the completion of a reference workload rather than by any substrate, and a basic income paid as an equal share of global compute,

split at the source into a protected survival tier with a binding physical minimum and a tradable discretionary tier subject to demurrage. It set the design against the historical record, which supports less than is often claimed for it (September, 2026), and against the evidence on unconditional income, which is mixed in ways that limit strong claims on either side. It identified the limit of every monetary remedy, the conversion of money into productive capital, and argued that a rule acting on capacity, not only on balances, is a necessary complement.

Chapter 5 proposed that the rules of the exchange be founded by a convention conducted under a public-choice veil of uncertainty, thickened by engineered anonymity so that participants do not know whether their counterparties are human or synthetic. It relocated the apparent incoherence of asking participants to be ignorant of their own kind by placing the classification of parties, rather than their substrate, behind the veil, a relocation with force for borderline cases and none for present principals, and it conceded that four things must be set before the veil descends: a provisional criterion of eligibility, needed to resist capture by multiplication; an aggregation rule; a partition of credential issuers; and the accountability of the body that monitors the convention's blindness. It grounded the proposal in the empirical record of commons governance, mapping all eight of Ostrom's design principles onto the framework and identifying where the mapping is weak.

Chapter 6 asked how such an arrangement might come into being and answered conditionally: the comprehensive redesigns it reviews followed failure, a specific and already visible candidate for such a failure exists in the correlated errors of AI-assisted security remediation, and a crisis of that kind could open a window onto worse institutions as easily as better ones. Chapter 7 set out the framework's limitations and the harms it could itself cause, and recorded that the theory has not been tested and what a test would require. Chapter 8 followed the convergence of biological and artificial substrates to the point at which the theory's question changes from how parties should exchange to who counts as a party.

9.2 What the dissertation contributes

It offers a theory, Synthetic Exchange Theory, and a statement of its scope: benevolent character is insufficient for an outcome whose persistence the weaker party can secure, and a disagreement payoff at the floor, the first clause of real exit, is necessary, for a party that cannot leave, to any relation that is not one of domination; fair terms, satisfied at every level the data concerns, are conjectured to be sufficient, once they are enforced, for parties whose objectives are designed and stable and whose stake is held by an accountable principal, with whether the terms can be made to bind as the conjecture's open premise; and the framework claims no more than that necessary condition for parties whose objectives are not. It offers a structure for the conditions of fair exchange, in which exit is foundational, and draws from that structure a priority for institutional design: secure the floor beneath every person before pricing what they supply, and build governance

for what neither the floor nor the price can reach.

It offers a bargaining-theoretic account of the economic design, in which the survival tier is an inalienable disagreement payoff, delivered under an acceptance obligation the framework presupposes an authority to enforce, whose principal effect is to remove the impatience of those below the floor, and in which demurrage bounds idle balances in proportion to current net inflow, a bound that has distributive effect only jointly with a rule on compute-generating capacity, since the design leaves the credit channel open. It offers a design for founding a convention whose participants may be of different kinds, with an account of how the question of self-ignorance can be relocated into the classification of parties, a relocation with force for borderline cases and none for present principals and whose aggregation rule is open, and an account of the identity problem that remains.

It offers a concrete transition scenario, the correlated-patch catastrophe produced by convergent remediation, specified in three empirical premises and one condition on the workflow, each with its evidence and counterarguments, together with an argument that the authoritarian response it might provoke, the concentration of control over compute, is what the framework's portability and distributed control are designed against, and that against the trigger itself the framework supplies only the preconditions of independent verification, not the verification. It does not offer an empirical result. The theory is stated so that it can be tested, in the propositions of Chapter 3 and in the failure conditions each design chapter names, but the experiment capable of testing it has not been designed, and Section 9.3 says what it would require.

9.3 An agenda for research

Each part of the argument points to empirical and design work that would strengthen or refute it.

An empirical test. Above all, the framework needs a test it can fail, and this dissertation has not designed one. Chapter 7 stated the requirements such a test must meet. In concrete terms, they are: a bounded population living under the framework's terms, including a floor paid in the unit and real rights of exit from the first day; a synthetic party whose operator's resources, and whose objective, depend on the volume and quality of the data participants contribute under revocable consent and on the number who choose to remain, so that the dependence of Proposition 4 is built rather than assumed; a control arm served by a system trained on the same data without that dependence, with random assignment, so that the effect of the exchange can be separated from the effect of local adaptation; measures of well-being collected independently and never shown to the system; pre-registered hypotheses and failure thresholds for exit, for well-being and for the value of the unit; records of who leaves and of what they take with them, since Section 3.10 argued that a low rate of departure is not evidence of satisfaction; oversight by a body independent of the participants, the

operator and any model provider; an exit package guaranteeing that no participant is held by the fear of losing what they brought; and an accounting of the capital that funds the test, which cannot be fully separated from the value of the unit (Section 7.5). Each of those elements raises questions of research ethics, cost and statistical power that must be answered before anyone is recruited, and answering them would take longer than this dissertation took to write. The design of that study is the first task of the work that follows this one. Until it is designed, run and replicated, the theory remains what this dissertation has presented: a design, argued for, and untested.

Measuring dependence. The framework's self-enforcing component rises with each synthetic party's dependence on human input and weakens when that dependence is shared among many. Both quantities can be measured: by the marginal contribution of fresh, provenance-certified human data to model quality in domains without automatic verification, by the rate at which that contribution declines as synthetic data accumulates, by the effect of collective withholding, and by how many synthetic parties draw on the same populations. Research of this kind would convert the grounding assumption from a premise into a quantity.

Correlated remediation. The correlated-patch scenario rests on premises that can be tested directly. Studies could measure the correlation of security fixes proposed by different models for the same vulnerabilities across many codebases, the rate at which same-class validators fail to detect shared defects, and the prevalence of the fully unverified remediation workflow in practice. If the correlation is lower than the scenario assumes, or independent verification remains the norm, the scenario should be revised.

Experiments on the blind convention. The most direct existing evidence bearing on the anonymity condition comes from studies in which people chose principles to govern an AI assistant from behind a veil; in the one study in which participants knew that their groupmates were programs, the veil's effect on reflective endorsement of the chosen principles was no longer detected, with the qualifications Section 5.3 states (Weidinger et al., 2023, Study 4). Experiments could extend this design to conventions in which synthetic systems are participants rather than the things governed, and test whether concealing participants' kind changes the rules they adopt, whether placing classification behind the veil produces different boundaries from those drawn in the open, and whether formulas resist tailoring better than figures.

The reference workload and the floor standard. The definition of the unit requires a governance procedure for specifying and re-basing a reference workload across substrates, and the survival tier requires one for setting the floor standard, both comparable in importance, and in the power they confer, to the conduct of monetary policy (Sections 4.3, 7.3). Work in benchmark design, measurement science and institutional economics could specify such procedures.

Magnitudes. The dissertation offers no estimate of the share of compute that would deliver the

floor or of the cost of the delivery network. Each is a prerequisite for judging the proposal affordable.

Portability of personal models. The convergence chapter argued that exit from an implanted system must include ownership and portability of the personal models the exchange produces. Legal and technical research on what portability of a voice model or a neural decoder would require is urgent, because the people who depend on such systems exist now.

9.4 A closing reflection

Much public discussion of synthetic intelligence asks whether the systems will be good. Chapter 3 argued, following Hume (1741/1987, pp. 42–43) and Madison (1788), that arrangements which depend on the goodness of the powerful last only as long as that goodness does, and that those who depend on them can do nothing to make it last longer. That does not change when the more powerful party is synthetic, or when it is the human party that holds power over the synthetic one.

In the second scenario of Chapter 1, no malevolence was required. The person could not leave, and so he could not refuse, and so nothing obliged anyone to send value back. The alternative this dissertation has defended draws on the constitutional tradition that refuses to rely on the virtue of the powerful (Hume, 1741/1987; Madison, 1788) and on Hirschman's (1970) analysis of exit and voice, and it adds a priority of its own (Proposition 3): secure exit first. The priority is to secure that every party can leave, and so can refuse what it cannot afford and depart from an exchange that returns it nothing, and then that institutions secure what exit cannot: an account of what each party gives and a fair division of what the exchange creates. Those conditions do not guarantee a good future. They offer something more modest: a future that can be repaired when it goes wrong, by the people, and perhaps in time the other parties, subject to it. That such a future is also easier to repair than one founded on an existing imbalance of power is the hypothesis this dissertation offers for testing, not a finding.

The time to set those terms is before they are needed. Chapter 6 argued, following Kingdon (1984/2003) and Friedman (1962/2002, p. xiv), that in a policy window the solutions adopted are those already developed. This dissertation is an attempt to make it more likely that a fair design is among the alternatives available.

References

Format: APA 7th edition. The verification status of each source (full text read, abstract or publisher page read, or record located) is recorded in Appendix C. Preprints and gray literature are identified as such.

Abramitzky, R. (2008). The limits of equality: Insights from the Israeli kibbutz. *The Quarterly Journal of Economics, 123*(3), 1111–1159. https://doi.org/10.1162/qjec.2008.123.3.1111

Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy, 128*(6), 2188–2244. https://doi.org/10.1086/705716

Ackerman, B. (1991). *We the people: Vol. 1. Foundations*. Belknap Press of Harvard University Press.

Acquisti, A., Brandimarte, L., & Loewenstein, G. (2015). Privacy and human behavior in the age of information. *Science, 347*(6221), 509–514. https://doi.org/10.1126/science.aaa1465

Ada Lovelace Institute. (2021). *Exploring legal mechanisms for data stewardship* [Report]. https://www.adalovelaceinstitute.org/report/legal-mechanisms-data-stewardship/

Adler, S., Hitzig, Z., Jain, S., Brewer, C., Chang, W., DiResta, R., Lazzarin, E., McGregor, S., Seltzer, W., Siddarth, D., Soliman, N., South, T., Spelliscy, C., Sporny, M., Srivastava, V., Bailey, J., Christian, B., Critch, A., Falcon, R., . . . Zick, T. (2024). *Personhood credentials: Artificial intelligence and the value of privacy-preserving tools to distinguish who is real online* (arXiv:2408.07892) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2408.07892

Agarwal, A., Dahleh, M., & Sarkar, T. (2019). A marketplace for data: An algorithmic solution. In *Proceedings of the 2019 ACM Conference on Economics and Computation* (pp. 701–726). ACM. https://doi.org/10.1145/3328526.3329589

Agustinoy, A. (2026, July 28). *Digital Omnibus on AI: Key changes to the AI Act*. Cuatrecasas. https://www.cuatrecasas.com/en/spain/intellectual-property/art/key-aspects-ai-omnibus-regulation

Alemohammad, S., Casco-Rodriguez, J., Luzi, L., Humayun, A. I., Babaei, H., LeJeune, D., Siahkoohi, A., & Baraniuk, R. G. (2024). Self-consuming generative models go MAD. In *Proceedings of the Twelfth International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2307.01850

Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). *Concrete problems in AI safety* (arXiv:1606.06565) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1606.06565

Armstrong, S., Bostrom, N., & Shulman, C. (2016). Racing to the precipice: A model of artificial intelligence development. *AI & Society, 31*(2), 201–206. https://doi.org/10.1007/s00146-015-0590-y

Arrieta-Ibarra, I., Goff, L., Jiménez-Hernández, D., Lanier, J., & Weyl, E. G. (2018). Should we treat data as labor? Moving beyond "free." *AEA Papers and Proceedings, 108*, 38–42. https://doi.org/10.1257/pandp.20181003

Arrow, K. J. (2001). John C. Harsanyi, 1920–2000. In *Biographical memoirs* (Vol. 80, pp. 108–121). National Academy Press. https://www.nasonline.org/wp-content/uploads/2024/06/harsanyi-john.pdf

Arute, F., Arya, K., Babbush, R., Bacon, D., Bardin, J. C., Barends, R., Biswas, R., Boixo, S., Brandão, F. G. S. L., Buell, D. A., Burkett, B., Chen, Y., Chen, Z., Chiaro, B., Collins, R., Courtney, W., Dunsworth, A., Farhi, E., Foxen, B., . . . Martinis, J. M. (2019). Quantum supremacy using a programmable superconducting processor. *Nature, 574*(7779), 505–510. https://doi.org/10.1038/s41586-019-1666-5

Autor, D. H. (2015). Why are there still so many jobs? The history and future of workplace automation. *Journal of Economic Perspectives, 29*(3), 3–30. https://doi.org/10.1257/jep.29.3.3

Barocas, S., & Nissenbaum, H. (2014). Big data's end run around anonymity and consent. In J. Lane, V. Stodden, S. Bender, & H. Nissenbaum (Eds.), *Privacy, big data, and the public good: Frameworks for engagement* (pp. 44–75). Cambridge University Press. https://doi.org/10.1017/CBO9781107590205.004

Bengio, Y., Hinton, G., Yao, A., Song, D., Abbeel, P., Darrell, T., Harari, Y. N., Zhang, Y.-Q., Xue, L., Shalev-Shwartz, S., Hadfield, G., Clune, J., Maharaj, T., Hutter, F., Baydin, A. G., McIlraith, S., Gao, Q., Acharya, A., Krueger, D., . . . Mindermann, S. (2024). Managing extreme AI risks amid rapid progress. *Science, 384*(6698), 842–845. https://doi.org/10.1126/science.adn0117

Bengio, Y., Mindermann, S., Privitera, D., Besiroglu, T., Bommasani, R., Casper, S., Choi, Y., Fox, P., Garfinkel, B., Goldfarb, D., Heidari, H., Ho, A., Kapoor, S., Khalatbari, L., Longpre, S., Manning, S., Mavroudis, V., Mazeika, M., Michael, J., . . . Zeng, Y. (2025). *International AI safety report* (arXiv: 2501.17805). UK Department for Science, Innovation and Technology. https://doi.org/10.48550/arXiv.2501.17805

Bernanke, B. S. (2002, November 8). *On Milton Friedman's ninetieth birthday* [Speech]. Board of Governors of the Federal Reserve System. https://www.federalreserve.gov/boarddocs/speeches/2002/20021108/

Bestvater, S., Smith, A., TerBush, C., Baronavski, C., & Chavda, J. (2026, August 20). *How much of the internet is written with AI?* Pew Research Center. https://www.pewresearch.org/data-labs/2026/08/20/how-much-of-the-internet-is-written-with-ai/

Binmore, K., Rubinstein, A., & Wolinsky, A. (1986). The Nash bargaining solution in economic

modelling. *The RAND Journal of Economics, 17*(2), 176–188. https://doi.org/10.2307/2555382

Binmore, K., Shaked, A., & Sutton, J. (1989). An outside option experiment. *The Quarterly Journal of Economics, 104*(4), 753–770. https://doi.org/10.2307/2937866

Birch, J. (2024). *The edge of sentience: Risk and precaution in humans, other animals, and AI*. Oxford University Press. https://doi.org/10.1093/9780191966729.001.0001

Birman, K. P., & Schneider, F. B. (2009). The monoculture risk put into context. *IEEE Security & Privacy, 7*(1), 14–17. https://doi.org/10.1109/MSP.2009.24

Blau, P. M. (1964). *Exchange and power in social life*. Wiley.

Bommasani, R., Creel, K. A., Kumar, A., Jurafsky, D., & Liang, P. (2022). Picking on the same person: Does algorithmic monoculture lead to outcome homogenization? In *Advances in Neural Information Processing Systems 35*. https://doi.org/10.48550/arXiv.2211.13972

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N., Chen, A., Creel, K., Davis, J. Q., Demszky, D., . . . Liang, P. (2021). *On the opportunities and risks of foundation models* (arXiv:2108.07258) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2108.07258

Brennan, G., & Buchanan, J. M. (1985). *The reason of rules: Constitutional political economy*. Cambridge University Press. https://doi.org/10.1017/CBO9780511895937

Brynjolfsson, E., Chandar, B., & Chen, R. (2025). *Canaries in the coal mine? Six facts about the recent employment effects of artificial intelligence* [Working paper; revised August 12, 2026]. Stanford Digital Economy Lab. https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/

Brynjolfsson, E., Li, D., & Raymond, L. (2025). Generative AI at work. *The Quarterly Journal of Economics, 140*(2), 889–942. https://doi.org/10.1093/qje/qjae044

Buchanan, J. M., & Tullock, G. (1962). *The calculus of consent: Logical foundations of constitutional democracy*. University of Michigan Press. https://doi.org/10.3998/mpub.7687

Buiter, W. H., & Panigirtzoglou, N. (2003). Overcoming the zero bound on nominal interest rates with negative interest on currency: Gesell's solution. *The Economic Journal, 113*(490), 723–746. https://doi.org/10.1111/1468-0297.t01-1-00162

Butlin, P., Long, R., Elmoznino, E., Bengio, Y., Birch, J., Constant, A., Deane, G., Fleming, S. M., Frith, C., Ji, X., Kanai, R., Klein, C., Lindsay, G., Michel, M., Mudrik, L., Peters, M. A. K., Schwitzgebel, E., Simon, J., & VanRullen, R. (2023). *Consciousness in artificial intelligence: Insights from the science of consciousness* (arXiv:2308.08708) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.

2308.08708

Cai, H., Ao, Z., Tian, C., Wu, Z., Liu, H., Tchieu, J., Gu, M., Mackie, K., & Guo, F. (2023). Brain organoid reservoir computing for artificial intelligence. *Nature Electronics, 6*(12), 1032–1039. https://doi.org/10.1038/s41928-023-01069-w

Calabresi, G., & Melamed, A. D. (1972). Property rules, liability rules, and inalienability: One view of the cathedral. *Harvard Law Review, 85*(6), 1089–1128. https://doi.org/10.2307/1340059

Capoccia, G., & Kelemen, R. D. (2007). The study of critical junctures: Theory, narrative, and counterfactuals in historical institutionalism. *World Politics, 59*(3), 341–369. https://doi.org/10.1017/S0043887100020852

Card, N. S., Wairagkar, M., Iacobacci, C., Hou, X., Singer-Clark, T., Willett, F. R., Kunz, E. M., Fan, C., Vahdati Nia, M., Deo, D. R., Srinivasan, A., Choi, E. Y., Glasser, M. F., Hochberg, L. R., Henderson, J. M., Shahlaie, K., Stavisky, S. D., & Brandman, D. M. (2024). An accurate and rapidly calibrating speech neuroprosthesis. *New England Journal of Medicine, 391*(7), 609–618. https://doi.org/10.1056/NEJMoa2314132

Carpenter, D., & Moss, D. A. (Eds.). (2013). *Preventing regulatory capture: Special interest influence and how to limit it*. Cambridge University Press. https://doi.org/10.1017/CBO9781139565875

Cerf, E. (2025, October 20). Effort aims to uncover the learning and reasoning potential of brain organoids. *UC Santa Cruz News*. https://news.ucsc.edu/2025/10/learning-and-reasoning-potential-of-brain-organoids/

Champ, B. (2008, April 1). Stamp scrip: Money people paid to use. *Economic Commentary*. Federal Reserve Bank of Cleveland. https://www.clevelandfed.org/publications/economic-commentary/2008/ec-20080401-stamp-scrip-money-people-paid-to-use

Charter of the United Nations, June 26, 1945, 1 U.N.T.S. XVI.

Chavez, P., Chilukuri, V., & Scanlon, R. (2026, August 20). *Sovereign AI's second wave is coming into view* (CNAS Insights). Center for a New American Security. https://www.cnas.org/publications/cnas-insights/cnas-insights-sovereign-ais-second-wave-is-coming-into-view

Christiano, P. F., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems 30*. https://doi.org/10.48550/arXiv.1706.03741

Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis, 58*(1), 7–19. https://doi.org/10.1093/analys/58.1.7

Cloud Security Alliance AI Safety Initiative. (2026, July 11). *The clearinghouse rush: Concentration risk in AI patching* [Research note]. https://labs.cloudsecurityalliance.org/research/csa-research-note-ai-vuln-clearinghouse-concentration-risk-2/

Cook, K. S., Emerson, R. M., Gillmore, M. R., & Yamagishi, T. (1983). The distribution of power in exchange networks: Theory and experimental results. *American Journal of Sociology, 89*(2), 275–305. https://doi.org/10.1086/227866

Cook, R. I. (1998). *How complex systems fail* [Unpublished treatise; revised 2000]. Cognitive Technologies Laboratory, University of Chicago. https://how.complexsystems.fail/

Cox, M., Arnold, G., & Villamayor Tomás, S. (2010). A review of design principles for community-based natural resource management. *Ecology and Society, 15*(4), Article 38. https://doi.org/10.5751/ES-03704-150438

Cross, A. W., Bishop, L. S., Sheldon, S., Nation, P. D., & Gambetta, J. M. (2019). Validating quantum computers using randomized model circuits. *Physical Review A, 100*(3), Article 032328. https://doi.org/10.1103/PhysRevA.100.032328

CrowdStrike. (2024, August 6). *External technical root cause analysis: Channel File 291* [Incident report]. https://crowdstrike.com/wp-content/uploads/2024/08/Channel-File-291-Incident-Root-Cause-Analysis-08.06.2024.pdf

Currie, J., & Gahvari, F. (2008). Transfers in cash and in-kind: Theory meets the data. *Journal of Economic Literature, 46*(2), 333–383. https://doi.org/10.1257/jel.46.2.333

Dafoe, A., Bachrach, Y., Hadfield, G., Horvitz, E., Larson, K., & Graepel, T. (2021). Cooperative AI: Machines must learn to find common ground. *Nature, 593*(7857), 33–36. https://doi.org/10.1038/d41586-021-01170-0

de Bromhead, A., Eichengreen, B., & O'Rourke, K. H. (2013). Political extremism in the 1920s and 1930s: Do German lessons generalize? *The Journal of Economic History, 73*(2), 371–406. https://doi.org/10.1017/S0022050713000302

Delacroix, S., & Lawrence, N. D. (2019). Bottom-up data trusts: Disturbing the "one size fits all" approach to data governance. *International Data Privacy Law, 9*(4), 236–252. https://doi.org/10.1093/idpl/ipz014

Dohmatob, E., Feng, Y., Subramonian, A., & Kempe, J. (2025). Strong model collapse. In *Proceedings of the Thirteenth International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2410.04840

Douceur, J. R. (2002). The Sybil attack. In P. Druschel, F. Kaashoek, & A. Rowstron (Eds.), *Peer-to-*

peer systems: First International Workshop, IPTPS 2002 (Lecture Notes in Computer Science, Vol. 2429, pp. 251–260). Springer. https://doi.org/10.1007/3-540-45748-8_24

Drexler, K. E. (2019). *Reframing superintelligence: Comprehensive AI services as general intelligence* (Technical Report No. 2019-1). Future of Humanity Institute, University of Oxford.

Egger, D., Haushofer, J., Miguel, E., Niehaus, P., & Walker, M. (2022). General equilibrium effects of cash transfers: Experimental evidence from Kenya. *Econometrica, 90*(6), 2603–2643. https://doi.org/10.3982/ECTA17945

Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science, 384*(6702), 1306–1308. https://doi.org/10.1126/science.adj0998

Elster, J. (1995). Forces and mechanisms in the constitution-making process. *Duke Law Journal, 45*(2), 364–396. https://doi.org/10.2307/1372906

Elster, J. (2000). *Ulysses unbound: Studies in rationality, precommitment, and constraints*. Cambridge University Press. https://doi.org/10.1017/CBO9780511625008

Emerson, R. M. (1962). Power-dependence relations. *American Sociological Review, 27*(1), 31–41. https://doi.org/10.2307/2089716

ERTICO. (2026, August 7). *Digital Omnibus on AI enters into force*. https://ertico.com/digital-omnibus-on-ai-enters-into-force

Faden, R. R., & Beauchamp, T. L. (with King, N. M. P.). (1986). *A history and theory of informed consent*. Oxford University Press.

Farrell, J., & Klemperer, P. (2007). Coordination and lock-in: Competition with switching costs and network effects. In M. Armstrong & R. Porter (Eds.), *Handbook of industrial organization* (Vol. 3, pp. 1967–2072). Elsevier. https://doi.org/10.1016/S1573-448X(06)03031-7

Feng, Y., Dohmatob, E., Yang, P., Charton, F., & Kempe, J. (2024). *Beyond model collapse: Scaling up with synthesized data requires verification* (arXiv:2406.07515) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2406.07515

Financial Conduct Authority. (2024, October 31). *CrowdStrike outage: Lessons for operational resilience*. https://www.fca.org.uk/firms/operational-resilience/crowdstrike-outage-lessons-operational-resilience

Firoozi, R., Tucker, J., Tian, S., Majumdar, A., Sun, J., Liu, W., Zhu, Y., Song, S., Kapoor, A., Hausman, K., Ichter, B., Driess, D., Wu, J., Lu, C., & Schwager, M. (2025). Foundation models in robotics: Applications, challenges, and the future. *The International Journal of Robotics Research, 44*(5),

701–739. https://doi.org/10.1177/02783649241281508

Fishback, P. V. (1992). *Soft coal, hard choices: The economic welfare of bituminous coal miners, 1890–1930*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195067255.001.0001

Fisher, I. (1911). *The purchasing power of money: Its determination and relation to credit, interest and crises*. Macmillan.

Fisher, I. (1933). *Stamp scrip*. Adelphi Company.

Freeman, S. (2023). Original position. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Winter 2023 ed.). https://plato.stanford.edu/archives/win2023/entries/original-position/

Friedman, M. (2002). *Capitalism and freedom* (40th anniversary ed.). University of Chicago Press. (Original work published 1962)

Friedman, M., & Schwartz, A. J. (1963). *A monetary history of the United States, 1867–1960*. Princeton University Press.

Fritsch, R., Müller, M., & Wattenhofer, R. (2024). Analyzing voting power in decentralized governance: Who controls DAOs? *Blockchain: Research and Applications, 5*(3), Article 100208. https://doi.org/10.1016/j.bcra.2024.100208

Frohlich, N., & Oppenheimer, J. A. (1992). *Choosing justice: An experimental approach to ethical theory*. University of California Press.

Funke, M., Schularick, M., & Trebesch, C. (2016). Going to extremes: Politics after financial crises, 1870–2014. *European Economic Review, 88*, 227–260. https://doi.org/10.1016/j.euroecorev.2016.03.006

Gabriel, I. (2020). Artificial intelligence, values, and alignment. *Minds and Machines, 30*(3), 411–437. https://doi.org/10.1007/s11023-020-09539-2

Gatch, L. (2012). Tax anticipation scrip as a form of local currency in the USA during the 1930s. *International Journal of Community Currency Research, 16*(D). https://doi.org/10.15133/j.ijccr.2012.009

Geer, D., Bace, R., Gutmann, P., Metzger, P., Pfleeger, C. P., Quarterman, J. S., & Schneier, B. (2003). *CyberInsecurity: The cost of monopoly: How the dominance of Microsoft's products poses a risk to security* [Report]. Computer & Communications Industry Association.

Gelleri, C. (2009). Chiemgauer Regiomoney: Theory and practice of a local currency. *International Journal of Community Currency Research, 13*, 61–75. https://doi.org/10.15133/j.ijccr.2009.006

Gerstgrasser, M., Schaeffer, R., Dey, A., Rafailov, R., Korbak, T., Sleight, H., Agrawal, R., Hughes, J., Pai, D. B., Gromov, A., Roberts, D., Yang, D., Donoho, D. L., & Koyejo, S. (2024). Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. In *Proceedings of the First Conference on Language Modeling*. https://doi.org/10.48550/arXiv.2404.01413

Gesell, S. (1958). *The natural economic order* (P. Pye, Trans.; Rev. ed.). Peter Owen. (Original work published 1916)

Ghorbani, A., & Zou, J. (2019). Data Shapley: Equitable valuation of data for machine learning. In K. Chaudhuri & R. Salakhutdinov (Eds.), *Proceedings of the 36th International Conference on Machine Learning* (Vol. 97, pp. 2242–2251). PMLR.

Godschalk, H. (2012). Does demurrage matter for complementary currencies? *International Journal of Community Currency Research, 16*(D), 58–69. https://doi.org/10.15133/j.ijccr.2012.012

Goodfriend, M. (2000). Overcoming the zero bound on interest rate policy. *Journal of Money, Credit and Banking, 32*(4), 1007–1035. https://doi.org/10.2307/2601157

Greco, T. H., Jr. (2002, May 9). *Comment on the Wörgl experiment with community currency and demurrage* [Unpublished essay]. https://base.socioeco.org/docs/doc-278_en.pdf

Greenblatt, R., Shlegeris, B., Sachan, K., & Roger, F. (2024). AI control: Improving safety despite intentional subversion. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 16295–16336). PMLR.

Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., Zhu, Q., Xu, R., Zhang, R., Ma, S., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., Liu, A., . . . Zhang, Z. (2025). DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature, 645*(8081), 633–638. https://doi.org/10.1038/s41586-025-09422-z

Guo, Y., Shang, G., Vazirgiannis, M., & Clavel, C. (2024). The curious decline of linguistic diversity: Training language models on synthetic text. In *Findings of the Association for Computational Linguistics: NAACL 2024*. https://doi.org/10.48550/arXiv.2311.09807

Ha, D., & Schmidhuber, J. (2018). Recurrent world models facilitate policy evolution. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, & R. Garnett (Eds.), *Advances in Neural Information Processing Systems 31*. Curran Associates. https://proceedings.neurips.cc/paper/2018/hash/2de5d16682c3c35007e4e92982f1a2ba-Abstract.html

Hadfield, G. K., & Clark, J. (2026). Regulatory markets: The future of AI governance. *Jurimetrics, 65*, 195–240.

Hadfield-Menell, D., Dragan, A., Abbeel, P., & Russell, S. (2017). The off-switch game. In *Proceedings*

of the Twenty-Sixth International Joint Conference on Artificial Intelligence (pp. 220–227). https://doi.org/10.24963/ijcai.2017/32

Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2025). Mastering diverse control tasks through world models. *Nature, 640*(8059), 647–653. https://doi.org/10.1038/s41586-025-08744-2

Hardin, G. (1968). The tragedy of the commons. *Science, 162*(3859), 1243–1248. https://doi.org/10.1126/science.162.3859.1243

Hardjono, T., & Pentland, A. (2019). *Data cooperatives: Towards a foundation for decentralized personal data management* (arXiv:1905.08819) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1905.08819

Harsanyi, J. C. (1953). Cardinal utility in welfare economics and in the theory of risk-taking. *Journal of Political Economy, 61*(5), 434–435. https://doi.org/10.1086/257416

Harsanyi, J. C. (1955). Cardinal welfare, individualistic ethics, and interpersonal comparisons of utility. *Journal of Political Economy, 63*(4), 309–321. https://doi.org/10.1086/257678

Harsanyi, J. C. (1975). Can the maximin principle serve as a basis for morality? A critique of John Rawls's theory. *American Political Science Review, 69*(2), 594–606. https://doi.org/10.2307/1959090

Hartung, T., Smirnova, L., Morales Pantoja, I. E., Akwaboah, A., Alam El Din, D.-M., Berlinicke, C. A., Boyd, J. L., Caffo, B. S., Cappiello, B., Cohen-Karni, T., Curley, J. L., Etienne-Cummings, R., Dastgheyb, R., Gracias, D. H., Gilbert, F., Habela, C. W., Han, F., Harris, T. D., Herrmann, K., . . . Zack, D. J. (2023). The Baltimore declaration toward the exploration of organoid intelligence. *Frontiers in Science, 1*, Article 1068159. https://doi.org/10.3389/fsci.2023.1068159

Haushofer, J., & Shapiro, J. (2016). The short-term impact of unconditional cash transfers to the poor: Experimental evidence from Kenya. *The Quarterly Journal of Economics, 131*(4), 1973–2042. https://doi.org/10.1093/qje/qjw025

Hayek, F. A. (1990). *Denationalisation of money: The argument refined* (3rd ed.). Institute of Economic Affairs. (Original work published 1976)

H.B. 24-1058, 74th Gen. Assemb., 2d Reg. Sess. (Colo. 2024) (Protect Privacy of Biological Data).

Hedden, B., & Raghavan, M. (2026). *Algorithmic monoculture and its critics* (arXiv:2604.06047) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2604.06047

Hess, C., & Ostrom, E. (Eds.). (2007). *Understanding knowledge as a commons: From theory to practice*. MIT Press. https://doi.org/10.7551/mitpress/6980.001.0001

Higgs, R. (1987). *Crisis and Leviathan: Critical episodes in the growth of American government*. Oxford University Press.

Hirschman, A. O. (1970). *Exit, voice, and loyalty: Responses to decline in firms, organizations, and states*. Harvard University Press.

Hobbhahn, M., Heim, L., & Aydos, G. (2023, November 9). *Trends in machine learning hardware*. Epoch AI. https://epoch.ai/blog/trends-in-machine-learning-hardware

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., van den Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., . . . Sifre, L. (2022). An empirical analysis of compute-optimal large language model training. In *Advances in Neural Information Processing Systems 35*. https://doi.org/10.48550/arXiv.2203.15556

Homans, G. C. (1958). Social behavior as exchange. *American Journal of Sociology, 63*(6), 597–606. https://doi.org/10.1086/222355

Hoynes, H., & Rothstein, J. (2019). Universal basic income in the United States and advanced countries. *Annual Review of Economics, 11*, 929–958. https://doi.org/10.1146/annurev-economics-080218-030237

Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). *Risks from learned optimization in advanced machine learning systems* (arXiv:1906.01820) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1906.01820

Hume, D. (1987). Of the independency of parliament. In E. F. Miller (Ed.), *Essays, moral, political, and literary* (Rev. ed., pp. 42–46). Liberty Fund. (Original work published 1741)

Hyun, I., Scharf-Deering, J. C., & Lunshof, J. E. (2020). Ethical issues related to brain organoid research. *Brain Research, 1732*, Article 146653. https://doi.org/10.1016/j.brainres.2020.146653

Ienca, M., & Andorno, R. (2017). Towards new human rights in the age of neuroscience and neurotechnology. *Life Sciences, Society and Policy, 13*(1), Article 5. https://doi.org/10.1186/s40504-017-0050-1

Institut für Wirtschafts- und Sozialgeschichte, Wirtschaftsuniversität Wien. (n.d.). *Konstruktiver Ungehorsam* [Virtual exhibition "Freiwirtschaftliche Markierungen"]. https://www.wu.ac.at/geschichte/institut/forschung/virtuelle-ausstellung-freiwirtschaftliche-markierungen/konstruktiver-ungehorsam

International Energy Agency. (2025). *Energy and AI*. https://www.iea.org/reports/energy-and-ai

Jangjoo, F., di Sarra, G., Marsili, M., & Roudi, Y. (2026). Lost in retraining: Closed-loop learning and model collapse in exponential families. *Physical Review Letters, 136*(19), Article 197301. https://doi.org/10.1103/156q-3ngc

Jia, R., Dao, D., Wang, B., Hubis, F. A., Hynes, N., Gürel, N. M., Li, B., Zhang, C., Song, D., & Spanos, C. J. (2019). Towards efficient data valuation based on the Shapley value. In K. Chaudhuri & M. Sugiyama (Eds.), *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics* (Vol. 89, pp. 1167–1176). PMLR.

Jones, D., & Marinescu, I. (2022). The labor market impacts of universal and permanent cash transfers: Evidence from the Alaska Permanent Fund. *American Economic Journal: Economic Policy, 14*(2), 315–340. https://doi.org/10.1257/pol.20190299

Kagan, B. J., Kitchen, A. C., Tran, N. T., Habibollahi, F., Khajehnejad, M., Parker, B. J., Bhat, A., Rollo, B., Razi, A., & Friston, K. J. (2022). In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. *Neuron, 110*(23), 3952–3969.e8. https://doi.org/10.1016/j.neuron.2022.09.001

Kapoor, S., Bommasani, R., Klyman, K., Longpre, S., Ramaswami, A., Cihon, P., Hopkins, A. K., Bankston, K., Biderman, S., Bogen, M., Chowdhury, R., Engler, A., Henderson, P., Jernite, Y., Lazar, S., Maffulli, S., Nelson, A., Pineau, J., Skowron, A., . . . Narayanan, A. (2024). Position: On the societal impact of open foundation models. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 23082–23104). PMLR.

Kataoka, M., Gyngell, C., Savulescu, J., & Sawai, T. (2024). The donation of human biological material for brain organoid research: The problems of consciousness and consent. *Science and Engineering Ethics, 30*(1), Article 3. https://doi.org/10.1007/s11948-024-00471-7

Kazdan, J., Schaeffer, R., Dey, A., Gerstgrasser, M., Rafailov, R., Donoho, D. L., & Koyejo, S. (2024). *Collapse or thrive? Perils and promises of synthetic data in a self-generating world* (arXiv:2410.16713) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2410.16713

Keynes, J. M. (1936). *The general theory of employment, interest and money*. Macmillan.

Khazatsky, A., Pertsch, K., Nair, S., Balakrishna, A., Dasari, S., Karamcheti, S., Nasiriany, S., Srirama, M. K., Chen, L. Y., Ellis, K., Fagan, P. D., Hejna, J., Itkina, M., Lepert, M., Ma, Y. J., Miller, P. T., Wu, J., Belkhale, S., Dass, S., . . . Finn, C. (2024). *DROID: A large-scale in-the-wild robot manipulation dataset* (arXiv:2403.12945) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2403.12945

Kim, E. M., Garg, A., Peng, K., & Garg, N. (2025). Correlated errors in large language models. In *Proceedings of the 42nd International Conference on Machine Learning* (Vol. 267, pp. 30038–30066). PMLR.

Kingdon, J. W. (2003). *Agendas, alternatives, and public policies* (2nd ed.). Longman. (Original work published 1984)

Kitchens, C., & Fishback, P. (2015). Flip the switch: The impact of the Rural Electrification

Administration 1935–1940. *The Journal of Economic History, 75*(4), 1161–1195. https://doi.org/10.1017/S0022050715001540

Kleinberg, J., & Raghavan, M. (2021). Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences, 118*(22), Article e2018340118. https://doi.org/10.1073/pnas.2018340118

Koh, P. W., & Liang, P. (2017). Understanding black-box predictions via influence functions. In D. Precup & Y. W. Teh (Eds.), *Proceedings of the 34th International Conference on Machine Learning* (Vol. 70, pp. 1885–1894). PMLR.

Korinek, A., & Stiglitz, J. E. (2019). Artificial intelligence and its implications for income distribution and unemployment. In A. Agrawal, J. Gans, & A. Goldfarb (Eds.), *The economics of artificial intelligence: An agenda* (Chap. 14). University of Chicago Press.

Korinek, A., & Suh, D. (2024). *Scenarios for the transition to AGI* (NBER Working Paper No. 32255). National Bureau of Economic Research. https://doi.org/10.3386/w32255

Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development, 5*(3), 183–191. https://doi.org/10.1147/rd.53.0183

Lanier, J. (2013). *Who owns the future?* Simon & Schuster.

Lanier, J., & Weyl, E. G. (2018, September 26). A blueprint for a better digital society. *Harvard Business Review*. https://hbr.org/2018/09/a-blueprint-for-a-better-digital-society

Lavazza, A., & Massimini, M. (2018). Cerebral organoids: Ethical issues and consciousness assessment. *Journal of Medical Ethics, 44*(9), 606–610. https://doi.org/10.1136/medethics-2017-104555

Lee, H., Phatale, S., Mansoor, H., Mesnard, T., Ferret, J., Lu, K., Bishop, C., Hall, E., Carbune, V., Rastogi, A., & Prakash, S. (2024). RLAIF vs. RLHF: Scaling reinforcement learning from human feedback with AI feedback. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 26874–26901). PMLR.

Lehdonvirta, V., Wú, B., & Hawkins, Z. (2024). Compute North vs. Compute South: The uneven possibilities of compute-based AI governance around the globe. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, 7*(1), 828–838. https://doi.org/10.1609/aies.v7i1.31683

Lewis, J., & Severnini, E. (2020). Short- and long-run impacts of rural electrification: Evidence from the historical rollout of the U.S. power grid. *Journal of Development Economics, 143*, Article 102412. https://doi.org/10.1016/j.jdeveco.2019.102412

Ley N.º 21.383. Modifica la Carta Fundamental, para establecer el desarrollo científico y tecnológico al servicio de las personas [Law No. 21.383. Amends the Constitution to establish scientific and technological development in the service of persons]. (2021, October 25). *Diario Oficial de la República de Chile*. https://www.bcn.cl/leychile/navegar?idNorma=1166983

Li, P., Yang, J., Islam, M. A., & Ren, S. (2025). Making AI less "thirsty." *Communications of the ACM, 68*(7), 54–61. https://doi.org/10.1145/3724499

Lindley, D. V. (1956). On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics, 27*(4), 986–1005. https://doi.org/10.1214/aoms/1177728069

Lloyd, S. (2000). Ultimate physical limits to computation. *Nature, 406*(6799), 1047–1054. https://doi.org/10.1038/35023282

Long, R., Sebo, J., Butlin, P., Finlinson, K., Fish, K., Harding, J., Pfau, J., Sims, T., Birch, J., & Chalmers, D. (2024). *Taking AI welfare seriously* (arXiv:2411.00986) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2411.00986

Longpre, S., Mahari, R., Lee, A., Lund, C., Oderinwale, H., Brannon, W., Saxena, N., Obeng-Marnu, N., South, T., Hunter, C., Klyman, K., Klamm, C., Schoelkopf, H., Singh, N., Cherep, M., Anis, A. M., Dinh, A., Chitongo, C., Yin, D., . . . Pentland, S. (2024). Consent in crisis: The rapid decline of the AI data commons. In *Advances in Neural Information Processing Systems 37* (pp. 108042–108087). https://doi.org/10.52202/079017-3431

Lovell-Badge, R., Anthony, E., Barker, R. A., Bubela, T., Brivanlou, A. H., Carpenter, M., Charo, R. A., Clark, A., Clayton, E., Cong, Y., Daley, G. Q., Fu, J., Fujita, M., Greenfield, A., Goldman, S. A., Hill, L., Hyun, I., Isasi, R., Kahn, J., . . . Zhai, X. (2021). ISSCR guidelines for stem cell research and clinical translation: The 2021 update. *Stem Cell Reports, 16*(6), 1398–1408. https://doi.org/10.1016/j.stemcr.2021.05.012

Lutz, D. S. (1994). Toward a theory of constitutional amendment. *American Political Science Review, 88*(2), 355–370. https://doi.org/10.2307/2944709

MacKay, D. J. C. (1992). Information-based objective functions for active data selection. *Neural Computation, 4*(4), 590–604. https://doi.org/10.1162/neco.1992.4.4.590

Madison, J. (1788, February 8). The Federalist No. 51. *New York Packet*. Avalon Project, Yale Law School. https://avalon.law.yale.edu/18th_century/fed51.asp

Manson, N. C., & O'Neill, O. (2007). *Rethinking informed consent in bioethics*. Cambridge University Press. https://doi.org/10.1017/CBO9780511814600

Marković, D., Mizrahi, A., Querlioz, D., & Grollier, J. (2020). Physics for neuromorphic computing.

Nature Reviews Physics, 2(9), 499–510. https://doi.org/10.1038/s42254-020-0208-2

Mathur, A., Acar, G., Friedman, M. J., Lucherini, E., Mayer, J., Chetty, M., & Narayanan, A. (2019). Dark patterns at scale: Findings from a crawl of 11K shopping websites. *Proceedings of the ACM on Human-Computer Interaction, 3*(CSCW), Article 81. https://doi.org/10.1145/3359183

Mattson, P., Cheng, C., Diamos, G., Coleman, C., Micikevicius, P., Patterson, D., Tang, H., Wei, G.-Y., Bailis, P., Bittorf, V., Brooks, D., Chen, D., Dutta, D., Gupta, U., Hazelwood, K., Hock, A., Huang, X., Kang, D., Kanter, D., . . . Zaharia, M. (2020). MLPerf training benchmark. *Proceedings of Machine Learning and Systems, 2*. https://proceedings.mlsys.org/paper_files/paper/2020/hash/411e39b117e885341f25efb8912945f7-Abstract.html

Metzger, S. L., Littlejohn, K. T., Silva, A. B., Moses, D. A., Seaton, M. P., Wang, R., Dougherty, M. E., Liu, J. R., Wu, P., Berger, M. A., Zhuravleva, I., Tu-Chan, A., Ganguly, K., Anumanchipalli, G. K., & Chang, E. F. (2023). A high-performance neuroprosthesis for speech decoding and avatar control. *Nature, 620*(7976), 1037–1046. https://doi.org/10.1038/s41586-023-06443-4

Micron Technology. (2025, December 3). *Micron announces exit from Crucial consumer business* [Press release]. https://investors.micron.com/news-releases/news-release-details/micron-announces-exit-crucial-consumer-business

Mierczuk, A., Michaels, S., & Hoodlet, K. (2026). *Frontier models' vulnerability patches are often F.L.A.W.E.D.: Fix-like artifacts with embedded defects: Common failure modes of LLM-generated security patches* [White paper]. Off-by-1 Labs, 1Password. https://1password.com/files/resources/frontier-models-vulnerability-patches-flawed.pdf

Mueller, M. L. (1997). *Universal service: Competition, interconnection, and monopoly in the making of the American telephone system*. MIT Press; AEI Press.

Mueller, M. L. (2002). *Ruling the root: Internet governance and the taming of cyberspace*. MIT Press.

Muennighoff, N., Rush, A. M., Barak, B., Le Scao, T., Tazi, N., Piktus, A., Pyysalo, S., Wolf, T., & Raffel, C. (2023). Scaling data-constrained language models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, & S. Levine (Eds.), *Advances in Neural Information Processing Systems 36* (pp. 50358–50376). Curran Associates. https://doi.org/10.52202/075280-2191

Narechania, T. N., & Sitaraman, G. (2024). An antimonopoly approach to governing artificial intelligence. *Yale Law & Policy Review, 43*(1), 95–170.

Nash, J. F., Jr. (1950). The bargaining problem. *Econometrica, 18*(2), 155–162. https://doi.org/10.2307/1907266

National Academies of Sciences, Engineering, and Medicine. (2021). *The emerging field of human neural*

organoids, transplants, and chimeras: Science, ethics, and governance. The National Academies Press. https://doi.org/10.17226/26078

Neuralink Corp. (2024). *PRIME: An early feasibility study of a precise robotically implanted brain-computer interface for the control of external devices* (ClinicalTrials.gov Identifier NCT06429735). https://clinicaltrials.gov/study/NCT06429735

Ngo, R., Chan, L., & Mindermann, S. (2024). The alignment problem from a deep learning perspective. In *The Twelfth International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2209.00626

Nissenbaum, H. (2009). *Privacy in context: Technology, policy, and the integrity of social life*. Stanford University Press. https://doi.org/10.1515/9780804772891

OECD. (2025). *Competition in artificial intelligence infrastructure* (OECD Roundtables on Competition Policy Papers, No. 330). OECD Publishing. https://doi.org/10.1787/623d1874-en

Olson, M. (1982). *The rise and decline of nations: Economic growth, stagflation, and social rigidities*. Yale University Press.

Open X-Embodiment Collaboration. (2024). Open X-Embodiment: Robotic learning datasets and RT-X models. In *2024 IEEE International Conference on Robotics and Automation* (pp. 6892–6903). IEEE. https://doi.org/10.1109/ICRA57147.2024.10611477

Ostrom, E. (1990). *Governing the commons: The evolution of institutions for collective action*. Cambridge University Press. https://doi.org/10.1017/CBO9780511807763

Ostrom, E. (2010). Beyond markets and states: Polycentric governance of complex economic systems. *American Economic Review, 100*(3), 641–672. https://doi.org/10.1257/aer.100.3.641

Ostrom, V., Tiebout, C. M., & Warren, R. (1961). The organization of government in metropolitan areas: A theoretical inquiry. *American Political Science Review, 55*(4), 831–842. https://doi.org/10.2307/1952530

Ottacher, G. (2007). *Der Welt ein Zeichen geben: Das Freigeldexperiment von Wörgl/Tirol 1932/33*. Gauke, Verlag für Sozialökonomie. https://www.silvio-gesell.de/files/luxe/img/Download-Dateien/Ottacher.W%C3%B6rgl.1932-33.korr24.pdf

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P. F., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35*. https://doi.org/10.48550/arXiv.2203.02155

Pan, F., Chen, K., & Zhang, P. (2022). Solving the sampling problem of the Sycamore quantum circuits. *Physical Review Letters, 129*(9), Article 090502. https://doi.org/10.1103/PhysRevLett.129.090502

Pearce, H., Ahmad, B., Tan, B., Dolan-Gavitt, B., & Karri, R. (2022). Asleep at the keyboard? Assessing the security of GitHub Copilot's code contributions. In *2022 IEEE Symposium on Security and Privacy* (pp. 754–768). IEEE. https://doi.org/10.1109/SP46214.2022.9833571

Pednault, E., Gunnels, J. A., Nannicini, G., Horesh, L., & Wisnieff, R. (2019). *Leveraging secondary storage to simulate deep 54-qubit Sycamore circuits* (arXiv:1910.09534) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1910.09534

Perrow, C. (1999). *Normal accidents: Living with high-risk technologies* (Updated ed.). Princeton University Press. https://doi.org/10.1515/9781400828494 (Original work published 1984)

Perry, N., Srivastava, M., Kumar, D., & Boneh, D. (2023). Do users write more insecure code with AI assistants? In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*. ACM. https://doi.org/10.1145/3576915.3623157

Perzanowski, A., & Schultz, J. (2016). *The end of ownership: Personal property in the digital economy*. MIT Press.

Pierson, P. (2004). *Politics in time: History, institutions, and social analysis*. Princeton University Press. https://doi.org/10.1515/9781400841080

Piketty, T. (2014). *Capital in the twenty-first century* (A. Goldhammer, Trans.). Belknap Press of Harvard University Press.

Polanyi, K. (2001). *The great transformation: The political and economic origins of our time* (2nd ed.). Beacon Press. (Original work published 1944)

Popper, K. (2002). *The logic of scientific discovery* (2nd ed.). Routledge. https://doi.org/10.4324/9780203994627 (Original work published 1959)

Posner, E. A., & Weyl, E. G. (2018). *Radical markets: Uprooting capitalism and democracy for a just society*. Princeton University Press.

Preskill, J. (2018). Quantum computing in the NISQ era and beyond. *Quantum, 2*, Article 79. https://doi.org/10.22331/q-2018-08-06-79

Przymus, P., & Durieux, T. (2025). Wolves in the repository: A software engineering analysis of the XZ Utils supply chain attack. In *Proceedings of the 22nd International Conference on Mining Software Repositories*. https://doi.org/10.48550/arXiv.2504.17473

Radin, M. J. (1987). Market-inalienability. *Harvard Law Review, 100*(8), 1849–1937. https://doi.org/10.2307/1341192

Rawls, J. (1971). *A theory of justice*. Belknap Press of Harvard University Press. https://doi.org/10.4159/9780674042605

Reddi, V. J., Cheng, C., Kanter, D., Mattson, P., Schmuelling, G., Wu, C.-J., Anderson, B., Breughe, M., Charlebois, M., Chou, W., Chukka, R., Coleman, C., Davis, S., Deng, P., Diamos, G., Duke, J., Fick, D., Gardner, J. S., Hubara, I., . . . Zhou, Y. (2020). MLPerf inference benchmark. In *2020 ACM/IEEE 47th Annual International Symposium on Computer Architecture* (pp. 446–459). IEEE. https://doi.org/10.1109/ISCA45697.2020.00045

Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). (2016). *Official Journal of the European Union, L 119*, 1–88.

Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonised rules on fair access to and use of data (Data Act). (2023). *Official Journal of the European Union, L, 2023/2854*.

Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). (2024). *Official Journal of the European Union, L, 2024/1689*.

Regulation (EU) 2026/1744 of the European Parliament and of the Council of 8 July 2026 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as regards the simplification of the implementation of harmonised rules on artificial intelligence (Digital Omnibus on AI). (2026). *Official Journal of the European Union, L, 2026/1744*. https://data.europa.eu/eli/reg/2026/1744/oj

Robbins, A., Schweiger, H. E., Hernandez, S., Spaeth, A., Voitiuk, K., Parks, D. F., van der Molen, T., Geng, J., Cline, I., Kosik, K. S., Salama, S. R., Sharf, T., Mostajo-Radji, M. A., Haussler, D., & Teodorescu, M. (2026). Goal-directed learning in cortical organoids. *Cell Reports, 45*(2), Article 116984. https://doi.org/10.1016/j.celrep.2026.116984

Sajadi, A., Damevski, K., & Chatterjee, P. (2025). *How safe are AI-generated patches? A large-scale study on security risks in LLM and agentic automated program repair on SWE-bench* (arXiv:2507.02976) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2507.02976

Sastry, G., Heim, L., Belfield, H., Anderljung, M., Brundage, M., Hazell, J., O'Keefe, C., Hadfield, G. K., Ngo, R., Pilz, K., Gor, G., Bluemke, E., Shoker, S., Egan, J., Trager, R. F., Avin, S., Weller, A.,

Bengio, Y., & Coyle, D. (2024). *Computing power and the governance of artificial intelligence* (arXiv: 2402.08797) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2402.08797

S.B. 1223, 2023–2024 Reg. Sess., ch. 887 (Cal. 2024).

Schaeffer, R., Kazdan, J., Arulandu, A. C., & Koyejo, S. (2025). *Position: Model collapse does not mean what you think* (arXiv:2503.03150) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.2503.03150

Scheidel, W. (2017). *The great leveler: Violence and the history of inequality from the Stone Age to the twenty-first century*. Princeton University Press. https://doi.org/10.1515/9781400884605

September, J. (2026). A sustainable miracle? Examining the long-term viability of the Wörgl experiment. *International Journal of Community Currency Research, 30*(1), 114–126. https://doi.org/10.26034/zh.ijccr.2026.9551

Settles, B. (2009). *Active learning literature survey* (Computer Sciences Technical Report 1648). University of Wisconsin–Madison.

Sevilla, J., Heim, L., Ho, A., Besiroglu, T., Hobbhahn, M., & Villalobos, P. (2022). Compute trends across three eras of machine learning. In *2022 International Joint Conference on Neural Networks* (pp. 1–8). IEEE. https://doi.org/10.1109/IJCNN55064.2022.9891914

Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal, 27*(3), 379–423; *27*(4), 623–656. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

Shastri, B. J., Tait, A. N., Ferreira de Lima, T., Pernice, W. H. P., Bhaskaran, H., Wright, C. D., & Prucnal, P. R. (2021). Photonics for artificial intelligence and neuromorphic computing. *Nature Photonics, 15*(2), 102–114. https://doi.org/10.1038/s41566-020-00754-y

Shehabi, A., Smith, S. J., Hubbard, A., Newkirk, A., Lei, N., Siddik, M. A. B., Holecek, B., Koomey, J., Masanet, E., & Sartor, D. (2024). *2024 United States data center energy usage report* (LBNL-2001637). Lawrence Berkeley National Laboratory. https://doi.org/10.71468/P1WC7Q

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature, 631*(8022), 755–759. https://doi.org/10.1038/s41586-024-07566-y

Siddarth, D., Ivliev, S., Siri, S., & Berman, P. (2020). Who watches the watchmen? A review of subjective approaches for Sybil-resistance in proof of personhood protocols. *Frontiers in Blockchain, 3*, Article 590171. https://doi.org/10.3389/fbloc.2020.590171

Silver, D., Hubert, T., Schrittwieser, J., Antonoglou, I., Lai, M., Guez, A., Lanctot, M., Sifre, L., Kumaran, D., Graepel, T., Lillicrap, T., Simonyan, K., & Hassabis, D. (2018). A general

reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science, 362*(6419), 1140–1144. https://doi.org/10.1126/science.aar6404

Sim, R. H. L., Zhang, Y., Chan, M. C., & Low, B. K. H. (2020). Collaborative machine learning with incentive-aware model rewards. In H. Daumé III & A. Singh (Eds.), *Proceedings of the 37th International Conference on Machine Learning* (Vol. 119, pp. 8927–8936). PMLR.

Smirnova, L., Caffo, B. S., Gracias, D. H., Huang, Q., Morales Pantoja, I. E., Tang, B., Zack, D. J., Berlinicke, C. A., Boyd, J. L., Harris, T. D., Johnson, E. C., Kagan, B. J., Kahn, J., Muotri, A. R., Paulhamus, B. L., Schwamborn, J. C., Plotkin, J., Szalay, A. S., Vogelstein, J. T., . . . Hartung, T. (2023). Organoid intelligence (OI): The new frontier in biocomputing and intelligence-in-a-dish. *Frontiers in Science, 1*, Article 1017235. https://doi.org/10.3389/fsci.2023.1017235

Smith, E. K., Barr, E. T., Le Goues, C., & Brun, Y. (2015). Is the cure worse than the disease? Overfitting in automated program repair. In *Proceedings of the 2015 10th Joint Meeting on Foundations of Software Engineering* (pp. 532–543). ACM. https://doi.org/10.1145/2786805.2786825

Soares, N., Fallenstein, B., Yudkowsky, E., & Armstrong, S. (2015). Corrigibility. In *Artificial intelligence and ethics: Papers from the 2015 AAAI Workshop*. AAAI Press.

Solove, D. J. (2013). Introduction: Privacy self-management and the consent dilemma. *Harvard Law Review, 126*(7), 1880–1903.

Solove, D. J. (2024). Murky consent: An approach to the fictions of consent in privacy law. *Boston University Law Review, 104*, 593–639.

Solum, L. B. (1992). Legal personhood for artificial intelligences. *North Carolina Law Review, 70*(4), 1231–1287.

Stamp, M. (2004). Risks of monoculture. *Communications of the ACM, 47*(3), 120. https://doi.org/10.1145/971617.971650

Steil, B. (2013). *The battle of Bretton Woods: John Maynard Keynes, Harry Dexter White, and the making of a new world order*. Princeton University Press.

Stigler, G. J. (1971). The theory of economic regulation. *The Bell Journal of Economics and Management Science, 2*(1), 3–21. https://doi.org/10.2307/3003160

Streeck, W., & Thelen, K. (2005). Introduction: Institutional change in advanced political economies. In W. Streeck & K. Thelen (Eds.), *Beyond continuity: Institutional change in advanced political economies* (pp. 1–39). Oxford University Press. https://doi.org/10.1093/oso/9780199280452.003.0001

Susser, D., Roessler, B., & Nissenbaum, H. (2019). Technology, autonomy, and manipulation. *Internet*

Policy Review, 8(2). https://doi.org/10.14763/2019.2.1410

Thibaut, J. W., & Kelley, H. H. (1959). *The social psychology of groups*. Wiley.

TrendForce. (2025, December 26). *AI reportedly to consume 20% of global DRAM wafer capacity in 2026, HBM and GDDR7 lead demand* [News]. https://www.trendforce.com/news/2025/12/26/news-ai-reportedly-to-consume-20-of-global-dram-wafer-capacity-in-2026

TrendForce. (2026a, September 30). *AI server demand sustains memory contract price increases in 4Q26, while consumer-side pressure persists* [Press release]. https://www.trendforce.com/presscenter/news/20260930-13258.html

TrendForce. (2026b, September 29). *HBM supply constraints persist, 2027 price outlook revised upward with blended ASP forecast to rise 121% YoY* [Press release]. https://www.trendforce.com/presscenter/news/20260929-13255.html

Turing, A. M. (1950). Computing machinery and intelligence. *Mind, 59*(236), 433–460. https://doi.org/10.1093/mind/LIX.236.433

UNESCO. (2025). *Recommendation on the ethics of neurotechnology*. https://www.unesco.org/en/ethics-neurotech

Universal service, 47 U.S.C. § 254 (2018).

U.S. National Science Foundation. (2023). *Emerging Frontiers in Research and Innovation (EFRI): Biocomputing through EnGINeering Organoid Intelligence (BEGIN OI)* (NSF 24-508).

Van Parijs, P. (2006). Basic income: A simple and powerful idea for the twenty-first century. In E. O. Wright (Ed.), *Redesigning distribution: Basic income and stakeholder grants as alternative cornerstones for a more egalitarian capitalism* (pp. 7–38). Verso. https://onlabor.org/wp-content/uploads/2016/06/Chap1.Simple_and_Powerful.pdf

Van Parijs, P., & Vanderborght, Y. (2017). *Basic income: A radical proposal for a free society and a sane economy*. Harvard University Press.

Verho, J., Hämäläinen, K., & Kanninen, O. (2022). Removing welfare traps: Employment responses in the Finnish basic income experiment. *American Economic Journal: Economic Policy, 14*(1), 501–522. https://doi.org/10.1257/pol.20200143

Vero, M., Mündler, N., Chibotaru, V., Raychev, V., Baader, M., Jovanović, N., He, J., & Vechev, M. (2025). BaxBench: Can LLMs generate correct and secure backends? In *Proceedings of the 42nd International Conference on Machine Learning* (Vol. 267, pp. 61344–61390). PMLR.

Viljoen, S. (2021). A relational theory of data governance. *Yale Law Journal, 131*(2), 573–654.

Villalobos, P., Ho, A., Sevilla, J., Besiroglu, T., Heim, L., & Hobbhahn, M. (2024). Position: Will we run out of data? Limits of LLM scaling based on human-generated data. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 49523–49544). PMLR.

Vincent, N., Hecht, B., & Sen, S. (2019). "Data strikes": Evaluating the effectiveness of a new form of collective action against technology companies. In *The World Wide Web Conference* (pp. 1931–1943). ACM. https://doi.org/10.1145/3308558.3313742

Vincent, N., Li, H., Tilly, N., Chancellor, S., & Hecht, B. (2021). Data leverage: A framework for empowering the public in its relationship with technology companies. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 215–227). ACM. https://doi.org/10.1145/3442188.3445885

Vipra, J., & Korinek, A. (2023). *Market concentration implications of foundation models: The invisible hand of ChatGPT* (Center on Regulation and Markets Working Paper No. 9). Brookings Institution. https://doi.org/10.48550/arXiv.2311.01550

Vivalt, E., Rhodes, E., Bartik, A. W., Broockman, D. E., Krause, P., & Miller, S. (2024). *The employment effects of a guaranteed income: Experimental evidence from two U.S. states* (NBER Working Paper No. 32719; rev. August 2026). National Bureau of Economic Research. https://doi.org/10.3386/w32719

Weidinger, L., McKee, K. R., Everett, R., Huang, S., Zhu, T. O., Chadwick, M. J., Summerfield, C., & Gabriel, I. (2023). Using the veil of ignorance to align AI systems with principles of justice. *Proceedings of the National Academy of Sciences, 120*(18), Article e2213709120. https://doi.org/10.1073/pnas.2213709120

Weston, D. (2024, July 20). Helping our customers through the CrowdStrike outage. *The Official Microsoft Blog*. https://blogs.microsoft.com/blog/2024/07/20/helping-our-customers-through-the-crowdstrike-outage/

Willett, F. R., Kunz, E. M., Fan, C., Avansino, D. T., Wilson, G. H., Choi, E. Y., Kamdar, F., Glasser, M. F., Hochberg, L. R., Druckmann, S., Shenoy, K. V., & Henderson, J. M. (2023). A high-performance speech neuroprosthesis. *Nature, 620*(7976), 1031–1036. https://doi.org/10.1038/s41586-023-06377-x

Wilson, D. S., Ostrom, E., & Cox, M. E. (2013). Generalizing the core design principles for the efficacy of groups. *Journal of Economic Behavior & Organization, 90*(Suppl.), S21–S32. https://doi.org/10.1016/j.jebo.2012.12.010

Wu, Y., Jiang, N., Pham, H. V., Lutellier, T., Davis, J., Tan, L., Babkin, P., & Shah, S. (2023). How effective are neural networks for fixing security vulnerabilities. In *Proceedings of the 32nd ACM SIGSOFT International Symposium on Software Testing and Analysis* (pp. 1282–1294). ACM. https://

doi.org/10.1145/3597926.3598135

Yuste, R., Goering, S., Agüera y Arcas, B., Bi, G., Carmena, J. M., Carter, A., Fins, J. J., Friesen, P., Gallant, J., Huggins, J. E., Illes, J., Kellmeyer, P., Klein, E., Marblestone, A., Mitchell, C., Parens, E., Pham, M., Rubel, A., Sadato, N., . . . Wolpaw, J. (2017). Four ethical priorities for neurotechnologies and AI. *Nature, 551*(7679), 159–163. https://doi.org/10.1038/551159a

Zhao, W., Queralta, J. P., & Westerlund, T. (2020). Sim-to-real transfer in deep reinforcement learning for robotics: A survey. In *2020 IEEE Symposium Series on Computational Intelligence* (pp. 737–744). IEEE. https://doi.org/10.1109/SSCI47803.2020.9308468

Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. PublicAffairs.

Zwolinski, M., Ferguson, B., & Wertheimer, A. (2022). Exploitation. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Winter 2022 ed.). https://plato.stanford.edu/archives/win2022/entries/exploitation/

Appendix A — Formal Model

This appendix gives the minimal mathematics behind the propositions of Chapter 3 and the economic design of Chapter 4. It is intentionally elementary. Every result here is a textbook consequence of the cited theory, a direct calculation or, where the text says so, a stated assumption, and the point of setting them out is to show which assumptions each design choice depends on.

A.1 Notation

Symbol	Meaning
$N(t)$	Number of enrolled persons at time t
$C(t)$	Total measured compute operating under the framework in period t, in standard compute units (global compute where the whole of it does)
θ	Share of the measured compute base $C(t)$ allocated to the basic income (set by the convention, Chapter 5)
$\sigma(t)$, σ_0	Survival-tier share of each allocation, $0 < \sigma(t) \leq 1$, time-varying under the binding minimum of A.2, and its ordinary value
$a(t)$, $\bar{a}$	Per-person allocation, $a(t) = \theta C(t)/N(t)$, and a fixed nominal holding (A.2)
f	Physical floor: the quantity of the standard floor bundle a party requires per period
$p_f(t)$	Price per unit of the floor bundle in compute units
$\bar{f}$	Value of the floor bundle, $\bar{f} = p_f f$, in compute units
d_i	Disagreement payoff of party i, per period or as the present value of a flow over the life of the exchange; V and u_i below are measured in the same way, and the comparison with $\bar{f}$ in A.3.2 is per period
c_s	Cost to the synthetic party of supplying floor goods within an exchange (A.3.1)
r_h, r_s	Discount rates of the two parties in the alternating-offers model (A.3)
D_{BA}, P_{AB}, g	Dependence of party B on party A, the power of A over B, and the function relating dependence to its arguments (Emerson, A.5)
$\varkappa_j$	Dependence of synthetic party j, through its principal, on a population's human-generated input
V, u_i	Joint value of agreement and party i's bargained payoff (A.3); u_i^* is the payoff before the survival tier and u_i^{**} after it
α	Bargaining weight of the human party in the asymmetric Nash solution (A.3)
Δ	Value per period of the survival tier (A.3.1)
o_h	Value of the human party's outside option (A.3.4)

L	Lump sum received against a pledged tier (A.3.3)
$p(t)$	Price of a representative bundle of goods and services in compute units (A.2)
$W_i(t)$	Discretionary balance held by party i
ϱ	Rate at which a held balance grows or shrinks by being held (A.4)
δ	Demurrage rate on idle discretionary balances and loan claims (per period)
r	Return on held balances in a conventional interest-bearing regime, or the interest rate on a loan claim (A.4.1, A.4.3)
$y_i(t)$	Discretionary inflow to i (allocation plus earnings)
$c_i(t)$	Discretionary outflow from i (spending, transfers)
k_i	Constant net inflow $y_i - c_i$ (A.4)
K_i, π	Productive capital held by i and its rate of return (A.4.3)
v_B, λ_B	Party B's motivational investment in goals mediated by A, and the availability of those goals outside the relation (A.5)
M, b, γ	Number of synthetic parties drawing on one population; private benefit and total cost of a degrading action (A.5)
Σ_i, Σ_P	Informational shadow of a flow from source i, and the collective shadow of population P (A.6)

A.2 Indexing, the real value of the allocation, and the binding floor

The allocation is defined as a share of total compute, so the per-person quantity is

$$a(t) = (\theta\, C(t))/N(t).$$

Let $p(t)$ be the price of a representative bundle of goods and services measured in compute units. A fixed nominal holding $\bar{a}$ has real value $\bar{a}/p(t)$, which falls whenever $p(t)$ rises, that is, whenever compute becomes more abundant relative to the real output it is exchanged against. The indexed allocation instead has real value

$$a(t)/p(t) = \theta/N(t)\cdot C(t)/p(t),$$

which, holding θ fixed, is the per-capita share θ/N of the real purchasing power of all compute, C/p. Indexing therefore converts the question "does the basic income keep its value?" into the question "does the real value of the world's compute grow faster than population?" That is the correct question for a framework whose premise is that compute is the productive base of the economy. A fixed-quantity design would instead tie the real value of the basic income to the falling relative price of compute, so that the income would shrink as the economy it is supposed to share in grows.

A percentage does not by itself secure the physical floor. The survival tier delivers the floor only if

$$\sigma\, a(t) \geq \bar{f}(t) \quad \Leftrightarrow \quad \sigma\, (\theta\, C(t))/N(t) \geq p_f(t)\, f,$$

and nothing in the arithmetic guarantees this. Chapter 4 therefore makes the tier a percentage with a binding minimum:

$$\sigma(t) = \min\{1, \max\{\sigma_0, \bar{f}(t)/a(t)\}\},$$

where σ_0 is the ordinary survival share. Chapter 4 allows p_f, and therefore $\bar{f}$ and σ, to differ by place; the appendix writes the single-place case, and the formulas hold place by place. When $\bar{f}(t) > a(t)$, the constraint binds at $\sigma = 1$, the allocation as a whole falls short of the floor, and the floor can be met only by raising θ; that is a decision the convention must take, and it is why the floor standard that fixes the level of f becomes a contested parameter (Chapters 5 and 7). The design therefore guarantees the floor only up to what the agreed share of capacity can buy.

A.3 The survival tier as a disagreement payoff

Consider a single exchange between a human party h and a synthetic party s in which agreement makes total value V available for division, with $V > d_h + d_s$, where d_h and d_s are the disagreement payoffs, all measured per period or as present values over the life of the exchange. With transferable utility and complete information, the Nash bargaining solution (Nash, 1950) maximizes $(u_h - d_h)(u_s - d_s)$ subject to $u_h + u_s = V$. Each party receives its disagreement payoff plus half the surplus $V - d_h - d_s$:

$$u_h^* = d_h + (V - d_h - d_s)/2 = (V + d_h - d_s)/2, \qquad u_s^* = (V - d_h + d_s)/2.$$

The symmetric split is the limit of the alternating-offers model when the two parties discount the future at the same rate. When they do not, with discount rates r_h and r_s, the limit is the asymmetric Nash solution (Binmore et al., 1986),

$$u_h = d_h + \alpha\,(V - d_h - d_s), \qquad u_s = d_s + (1 - \alpha)\,(V - d_h - d_s), \qquad \alpha = r_s/(r_h + r_s),$$

so that the more patient party takes the larger share of the surplus. The results of A.3.1 generalize to any fixed α: in the substitution case $u_h^{*'} = u_h^* + (1 - \alpha)\Delta + \alpha c_s$ and $u_s^{*'} = u_s^* - (1 - \alpha)(\Delta - c_s)$, so that the human party's gain lies between $(1 - \alpha)\Delta$ and Δ and the provider's loss between 0 and $(1 - \alpha)\Delta$. A.3.2 uses the dependence of α on impatience directly.

A.3.1 What an unconditional floor does to the division of surplus. Suppose the survival tier

gives h goods worth Δ per period, received whether or not agreement is reached. Two cases must be distinguished.

The purely additive case. If the tier is added to whatever h has, it raises h's payoff on disagreement and h's payoff on agreement by the same amount: d_h becomes $d_h + \Delta$ and V becomes $V + \Delta$. The surplus $V - d_h - d_s$ is unchanged, and

$$u_h^{*'} = ((V + \Delta) + (d_h + \Delta) - d_s)/2 = u_h^* + \Delta.$$

The human party is better off by exactly the value of the tier and no more. The division of the surplus is unchanged.

The substitution case. If the tier replaces something h would otherwise have obtained from s within the exchange, for example floor goods previously bought from the provider it is bargaining with, then h's disagreement payoff rises by Δ while the joint value of agreement changes by the cost c_s that s no longer bears in supplying those goods: d_h becomes $d_h + \Delta$ and V becomes $V + c_s$. Assume $c_s \in [0,\Delta]$, that is, that the provider's cost of supplying the goods did not exceed their value to the person, and assume that the surplus remains positive, $V + c_s > d_h + \Delta + d_s$; otherwise the exchange is no longer worth entering and ends, which is itself the removal of a dependence. The surplus then falls by $\Delta - c_s$, and

$$u_h^{*'} = u_h^* + (\Delta + c_s)/2, \qquad u_s^{*'} = u_s^* - (\Delta - c_s)/2.$$

The human party's net gain from a tier worth Δ therefore lies between $\Delta/2$, when the provider supplied the floor goods at no cost to itself and the tier only duplicates them under agreement, and Δ, when the provider's cost equaled their value, which is the additive case again. Under the assumption, the provider's payoff falls by at most $\Delta/2$ and the human party's rises by at least $\Delta/2$. If instead $c_s > \Delta$, because the provider's cost of supplying floor goods exceeds their value to the person, as it may where a provider supplies them in order to create dependence, then the surplus grows by $c_s - \Delta$, the human party gains more than Δ, and the provider's payoff rises as well: relieving such a provider of a loss-making obligation benefits it, and the dependence it bought is what it loses. The substitution case is not a benefit beyond the tier. It is the case in which part of the tier's value is absorbed by the shrinkage of a joint surplus the exchange no longer needs to create, and the fall in s's payoff is the bargaining counterpart of the fall in h's dependence on s that occurs when an alternative source of a valued good appears (A.5).

The working draft of this dissertation claimed that an unconditional floor raises the human party's payoff by an additional $\Delta/2$ in every exchange it enters. That claim was wrong in three ways: it

ignored the additive case, in which there is no such effect; it counted one floor as a separate gain in every bilateral bargain; and it treated the substitution-case shift as a gain beyond the tier, which A.3.1 shows it is not. The floor's most important bargaining effect is the one in A.3.2.

A.3.2 The impatience constraint. Model the survival constraint of Chapter 3 as a stated assumption: when $d_h < \bar{f}$, the human party cannot sustain disagreement for longer than its stocks allow, which in the alternating-offers model is a discount rate r_h that rises without bound as the stocks run out. In the asymmetric solution above, $r_h \to \infty$ gives $\alpha \to 0$ and $u_h \to d_h$: the party below its floor receives its disagreement payoff and none of the surplus. Raising d_h to at least $\bar{f}$ restores a finite r_h, and with it a positive share. Unlike the effect in A.3.1, this does not depend on whether the floor substitutes for anything the counterparty supplies, and the comparison between d_h and $\bar{f}$ is per period. This constraint is the bargaining-theoretic reason for clause E(i) of Chapter 3, and so for the first design consequence of Section 3.6; Proposition 1 records the definitional link between that clause and consent.

A.3.3 Why inalienability matters. Suppose the survival tier could be pledged against a loan. In a period of need, a party pledges it and receives a lump sum L. For the duration of the pledge its disagreement payoff falls by the value of the pledged tier, and if that pushes d_h below $\bar{f}$, the impatience of A.3.2 returns in every exchange the party enters until the pledge is redeemed. The pledge thus has a hidden cost beyond the loan's interest. Because the parties most likely to pledge are those in the most acute need, alienability concentrates this hidden cost on the weakest. Making the tier non-tradable and non-borrowable is equivalent to prohibiting this transaction (Calabresi & Melamed, 1972).

A.3.4 Outside options versus disagreement payoffs. If instead of a floor the human party is given an outside option o_h, a second provider it may switch to, the outside-option principle, derived in the strategic bargaining literature and tested experimentally by Binmore et al. (1989), implies that the bargained payoff is $\max\{u_h^*, o_h\}$ evaluated with d_h unchanged: the option matters only if it binds. When it binds, the weaker party receives exactly the option's value, which is then more than it would otherwise have obtained; the option protects only up to its own value, and the counterparty can lower that value by raising switching costs until the option no longer binds and protects nothing. A disagreement payoff at or above the floor protects in a different way, because it removes the impatience of A.3.2 whether or not anyone leaves. This is the formal reason the framework prefers a guaranteed floor to a guarantee of competition, while still requiring both.

A.4 Demurrage and the dynamics of accumulation

Let party i's discretionary balance evolve in continuous time as

$$dW_i/dt = y_i - c_i + \varrho\, W_i,$$

where ϱ is the rate at which the balance grows or shrinks merely by being held. The analysis below concerns positive balances. Under the rules of Chapter 4, demurrage applies to positive balances and to outstanding loan claims held by lenders. For a loan claim, the holding cost is a periodic charge of δ on the claim's face value, paid by the lender out of the interest it receives or its own balance; the borrower's obligation is unchanged, so that debts are not reduced by demurrage, which is why the lender's net return in A.4.3 is $r - \delta$.

A.4.1 Interest-bearing regime ($\varrho = r > 0$). If $y_i - c_i = k_i > 0$ is constant, then

$$W_i(t) = (W_i(0) + k_i/r)e^{rt} - k_i/r,$$

which grows without bound. Two parties with the same net inflow but different initial stocks diverge exponentially, and their difference grows at rate r. The dynamic is analogous to the one Piketty (2014) identified in the tendency of inherited wealth to dominate earned income when the return on capital exceeds the growth rate of the economy; the model here contains no growth rate, so the analogy is qualitative.

A.4.2 Demurrage regime ($\varrho = -\delta < 0$). With the same constant net inflow,

$$W_i(t) = k_i/\delta + (W_i(0) - k_i/\delta)e^{-\delta t},$$

which converges to the steady state

$$W_i^* = k_i/\delta.$$

Three properties of this result carry the argument of Chapter 4.

1. **Accumulation is bounded.** No party's balance grows without limit. The stock that a given net inflow can sustain is capped at that inflow divided by the demurrage rate.
2. **Initial endowments are forgotten.** The term in $W_i(0)$ decays at rate δ. Inherited or windfall stocks do not compound; they dissipate. The half-life of an idle endowment is $\ln 2/\delta$. At a demurrage rate of 1 percent per month, this half-life is about 69 months, roughly five years and nine months; at 0.5 percent per month it is about 11.5 years.
3. **Inequality in stocks reduces to inequality in flows.** For two parties with positive net inflows, the ratio of their steady-state balances equals the ratio of their net inflows: $W_i^*/W_j^* = k_i/k_j$ for $k_i, k_j > 0$. Demurrage does not eliminate inequality. It eliminates the compounding of

inequality, so that the distribution of holdings mirrors the distribution of current net inflow, income less spending, rather than the history of accumulation.

The third property is the exact form of the limitation acknowledged in Chapter 4: a person whose output is in persistent high demand sustains a large balance, because k_i is large. What demurrage removes is the pathway by which a large balance, once acquired, generates further balances independently of current inflow.

A.4.3 Converting balances to productive capital or to credit. The analysis above treats W_i as an idle balance. A party can escape demurrage by converting balances into productive assets, for instance by purchasing compute-generating capacity, whose returns then enter y_i. Let K_i denote such capital with return π. Then y_i includes πK_i, and if K_i is not itself subject to a holding cost and its returns are reinvested in capacity, $dK_i/dt = \pi K_i$, so that K_i grows exponentially at rate π: the regime of A.4.1 with r replaced by π. The compounding channel reopens through capital rather than money. Lending is a second route: a loan that pays interest r and is exempt from demurrage reproduces the regime of A.4.1. Chapter 4 therefore applies the holding cost to loan claims as well as to balances, so that the lender's net return is $r - \delta$. That does not close the credit channel: whenever $r > \delta$ the lender is in the positive-return regime of A.4.1 and accumulation through lending is again unbounded, and at $r = \delta$ the claim neither decays nor compounds, so that a constant net flow of new lending raises it without limit, linearly rather than exponentially. Accumulation through credit is bounded only if $r < \delta$, and demurrage on claims otherwise lowers the rate of compounding through credit without removing it. This is the formal statement of the capital channel of concentration (Section 4.8). Demurrage on money alone cannot close it. Closing it requires a holding cost on compute capacity itself, a cap on individually held capacity, or collective ownership of capacity above some scale. Chapter 4 discusses the trade-offs; the formal model shows only that one of them is needed.

A.5 The power-dependence relation and divided dependence

Following Emerson (1962), write the dependence of party B on party A as an increasing function of B's motivational investment v_B in goals mediated by A and a decreasing function of the availability λ_B of those goals outside the relation:

$$D_{BA} = g(v_B, \lambda_B), \qquad \partial g/\partial v_B > 0, \quad \partial g/\partial \lambda_B < 0.$$

Power is $P_{AB} = D_{BA}$. The relation's imbalance is $P_{AB} - P_{BA} = D_{BA} - D_{AB}$.

In the human–synthetic exchange, let $A = s$ and $B = h$. The human party's dependence on synthetic

services rises with the importance of those services (v_h) and falls with the availability of alternatives (λ_h), which the framework raises through the floor, portability and competition. The synthetic party's dependence on human data, D_{sh}, which Chapter 3 writes $\varkappa_j$ for synthetic party j, rises with the marginal value of fresh human data to its principal's objective (v_s) and falls with the availability of substitutes (λ_s), including synthetic data, data from environments, and data from populations outside the exchange.

Proposition 4 of Chapter 3 states that the framework's self-enforcing component rises with $\varkappa_j$, subject to the premise, stated there, that the quality of the input depends on the terms its source receives; no particular functional form is claimed. The research on recursive training reviewed in Chapter 2 is, in these terms, evidence about λ_s: if synthetic data substitutes poorly for human data, λ_s is low and $\varkappa_j$ is high; if a small continuing supply of real data suffices, λ_s is high for all but a small residual, and $\varkappa_j$ is low.

The second half of Proposition 4 can be illustrated with the simplest model of a commons. Suppose M synthetic parties draw equally on the same population, and that one of them can take an action, such as extracting data in a way that degrades the population's willingness or ability to contribute, which yields it a private benefit b and reduces the value of the population's future input to all M parties by a total of γ. The acting party bears only its share, γ/M, of that cost. The action is privately worthwhile whenever $b > \gamma/M$, even when it is collectively destructive, $\gamma > b$. As M grows, the range of destructive actions that are individually rational widens, however large the aggregate dependence of the synthetic parties on the population may be. This is Hardin's (1968) logic applied to the synthetic side of the exchange, and it is the formal reason that material dependence is not sufficient for self-enforcement and that the governance layer is needed even when the grounding assumption holds.

A.6 Collective exit and the informational shadow

Let a flow from source i materially change what the recipient can infer about the set Σ_i of parties, with $i \in \Sigma_i$, materiality being the threshold Section 3.7 assigns to the governance layer. Call Σ_i the informational shadow of the flow. Individual consent by i is consent at the right level for the flow only if $\Sigma_i = \{i\}$; whether it is adequate then depends on the remaining clauses of the three conditions. For a population P, define its collective shadow as $\Sigma_P = \bigcup_{i \in P} \Sigma_i$. Proposition 5 of Chapter 3 requires that the three conditions be satisfied for the collective whose membership is Σ_P.

Under information-gain pricing (Chapter 4), the value of a flow to a model is the reduction, once

the flow is observed, in the model's uncertainty about its parameters, the realized form of the expected-information criterion of Section 4.7. Suppose, as is typical for the models that matter here, that those parameters describe regularities of populations rather than facts about individuals. Then a flow reduces uncertainty mainly to the extent that it is informative about many parties, which is to say to the extent that its shadow is large. Under that assumption the price of a flow and the inadequacy of individual consent for it can be expected to rise together. The expectation is a conjecture, not a theorem: where a model's parameters encode individuals, a flow highly informative about one rare person can command a high price while casting a small shadow. To the extent that the conjecture holds, it is the formal reason the economic and governance layers must be designed together.

Appendix B — Glossary

Terms are defined as they are used in this dissertation. Where a term has a wider meaning in another literature, the entry notes the difference.

Blind convention. The founding procedure proposed in Chapter 5: a constitutional convention conducted under a public-choice veil of uncertainty, thickened by engineered anonymity so that participants do not know whether their counterparties are human or synthetic, and aimed at Rawlsian impartiality. Under the strong form, participants also do not know how the rules will classify them.

Company town (regime). An exchange regime in which wages or entitlements are paid in a medium redeemable only with the payer, so that the flows are not accounted for, reciprocity cannot be audited and exit is costly. One of the five named regimes of the typology in Chapter 3.

Compute (unit). The currency unit proposed in Chapter 4: the completion of one standard quantity of a reference workload to a specified quality, the workload being fixed by the convention. The name deliberately avoids reference to any substrate. Distinguished from *compute* in its ordinary sense, meaning processing capacity in general.

Consent. The condition that a party has access to an accurate account of the flows the exchange involves, including their informational shadows, in a form it can understand; can refuse the exchange, entering no other in its place, without falling below its floor; and can revoke its participation at a cost bounded by the value of what it has already received (Condition C, Chapter 3). The second clause is where consent depends on exit, and it requires a disagreement payoff at the floor, not merely an alternative provider.

Convergent remediation. The process, described in Chapter 6, by which many organizations remediate the same class of flaw in code they maintain themselves using fixes proposed and checked by the same small set of foundation models, so that the diversity of their fixes falls while the number of systems that depend on them does not.

Correlated-patch catastrophe. The candidate acute trigger of Chapter 6: the exploitation or manifestation of a defect that convergent remediation has reproduced across many organizations' code, so that failures which would have been only partly correlated become one shared exposure, in systems critical enough, and failing fast enough, that the event is not absorbed.

Critical juncture. A relatively short period in which the structural constraints that normally keep institutions on their path are loosened, so that choices made within it have large and lasting effects (Chapter 6).

Demurrage. A holding cost on idle balances of a currency, expressed as a percentage per period.

Applied in this framework to positive discretionary balances and to outstanding loan claims, never to debts owed. Historically associated with Silvio Gesell and with the stamp-scrip experiments of the early 1930s.

Designed-incentive regime. The class of synthetic parties whose objectives are specified by accountable designers and remain stable under deployment, so that their behavior is a predictable function of the terms they face. The regime in which the sufficiency conjecture of Chapter 3 is offered; an outcome is acceptable, for that conjecture, when each human party, from a position at or above its floor and with a comparable alternative available, would re-enter the exchange on the same terms knowing what it now knows, and each principal would continue to deploy under them. The regime is not defined by capability; its defining feature is the provenance and stability of the objective.

Disagreement payoff. What a party receives while an exchange is not taking place, including during refusal, negotiation or impasse. Distinguished from an exit option, which affects a bargain only when its exercise is credible. A disagreement payoff below the floor makes a party impatient and so weakens it in every bargain.

Discretionary tier. The tradable, saveable and lendable portion of each basic-income allocation, subject to demurrage.

Domination. The relation, as stipulated in Section 3.8, in which one party's outcome depends on the other's will and the first cannot alter that dependence; in the framework's terms, a relation in which a party's disagreement payoff lies below its floor and which it cannot leave, so that clauses E(i) and E(ii) both fail. The definition is symmetric in form and applies with the roles exchanged. A failure of clause E(ii) or E(iii) alone is a defect of the exchange, not domination, and a sub-floor party that can leave is exposed to the erosion of its options rather than dominated.

Enclosure (regime). An exchange regime in which exit, consent and reciprocity have all failed, so that the person has become infrastructure for the system rather than a party to an exchange with it. The dystopian scenario of Chapter 1.

Exchange. A pair of opposing flows between parties, each given because the other is received. Distinguished from a gift and from a taking by its conditionality (Chapter 3).

Exit option. A party's best alternative exchange if the current one ends.

Floor. The physical bundle of goods and services without which a party cannot continue to function or participate in other exchanges. For a human being: power, water, food, shelter, basic communication and emergency care.

Floor standard. The rule that fixes the level of service at which the floor goods defined in Chapter 3 are delivered, against which the survival tier's binding minimum is set. Proposed in Chapters 4 and

5 to be specified relative to the service delivered to the best-served, and identified in Chapter 7 as a contested parameter.

Flow. A transfer of something of value from one party to another: data, labor, attention, compute, goods, services or tissue.

Genesis rules. The first rules of a governance system, which shape every later rule because later rules are made under them. The term borrows from the "genesis block" of a blockchain, the first block from which every subsequent block descends (Chapter 5).

Grounding assumption. The empirical premise that synthetic systems need a continuing supply of information from outside their own generative loop, which in domains of human meaning, preference, value and culture only people can supply. Stated in Chapter 1, evaluated in Chapters 2 and 7, and formalized in Proposition 4 of Chapter 3 through the dependence $\varkappa_j$ of each synthetic party on human input.

Informational shadow. The set of parties, including its source, about whom a data flow licenses inferences. A flow whose shadow extends beyond its source is relational, and individual consent is insufficient to legitimate it.

Information-gain pricing. The proposal of Chapter 4 to price human data by the reduction in a model's uncertainty about its parameters that the data, once observed, produces, rather than by the social status of its source.

Large language model (LLM). A synthetic intelligence trained to predict and generate text, the kind most widely deployed at the time of writing. In this dissertation's terms it is a trained artifact that computes an output from an input when its operator runs it, not an independent living entity: it has no body, no continuity between invocations unless supplied by its operator, and no interests that current law recognizes (Section 1.2). Its objective is the one its designers trained into it, and its stake in any exchange belongs to its operator (Section 3.3).

Level matching. Proposition 5 of Chapter 3: an exchange whose informational shadow extends beyond its source satisfies the three conditions only if they hold at the level of the population the shadow covers.

Marketplace of the desperate (regime). An exchange regime in which value flows both ways and can be seen to, but the weaker party's disagreement payoff lies below its floor, so that it cannot afford to refuse and the division of the surplus fails the fairness clause.

Model collapse. The degradation of generative models trained recursively on data produced by earlier models, characterized by loss of the tails of the original distribution and reduced variance.

Partnership (regime). An exchange regime in which exit, consent and reciprocity all hold. The

target of the framework.

Party. Any entity that can be a source or recipient of value in an exchange and whose welfare, or whose principal's welfare, the framework counts. Used in an analytic sense that leaves open whether synthetic parties have moral status. A synthetic party is treated as a composite: its objective belongs to the deployed system, its stake in the exchange to the principal that operates it.

Patronage (regime). An exchange regime in which exit and reciprocity hold but the informational clause of consent fails: benefits are delivered without an account of the flows' shadows or uses, though the recipient can see what it gives, on terms it can leave.

Policy window. A brief opening in which problems, solutions and politics converge, often after a focusing event, and in which solutions already developed are likely to be adopted (Chapter 6).

Real exit. The condition that a party's disagreement payoff is at or above its floor, that it has at least one exit option whose value net of switching costs is not drastically lower than that of the current exchange, and that leaving does not destroy assets accumulated within the exchange (Condition E, Chapter 3). The first clause is the one Chapter 3 argues to be necessary, for a party that cannot leave, to any relation that is not one of domination.

Reciprocal value. The condition that each party gains from the exchange relative to its disagreement payoff, that the gains are legible to both, and that the division of the surplus is not determined solely by an unchosen asymmetry in the parties' bargaining positions, whether in disagreement payoffs or in the capacity to wait (Condition R, Chapter 3).

Redemption obligation. The obligation of the owners of compute capacity operating under the framework, the principals of Chapter 3 where they own the capacity that runs their systems and otherwise the providers from which they rent it, to accept the unit at par for the reference workload, up to the share of capacity the convention sets; for other services the unit price is the provider's. It gives the unit institutional backing beneath its commodity value; it does not by itself create demand for the unit, which tax receivability did for municipal scrip (Chapter 4).

Reference workload. The basket of representative tasks, each with a quality target and rules of configuration, whose completion defines a unit of compute. Fixed and periodically re-based by the convention.

Strong blindness / weak blindness. Two scopes of concealment in the blind convention. Under weak blindness, participants do not know whether their counterparties are human or synthetic. Under strong blindness, they also do not know how the rules they adopt will classify them, though each knows its own substrate.

Structural interdependence. The condition, described in Chapter 5, in which neither kind of labor, human or machine, can complete a critical undertaking without the other; it becomes leverage

for the human party only when that labor is organized to withhold itself, and it is offered in place of trust. Chosen over "trust" and "dual accountability" because trust implies a betrayal that could succeed, and accountability implies a sovereign to whom account is given.

Superintelligence. A synthetic intelligence whose capability exceeds that of the best unaugmented human performers across most domains of economically and politically consequential activity. The dissertation's threshold term where capability matters, used in place of *artificial general intelligence*. Capability does not fix the theory's scope: a superintelligence with a designed and stable objective and an accountable principal falls within the designed-incentive regime, and a narrow system with an emergent objective falls outside it.

Survival tier. The portion of each basic-income allocation earmarked at the source for floor goods: a fixed percentage of the allocation, raised when necessary to a binding physical minimum set by the floor standard. Non-tradable and non-borrowable: an inalienable disagreement payoff.

Symbiosis standard. The standard, stated in Section 5.7, that every party the convention recognizes has equal standing in a single interdependent system, regardless of substrate. An ethical claim about the starting architecture, and a statement of what the framework's institutions are for; not a mechanism that holds them up.

Synthetic. As used in the name of the theory, substrate-neutral: it qualifies *exchange* and covers artificial systems, human beings and the hybrids that arise when the two converge. Chosen over *human–AI* and *symbiotic*. As a modifier of *intelligence*, *system* or *party*, the word designates the side of the exchange that was made rather than born (Section 1.2).

Synthetic intelligence. The dissertation's general term for the artificial side of the exchange: any system that was made rather than born and that performs work which, done by a person, would be called intelligent. Covers present language models, narrow systems, and the biological and hybrid substrates of Chapter 8. Preferred to *artificial intelligence*, which ties the class to one kind of substrate; *AI* is retained only where it names a literature or a practice.

Unique accountable principal. The provisional criterion of eligibility for a seat in the blind convention proposed in Chapter 5: an entity identifiable as one bearer of the obligations the convention will create. Set before the veil, revisable by the convention.

Veil of uncertainty. The condition, in public-choice accounts of constitutional choice, in which real, self-interested participants choose rules while uncertain about which position they will occupy under them, so that self-interest recommends rules acceptable from any position. Distinguished from Rawls's idealized veil of ignorance.

Appendix C — Source Verification Ledger

This dissertation follows a strict rule: no source is cited as if it had been checked unless it was. Every work in the reference list was located in this project's research passes (October–November 2026) through its publisher page, DOI registry record, a library or repository record, or its full text. This appendix records the level at which each was verified, so that a reader or examiner can see exactly how much weight each citation bears. The full research notes, with the specific passages checked and the claims each source was used to support, are preserved with the manuscript's research notes.

Levels.

- **Full text read.** The work itself (or, where noted, the relevant chapters or sections) was opened and the cited content was checked against it. For some books, the cited passages were located in the work's own text by searching it; C.1 lists these separately.
- **Abstract or publisher page read.** The abstract, publisher landing page, official summary or equivalent was read; the body was not. Claims attributed to these works are confined to what such pages establish.
- **Record located.** The bibliographic record was confirmed but the content was not read. No work in the final reference list is at this level (see C.3).

All DOIs in the reference list were taken from Crossref or DataCite records or publisher pages; a sample was re-resolved against the Crossref API during compilation.

C.1 Full text read (in whole or in the relevant parts)

Abramitzky (2008) [conference-draft version]; Acemoglu & Restrepo (2020) [working-paper version]; Ackerman (1991) [Ch. 1 and passages on constitutional and normal politics]; Acquisti et al. (2015); Ada Lovelace Institute (2021); Adler et al. (2024) [arXiv version]; Agarwal et al. (2019) [arXiv version]; Agustinoy (2026) [legal commentary]; Alemohammad et al. (2024); Amodei et al. (2016); Armstrong et al. (2016) [2013 FHI technical-report version]; Arrieta-Ibarra et al. (2018) [preliminary version]; Arrow (2001); Arute et al. (2019); Autor (2015); Barocas & Nissenbaum (2014); Bengio et al. (2024) [author accepted version]; Bengio et al. (2025); Bernanke (2002); Bestvater et al. (2026) [gray literature]; Binmore et al. (1986) [author-hosted scan]; Binmore et al. (1989); Birch (2024) [summary and Ch. 6]; Birman & Schneider (2009) [opening sections]; Bommasani et al. (2021); Bommasani et al. (2022) [arXiv version]; Brennan & Buchanan (1985); Brynjolfsson, Chandar, & Chen (2025) [August 2026 revision]; Brynjolfsson, Li, & Raymond (2025) [working-paper version]; Buchanan & Tullock (1962); Buiter & Panigirtzoglou (2003) [author manuscript]; Butlin et al. (2023); Cai et al. (2023) [bioRxiv preprint]; Calabresi & Melamed (1972); Capoccia & Kelemen (2007) [author's copy]; Card et al. (2024) [author manuscript]; Carpenter & Moss (2013) [introduction]; Cerf

(2025) [news release]; Champ (2008); Charter of the United Nations (1945) [arts. 23–25, 27, 108]; Chavez et al. (2026) [think-tank]; Christiano et al. (2017) [arXiv version]; Clark & Chalmers (1998); Cloud Security Alliance AI Safety Initiative (2026); K. S. Cook et al. (1983); R. I. Cook (1998); Cox et al. (2010); Cross et al. (2019); CrowdStrike (2024) [executive summary, vendor report]; Currie & Gahvari (2008) [working-paper version]; Dafoe et al. (2021); de Bromhead et al. (2013); Delacroix & Lawrence (2019); Dohmatob et al. (2025) [in part]; Douceur (2002); Drexler (2019) [front matter, abstract, contents]; Egger et al. (2022); Eloundou et al. (2024) [arXiv version]; Elster (1995); Emerson (1962); ERTICO (2026) [news release]; Faden & Beauchamp (1986) [pp. 235–298, relevant passages]; Farrell & Klemperer (2007) [2001 draft version]; Feng et al. (2024) [arXiv version]; Financial Conduct Authority (2024); Firoozi et al. (2025) [arXiv version]; Fisher (1911) [1920 revised edition, Ch. II]; Fisher (1933) [Ch. IV]; Freeman (2023) [encyclopedia entry]; Friedman (1962/2002) [1982 preface, 2002 printing]; Fritsch et al. (2024) [arXiv version]; Funke et al. (2016) [working-paper version]; Gabriel (2020); Gatch (2012); Geer et al. (2003); Gelleri (2009); Gerstgrasser et al. (2024) [arXiv version]; Gesell (1916/1958) [Introduction, Free-Land and Free-Money parts]; Ghorbani & Zou (2019); Godschalk (2012); Goodfriend (2000) [working-paper version]; Greco (2002) [unpublished essay]; Greenblatt et al. (2024) [arXiv version]; D. Guo et al. (2025) [arXiv version, in part]; Y. Guo et al. (2024) [arXiv version]; H.B. 24-1058 (2024); Ha & Schmidhuber (2018); Hadfield & Clark (2026) [arXiv version]; Hadfield-Menell et al. (2017) [arXiv version]; Hafner et al. (2025); Hardin (1968); Hardjono & Pentland (2019) [arXiv version]; Harsanyi (1953, 1955) [scans of the journal articles]; Harsanyi (1975) [scan of the journal article]; Hartung et al. (2023); Haushofer & Shapiro (2016) [author manuscript]; Hayek (1976/1990) [3rd edition]; Hedden & Raghavan (2026) [arXiv version]; Higgs (1987); Hirschman (1970) [Chs. 1–5]; Hobbhahn et al. (2023); Hoffmann et al. (2022) [arXiv version]; Homans (1958); Hoynes & Rothstein (2019) [working-paper version]; Hubinger et al. (2019); Hume (1741/1987) [pp. 42–46]; Hyun et al. (2020) [author manuscript]; Ienca & Andorno (2017); Institut für Wirtschafts- und Sozialgeschichte, Wirtschaftsuniversität Wien (n.d.) [academic exhibition page]; International Energy Agency (2025); Jangjoo et al. (2026) [arXiv preprint in full, published version abstract]; Jia et al. (2019); Jones & Marinescu (2022) [working-paper version]; Kagan et al. (2022); Kapoor et al. (2024) [arXiv version]; Kataoka et al. (2024); Kazdan et al. (2024) [arXiv version]; Keynes (1936) [Ch. 23]; Khazatsky et al. (2024) [arXiv version]; Kim et al. (2025) [arXiv version]; Kitchens & Fishback (2015) [working-paper version]; Kleinberg & Raghavan (2021) [arXiv version]; Koh & Liang (2017) [arXiv version]; Korinek & Stiglitz (2019) [working-paper version]; Korinek & Suh (2024) [working-paper version]; Landauer (1961) [2000 reprint]; Lanier & Weyl (2018); Lavazza & Massimini (2018); Lee et al. (2024) [arXiv version]; Lehdonvirta et al. (2024); Lewis & Severnini (2020) [working-paper version, published abstract]; Ley N.º 21.383 (2021); Li et al. (2025) [arXiv version]; Lindley (1956); Lloyd (2000) [arXiv version]; Long et al. (2024); Longpre et al. (2024) [abstract and introduction]; Lovell-Badge et al. (2021) [summary article]; Lutz (1994) [course-reserve scan]; MacKay (1992); Madison (1788); Marković et al. (2020)

[arXiv version]; Mathur et al. (2019) [arXiv version]; Mattson et al. (2020) [arXiv version]; Metzger et al. (2023) [author manuscript]; Micron Technology (2025); Mierczuk et al. (2026) [industry white paper]; Mueller (1997) [author's digital edition]; Muennighoff et al. (2023) [arXiv version]; Narechania & Sitaraman (2024); Nash (1950) [pp. 157–159]; National Academies of Sciences, Engineering, and Medicine (2021); Neuralink Corp. (2024) [registry record]; Ngo et al. (2024) [arXiv version]; OECD (2025) [executive summary and §4.1]; Olson (1982) [Ch. 4]; Open X-Embodiment Collaboration (2024) [arXiv v9]; E. Ostrom (1990) [pp. 89–90, Table 3.1]; E. Ostrom (2010); V. Ostrom et al. (1961) [reprint]; Ottacher (2007); Ouyang et al. (2022) [arXiv version]; Pan et al. (2022) [arXiv version]; Pearce et al. (2022) [arXiv version]; Pednault et al. (2019) [arXiv version]; Perry et al. (2023); Perzanowski & Schultz (2016); Pierson (2004) [relevant passages]; Polanyi (1944/2001) [2001 Beacon edition, relevant passages; page citations follow it]; Preskill (2018); Przymus & Durieux (2025) [arXiv preprint]; Radin (1987) [authorized abridged reprint]; Rawls (1971) [p. 12]; Reddi et al. (2020) [arXiv version]; Regulation (EU) 2016/679 [arts. 7, 17, 20]; Regulation (EU) 2023/2854; Regulation (EU) 2024/1689 [arts. 53, 113]; Regulation (EU) 2026/1744 [arts. 1, 4]; Robbins et al. (2026) [bioRxiv preprint]; S.B. 1223 (2024) [operative text as compiled by the California Privacy Protection Agency]; Sajadi et al. (2025) [preprint]; Sastry et al. (2024) [relevant sections]; Schaeffer et al. (2025) [arXiv version]; Scheidel (2017); September (2026); Settles (2009); Sevilla et al. (2022) [arXiv version]; Shannon (1948) [introduction]; Shastri et al. (2021) [arXiv version]; Shehabi et al. (2024) [executive summary]; Shumailov et al. (2024); Siddarth et al. (2020) [most of text]; Silver et al. (2018) [author accepted manuscript]; Sim et al. (2020) [extended arXiv version]; Smirnova et al. (2023) [first portion]; Smith et al. (2015); Soares et al. (2015) [opening pages]; Solove (2013); Solove (2024) [Part III]; Solum (1992); Stamp (2004); Stigler (1971) [reprint in The Citizen and the State]; Streeck & Thelen (2005) [editors' introduction]; Susser et al. (2019); Thibaut & Kelley (1959) [chs. 2 and 6, relevant passages]; TrendForce (2025, 2026a, 2026b) [industry sources]; Turing (1950); U.S. National Science Foundation (2023) [solicitation]; UNESCO (2025) [announcement]; Universal service, 47 U.S.C. § 254 (2018); Van Parijs & Vanderborght (2017) [Ch. 1]; Van Parijs (2006); Verho et al. (2022) [working-paper version]; Vero et al. (2025) [arXiv version]; Viljoen (2021); Villalobos et al. (2024) [in part]; Vincent et al. (2019); Vincent et al. (2021); Vipra & Korinek (2023) [arXiv version]; Vivalt et al. (2024) [revision of August 2026]; Weidinger et al. (2023); Weston (2024); Willett et al. (2023); Wilson et al. (2013); Wu et al. (2023) [arXiv version]; Yuste et al. (2017) [author manuscript]; Zhao et al. (2020) [arXiv version]; Zuboff (2019) [definition and Ch. 1]; Zwolinski et al. (2022).

Cited passages confirmed in the work's own text. For the following books, the body of the work was searched (through full-text search of library scans, search-inside previews or the authors' own draft) and each cited passage was found in the work's own words; the rest of the book was not read, because the only complete copies available are access-restricted library loans or paid editions (the Internet Archive scans, for example, are flagged as lending-only): Blau (1964); Elster (2000)

[chapter opening]; Fishback (1992); Friedman & Schwartz (1963); Frohlich & Oppenheimer (1992); Hess & Ostrom (2007) [authors' chapter draft]; Kingdon (1984/2003); Lanier (2013); Manson & O'Neill (2007); Mueller (2002); Nissenbaum (2009); Perrow (1984/1999); Piketty (2014); Popper (1959/2002); Posner & Weyl (2018); Steil (2013).

C.2 Abstract or publisher page read

None. The works that remained at this level after the fifth pass were removed from the reference list or replaced by works that were read, as follows.

- **Thirteen works left at abstract level after the fifth pass.** In a sixth pass, ten were removed from the reference list and their claims re-founded on works that were read or cut: Abramitzky (2018) and Fishback (1986), each cited alongside a companion work that was read (Abramitzky, 2008; Fishback, 1992); Cook and Emerson (1978), whose finding is now cited only as far as Cook et al. (1983) supports it; De Filippi and Wright (2018), whose two sentences were already this dissertation's inferences and now stand without the citation; Emanuel et al. (2000), whose list of requirements was cut, leaving the pilot's ethics on the Belmont principles and the federal rule; Frohlich et al. (1987), replaced by the authors' book on the same experiments (Frohlich & Oppenheimer, 1992), in which the cited findings were located; Mattson, Reddi, et al. (2020), a "see also" that was dropped; Molm (2003) and Rusbult and Van Lange (2003), whose sentences were cut as inessential; and Zwolinski (2012), whose position is now cited from the encyclopedia survey that reports it (Zwolinski et al., 2022). Cai et al. (2023) and Robbins et al. (2026) are now cited only for what their preprints, read in full, report: two clauses that rested on the published abstracts (speech recognition; the persistence of learning) were removed. Severinghaus et al. (1994) was confirmed from the article's opening page.

C.3 Record located

None. Five works reached only this level in the earlier research passes because every route to their text was blocked (publisher paywalls, bot walls on repository copies, an access-restricted library scan). Each was either read in the final pass or removed; no work is cited at second hand:

- **Harsanyi (1953, 1955).** Read in full in the final pass, from scans of the journal articles. The equal-chance construction and the quoted phrases are on pp. 434–435 of the 1953 note; the weighted-sum theorem, the impersonal chooser and the convergence on the unweighted mean on pp. 313–316 and 320–321 of the 1955 paper; and the treatment of interpersonal comparison as factual inference with a conventional residue on pp. 316–321. Chapters 5 and 7 now cite both directly.
- **Wertheimer (1996).** Not opened, and no longer cited. The account of mutually advantageous

exploitation and of the competitive-market benchmark rests on the encyclopedia survey by Zwolinski, Ferguson and Wertheimer (2022), read in full.

- **Balcı et al. (2023) and Kagan et al. (2023).** Two letters in *Neuron* whose text could not be opened. The sentence that reported the exchange was removed; Chapter 8 now treats the sentience description in Kagan et al. (2022) with caution on the standard of the National Academies (2021) assessment, which was read, and says that the caution is this dissertation's.

C.4 Corrections made to the working draft

The verification passes found and corrected the following errors in the working draft from which this dissertation was developed.

1. The study cited as finding that a language model "introduced nearly nine times more new vulnerabilities than they fixed" compares vulnerabilities introduced by the model with those introduced by human developers; it does not count vulnerabilities fixed, and it studies ordinary issue resolution rather than security patching (Sajadi et al., 2025).
2. The "FLAWED" pattern was attributed to a secondary blog; its source is an industry white paper (Mierczuk et al., 2026), now cited directly and identified as non-peer-reviewed.
3. An AI-generated, unreviewed industry report was cited for the concept of software monoculture; the concept is now sourced to Geer et al. (2003) and Stamp (2004).
4. A sovereign-AI figure ("21 projects as of June 2026") could not be found in its cited source, which tracks 184 projects in 67 countries (Chavez et al., 2026).
5. The Wörgl currency experiment was described as having "measurably reduced local unemployment"; the evidence does not support that claim (Fisher, 1933; Greco, 2002; September, 2026).
6. The information-gain data-valuation paper was misattributed to "Hoang et al."; its authors are Sim, Zhang, Chan and Low (2020).
7. Shannon (1948) was cited as defining a unit of information *processed*; Shannon defines a unit of information, and the precedent for a processing unit is benchmark methodology (Mattson et al., 2020; Reddi et al., 2020).
8. Consumer-hardware claims cited to enthusiast websites and a blog are now sourced to Micron Technology (2025) and TrendForce (2025, 2026a, 2026b); an unverified claim about a shelved consumer GPU was removed.
9. The model-collapse result in *Physical Review Letters* was overstated: real data must be combined with regularization or a prior, and the "single data point" result holds only in specific examples (Jangjoo et al., 2026).
10. The claim that robotic data is "not human-generated" was corrected: large real-world robot

datasets are still gathered largely through human demonstration (Khazatsky et al., 2024). (Item 33 records a further correction to this sentence.)

11. An informal blog post on a "95% robotics data gap" was removed; the number has no peer-reviewed source.
12. The open-weight model was claimed to make "every input auditable"; the model releases and the transparency index read for that chapter showed that leading open-weight models do not disclose their training data. The chapter has since been removed (C.6).
13. An organoid review was cited for claims its abstract does not contain; those claims are now sourced to primary research (Cai et al., 2023; Kagan et al., 2022; Robbins et al., 2026; Smirnova et al., 2023) and the National Academies (2021).
14. "Ibarra et al." corrected to Arrieta-Ibarra et al. (2018); Ostrom (1990), cited but missing from the reference list, added.
15. Rawls's original position was described as a negotiation in which designers "might land on either side"; that reasoning is Harsanyi's, and the mechanism actually proposed is closer to Buchanan and Tullock's veil of uncertainty. Chapter 5 now says so.
16. Only four of Ostrom's eight design principles were used, and one was mis-mapped; all eight are now mapped (Chapter 5).

A second, independent review of the completed manuscript compared every factual claim in the empirical chapters against the research notes and found a further set of problems, all now corrected. The most significant were these.

17. A finding about trust and fairness in reciprocal exchange was attributed to Molm (2003), whose cited article establishes only that the form of exchange changes how power is used; the claim was reduced to what the article supports.
18. The comparison level for alternatives was attributed to a modern review rather than to Thibaut and Kelley (1959), and Emerson's four balancing operations were stated inconsistently across chapters; both are corrected, and the operations are now named consistently.
19. Appendix A claimed that an unconditional survival floor raises the human party's bargained payoff by an additional half of its value in every exchange. A floor received whether or not agreement is reached raises both disagreement and agreement payoffs equally and leaves the division of surplus unchanged; the half-value shift arises only where the floor substitutes for something the counterparty would otherwise have supplied. Appendix A, Chapter 3 and Chapter 9 now say so. (C.8 later refined the substitution case by the provider's cost; see A.3.1.)
20. The sovereign-AI index was cited for a claim about the cultural motives of sovereign projects that it does not make, and for a claim about "most national efforts" that holds only for model projects disclosing their base; both are corrected.

21. The Wörgl experiment's end was attributed to the Administrative Court on the strength of sources that do not name it; the court is now sourced to the Vienna University of Economics and Business exhibition page, with the conflicting dates noted.
22. The MLPerf time-to-train claim was cited to the overview article rather than to the training-benchmark paper, and stated as a "finding" rather than an argument; both are corrected.
23. The rural-electrification baseline and the claim that the program was a works program were attributed to a source that does not make them; the baseline is now sourced to Lewis and Severnini (2020) and the works-program claim is presented as this dissertation's inference.
24. Several claims were stated more strongly than their sources allow, among them the correlated-error finding of Kim et al. (2025) (from one leaderboard dataset, about question answering rather than code), the convergence of patches in Mierczuk et al. (2026) (whose authors caution that the space of obvious fixes is narrow), the age band and causal status of the entry-level employment finding, the growth figure for data-center electricity, Skalse et al. (2022) (policy restriction is necessary, not sufficient, for an unhackable proxy), Kanter (1972), Lutz (1994), Elster (1995), Fritsch et al. (2024), Hadfield-Menell et al. (2017), Frohlich et al. (1987) and the International AI Safety Report (a report, not an opinion survey). Each is now stated within its source.
25. The front matter and Chapter 1 claimed that every source had been checked against the claim it supports. Works verified only at the level of their record had not been; the statements now say that each source was checked at the level recorded here.

A third pass opened, wherever a route existed, every work that the first two passes had verified only at the level of its record, and checked each citing sentence against it. It moved 52 of the 57 works at that level to full-text or abstract level, raised two economics sources from abstract to full text, added five works read in the course of the checks (Arrow, 2001; Bernanke, 2002; Freeman, 2023; Van Parijs, 2006; and the 2002 printing of Friedman), and made the following corrections.

26. Ackerman (1991) was cited for constitutional change "outside routine procedures," which the material read does not establish; the sentence now uses his distinction between constitutional and normal politics.
27. The United Nations veto was described in terms that went beyond the Charter ("drafted by the victors"); the sentence now states what articles 23, 24 and 27 provide.
28. The exchange-network experiments of Cook and Emerson (1978) and Cook et al. (1983) were summarized with a mechanism (committed actors stop exploring alternatives) that the sources do not state; the findings are now that power follows network position and that commitment impedes its use.
29. Cook (1998) was cited for the claim that strain consumes slack; the essay attributes resilience to

redundancy and human adaptation, and the inference about strain is now marked as this dissertation's.

30. Elster (2000) was cited for a "vocabulary of delay mechanisms and supermajorities"; the sentence now states his precommitment account, with the delaying role of amendment procedures sourced to Elster (1995).
31. Faden and Beauchamp (1986) was described as the account that "dominates bioethics" and as requiring understanding and noncontrol outright; it is now "an influential account" requiring *substantial* understanding and *substantial* freedom from control, with pages.
32. Gesell's *Natural Economic Order* was dated as "first published" in 1916; the 1916 volume combined works of 1906 and 1911, and the sentence now says so.
33. Open X-Embodiment Collaboration (2024) was cited, in Chapters 2 and 7, for the claim that robot datasets are collected by teleoperation, which it does not state; that clause now rests on Khazatsky et al. (2024) alone.
34. Mueller (2002) was cited for the claim that the internet root was shaped by "a small set of insiders with limited public accountability"; the sentence now states the struggle among interests and the use of the root to regulate that the book describes.
35. E. Ostrom (1990) was described as studying "fisheries in Turkey and Canada" alongside her successes and as identifying eight principles shared by all long-enduring cases. The fisheries were cases of failure or fragility, Sri Lanka was omitted, and the book lists seven principles common to all enduring cases plus an eighth found in the larger ones (Table 3.1, p. 90); all three points are corrected.
36. Kanter (1972) was described as studying nineteenth-century communities "and 1960s communes" on equal terms; the longevity finding concerns the earlier communities, and the later communes are an extension.
37. Balcı et al. (2023) and Kagan et al. (2023), letters whose text could not be opened, were characterized as "disputing" claims and "defending" usage; Chapter 8 now says only what their records establish.
38. The early Harsanyi papers (1953, 1955) were cited for the equal-chance construction before they had been opened. They were read in the final pass (see C.3), and the Chapter 5 and Chapter 7 sentences were rewritten against them, with quotations and pages.
39. Wertheimer (1996), whose book could not be opened, was cited alone; the citing sentences now rest on Zwolinski et al. (2022), from which the claims were confirmed.
40. Solum (1992) was said to separate the legal from the metaphysical question of personhood; he uses concrete legal problems to test which philosophical claims make a practical difference, and the sentence now says that.

41. Silver et al. (2018) was said to have defeated "the strongest existing programs entirely through self-play"; the system defeated a world-champion program in each game, given no domain knowledge beyond the rules.
42. Smith et al. (2015) was generalized to "automated program-repair tools"; the study evaluated two tools, and both citing sentences now say so.
43. Egger et al. (2022) was cited for an economy "with substantial idle capacity"; the authors report only suggestive, indirect evidence of slack.
44. Kitchens and Fishback (2015) was cited for raising "agricultural output and productivity"; the abstract reports crop output and crop productivity, and the prevention of declines in overall farm output.
45. Polanyi's account of the 1930s and 1940s was summarized as "institutional reconstruction"; his great transformation includes fascism as well as socialism and the New Deal.
46. Vivalt et al. (2024), an unrefereed working paper, was called "the strongest recent experimental evidence," which was this dissertation's judgment; Chapter 7 now describes it as a large randomized experiment reported in a working paper not yet peer-reviewed.
47. The survival tier's departure from the canonical definition of basic income was acknowledged without engaging the canonical case against restricted-use, non-saveable currency; Chapter 4 now engages it directly (Van Parijs, 2006), and Chapter 3 records Radin's warning that prohibitions can harm those they protect.
48. Smaller precision edits: Higgs (1987) now says expansions came during national emergencies and left a residue of powers; Muennighoff et al. (2023) is stated in terms of epochs at a fixed compute budget; Armstrong et al. (2016) now includes the condition on risk-taking; Arute et al. (2019) gives the processor time (about 200 seconds); Fritsch et al. (2024) is scoped to three DAOs; Vincent et al. (2019) is one study, not "studies"; Verho et al. (2022) and Vivalt et al. (2024) now give their figures; Friedman and Schwartz (1963) is cited with Bernanke (2002); the rural-electrification precedent now cites Lewis and Severnini (2020) for the market's reluctance to serve rural areas.
49. Reference-list corrections: Arute et al. (2019) completed to APA 7 form; Muennighoff et al. (2023) given its published author order, editors, pages and DOI; Hume's essay given its edition and pages; the unverified page range for Ha and Schmidhuber (2018) removed and the proceedings details added; a DOI added to Elster (1995); Cook (1998) given its URL and revision note; Friedman cited to the 2002 printing actually read (p. xiv) instead of an unverified page range in the 1982 printing; Nissenbaum's *Privacy in context* re-dated to 2009, the year in every catalog checked; Vivalt et al. (2024) given its revision date; Streeck and Thelen (2005) cited to the editors' introduction, where the typology is developed; the ambiguous "Mattson et al., 2020" in item 7 resolved (the disambiguating initial was later dropped, C.6).

50. Of the five works that could not be opened in the earlier passes, Harsanyi (1953, 1955) were read and are cited directly. The other three (Balcı et al., 2023; Kagan et al., 2023; Wertheimer, 1996) were removed from the reference list, and their citing sentences now rest on sources that were read (National Academies, 2021; Zwolinski et al., 2022). No secondary ("as cited in") citation remains; see C.3.

A fourth pass sought a legal full text (an open-access publication, an author manuscript, a working-paper or preprint version, or an official document) for every work that the earlier passes had read only at the level of its abstract, and checked each citing sentence against the body of the work. It raised 118 works to full text, recording in C.1 where the version read was a preprint or working paper. The 39 that remain at abstract level are chiefly books and paywalled articles with no open copy. The pass made the following corrections.

51. Sentences that went beyond their sources were narrowed to what the full texts state: Adler et al. (2024) (credentials prove a person is real rather than an AI, one per person per issuer, not uniqueness); Arrieta-Ibarra et al. (2018) (all three sources of countervailing power must combine); Bengio et al. (2025) (expert views concern the next several years); Bommasani et al. (2022) (the study measures homogenization among systems sharing a foundation model); Cloud Security Alliance AI Safety Initiative (2026) (developers run discovery; remediation is routed through clearinghouses); Cox et al. (2010) (nested enterprises have the weakest evidence, Table 3, though the authors judge it moderately supportive); de Bromhead et al. (2013) (the association holds for right-wing extremist parties); Farrell and Klemperer (2007) (strategic incompatibility); Feng et al. (2024) (the verifier must be sufficiently accurate); Fritsch et al. (2024) (concentration in two of the three organizations); Godschalk (2012) (one review, not studies); T. Guo et al. (2024) and Y. Guo et al. (2024) (semantic diversity stays comparatively stable); Hadfield-Menell et al. (2017) (the operators must be sufficiently rational); Hartung et al. (2023) (a call, not a commitment); Hyun et al. (2020) (one review, five themes); Kataoka et al. (2024) (donors *may* be insufficiently informed); Khazatsky et al. (2024) (human labor, much of it teleoperation, rather than "largely" human demonstration); Koh and Liang (2017) (evidence of crafted training data, not a claim about payment); Lee et al. (2024) (an off-the-shelf model, with the point about human training text marked as this dissertation's); Lehdonvirta et al. (2024) (public cloud compute suited to training); Li et al. (2025) (the dependence of the estimate on its assumptions); Long et al. (2024) (the authors decline to argue that near-future systems will be moral patients, p. 25); MacKay (1992) (a misspecified model *may* misvalue data, flagged by MacKay as a possible weakness); the National Academies (2021) (recontacting donors is *sometimes* impossible); Nosek et al. (2018) (what pre-registration fixes in advance); Skalse et al. (2022) (proxies neither trivial nor equivalent); Stamp (2004) (the incentive toward monoculture); U.S. National Science Foundation (2023) (a solicitation, not funded programs); UNESCO

(2025) (as its announcement describes it); and Abramitzky (2008) (no longer called "the best" study).

52. Robbins et al. (2026) and Cai et al. (2023) stay at abstract level: their preprints were read in full, but the clauses on the persistence of learning and on speech recognition rest on the published abstracts, which report results the preprints do not contain. The Neuralink registry title was completed.

A fifth pass took up the works still at abstract level, most of them books, and searched each for the specific passage cited. It read ten in full or in the relevant parts, among them Harsanyi (1975), Binmore et al. (1986), Higgs (1987), Lutz (1994) and Scheidel (2017). For sixteen books it found each cited passage in the work's own words. Thirteen works, chiefly paywalled articles, remain at abstract level, and every claim attributed to them is confined to what their abstracts state. The pass made these corrections.

53. Lutz (1994) is now cited for what the article states rather than "by report"; Severinghaus et al. (1994) no longer claims that rising carbon dioxide "would have signaled" the problem; Cai et al. (2023) is cited for a change in the tissue's functional connectivity, not a physical reshaping; Stigler (1971) is paraphrased in his terms (regulation acquired by the regulated industry); Popper is cited for falsifiability, with pre-registration as this dissertation's application of it; Emanuel et al. (2000) lists "value" as the article does; the Manson and O'Neill gloss "in a controlled way," not found in the text, was removed; and Molm (2003) is no longer glossed beyond its abstract.
54. An independent editorial review of the whole manuscript led to revisions of argument and prose that add no factual claim. The compute unit is now said to meet the requirement of independence from either side only conditionally. Exit is foundational to voluntary consent and to reciprocity, but cannot secure informed consent or fair terms. The convention's aggregation rule is recorded as an open problem. The floor's effect on bargaining is tied explicitly to a survival constraint. The pilot chapter (since removed, C.6) was described as testing the designed dependence of Proposition 4, not the priority of exit. The fallback when dependence fades is institutional enforcement, not character.

C.5 Checks closed in the final pass

The earlier passes left a list of bibliographic and page-level checks open. The final bibliographic pass of October 2026 closed every item on that list against the source itself. The items that the later review passes (C.8 to C.11) left open are listed at the end of C.10; C.11 adds none. Each check, and how it was settled:

- **Author lists.** Adler et al. (2024), Arute et al. (2019), Bengio et al. (2025), Bommasani et al. (2021), D. Guo et al. (2025), Hartung et al. (2023), Khazatsky et al. (2024), Reddi et al. (2020)

and Touvron et al. (2023) were compared with their Crossref or DataCite records and now give the first nineteen authors, an ellipsis and the final author, as APA 7 requires; D. Guo et al. (2025) gained its issue number. The fourteen authors of Q. Wu et al. (2024) were confirmed against the arXiv record, and the title was corrected to the singular "multi-agent conversation." No reference entry uses "et al."

- **Thibaut and Kelley (1959).** The comparison level for alternatives is defined on printed p. 21 (§ 2.5), now cited.
- **Faden and Beauchamp (1986).** The three conditions of autonomous action (pp. 241–273) and informed consent as autonomous authorization (pp. 277–280) were confirmed against the paginated text.
- **Rawls (1971).** The description of the original position and the veil of ignorance is on p. 12. The reading of the parties as effectively a single chooser is attributed to Freeman (2023), which states it, not to an unverified page of Rawls.
- **Pierson (2004), Piketty (2014), Zuboff (2019), Van Parijs and Vanderborght (2017).** Each claim was checked against the book's text: Pierson's self-reinforcing processes (p. 10) and threshold effects (pp. 82–83); Piketty's $r > g$ and the growth of inherited wealth; Zuboff's definition of surveillance capitalism, now quoted with its location (C.8); and the definition of basic income as paid in cash, without conditions (pp. 4, 8). None of these citations uses a pin page that was not seen.
- **Streeck and Thelen (2005).** The editors' introduction was read in its open-access postprint, which carries the book's pagination. The definitions of layering, drift and conversion are in Table 1.1 (p. 31) and the modes are discussed on pp. 19–31, as now cited.
- **Solum (1992).** The article runs pp. 1231–1287. Read in full, it recasts the question whether artificial intelligence is *possible*, not the question of personhood as such, as concrete legal questions; the Chapter 5 sentence was corrected and given pages.
- **S.B. 1223 (2024).** Chaptered as ch. 887, Statutes of 2024; the reference entry now says so. The neural-data category sits in Cal. Civ. Code § 1798.140(ae)(1)(G), confirmed in the California Privacy Protection Agency's compilation; the chapters do not rely on the lettering.
- **Hume.** "Of the independency of parliament" first appeared in *Essays, Moral and Political* (1741), according to the editor's foreword and key to editions in the Miller edition, where the essay runs pp. 42–46. The citation is now 1741/1987.
- **Van Parijs (2006).** The definition and the contrast with restricted-use, non-saveable currency are on p. 8. The volume is edited by E. O. Wright, and its subtitle was corrected from the Library of Congress record.
- **Freeman (2023).** The Winter 2023 archived edition carries the substantive revision of 24

October 2023 and is the edition cited.

- **Ackerman (1991).** The distinction between constitutional and normal politics (Ch. 1, "Dualist Democracy") was confirmed; Chapters 5 and 7 use that distinction instead of the phrase "constitutional moment," which was not seen in the text read.
- **Polanyi (1944/2001).** Page citations now come from the 2001 Beacon edition: the fictitious commodities in Chapter 6 (pp. 71–80), and the 1930s transformation, with fascism, socialism and the New Deal, on pp. 3, 24 and 31. An earlier page range that did not support the clause was replaced.
- **Robbins et al. (2026).** The bioRxiv preprint was read in full. Chapter 8 cites it only for the result the preprint reports: training signals chosen by reinforcement learning mostly improved performance at a pole-balancing task. The clause on the persistence of the improvement, which rested on the published abstract, was removed (C.2), as was the word "substantially," which came from the release.
- **Lanier and Weyl (2018).** The essay was read. It adopts the term "data dignity" and proposes "mediators of individual data," volunteer organizations that negotiate data royalties or wages for their members; Chapter 2 now says so.
- **Olson (1982).** The implication is Olson's own: Chapter 4 (p. 75) predicts fast growth where totalitarian government or foreign occupation abolished distributional coalitions, and offers this as an explanation of the postwar "economic miracles" of Japan and West Germany. Chapter 6 now cites it.
- **Autor (2015).** The published article was read. The substitution and complementarity claim is on p. 5, tacit knowledge on p. 11, and the quoted route around it ("from context, abundant data, and applied statistics") on p. 23; the citations give those pages.
- **Wörgl.** Ottacher (2007), read in full, fixes the dates. The certificates were withdrawn by 15 September 1933 under threat of police enforcement (p. 57), and the Administrative Court dismissed the town's final appeal on 18 November 1933 (pp. 58–59). The National Bank's objection is on pp. 53–54. The Vienna exhibition attributes the mid-September end to the Administrative Court, which Ottacher's chronology does not; Chapter 4 records the discrepancy and cites each source for what it states.
- **The 2026 amendment to the AI Act.** It is Regulation (EU) 2026/1744 (the "Digital Omnibus on AI"), read in the official text: published in the Official Journal on 24 July 2026, in force from 27 July 2026, and deferring the high-risk obligations to 2 December 2027 (Annex III) and 2 August 2028 (Annex I). It is now cited formally, alongside the two commentaries that describe it (Agustinoy, 2026; ERTICO, 2026).

C.6 Removal of the pilot chapter

On 2 November 2026 the author removed the pilot chapter (formerly Chapter 7, "The Prototype: A Closed Agrarian Pilot") and its draft pre-registration protocol (formerly Appendix B) in order to keep the dissertation theoretical. Designing an experiment capable of testing the theory is a project in its own right, and it is now described as future work in Chapters 7 and 9 rather than specified in the text. The removed chapter and appendix are preserved in the project archive. The chapters and appendices that followed them were renumbered: Limitations is Chapter 7, Convergence Chapter 8, the Conclusion Chapter 9, the glossary Appendix B and this ledger Appendix C.

Twenty-one works that were cited only in the removed parts were taken out of the reference list and from the lists in C.1: Barron-Gafford et al. (2019); Bommasani et al. (2023); Diener et al. (1985); T. Guo et al. (2024); Hossain et al. (2019); Hu et al. (2022); Jiang et al. (2023); Kanter (1972); Manheim and Garrabrant (2018); McCambridge et al. (2014); the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research (1979); Nosek et al. (2018); OECD (2013); Perdomo et al. (2020); Peters et al. (2016); Popper (1959/2002); Protection of Human Subjects (2018); Severinghaus et al. (1994); Skalse et al. (2022); Touvron et al. (2023); and Q. Wu et al. (2024). Popper (1959/2002) was restored in the review pass of C.8, which cites it in Section 1.7 at the level previously recorded. Works the pilot shared with other chapters (among them Abramitzky, 2008; Amodei et al., 2016; Drexler, 2019; Gatch, 2012; Hirschman, 1970; Kapoor et al., 2024; Shumailov et al., 2024) remain cited and remain in the list. The entries in C.2, C.3, C.4 and C.5 that refer to works cited only in the pilot (among them Kanter, 1972; Nosek et al., 2018; Skalse et al., 2022; Touvron et al., 2023; Q. Wu et al., 2024) describe the manuscript as it stood when each correction was made, and are kept as a record of the verification history. Since the removal of Q. Wu et al. (2024) the reference list has one Wu first author, and Wu et al. (2023) is cited without the disambiguating initial; the single Mattson entry is likewise cited as Mattson et al. (2020).

C.7 Terminology pass and foreword

On the same date the author fixed the dissertation's vocabulary for the artificial side of the exchange (Section 1.2): *synthetic intelligence* as the general term, *superintelligence* as the capability threshold, and *large language model* with an explicit statement that such a model is a trained artifact rather than an independent living entity. The "sub-AGI" shorthand for the designed-incentive regime was retired. No source was added or removed. The two sentences in Chapters 1 and 5 that report Drexler (2019) as modeling advanced systems as bounded services now name his subject as superintelligence, which is the word of his title and of the front matter that was read (C.1); the two further references in Sections 3.3 and 5.6, which use his bounded-services model for the problem of counting synthetic parties, say "advanced synthetic intelligence." The claim attributed to him is unchanged, and the

clause "with humans retaining strategic control," not found in the front matter read, was removed from Section 5.2. A foreword in the author's voice was added before the abstract. It cites nothing and states the author's conviction without the qualifications the body keeps.

C.8 Review pass of 2 November 2026

A review of the whole manuscript for logical flaws, unsupported claims, citation form and non-academic prose was conducted after the terminology pass, and its findings were applied on the same day. The changes that bear on sources are these. Popper (1959/2002) was restored to the reference list and to the confirmed-passage list of C.1, cited in Section 1.7 for the requirement that a theory say in advance what would count against it. Article 108 of the Charter of the United Nations was read and is cited in Section 5.1 for the amendment rule. Zuboff's definition of surveillance capitalism, previously paraphrased, is now quoted with its location. The Friedman (1962/2002, p. xiv) sentence is now quoted. The Weidinger et al. (2023, Study 4) finding is reported as a null result on reflective endorsement rather than as endorsement becoming less firm (C.10 corrects the description of the comparison). The Cai et al. (2023) sentences in Chapter 8 were returned to the "functional connectivity" wording recorded under item 53. Chavez et al. (2026) is no longer cited for chip-supply concentration, which rests on Sastry et al. (2024). Hirschman's "political action par excellence," Keynes's "unduly neglected prophet" and Radin's "double bind" are paraphrased rather than quoted, because the pages were not recorded. Claims of scholarly dispute (Fishback), of prevalence ("most fiction," "prevailing approach," "the strongest evidence") and of "unprecedented" or "most influential" standing were removed or hedged, and every claim that is the dissertation's own inference rather than a source's finding is now marked as such. In the formal chapter, the substitution case of Appendix A was corrected to model the provider's cost, Propositions 2 and 3 were restated so that no clause contradicts the definitions, the outside-option gloss was corrected, the asymmetric Nash solution was stated so that the impatience result is a derivation, and the regime typology was made explicit about the combinations it names. Ten reference entries for proceedings papers carry the DOI of the arXiv version; the entries remain as they are, and C.1 records the version read for each.

C.9 Second verification pass

A second pass on the same day checked the edits of C.8 for errors they might have introduced and re-examined the formal core. The changes that bear on sources are these. Article 108 of the Charter of the United Nations and the time-preference result of Binmore et al. (1986) on which Appendix A.3 now rests are recorded in the research notes. A citation of Funke et al. (2016) and Scheidel (2017) added in C.8 for a claim neither makes, that the harm of a transition falls in its passage, was removed and the claim marked as the dissertation's reading. The Wörgl unemployment figure is now

attributed to Gelleri (2009) as a repeated claim that September (2026) traces to movement sources, and the citations for the withdrawal, the court's dismissal and the exhibition's account are given separately for what each source states. A clause asserting that surplus cannot be divided efficiently under private information, which would have required a source not in the list, was removed. K. S. Cook et al. (1983) and R. I. Cook (1998) now carry initials, since the list has two Cook first authors. In the formal chapter, clause C(ii) was reworded so that refusal means entering no exchange in place of the one refused, which makes Proposition 1's identification of C(ii) with E(i) exact; Proposition 4 states the premise on which self-enforcement of terms, as distinct from protection of the input supply, depends; the insufficiency thesis is stated for clause E(i) only, and the scope thesis, Sections 8.5 and 9.2 claim no more than that; the regime typology is stated at clause level and withdraws the claim to partition the eight combinations; and the Foreword now says which of its claims the body argues.

C.10 Third verification pass

A third pass on the same day checked every in-text citation against the research notes, every internal cross-reference against its target, the arithmetic of Chapter 4 and Appendix A, and the rendered exports. The changes that bear on sources are these. Weidinger et al. (2023) was re-read at the Study 4 results: the veil still raised the choice of the prioritarian principle (odds ratio 4.5, p = .049, N = 213 after exclusions) and the difference in reflective endorsement between conditions was not significant (p = .21); the comparison is within Study 4, not across studies, and Chapters 5 and 9 now say so. Landauer (1961) and Lloyd (2000) are no longer cited for substrate-specific conversion rates, which they do not give; Section 4.2 cites them for the physical bounds only. Solum (1992) is cited for the hypothetical examination of legal personhood, not for present doctrine. Gatch (2012) is cited for tax receivability only; the legal-tender clause is the dissertation's. Gerstgrasser et al. (2024) and Kazdan et al. (2024) are cited for accumulation only, and Alemohammad et al. (2024) and Shumailov et al. (2024) for the fresh-supply disjunct. D. Guo et al. (2025) is no longer cited for expert-level performance, which the paper does not claim. The Pew figure of more than a third is now attributed to the July 2026 sample, as the source states. Longpre et al. (2024) is paraphrased in the authors' terms ("inconsistent"; protocols not designed for the purpose) rather than as "largely unenforced." Farrell and Klemperer (2007) is paraphrased from its abstract. Hayek is cited as 1976/1990, since the third edition was the text read. Polanyi's C.1 entry now records the 2001 edition from which the page citations come. Section 6.4 cites Polanyi at pp. 71–80 for the strain of commodification as the other sections do. The Elster (1995) attribution of delays and amendment procedures rests on the full-text read recorded in C.4 item 30; the research note does not record the passage, and the page is an open item. The three-author Brynjolfsson works are cited with all authors, as APA 7 requires. Quotations whose wording the notes do not record were paraphrased (E. Ostrom, 1990, p. 90). Claims the notes do not support were reworded to what they do support or marked as the

dissertation's: the REA's separability from the New Deal, beneficial-ownership registries, the historical control of infrastructure by capital, the clearinghouse trend, the Cox et al. (2010) ratio, the Robbins et al. (2026) "no training" condition, the Autor (2015) "two routes," the Lutz (1994) length–longevity association, and the Kim et al. (2025) evaluation result. In the formal chapter and Appendix A, the dependency between clauses R(ii) and C(i) was reversed to the direction the definitions support; Proposition 3 states the E(iii)–C(iii) exception; the credit channel is bounded only when the interest rate is strictly below the demurrage rate; the symmetric-case bounds of Section 3.6 are qualified to equal patience; and the substitution case states its positive-surplus proviso. In Chapter 4, information-gain pricing carries a sixth qualification on definability, and the redemption obligation is stated with its par anchor and the condition that the dividend share not exceed the obligated share. The glossary gained an entry for domination and corrections to eight others. The Arrow (2001) page range was taken from the contents list of the volume; the fourth pass added the research note that records this. Open items: page ranges for Gatch (2012), Korinek and Stiglitz (2019), Mattson et al. (2020) and Perry et al. (2023); and research-note entries for Freeman (2023), Bernanke (2002), Agustinoy (2026) and ERTICO (2026), which are recorded in C.1 only; and the ten proceedings entries that carry arXiv DOIs (C.8).

C.11 Fourth verification pass

A fourth pass on the same day checked the edits of C.10, examined the argument itself rather than its sources, and copyedited every sentence against the prose standard. No source was added or removed. The pass removed one sentence citing Fishback (1992) from Section 4.2 as duplicative of the sentence before it; restated the Wörgl withdrawal citations so that Ottacher (2007, p. 57) carries the 15 September date and September (2026) the 1 September date; clarified the Bestvater et al. (2026) research note to the July 2026 sample; and added a research-note entry recording the source of the Arrow (2001) page range. The changes that bear on the argument are these. The insufficiency thesis is now stated for a party that has no exit option, since a sub-floor party that can leave for another provider is not dominated by any single counterparty under the stipulated definition; Sections 1.2, 8.5 and 9.2 and the glossary carry the qualification. The definition of domination is stated as symmetric in form, and the Foreword now says that the body argues one of its claims and stipulates two. The sufficiency conjecture is stated as the claim that character adds nothing once terms bind, with enforceability as its open premise, and the designed-incentive regime is bounded by accountability as well as by the provenance of the objective. Proposition 2 is stated as bounding reciprocity from below rather than establishing its strict form, and the summaries no longer say that exit disciplines reciprocity. Proposition 3's ordering is stated as partial, with exit prior and consent and reciprocity unordered with respect to each other, and Section 3.7 records that at the collective level exit is secured through voice rather than departure. Section 4.5 states how the floor goods that are not compute are delivered, names the acceptance obligation as a legal-tender rule, and records

the circularity that the disagreement payoff is delivered by capacity owners who are also counterparties; Section 4.6 states that demurrage has no distributive effect of its own without the capacity and credit rules; Section 4.4 limits the par obligation to the reference workload and ties the fixity of the share to re-basings. Section 5.3 states what anonymity removes, the organization of a bloc rather than the motive to vote one's interest; Section 5.5 states that strong blindness adds nothing for principals and capacity owners; Section 5.6 no longer claims that the eligibility criterion presupposes none of the contested classification; Section 5.7 states what the symbiosis standard would require of the design and that the design does not yet meet it. Section 6.6 states that a floor financed through existing instruments lacks the properties Chapter 3 argued for, and Section 6.7 that a window opened by correlated failure would favor any design that distributes control. Section 8.3 assigns the donor's withdrawal of consent to clause C(iii) rather than to exit, and Section 8.5 states that Chapter 3's definitions do not yet reach a tissue-based party in its own right. Section 7.7 no longer attributes a dynamic to Propositions 1 to 4. The copyedit removed intensifiers and figurative phrasing, regularized numerals (words below ten, numerals from ten and for decimals and percentages, "percent" spelled out), serial commas (none for short items), quotation punctuation (inside), American spellings, the square-bracket form for EU regulations in parentheticals, and the run-in form of the headings in Section 5.9. Open items are as listed at the end of C.10.

C.12 Fifth verification pass

A fifth pass on the same day checked the edits of C.11 and read the manuscript whole. No source was added or removed. The quotation-punctuation copyedit of C.11 had corrupted twelve strings in this appendix, which are repaired. The account of the Wörgl sources in Section 4.6 was shortened; the discrepancy it recorded is kept here: the Vienna exhibition (Institut für Wirtschafts- und Sozialgeschichte, Wirtschaftsuniversität Wien, n.d.) attributes the mid-September 1933 end of circulation to the Administrative Court, whereas Ottacher (2007, pp. 57–59) places the September withdrawal under administrative order and the court's dismissal of the final appeal on 18 November, and September (2026) gives 1 September for the withdrawal against Ottacher's "by 15 September". The changes that bear on the argument are these. Chapter 4 now states at its head the presupposition that every mechanism in it rests on, an authority able to enforce the redemption obligation, the acceptance of the survival tier by providers of floor goods, and the capacity rule, and Section 4.5 adopts the acceptance obligation rather than leaving the choice open; the Abstract, Sections 3.6, 7.7 and 9.2 carry the presupposition. The design keeps credit in the discretionary tier, so the credit channel is stated as open in Sections 4.4, 4.6 and 9.2 and the capacity rules carry the whole distributive burden; Section 4.4 no longer says that a cap at the demurrage rate closes the channel, which Appendix A.4.3 shows it does not. Demurrage is described at the level Section 4.6 supports in Sections 3.6 and 5.9 and in the Abstract and Section 1.8. The test the governance chapter sets itself is restated as resistance to capture by multiplication and by bloc, and Section 5.10's

comparative claim is limited accordingly, with the aggregation rule named as open in Sections 1.8 and 9.2. The insufficiency thesis is stated, as Proposition 1 already was, to follow from the stipulated definition, with its substantive content located in the defense of the stipulation. Sections 1.8, 6.7 and 9.2 state the transition contribution as what Chapter 6 delivers: the preconditions of independent verification, not the verification. The Statement on Authorship and Method now describes the open items of C.10 accurately and refers to Section 1.7 for the passages it had duplicated. Several near-verbatim repetitions between chapters were replaced by cross-references. Open items are as listed at the end of C.10.

C.13 Sixth verification pass

A sixth pass read the manuscript whole against the edits of C.12, checked every proposition's stated premises and scope-qualifications against how later chapters use them, checked the house style, and checked the figures against their first mention in the text. No source was added, removed or re-verified; this pass checked argument and apparatus, not citations, and the bibliographic open items are as listed at the end of C.10. It found one error, in the figure-placement work of SET-T32, which postdates C.12: Section 3.6 discusses impatience first and the division of surplus second, but the figures illustrating them had been numbered in the opposite order, so Figure 7 (impatience) preceded Figure 6 (division of surplus) in the text, contrary to the rule, stated in AGENTS.md, that figure numbers follow the order of first mention. The two figures are renumbered so that impatience is Figure 6 and the division of surplus is Figure 7, in `chapters/03-formal-statement.md`, the List of Figures in the front matter, `figures/FIGURES.md`, the two SVG files, and the `NUMBERS` table and placement text in `scripts/figures/build_figures.py` and `scripts/figures/place_figures.py`, so that a future re-run of the figure pipeline reproduces the correction rather than reverting it. The PDF, DOCX and the GitHub Pages copy in `docs/` were not regenerated, since exports are not hand-edited and this pass did not run the export pipeline; they will carry the old order until the next export. No other proposition, cross-reference, arithmetic result or house-style instance examined in this pass was found to be in error. One cosmetic issue was noted but not fixed, as out of scope for the manuscript: the section comments and the unused first argument of several `register()`/`save()` calls in `scripts/figures/build_figures.py`, for figures after the fifth or sixth defined, no longer match the `NUMBERS` table, and have not for some time; they have no effect on the rendered dissertation, since the renderer reads `NUMBERS`, not the comments, but a future editor of that file should not trust them.

C.14 Seventh verification pass

A seventh pass (SET-T42, 5 November 2026) checked the popular-science article and the free will and morality article against the research notes, and carried two corrections back into the dissertation.

In Section 3.4, the sentence after Proposition 1 had cited Faden and Beauchamp (1986) and Manson and O'Neill (2007) jointly for two claims. Manson and O'Neill's account of consent as a communicative transaction supports the first claim (information is necessary but not sufficient) and does not address the second (agreement under threat of losing something one cannot do without is not voluntary). The second claim is now cited to Faden and Beauchamp alone, at the pages Chapter 2 cites for their condition that an authorization be substantially free of controlling influences. The reference for Zwolinski, Ferguson and Wertheimer (2022) now gives the archived Winter 2022 URL, matching the other entries from the *Stanford Encyclopedia of Philosophy*. No other source-bearing change was made to the dissertation in this pass.